

VU MEDICAL ZONE

Admin: Amaan Khan

BT 504 Midterm ppt

Lecture 1 to 75

Genetics and Genomics

Study of Omics

A close-up photograph of a petri dish containing numerous bacterial colonies. The colonies are mostly yellow and orange, with some smaller red ones. A gloved hand is visible on the right side, holding a pipette and adding liquid to the dish. The background is a soft, out-of-focus blue.

Study of Omics

Study of Omics

- Genomics: all the genes
- Pharmacogenomics
choice of personalized medicine
- Nutrigenomics
choice of best diet
- Toxicogenomics
prediction of toxicity

Study of Omics

Study of Omics

- Epigenomics: all epigenetic changes in genome
- Transcriptomics: all the mRNAs
- Proteomics : all the proteins

Study of Omics

Study of Omics

- Interactomics : all interactions between all proteins
- Metabolomics (or metabonomics): all metabolites

Genetics and Genomics

**Genomics,
Proteomics and
Metabolomics**

Genomics, Proteomics, Metabolomics

Genome and Genomics

- The complete set of DNA found in each cell is known as the genome and study is called as genomics.

Genomics, Proteomics, Metabolomics

Proteome and Proteomics

- The complete set of proteins found in each cell is known as the proteome.

Genomics, Proteomics, Metabolomics

Proteome and Proteomics

- Proteins concentration (and activity) may be different than gene expression due to post-translational modification

Genomics, Proteomics, Metabolomics

Metabolomics

- The complete set of metabolites found in each cell is known as the metabolome.
- Use of high-throughput mass spectrometry to analyze the metabolic components of cell.

Genomics, Proteomics, Metabolomics

Metabolomics

- Useful for determining the effects of the environment or gene transformation on the metabolism of the plants/animals.

Genomics, Proteomics, Metabolomics

Conclusion

- Genomics, proteomics and metabolomics will give an integrated, wholistic view of the cell.
- Can be used to monitor or modify organisms in a comprehensive way.

Genomics, Proteomics, Metabolomics

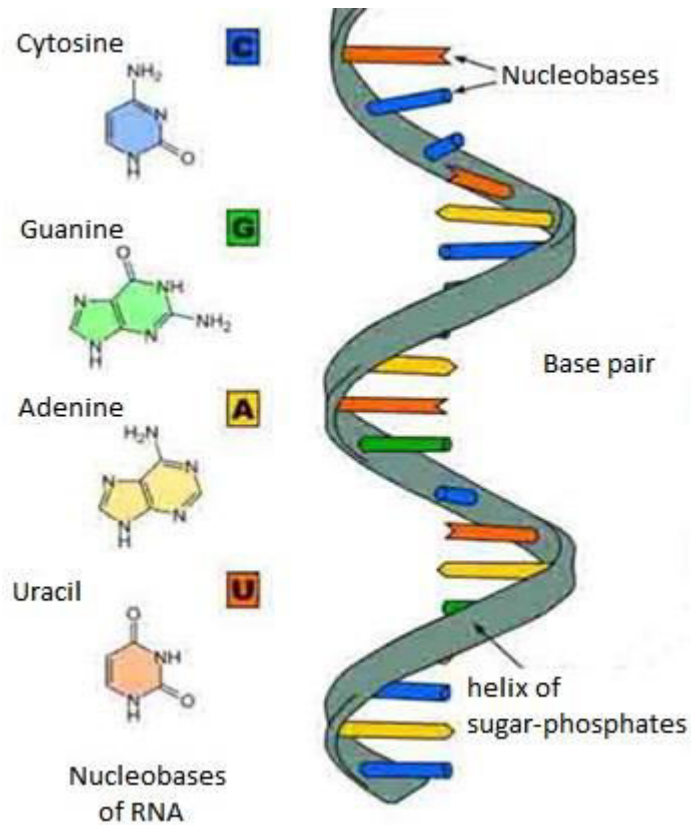
Conclusion

- Bioinformatics - the key to understand the plethora of information and modeling the cell.

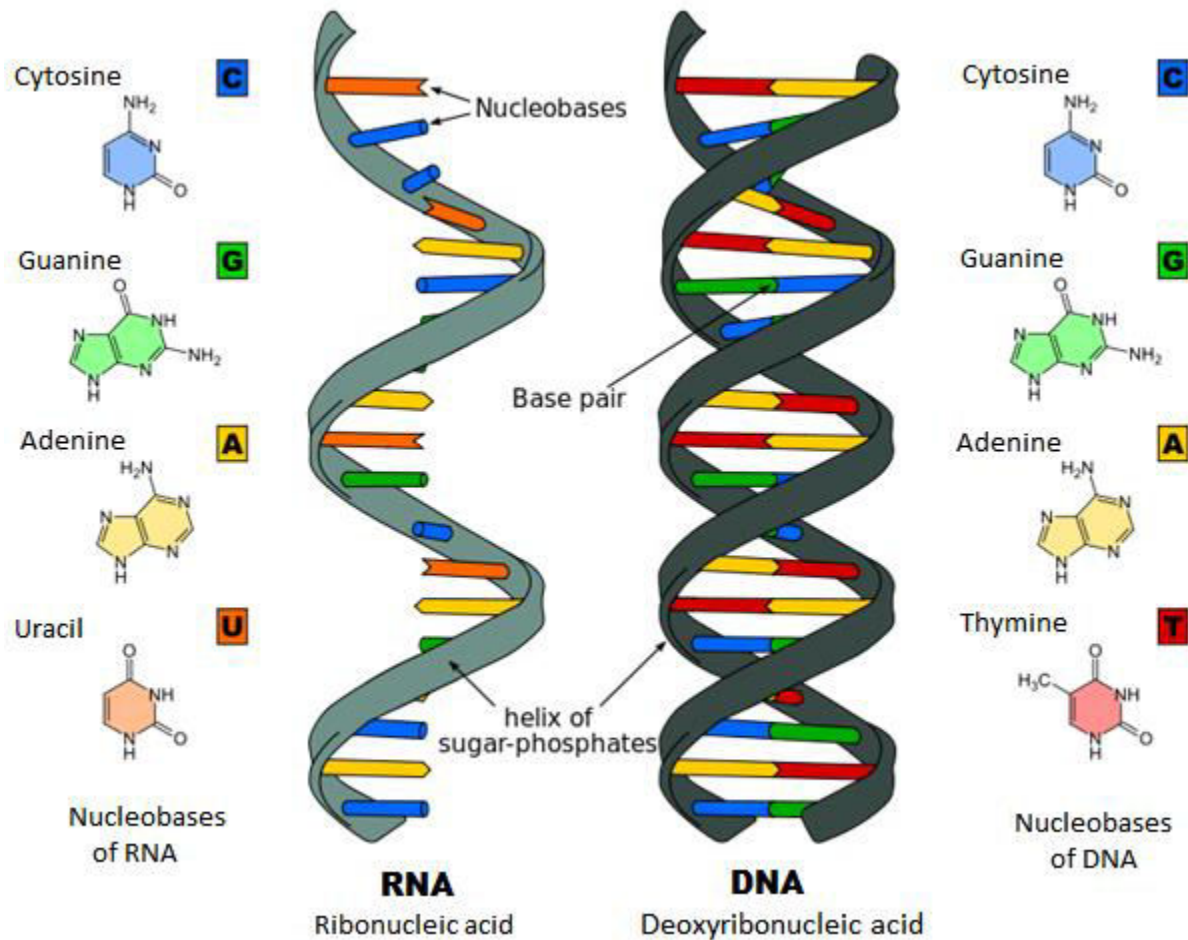
Advanced Computing Applications

Structure of RNA

Structure of RNA



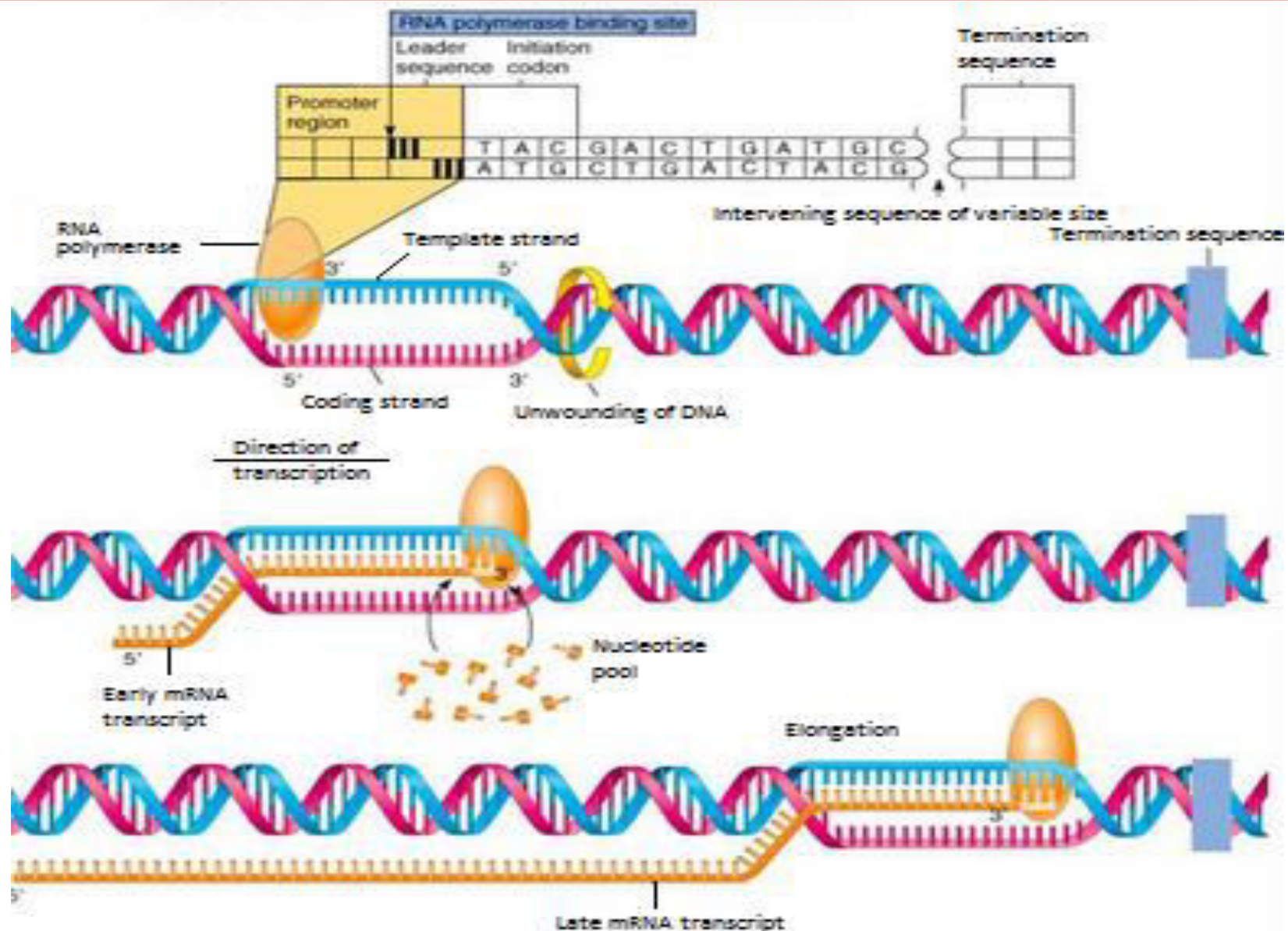
Structure of RNA



DNA Transcription

Transcription

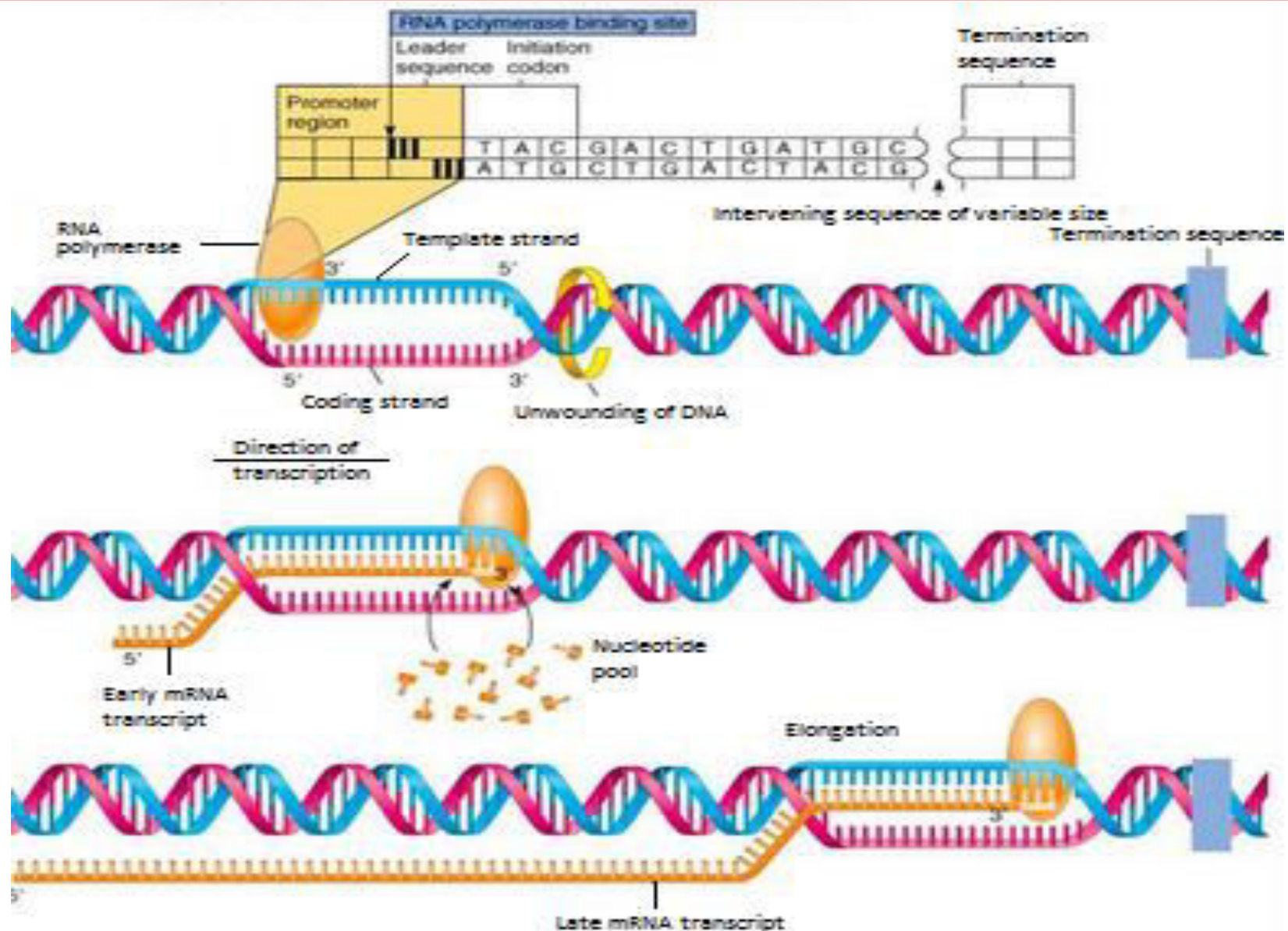
DNA Transcription



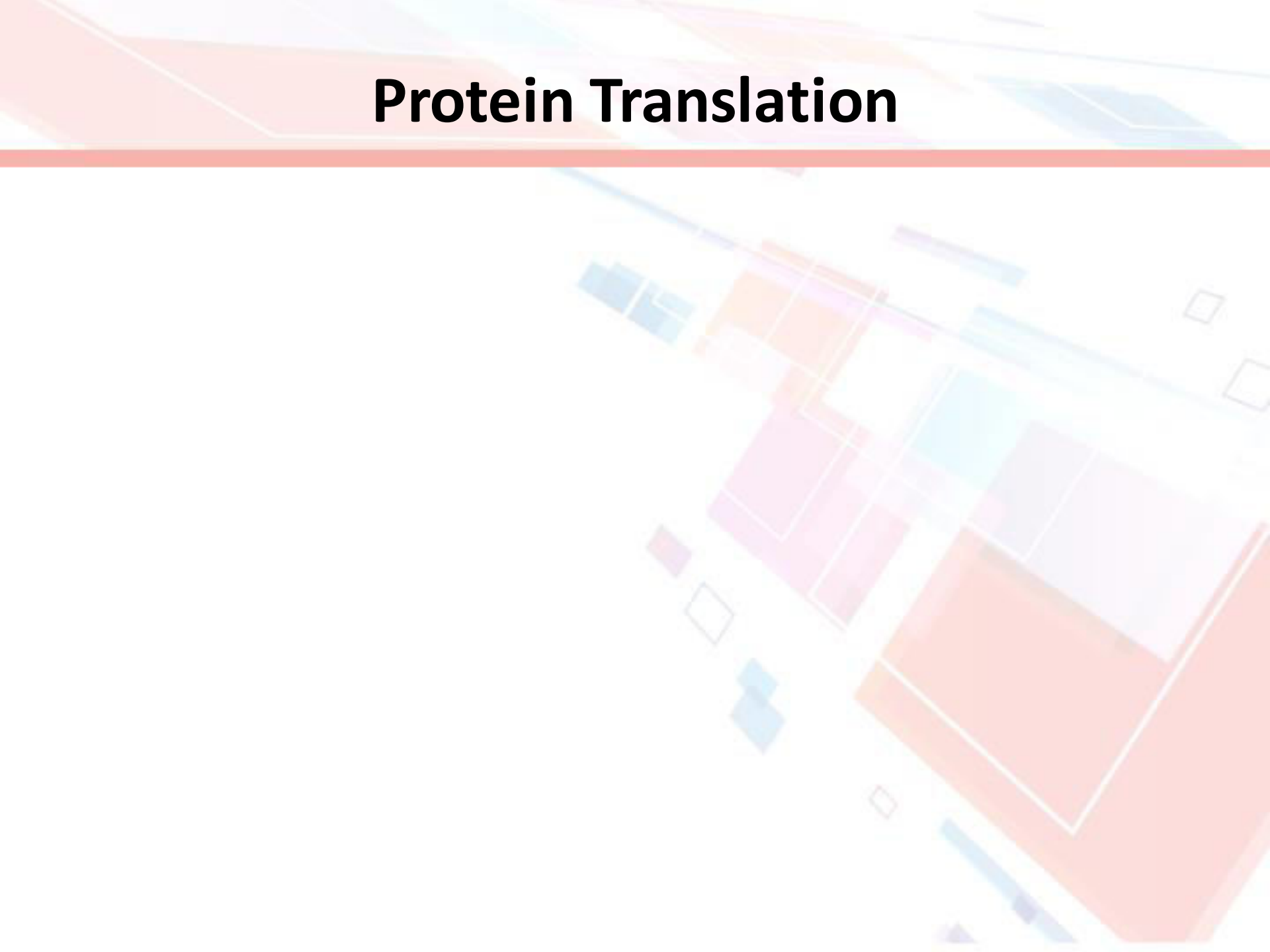
DNA Transcription

Transcription

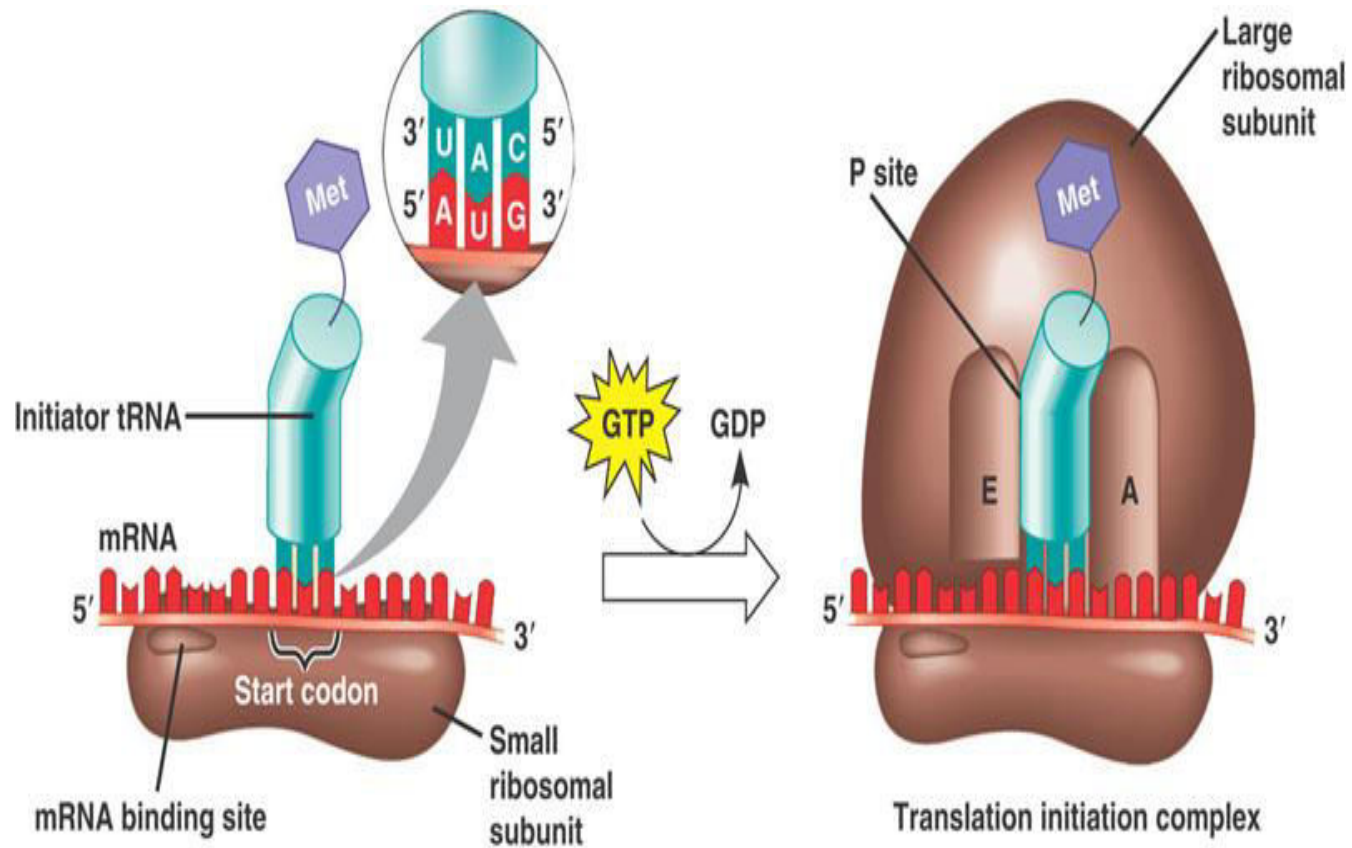
DNA Transcription



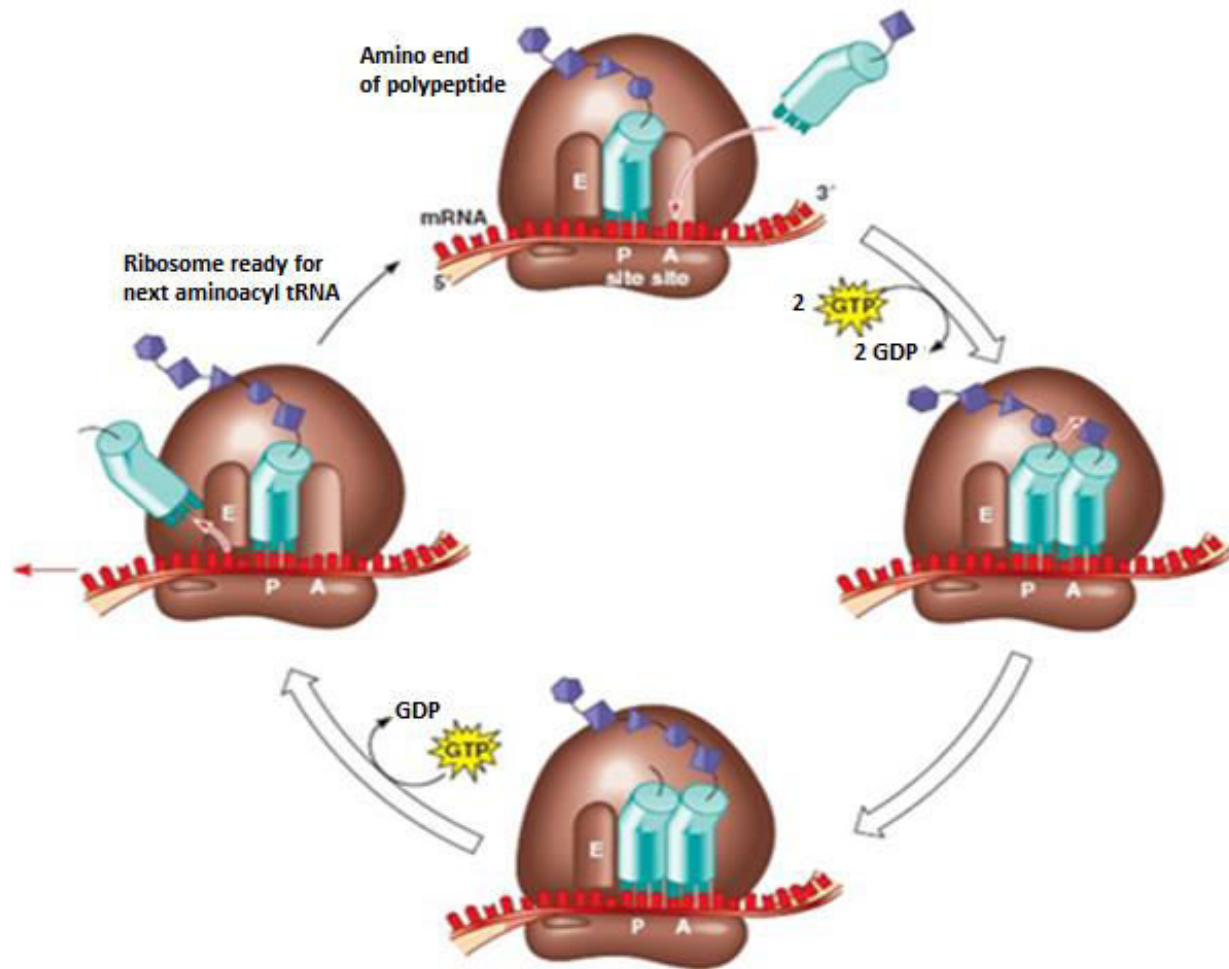
Protein Translation



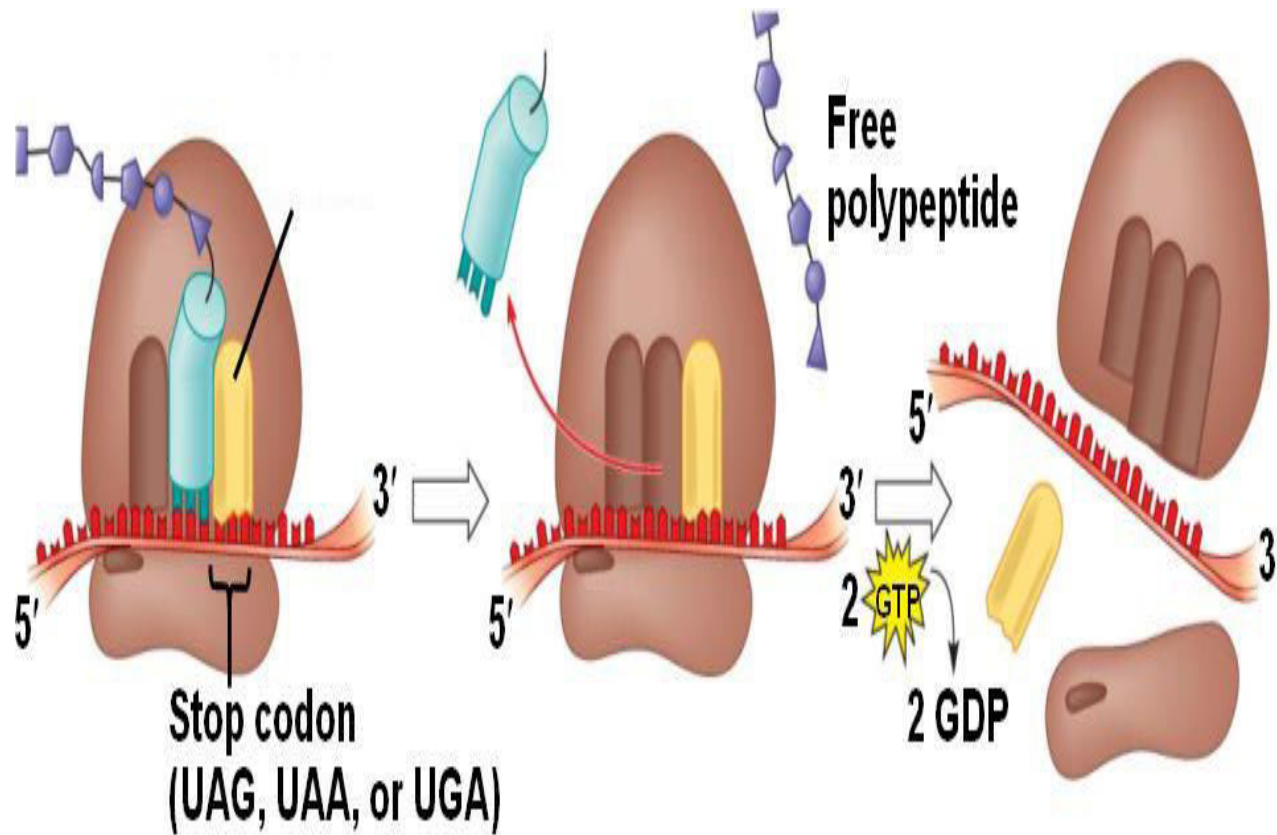
Initiation



Elongation

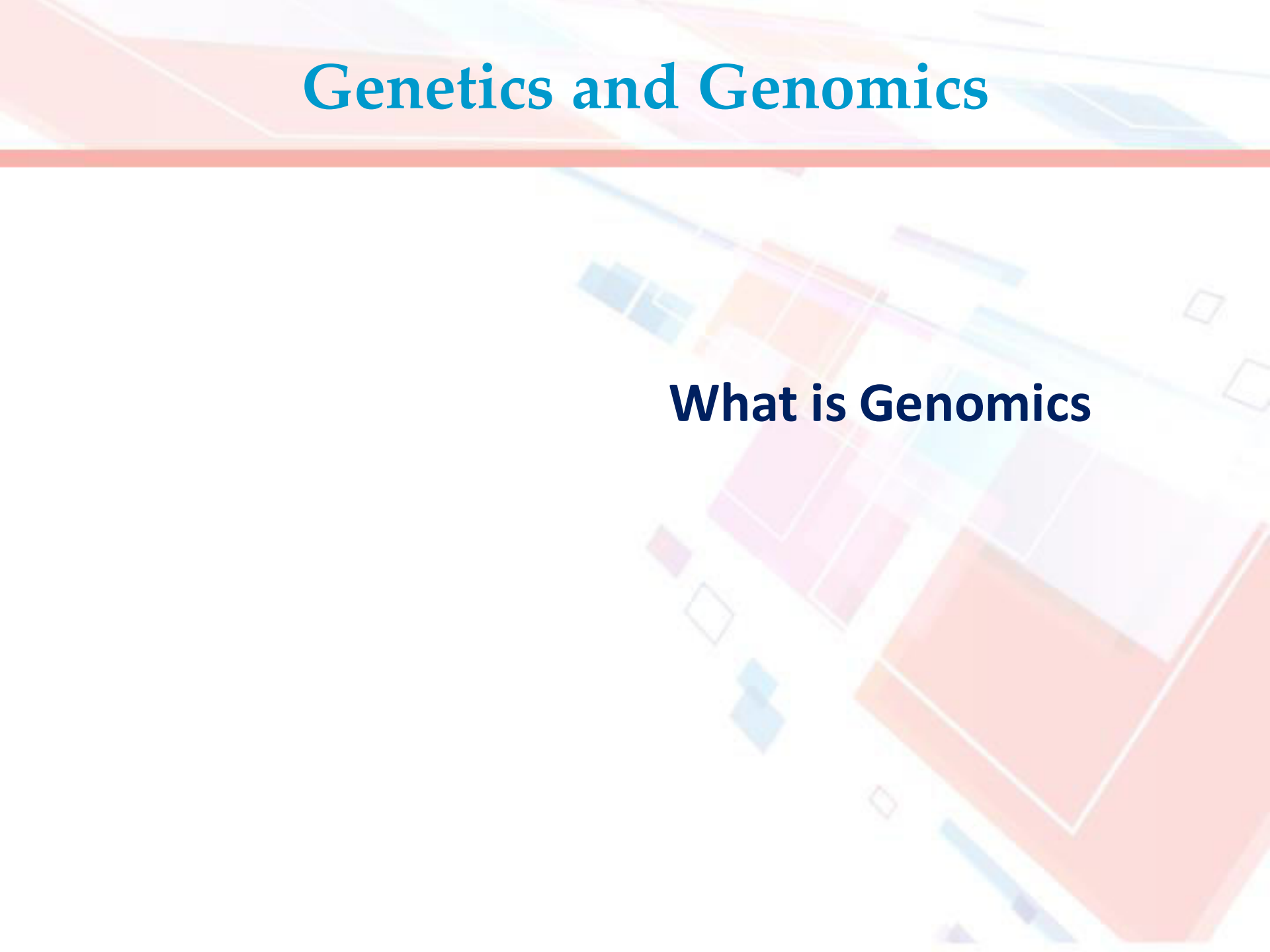


Termination



		Second base					
		U	C	A	G		
First base (5' terminus)	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U	C
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U	C
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U	C
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U	C
						Third base (3' terminus)	
						U	A
						C	G

Genetics and Genomics



What is Genomics

What is Genomics

Genome

- Total amount of DNA of a single cell of an organism (haploid cell in the case of a diploid).
- The whole hereditary information of an organism encoded by DNA.

What is Genomics

Genomics and Biology of Organisms

- Determination of entire genome sequence is a prerequisite to understand the complete biology of an organism.

What is Genomics

Genomics includes

- To generate physical, genetic, and sequence maps of different genomes.
- To sequence the genomes of model organisms.
- To develop new technologies for mapping/sequencing.

What is Genomics

Genomics helps in different ways

- Functions of genes
- Organizations of genomes.
- Structural make-up of genomes.
- Functions of coding and non-coding DNA

What is Genomics

**Study of Genomics
helps to understand**

- The ethical, social, and legal issues and challenges posted by genomic information.

What is Genomics

Genomics Sub-disciplines

- Comparative genomics
- Structural genomics
- Functional genomics
- Population genomics

What is Genomics

Genomics Sub-disciplines

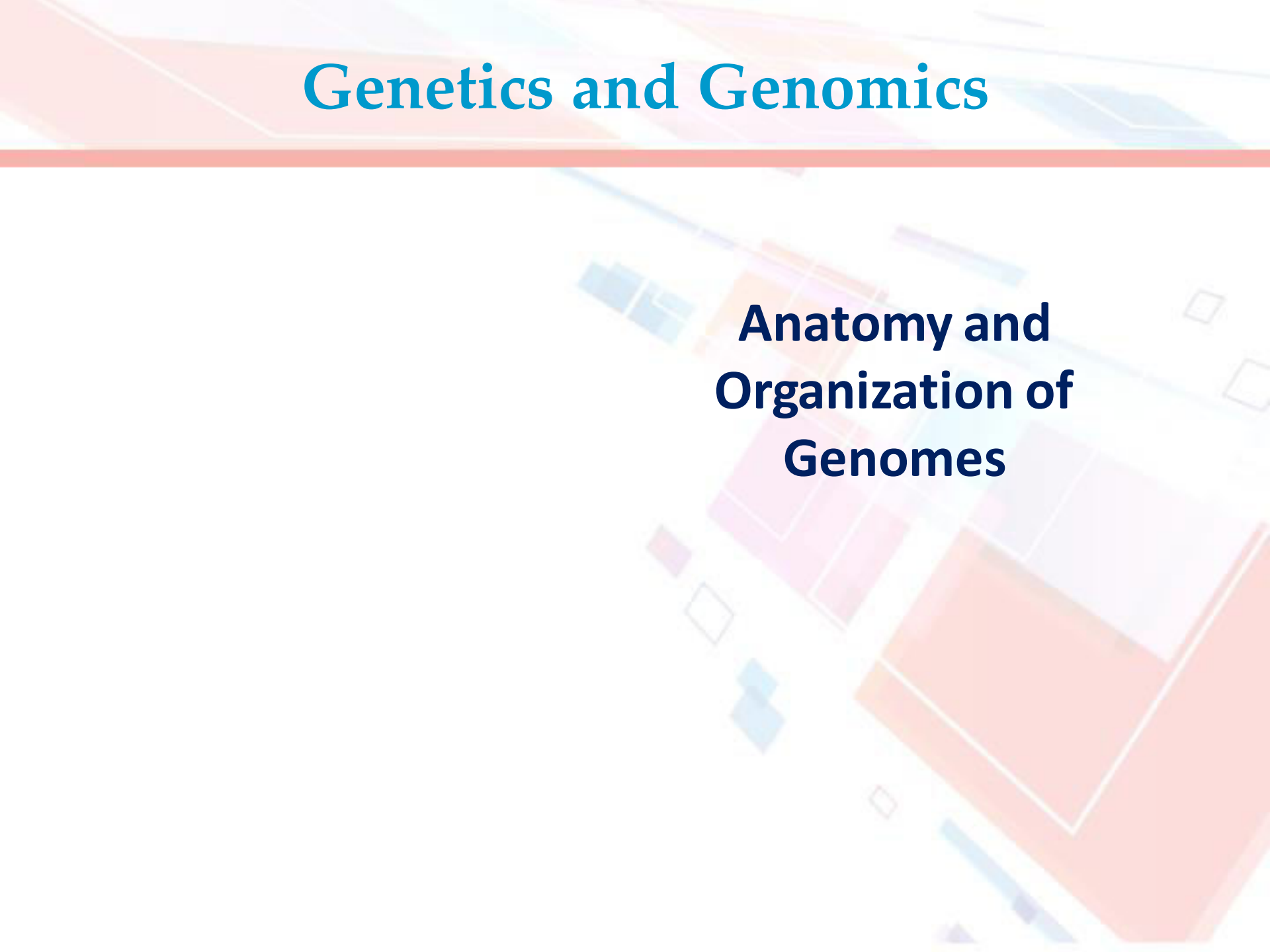
- Metagenomics
- Microbial genomics

What is Genomics

Genomics - Conclusion

- Total amount of DNA of a single cell of an organism – Genome and study is called as Genomics.

Genetics and Genomics



Anatomy and Organization of Genomes

Genomes Anatomy/Organization

Genome Anatomy

- Anatomy of different genomes differ from each other.
- Eukaryotes and prokaryotes genomes differ very significantly.

Genomes Anatomy/Organization

Genome Anatomy

- Size of genomes - 1000 fold difference between eukaryotes and prokaryotes.
- ~ 30 fold between genomes of different eukaryotes.

Genomes Anatomy/Organization

Genome Anatomy

- In humans ~ 23,000
- Bacterial genomes ~ 1,500 – 2,000 genes.

Genomes Anatomy/Organization

Genome Anatomy - Eukaryotes

- Eukaryotic genomes are full of simple repeats, numerous types of transposable elements and other sequences.

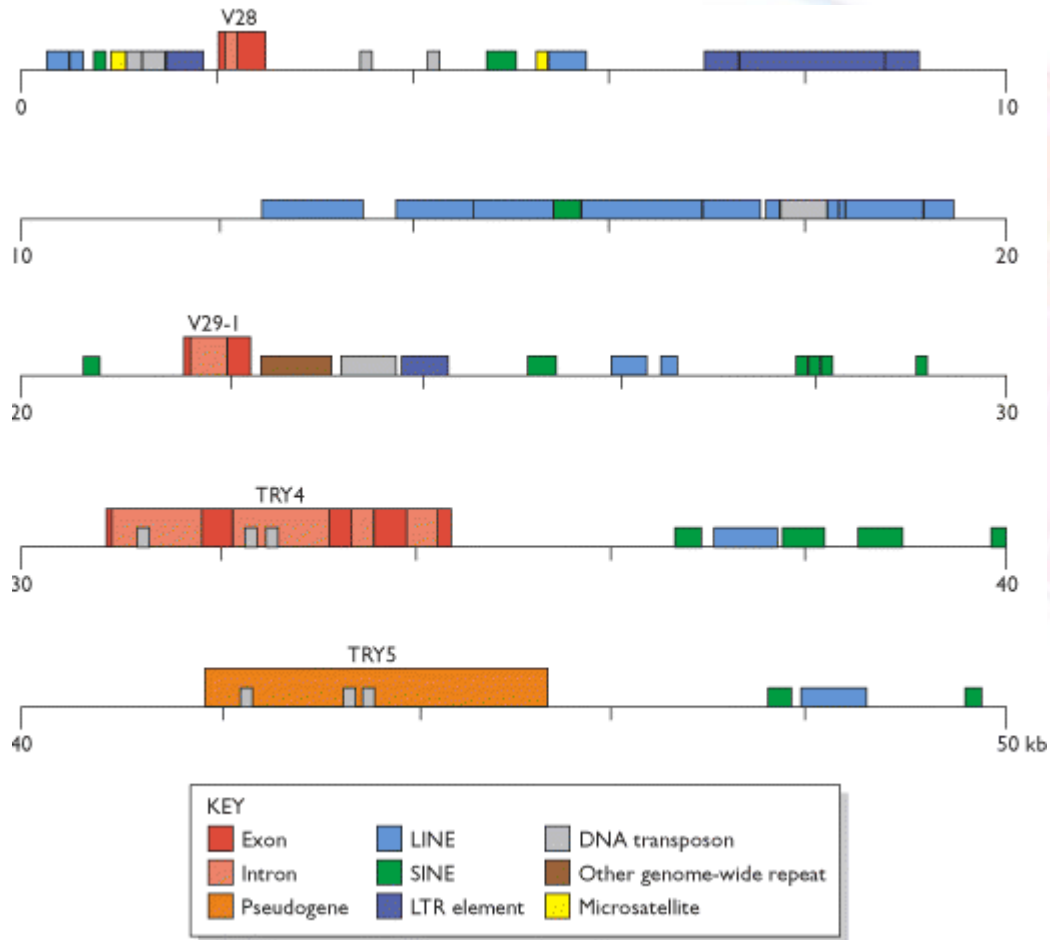
Genomes Anatomy/Organization

Genome Anatomy - Prokaryotes

- Prokaryotes have a few repeats and transposable elements and their genomes consist of mainly the genes.

Genomes Anatomy/Organization

Genome Anatomy: Segment of human chromosome 7



1 Gene: TRY4

2 Gene Segments: V28 & V29-1

1 Pseudogene: TRY5

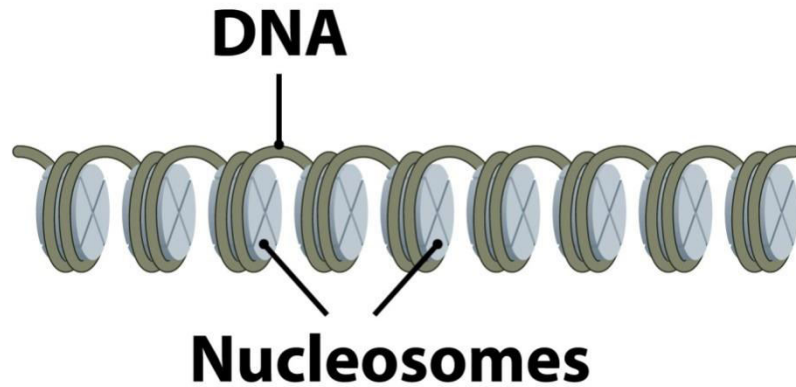
52 genome wide repeat sequences:

LINE, SINE, LTR, & DNA transposon.

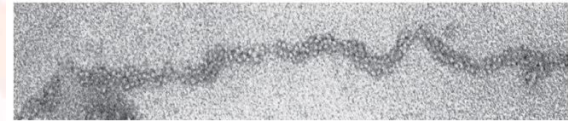
Two Microsatellites

Genomes Anatomy/Organization

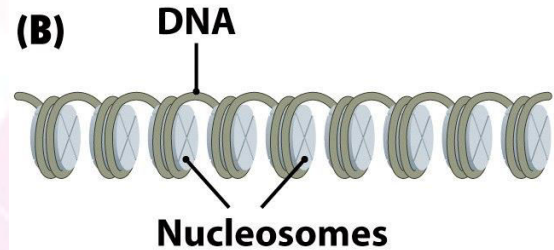
Genome Organization



(A)

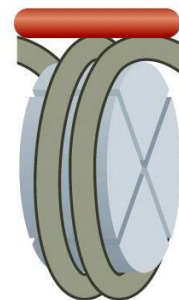


(B)

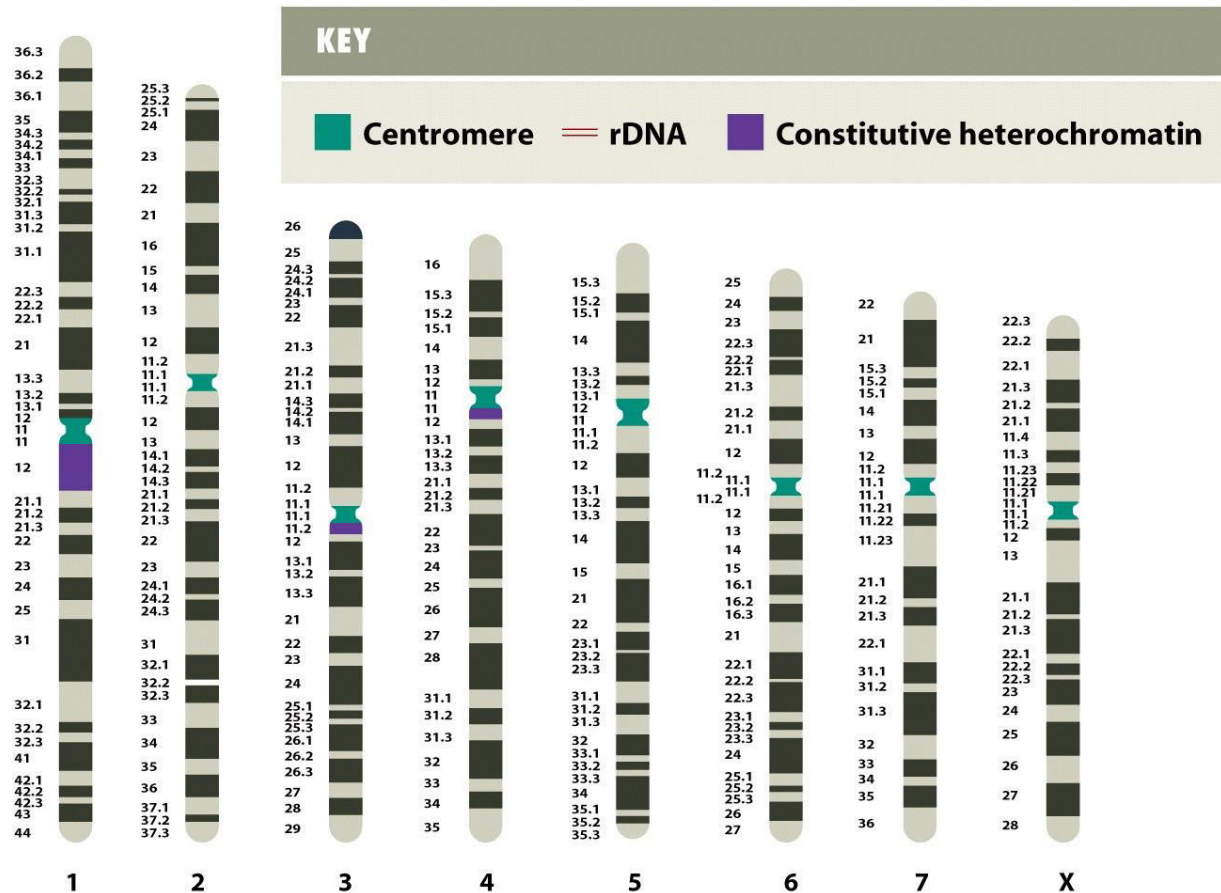


(C)

Linker histone

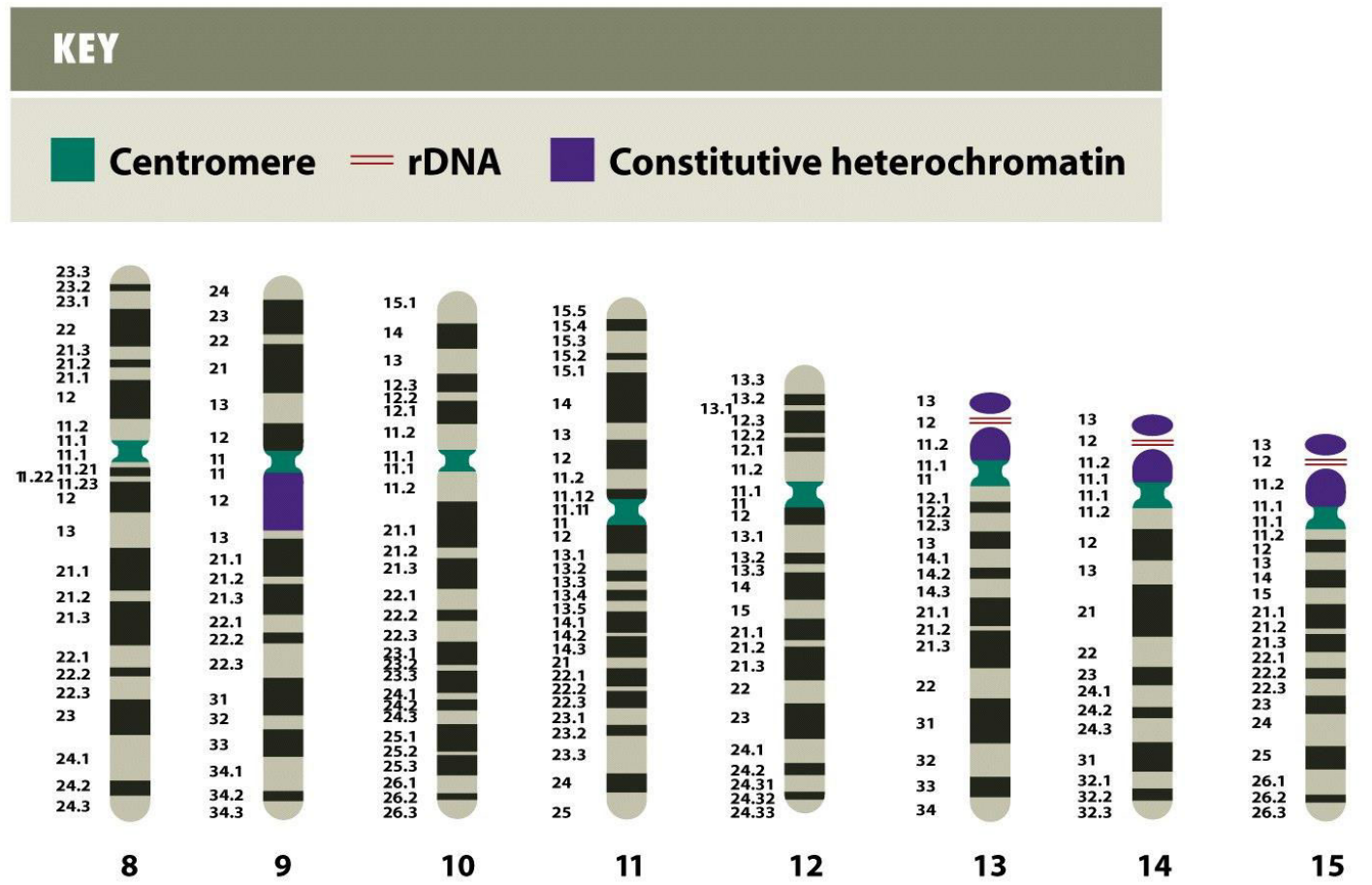


Genome Organization



Genomes Anatomy/Organization

Genome Organization



Genomes Anatomy/Organization

Genome Organization

KEY



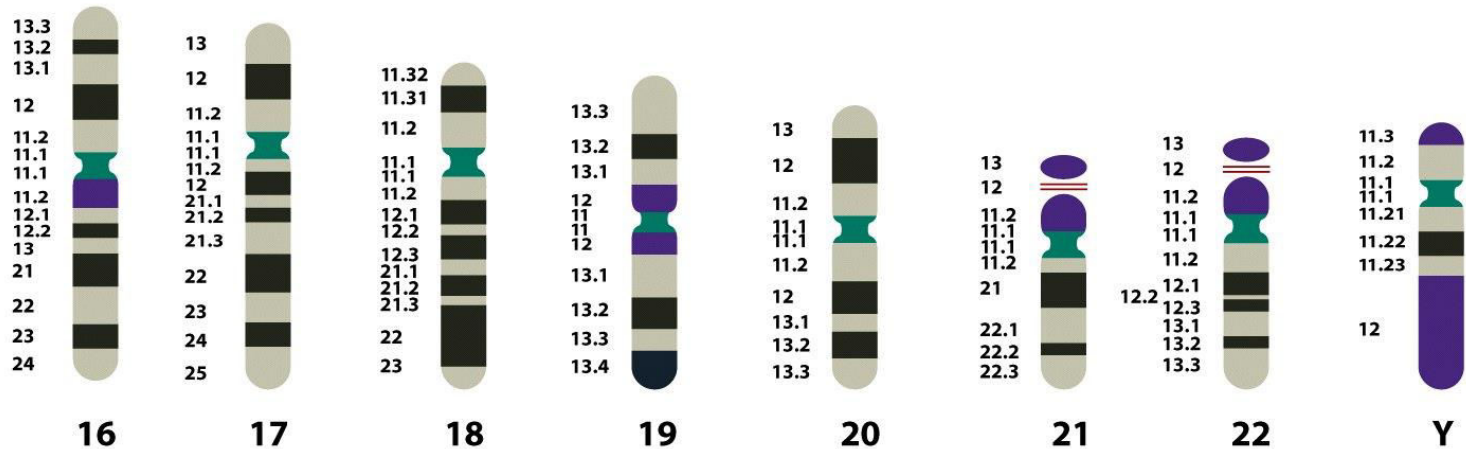
Centromere



rDNA



Constitutive heterochromatin



Genomes Anatomy/Organization

Genome Organization in Prokaryotes

Species	Genome organization		
	DNA molecules	Size (Mb)	Number of genes
<i>Escherichia coli</i> K12	One circular molecule	4.639	4405
<i>Vibrio cholerae</i> El Tor N16961	Two circular molecules		
	Main chromosome	2.961	2770
	Megaplasmid	1.073	1115
<i>Deinococcus radiodurans</i> R1	Four circular molecules		
	Chromosome 1	2.649	2633
	Chromosome 2	0.412	369
	Megaplasmid	0.177	145
	Plasmid	0.046	40

Genomes Anatomy/Organization

Genome Organization in Prokaryotes

Species	Size of genome (Mb)	Approximate number of genes
Bacteria		
<i>Mycoplasma genitalium</i>	0.58	500
<i>Streptococcus pneumoniae</i>	2.16	2300
<i>Vibrio cholerae</i> El Tor N16961	4.03	4000
<i>Mycobacterium tuberculosis</i> H37Rv	4.41	4000
<i>Escherichia coli</i> K12	4.64	4400
<i>Yersinia pestis</i> CO92	4.65	4100
<i>Pseudomonas aeruginosa</i> PA01	6.26	5700
Archaea		
<i>Methanococcus jannaschii</i>	1.66	1750
<i>Archaeoglobus fulgidus</i>	2.18	2500

Genomes Anatomy/Organization

Genome Organization: Comparisons

Species	Genome size (Mb)
Fungi	
<i>Saccharomyces cerevisiae</i>	12.1
<i>Aspergillus nidulans</i>	25.4
Protozoa	
<i>Tetrahymena pyriformis</i>	190
Invertebrates	
<i>Caenorhabditis elegans</i>	97
<i>Drosophila melanogaster</i>	180
<i>Bombyx mori</i> (silkworm)	490
<i>Strongylocentrotus purpuratus</i> (sea urchin)	845
<i>Locusta migratoria</i> (locust)	5000

Genomes Anatomy/Organization

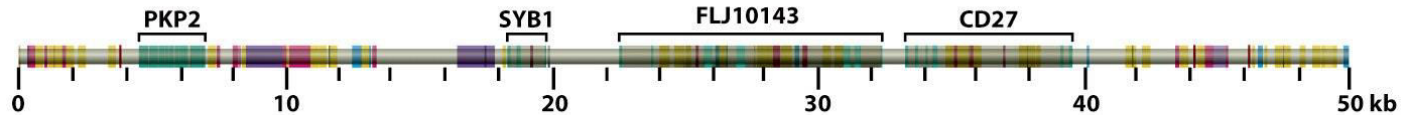
Genome Organization: Comparisons

Species	Genome size (Mb)
Vertebrates	
<i>Takifugu rubripes</i> (pufferfish)	400
<i>Homo sapiens</i>	3200
<i>Mus musculus</i> (mouse)	3300
Plants	
<i>Arabidopsis thaliana</i> (vetch)	125
<i>Oryza sativa</i> (rice)	466
<i>Zea mays</i> (maize)	2500
<i>Pisum sativum</i> (pea)	4800
<i>Triticum aestivum</i> (wheat)	16,000
<i>Fritillaria assyriaca</i> (fritillary)	120,000

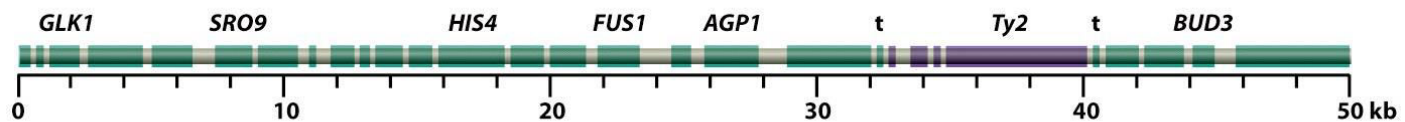
Genomes Anatomy/Organization

Genome Organization: Human, Yeast, Fruit Fly , Maize

(A) Human



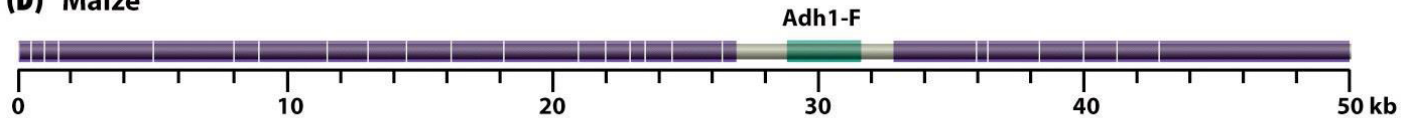
(B) *Saccharomyces cerevisiae*



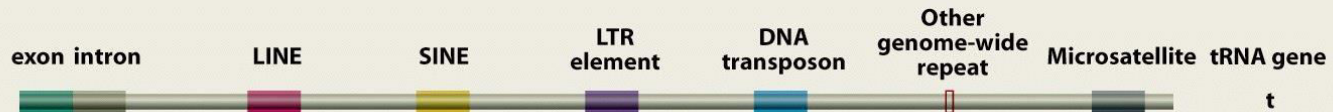
(C) *Drosophila melanogaster*



(D) Maize



KEY



Genomes Anatomy/Organization

Compactness of Genomes

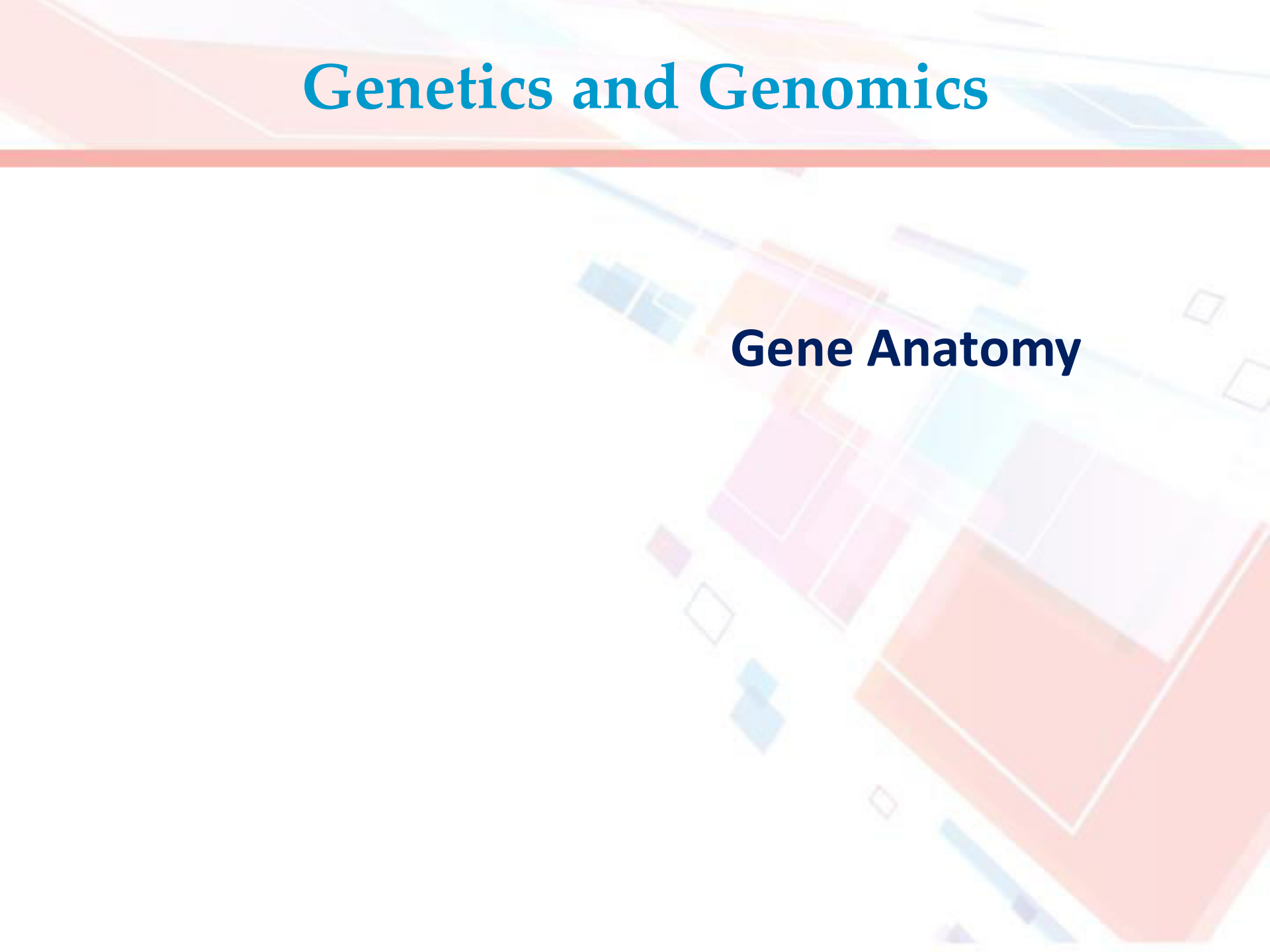
Feature	Yeast	Fruit fly	Human
Gene density (average number per Mb)	496	76	11
Introns per gene (average)	0.04	3	9
Amount of the genome that is taken up by genome-wide repeats	3.4%	12%	44%

Genomes Anatomy/Organization

Conclusion

- Anatomy of different genomes differ from each other.
- Eukaryotes and prokaryotes genomes differ very significantly.

Genetics and Genomics



Gene Anatomy

Gene Anatomy

What is Gene

- A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

Gene Anatomy

What is Gene

- Entire nucleic acid sequence necessary for the synthesis of a functional polypeptide (protein chain) or functional RNA

Gene Anatomy

Genetic information is stored in DNA

- Segments of DNA that encode proteins or other functional products are called genes.
- Gene sequences are transcribed into messenger RNA (mRNA).

Gene Anatomy

mRNA is translated into Proteins

- mRNA intermediates are translated into proteins that perform most of the life functions.

Gene Anatomy

Gene Anatomy

- Three components
- Open Reading Frame: From start codon (ATG) to stop (TGA, TAA, TAG)
- Upstream region with binding site. (e.g. TATA box, GC box, CAAT box).
- Poly-a tail

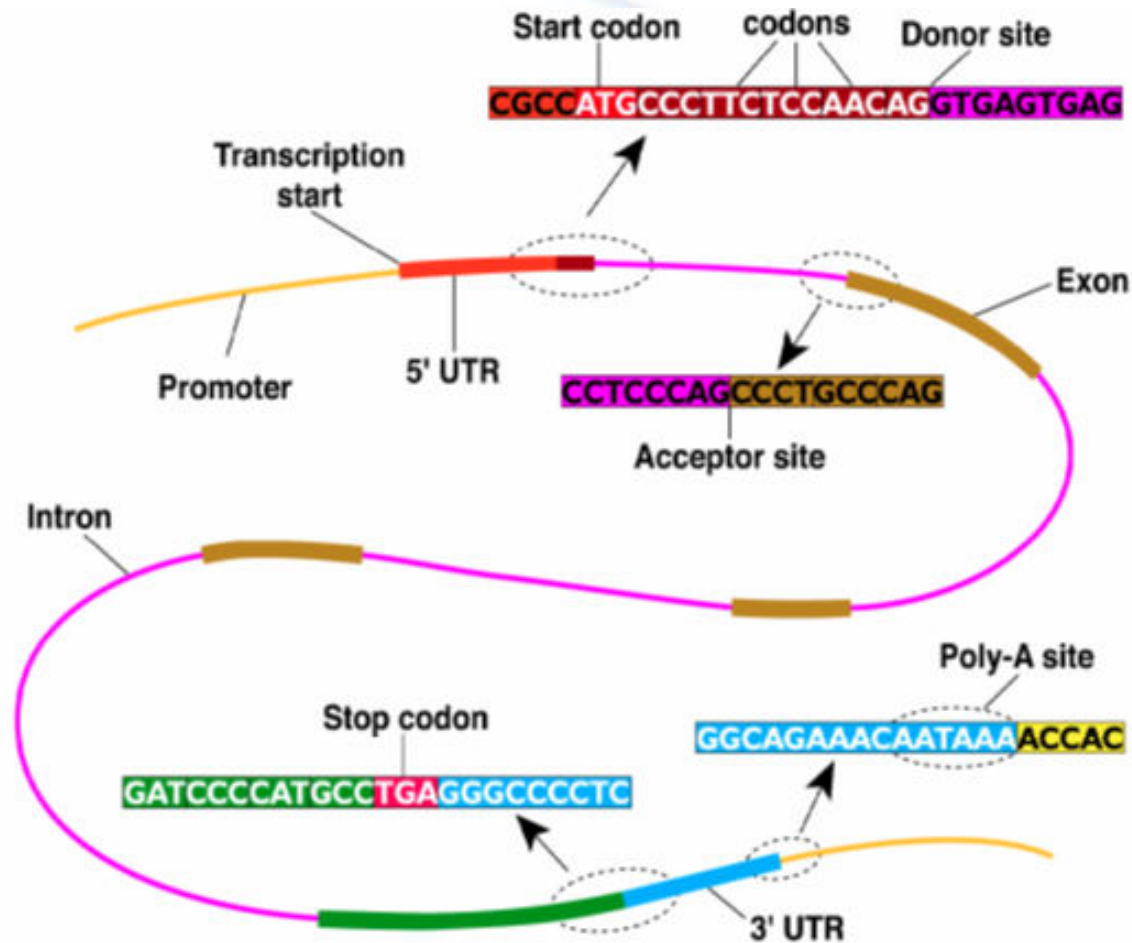
Gene Anatomy

Gene Anatomy – Typical Prokaryotic Gene



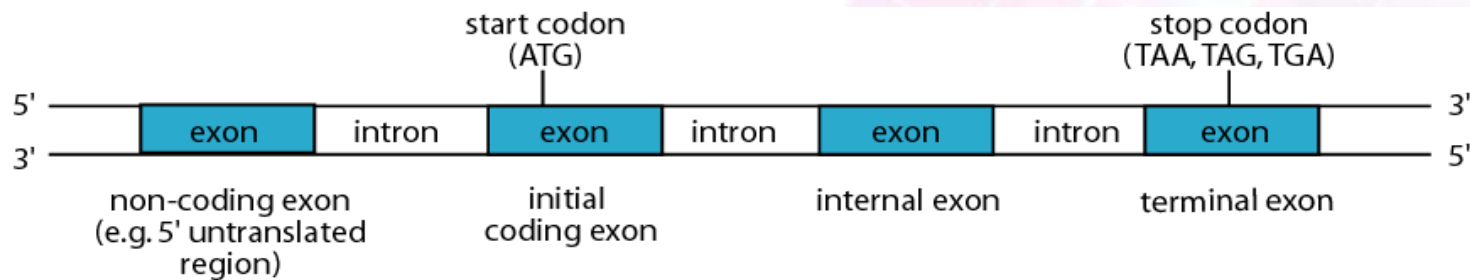
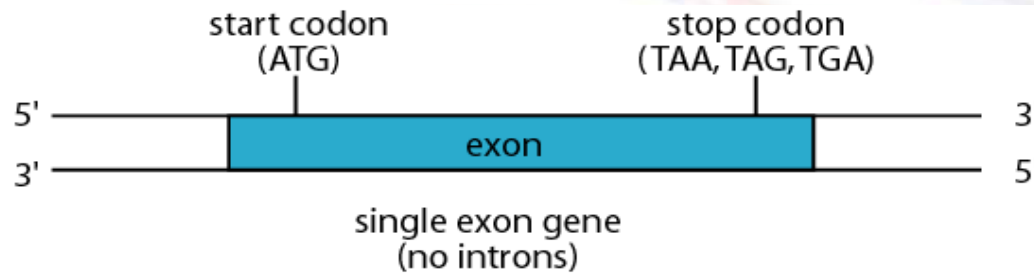
Gene Anatomy

Gene Anatomy – Typical Eukaryotic Gene



Gene Anatomy

Single Exon Gene and Multiple Exons Gene



Gene Anatomy

What is Gene

- A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

Genetics and Genomics

Prokaryotic Gene and Eukaryotic Gene

Prokaryotic Gene/Eukaryotic Gene

Bacterial Gene

- Most do not have introns
- Many are organized in operons: contiguous genes, transcribed as a single polycistronic mRNA, that encode proteins with related functions

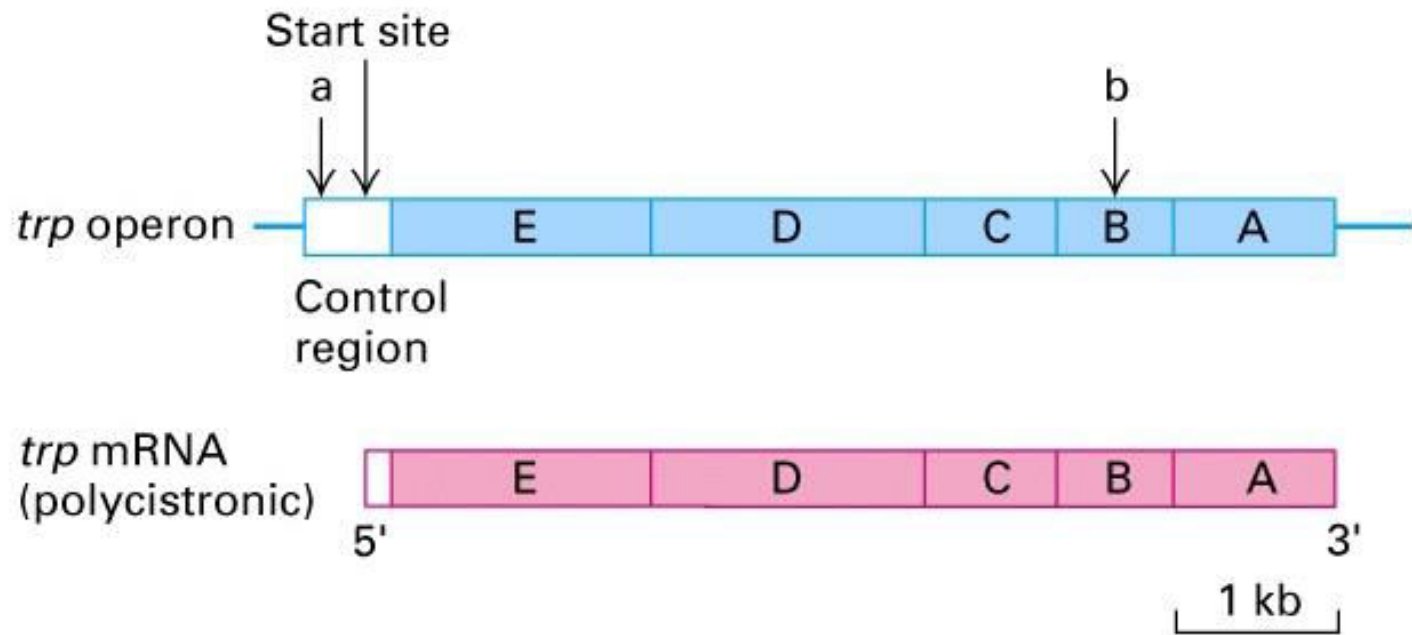
Prokaryotic Gene/Eukaryotic Gene

Bacterial Gene

- Polycistronic mRNA encodes several proteins

Prokaryotic Gene/Eukaryotic Gene

Bacterial Gene: Polycistronic mRNA encodes several proteins



Prokaryotic Gene/Eukaryotic Gene

Eukaryotic Gene: Exons and Introns

- Introns: intervening sequences within a gene that are not translated into a protein sequence.
- Exons: sequences within a gene that encode protein sequence.

Prokaryotic Gene/Eukaryotic Gene

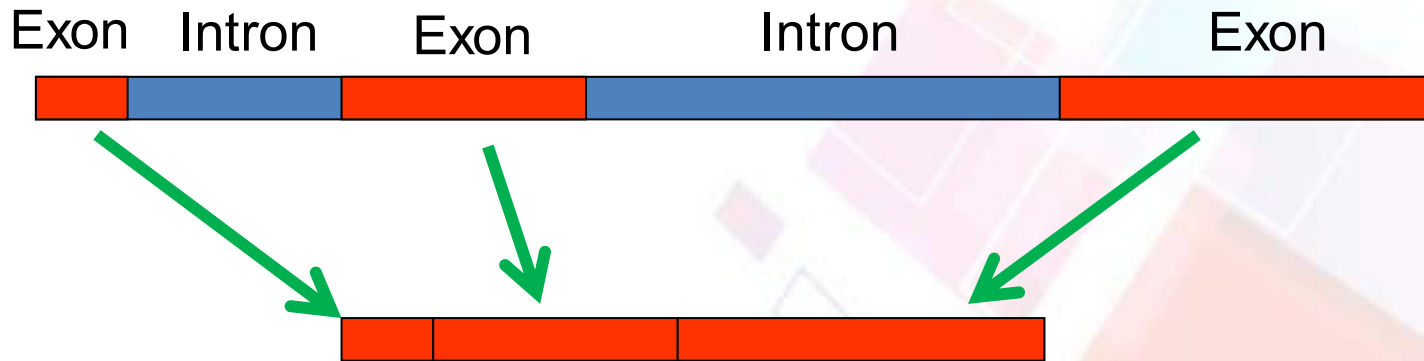
Eukaryotic Gene

- Splicing: Removal of introns from the mRNA molecule

Prokaryotic Gene/Eukaryotic Gene

Eukaryotic Gene

- Splicing: Removal of introns from the mRNA molecule



Prokaryotic Gene/Eukaryotic Gene

Eukaryotic Gene

- Organize expression of genes' (function calls)
- Promoter region (binding site), usually near coding region
- Binding can block (inhibit) expression

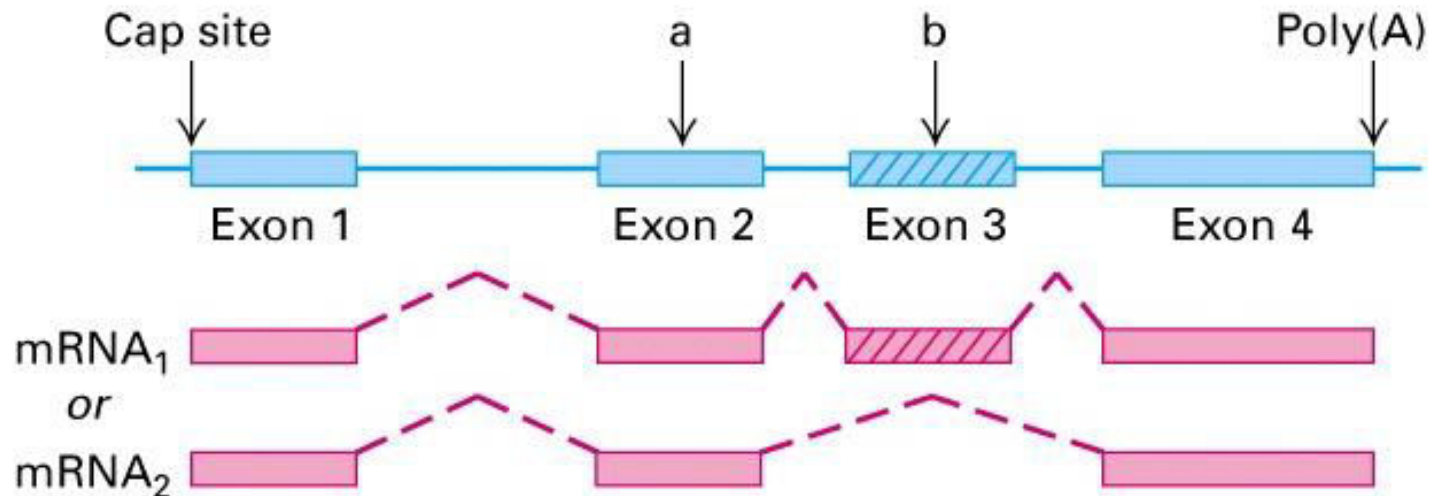
Prokaryotic Gene/Eukaryotic Gene

Alternative Splicing in Eukaryotic Genes

- Most have introns
- Produce monocistronic mRNA
- Large in size

Prokaryotic Gene/Eukaryotic Gene

Eukaryotic Gene: Alternative Splicing



Prokaryotic Gene/Eukaryotic Gene

Eukaryotic Gene

Computational
challenges

Identify binding sites

Correlate sequence to
expression

Prokaryotic Gene/Eukaryotic Gene

Conclusion

- Most of prokaryotic genes are without introns and in the are polycistronic.
- Eukaryotic genes have introns and alternative splicing.

Genetics and Genomics

Prokaryotic and Eukaryotic Gene Expression - Difference

Prokaryotic/Eukaryotic Gene Expression-Difference

Prokaryotic and Eukaryotic Gene Expression

- Bacterial genetics are different.
- Prokaryote genes are grouped in operons
- Prokaryotes have one type of RNA polymerase for all types of RNA

Prokaryotic/Eukaryotic Gene Expression-Difference

Difference between Prokaryotic and Eukaryotic gene Expression

- mRNA is not modified
- The existence of introns in prokaryotes is extremely rare

Prokaryotic/Eukaryotic Gene Expression-Difference

Difference

- Transcription in bacteria, sigma factors bind to RNA polymerases.
- RNA polymerases/sigma factors complex bind to promoter about 40 bases prior to coding region of gene.

Prokaryotic/Eukaryotic Gene Expression-Difference

Difference between Prokaryotic and Eukaryotic gene Expression

- In prokaryotes, the newly synthesized mRNA is polycistronic (code for more than one polypeptide chain).

Prokaryotic/Eukaryotic Gene Expression-Difference

Difference

- In prokaryotes, transcription of a gene and translation of the resulting mRNA occur simultaneously.
- Many polysomes are found associated with an active gene.

Prokaryotic/Eukaryotic Gene Expression-Difference

Difference - Prokaryotic and Eukaryotic Gene Expression

Prokaryotes	Eukaryotes
• Operons • 27% of E. coli genes (Housekeeping genes not in operons) • Simultaneous transcription and translation	• No operons, but they still need to coordinate regulation • More kinds of control elements • RNA processing • Chromatin remodeling • Histones must be modified to loosen DNA

Prokaryotic/Eukaryotic Gene Expression-Difference

Prokaryotic/Eukaryotic Gene Expression

- Prokaryotes-operons simultaneously transcription and translation
- Eukaryotes – no operons, RNA processing, chromatin remodeling

Genetics and Genomics

Genes Families and Genes Clusters

Genes Families and Genes Clusters

Genes Clusters

- Many genes are arranged in groups along a chromosome
 - Genes Clusters
- Eukaryotic ribosomal RNA genes: tandem repeats

Genes Families and Genes Clusters

Genes Families

- Related genes may be organized as several clusters at different locations
- These are known as gene families e.g. globin genes family

Genes Families and Genes Clusters

Single Copy Genes and Genes Families

- Many eukaryotic genes are present in one copy per haploid set of chromosomes
- The rest of the genes occur in multigene families, collections of identical or very similar genes

Genes Families and Genes Clusters

Gene Family and Family members -

- Peroxiredoxin family
- *PRDX* is the root symbol
- Family members are *PRDX1*, *PRDX2*, *PRDX3*, *PRDX4*, *PRDX5* and *PRDX6*

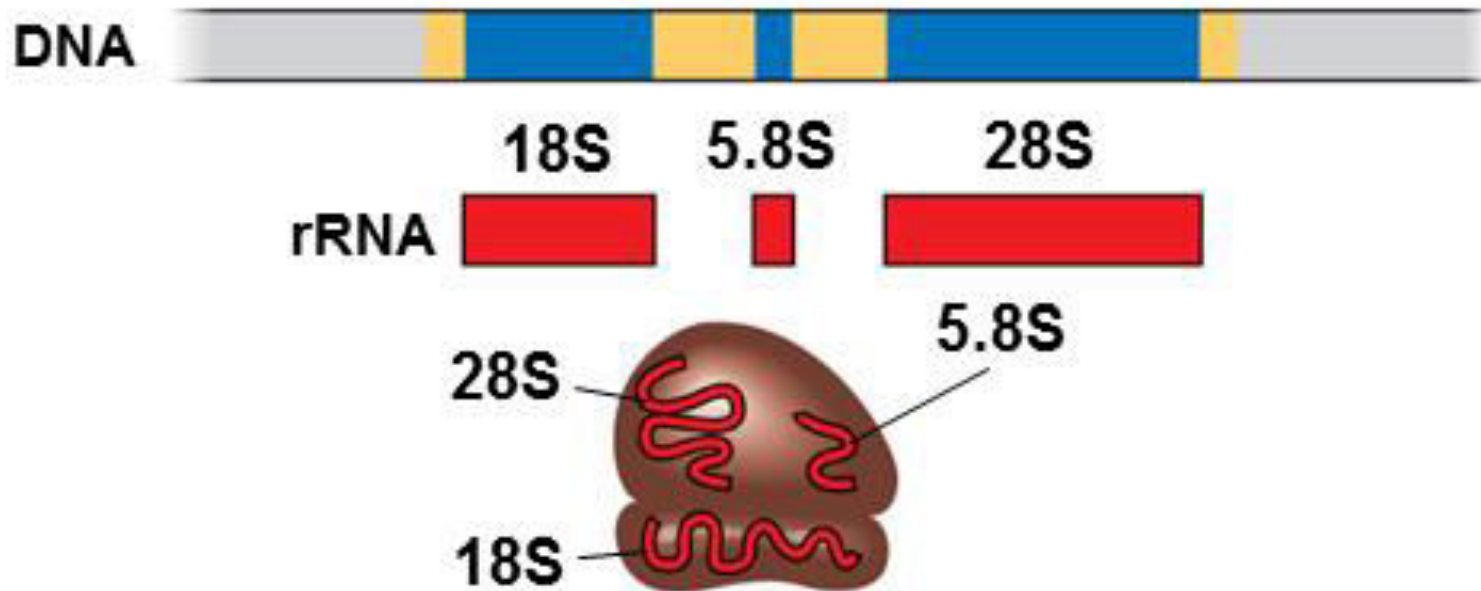
Genes Families and Genes Clusters

Genes Families

- Some multigene families consist of identical DNA sequences, usually clustered tandemly, such as those genes that code for rRNA products

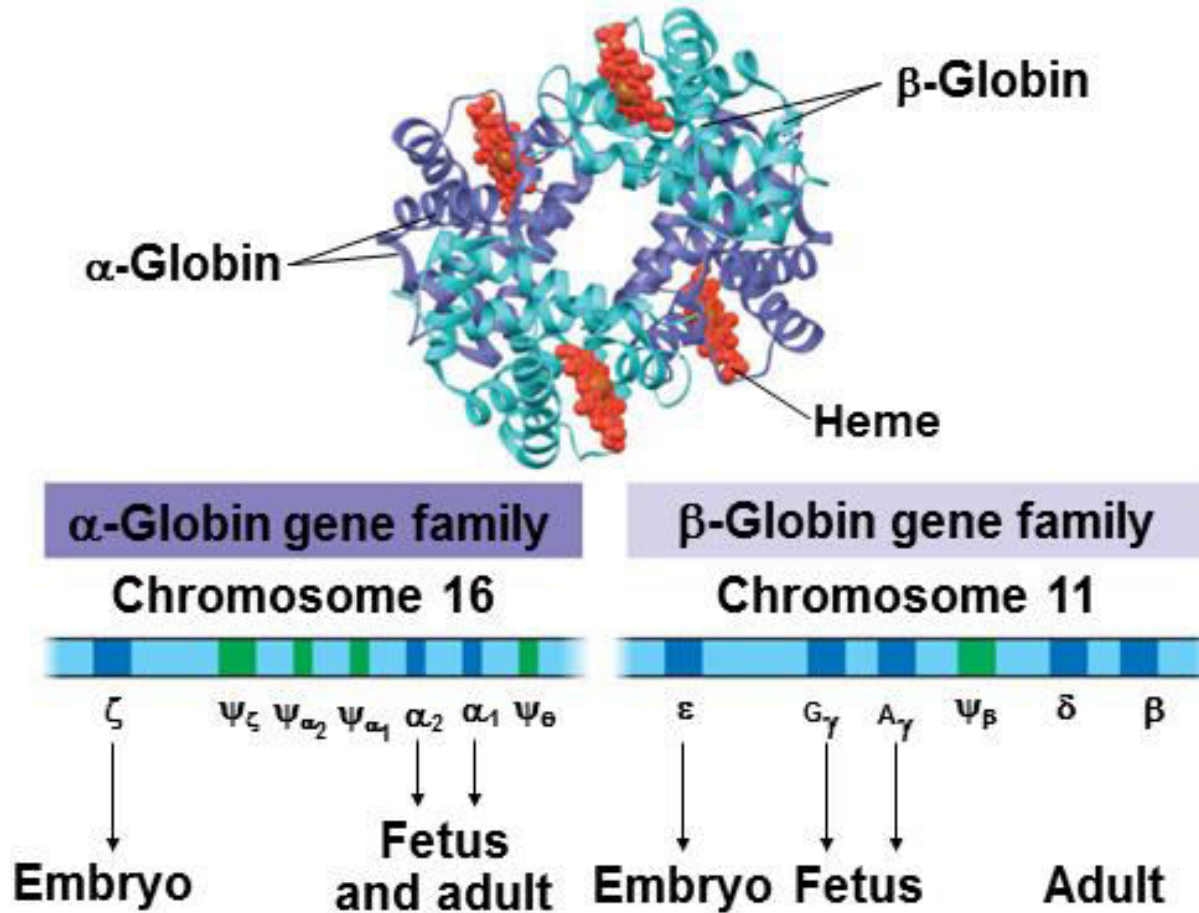
Genes Families and Genes Clusters

Ribosomal RNA Gene Family



Genes Families and Genes Clusters

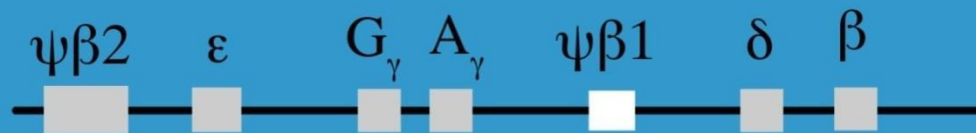
Human Alpha & Beta Globin Gene Families



Genes Families and Genes Clusters

Genes Families on Different Chromosomes

Chromosome 11



Chromosome 16



Chromosome 22



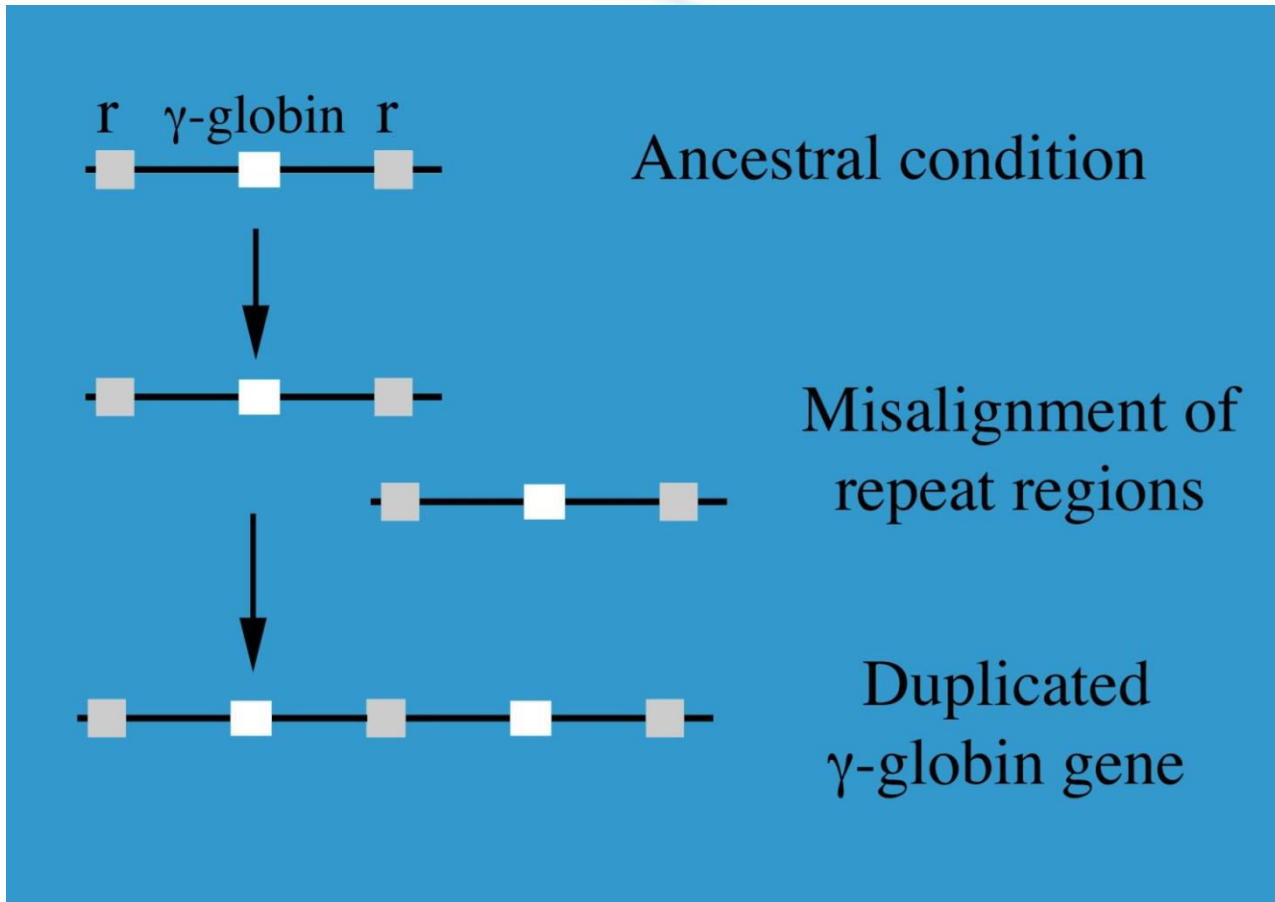
Genes Families and Genes Clusters

Genes Families

- The misalignment of genes during recombination is the most likely cause of gene duplication and clustering.
- Once repeats exist the probability of misalignment increases.

Genes Families and Genes Clusters

Genes Families: The misalignment - Globins



Genes Families and Genes Clusters

Super Families

- Super families are much larger than single multigene families.
- Super families contain hundreds of genes, including multiple multigene families and single individual genes

Genes Families and Genes Clusters

Super Families

- The large number of members allows super families to be widely dispersed with some genes clustered and some spread far apart.

Genes Families and Genes Clusters

Genes Families and Genes Clusters - Conclusion

- Single individuals
Genes
- Gene families
- Gene families in
form of clusters
- Gene families as
Super families

Genetics and Genomics

**Type of Proteins/
Families**

Type of Proteins/Families

Protein Structure

- Amino acids
 - 20 amino acids
 - Hydrophobic / hydrophylic
 - Charged / neutral
- Functions
 - Enzymes
 - Structure protein
 - Channel
 - Other functions

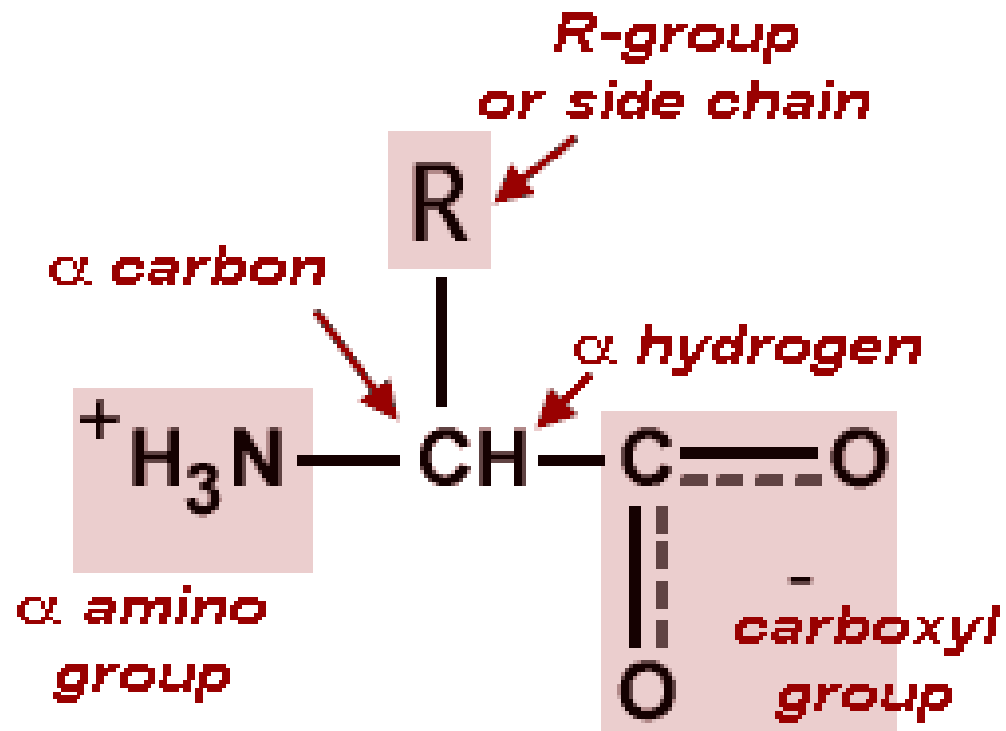
Type of Proteins/Families

Amino Acids: Building Blocks of Proteins

$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ (\text{CH}_2)_3 \\ \\ \text{NH} \\ \\ \text{C}=\text{NH}_2 \\ \\ \text{NH}_2 \end{array} $ Arginine (Arg / R)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{NH}_2 \end{array} $ Glutamine (Gln / Q)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5 \end{array} $ Phenylalanine (Phe / F)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{OH} \end{array} $ Tyrosine (Tyr / Y)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{C}_8\text{H}_6\text{N}_2 \end{array} $ Tryptophan (Trp, W)
$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ (\text{CH}_2)_4 \\ \\ \text{NH}_2 \end{array} $ Lysine (Lys / L)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{H} \end{array} $ Glycine (Gly / G)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_3 \end{array} $ Alanine (Ala / A)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{C}_3\text{H}_3\text{N}_2 \end{array} $ Histidine (His / H)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{OH} \end{array} $ Serine (Ser / S)
$ \begin{array}{c} \text{H}_2 \\ \\ \text{C} \\ / \quad \backslash \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \quad \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \end{array} $ Proline (Pro / P)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{COOH} \end{array} $ Glutamic Acid (Glu / E)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{COOH} \end{array} $ Aspartic Acid (Asp / D)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_3 \end{array} $ Threonine (Thr / T)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{SH} \end{array} $ Cysteine (Cys / C)
$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{S} \\ \\ \text{CH}_3 \end{array} $ Methionine (Met / M)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{CH} \\ / \quad \backslash \\ \text{CH}_3 \quad \text{CH}_3 \end{array} $ Leucine (Leu / L)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{NH}_2 \end{array} $ Asparagine (Asn / N)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{HC}-\text{CH}_3 \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array} $ Isoleucine (Ile / I)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{C} \begin{array}{l} \diagup \text{O}^- \\ \diagdown \text{O}^- \end{array} \\ \\ \text{CH} \\ / \quad \backslash \\ \text{CH}_3 \quad \text{CH}_3 \end{array} $ Valine (Val / V)

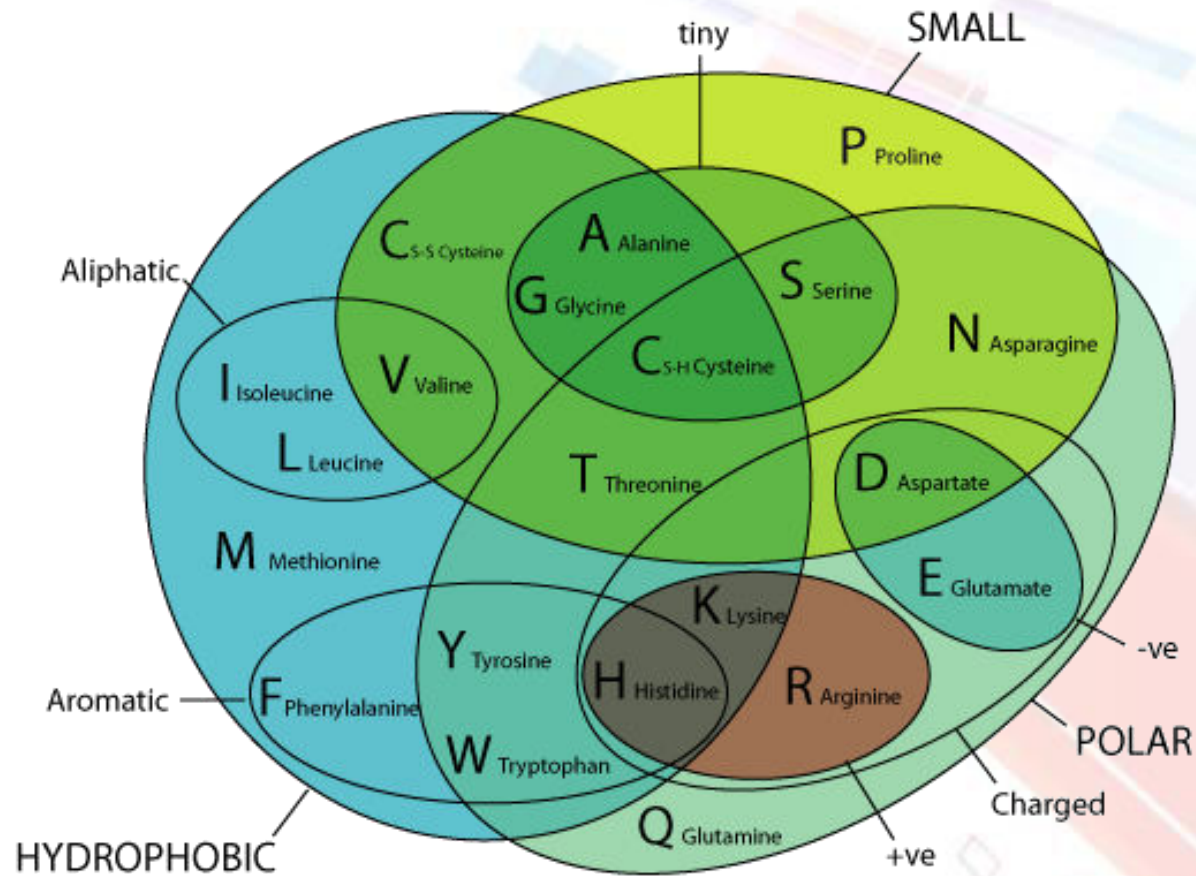
Type of Proteins/Families

Protein Structure: Amino Acid



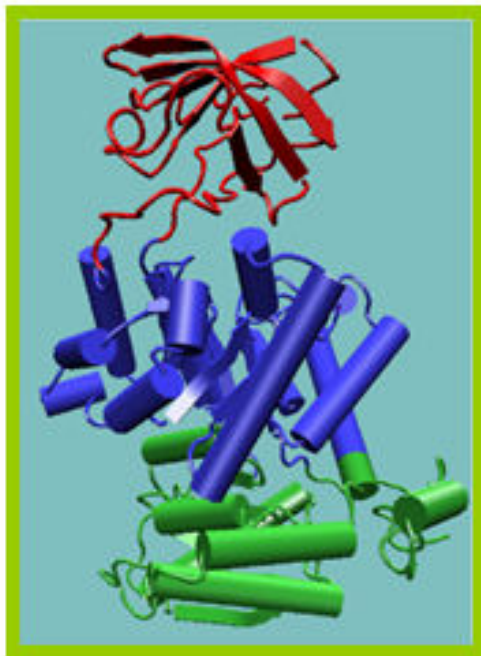
Type of Proteins/Families

Protein Structure: Types of Amino Acids



Type of Proteins/Families

Protein Domains: Domains are units of compact structure, function and evolution and folding



SKSHSEAGSAFIQTQQLHAAMADTFLEHMCRLDIDSAP
ITARNTGIICTIGPASRSVETLKEMIKSGMNVARMNFS
HGTHEYHAETIKNVRTATESFASDPILYRPVAVALDTK
GPEIRTGLIKSGTAEVELKKGATLKITLDNAYMAACD
ENILWLDYKNICKVVEVGSKVYVDDGLISLQVKQKGPD
FLVTEVENGGFLGSKKGVNLPGAAVDLPVSEKDIQDL
KFGVDEDEDVDMVFASFIRKAADVHEVRKILGEKGKNIKI
ISKIENHEGVRRFDEILEASDGIMVARGDLGIEIPA EK
VFLAQKMIIGRCNRAGKPVICATQMLESMIKKPRPTRA
EGSDVANAVLDGADCIMLSGETAKGDYPLEAVRMQH LI
AREAEAAMFHRKLFEE LARSSSHSTDLM EAMAMGSVEA
SYKCLAAALIVLTESGRSAHQVARYRPRAPIIAVTRNH
QTARQAHLYRGI FPVVCKDPVQEAWAEDVDLRVNLAMN
VGKAAGFFKKGDVVIVLTGWRPGSGFTNTMRVVPVP

Type of Proteins/Families

Proteins

- Different method of categorizing proteins
- Three major categories
- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Type of Proteins/Families

Fibrous Proteins – Cytoskeletal Proteins

- Actin
- Coronin
- Dystrophin
- Keratin
- Tubulin

Type of Proteins/Families

Fibrous Proteins – Extracellular Matrix Proteins

- Collagen
- Elastin
- Fibronectin

Type of Proteins/Families

Globular Proteins - Major Types

- Plasma proteins
- Hemoproteins
- Cell adhesion
- Transmembrane transport proteins
- Hormones and growth factors
- Receptors

Type of Proteins/Families

Globular Proteins - Major Types

- DNA-binding protein
- Immune system proteins
- Nutrient storage/transport
- Chaperone proteins
- Enzymes

Type of Proteins/Families

Complexes with multiple components including proteins

- Nucleosome
- Ribonucleoprotein (generic)
- Signal recognition particle
- Spliceosome

Type of Proteins/Families

Types of Proteins - Conclusion

- Three major categories
- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Genome Informatics

Introduction

- Genome sequencing provides the sequences of all the genes of an organism
- A major application of Bioinformatics is analysis of full genomes that have been sequenced

Genome Informatics

Introduction

- Challenge is to identify those genes that are predicted to have a particular biological function

Genome Informatics

Genomics

Study of all of a person's genes (the **Genome**), including interactions of those genes with each other and with the person's environment (**NHGRI**)

Genome Informatics

Genome Informatics

Genome informatics is the field in which computational and statistical techniques are applied to derive biological information from genome sequences

Genome Informatics

Genome Informatics

Genome informatics includes methods to analyze DNA sequence information and to predict protein sequence and structure

(lossifov, et al. 2014)

Genome Informatics

Genome Sequences

Availability of genome sequences facilitates;

- The discovery and utilization of sequence polymorphisms
- Opportunity to explore genetic variability both between and within the organisms

Genome Informatics

Genome Analysis

Following tasks;

- Sequencing
- Assembly
- Repeat identification and masking out
- Gene prediction
- Looking for EST and cDNA sequences
- Genome annotation

Genome Informatics

Genome Analysis

- Expression analysis
- Metabolic pathways and regulation studies
- Functional genomics
- Gene location/gene map identification

Genome Informatics

Genome Analysis

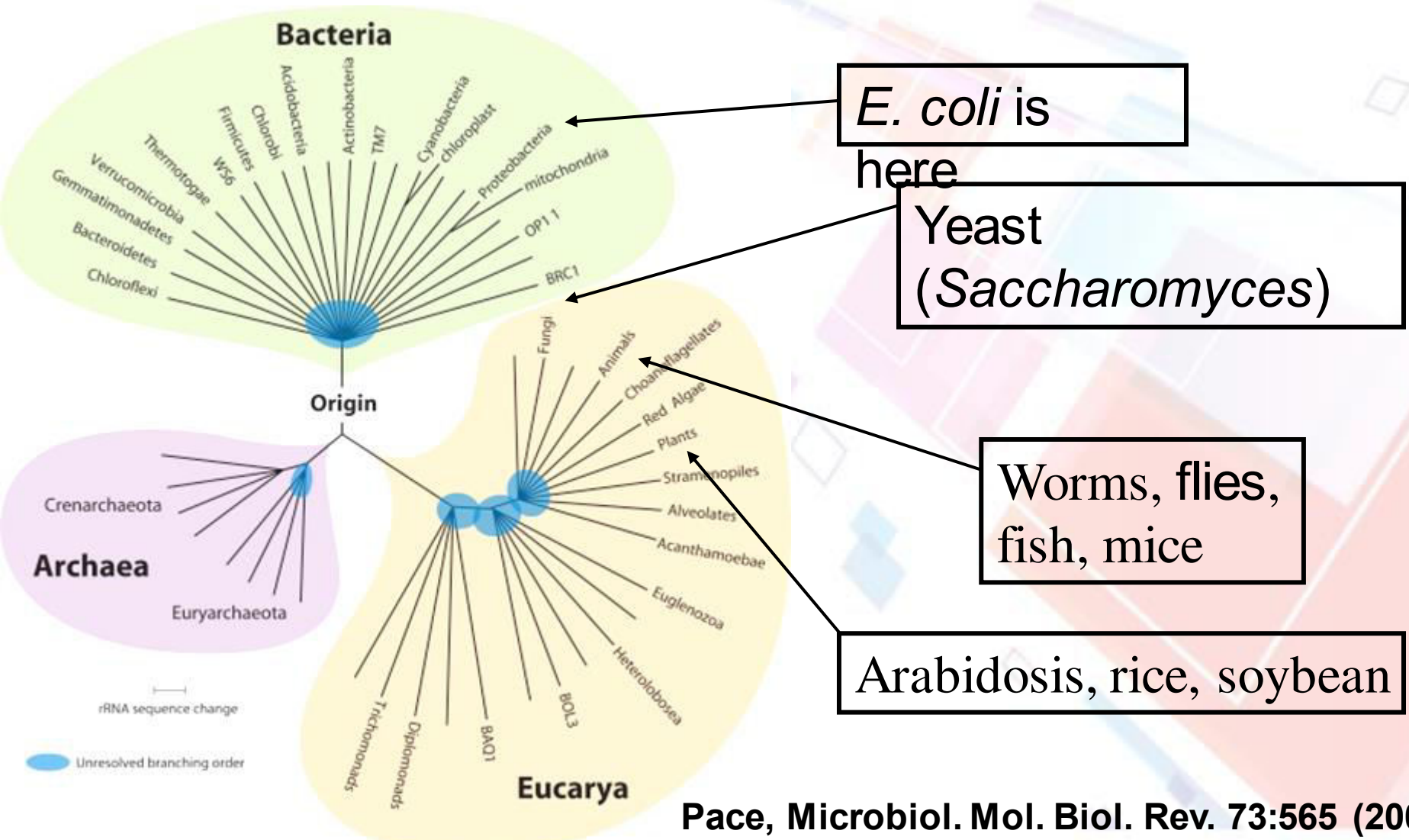
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling
- Self-comparison of proteome

Genome Informatics

Model organisms

- E. coli – bacteria
- S. cerevisiae – yeast
- C. elegans – worm
- D.melanogaster – fly
- Danio rerio – zebrafish
- Mus musculus - mouse
- Homo sapiens – you and me
- Arabidopsis - plant

Genome Informatics



Genome Informatics

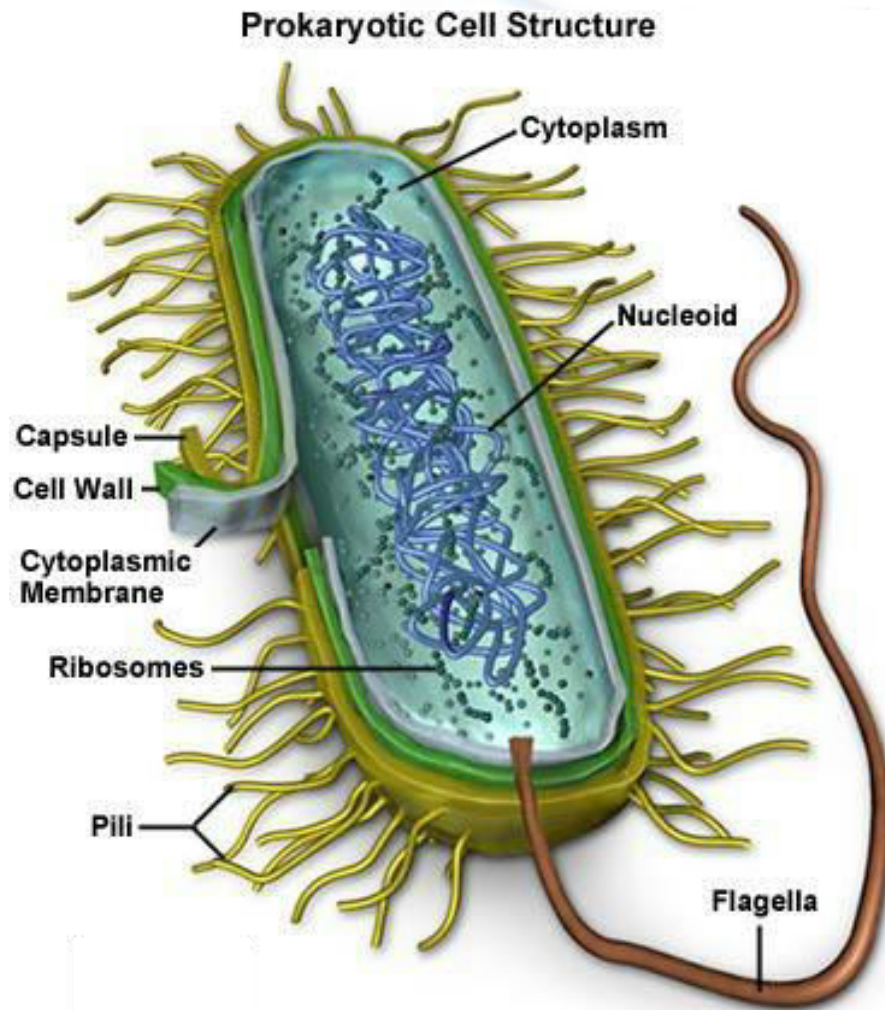
Conclusions

- Sequencing and analysis of full genomes paves the way for future discoveries
- Different model organisms can help explore our Genome and what matters most for us

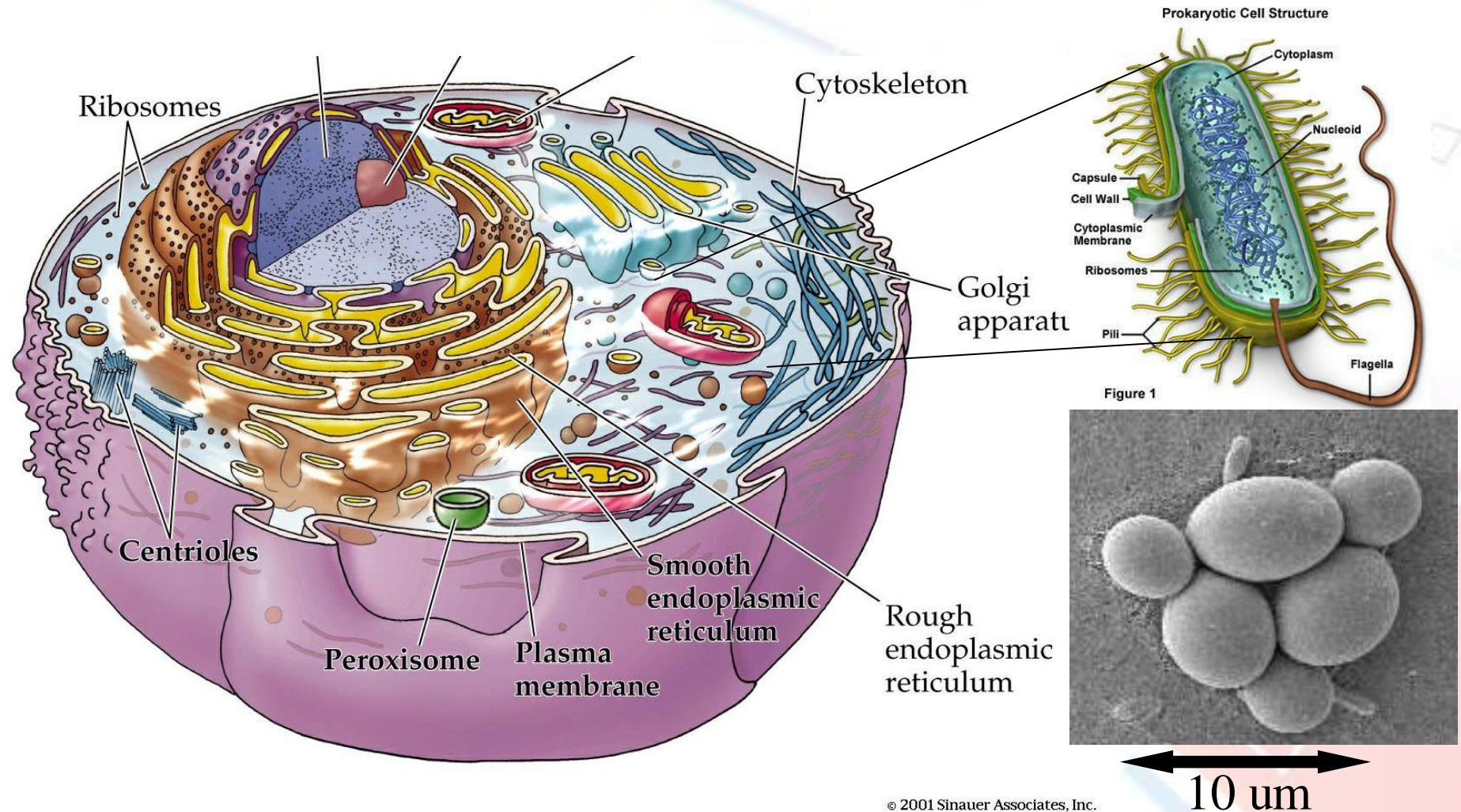
Prokaryotic Genome

- Prokaryotes are the organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane
- No membrane bound organelles

Prokaryotic Genome



Prokaryotic Genome



Mitochondria evolved from a bacterial endosymbiont

Prokaryotic Genome

- First prokaryotic Genome sequenced was that of *Hemophilus influenzae*
- Paved the way for sequencing of many other organisms

Prokaryotic Genome

Selection criteria

Following are the criteria for organisms selected for sequencing;

- They had been subjected to a detailed biological analysis and thus were model organisms

Prokaryotic Genome

Selection criteria

- important human pathogens
- They were of phylogenetic interest
- Sequences were annotated as they were sequenced

Prokaryotic Genome

Table 10.2. *Features of representative prokaryotic genomes*

Organism (reference)	Phylogenetic group	Genome size (Mbp) (no. protein- encoding genes)	Novel functions
<i>Escherichia coli</i> (Blattner et al. 1997)	Bacteria	4.6 (4288)	model organism
<i>Methanococcus jannaschii</i> (Bult et al. 1996)	Archaea	1.66 (1682) ^a	grows at high temperature and pressure and produces methane
<i>Hemophilus influenzae</i> (Fleischmann et al. 1995)	Bacteria	1.83 (1743)	human pathogen
<i>Mycoplasma pneumoniae</i> (Himmelreich et al. 1996)	Bacteria	0.82 (676)	human pathogen that grows inside cells; metabolically weak
<i>Bacillus subtilis</i> (Kunst et al. 1997)	Bacteria	4.2 (4098)	model organism
<i>Aquifex aeolicus</i> (Deckert et al. 1998)	Bacteria	1.55 (1512) ^b	ancient species, grows at high temperature and can grow in a hydrogen, oxygen, carbon dioxide atmosphere in the presence of only mineral salts
<i>Synechocystis</i> sp. (Kaneko et al. 1996a,b)	Bacteria	3.57 (3168)	ancient organism that produces oxygen by light-harvesting; may have oxygenated atmosphere

Prokaryotic Genome

Conclusions

- Prokaryotes are simple Genomes
- Easy models to study Biochemistry and Molecular biology of life processes
- Sequencing is done on economically important organisms

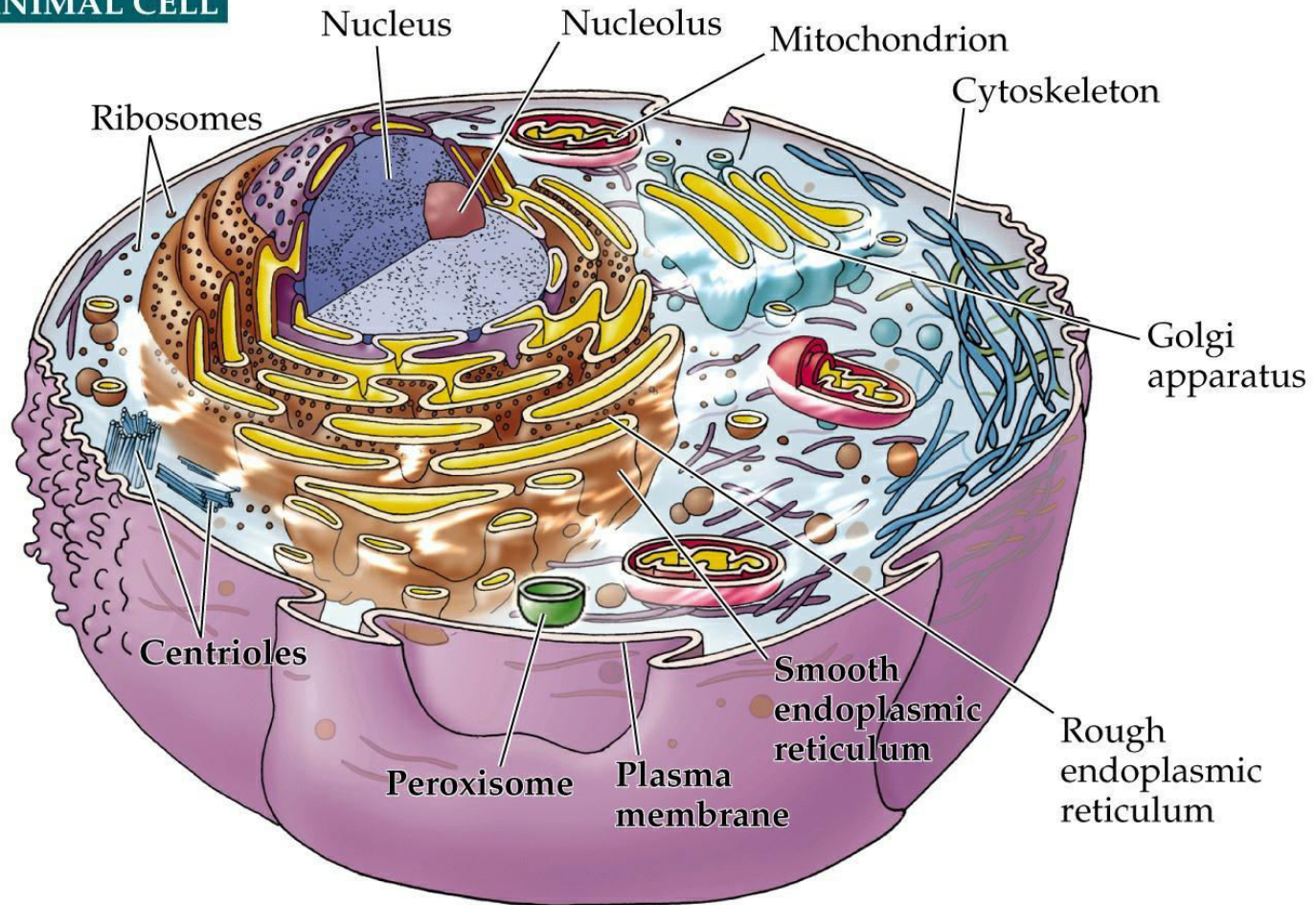
Eukaryotic Genome

Introduction

- Eukaryotes have larger genomes
- Have tandem repeats
- Have introns in their protein-coding genes
- Heterochromatin and euchromatin region

Eukaryotic Genome

AN ANIMAL CELL



Peroxisomes are organelles in all eukaryotes that participate in the metabolism of fatty acids and other metabolites. Peroxisomes have enzymes that rid the cell of toxic peroxides. They have a single lipid bilayer membrane that separates their contents from the cytosol (the internal fluid of the cell) and contain membrane proteins critical for various functions, such as importing proteins into the organelles and aiding in proliferation. Like lysosomes, peroxisomes are part of the secretory pathway of a cell, but they are more dynamic and can replicate by enlarging and then dividing. Peroxisomes were identified as organelles by the Belgian cytologist Christian de Duve in 1967[1] after they had been first described in a Swedish PhD thesis a decade earlier.[2]

Eukaryotic Genome

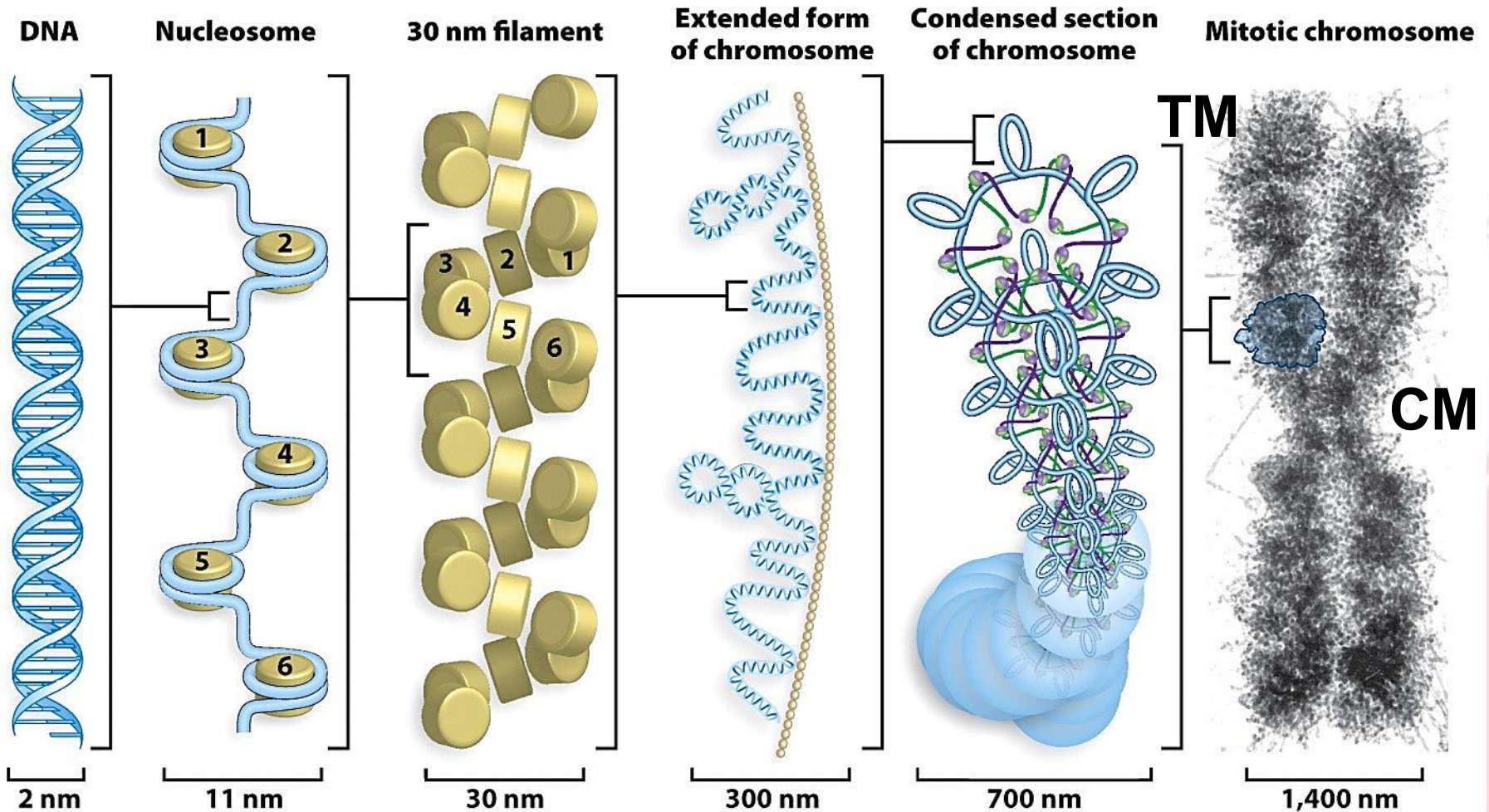


Figure 10-16
Molecular Biology: Principles and Practice

CM: Centromere

TM: Telomere

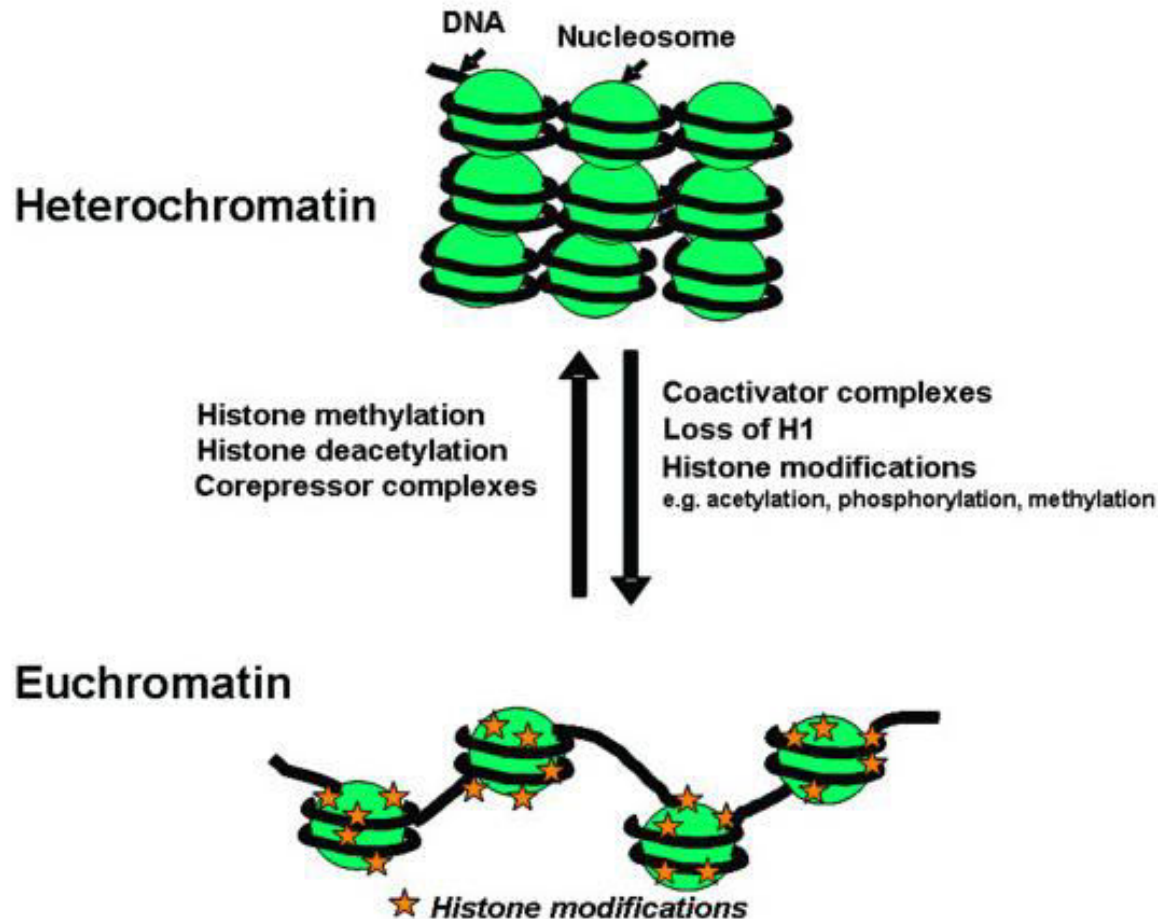
Peroxisomes are organelles in all eukaryotes that participate in the metabolism of fatty acids and other metabolites. Peroxisomes have enzymes that rid the cell of toxic peroxides. They have a single lipid bilayer membrane that separates their contents from the cytosol (the internal fluid of the cell) and contain membrane proteins critical for various functions, such as importing proteins into the organelles and aiding in proliferation. Like lysosomes, peroxisomes are part of the secretory pathway of a cell, but they are more dynamic and can replicate by enlarging and then dividing. Peroxisomes were identified as organelles by the Belgian cytologist Christian de Duve in 1967[1] after they had been first described in a Swedish PhD thesis a decade earlier.[2]

Eukaryotic Genome

Staining with dyes

- Dense heterochromatin
- Light Euchromatin
- Heterochromatic regions are poorly transcribed (expressed)
- Euchromatic regions are highly expressed

Eukaryotic Genome



Eukaryotic Genome

Conclusions

- Eukaryotes are distinguished by the presence of prominent nuclei
- have larger genomes, tandem repeats and introns in their protein-coding genes

Epichromosomal Elements (EEs)

Introduction

- Genome is the total collection of genetic material and is made up of
 - Chromosomes
 - Epichromosomal Elements

Epichromosomal Elements (EEs)

Prokaryotic EEs

- Plasmids
 - Self replicating
 - Additional rings
- Bacteriophages
 - Host colonization
- Transposons
 - Parasitic DNA elements

Epichromosomal Elements (EEs)

Eukaryotic EEs

- Organellar DNA
 - Mitochondria
 - Chloroplasts
- Plasmids
 - yeast
- Transposons
- Viral genomes
 - retroviruses

Epichromosomal Elements (EEs)

Eukaryotic EEs

- membrane-bound organelles
- hundreds to thousands of them
- Site of respiration (mitochondria) or photosynthesis (chloroplast)
- mtDNA
- cpDNA

Epichromosomal Elements (EEs)

Endosymbiont hypothesis

- originally proposed in 1883 by Andreas Schimper, but extended by Lynn Margulis in the 1980s
- Mitochondria and Chloroplast are derived from endosymbiotic bacteria

Epichromosomal Elements (EEs)

Organelle Genome

- Circular
- Double stranded
- Supercoiled
- No histones
- Multiple copies
of mtDNA/cell or
cpDNA/cell

Epichromosomal Elements (EEs)

Genome	Size & Organization
Plant plastid	150 kb circle
Plant mitochondria	150 – 2000 kb multipartite
Human mitochondria	17 kb circle
saccharomyces mitochondria	75 kb circle

Epichromosomal Elements (EEs)

Organelle Genome

- Encode only a subset of genes required to elaborate a functional organelle
 - rRNAs, tRNAs, ribosomal proteins, membrane-associated respiratory or photosynthetic components

Epichromosomal Elements (EEs)

Organelle Genome

- Other components encoded by nuclear genome, translated in the cytosol and imported into the organelle
- 10% of nuclear genes devoted to mitochondrial function; 15% to plastid function

Epichromosomal Elements (EEs)

Conclusions

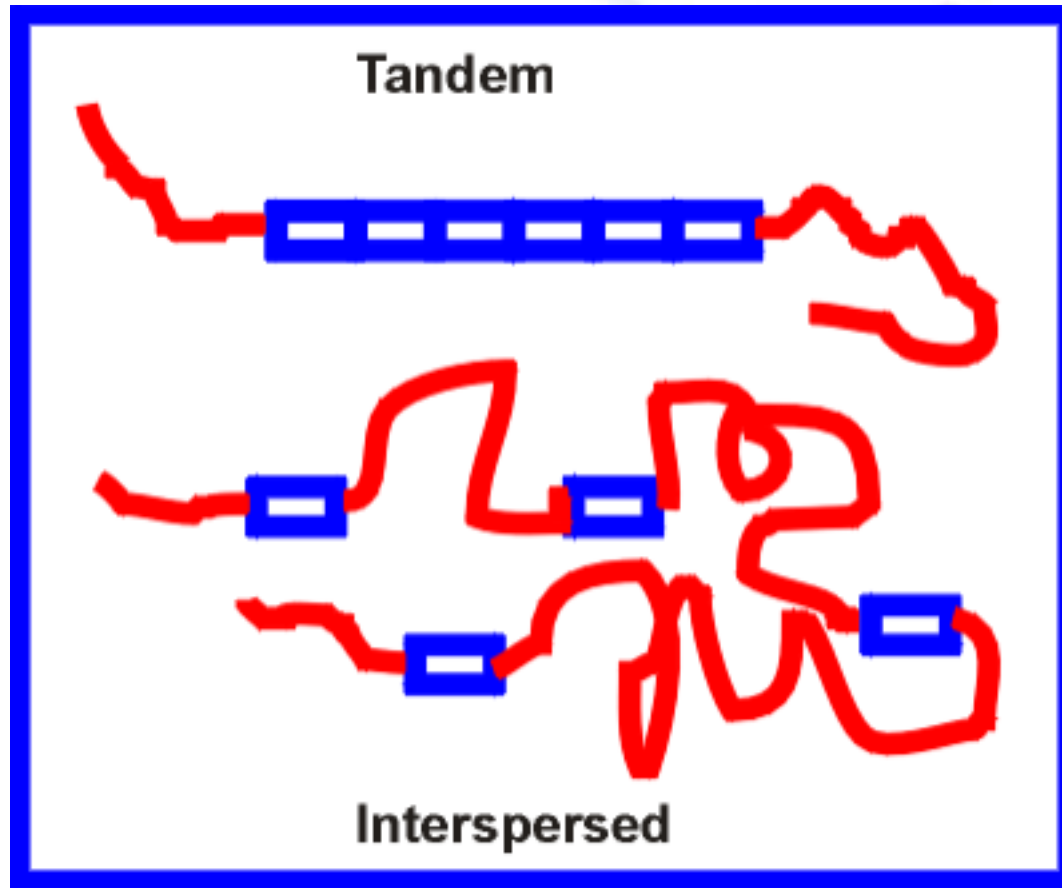
- Organelle genome is similar to prokaryotes
- High copy number
- mtDNA and cpDNA depends on Nuclear DNA

Genome Repeats

Sequence Repeats

- Repeats skew the base composition
- Contribute to differences in buoyant densities
- Repeat containing DNA can be separated as **satellite DNA** on the bases of these densities

Genome Repeats



<http://mcb1.ims.abdn.ac.uk/djs/web/lectures/repeats1.html#anchor10305>

Genome Repeats

Satellite DNA

- One to several thousand bp long
- Tandem array of 100 million bases long
- Near centromere and telomere
- Minisatellite and Nicrosatellite

Genome Repeats

Minisatellite

- 15 bases long in array of several hundred to thousands kb
- Typically in euchromatin
- VNTR e.g, used to identify human individuals in forensics

Genome Repeats

Microsatellite

- 2-6 bases long in arrays of 10-100 bases
- Inherited to offsprings
- Useful markers for genetic analysis and evolutionary studies
- Found in telomeres
TTAGGG
- SSRs and STRs

Genome Repeats

Transposable Elements (TEs)

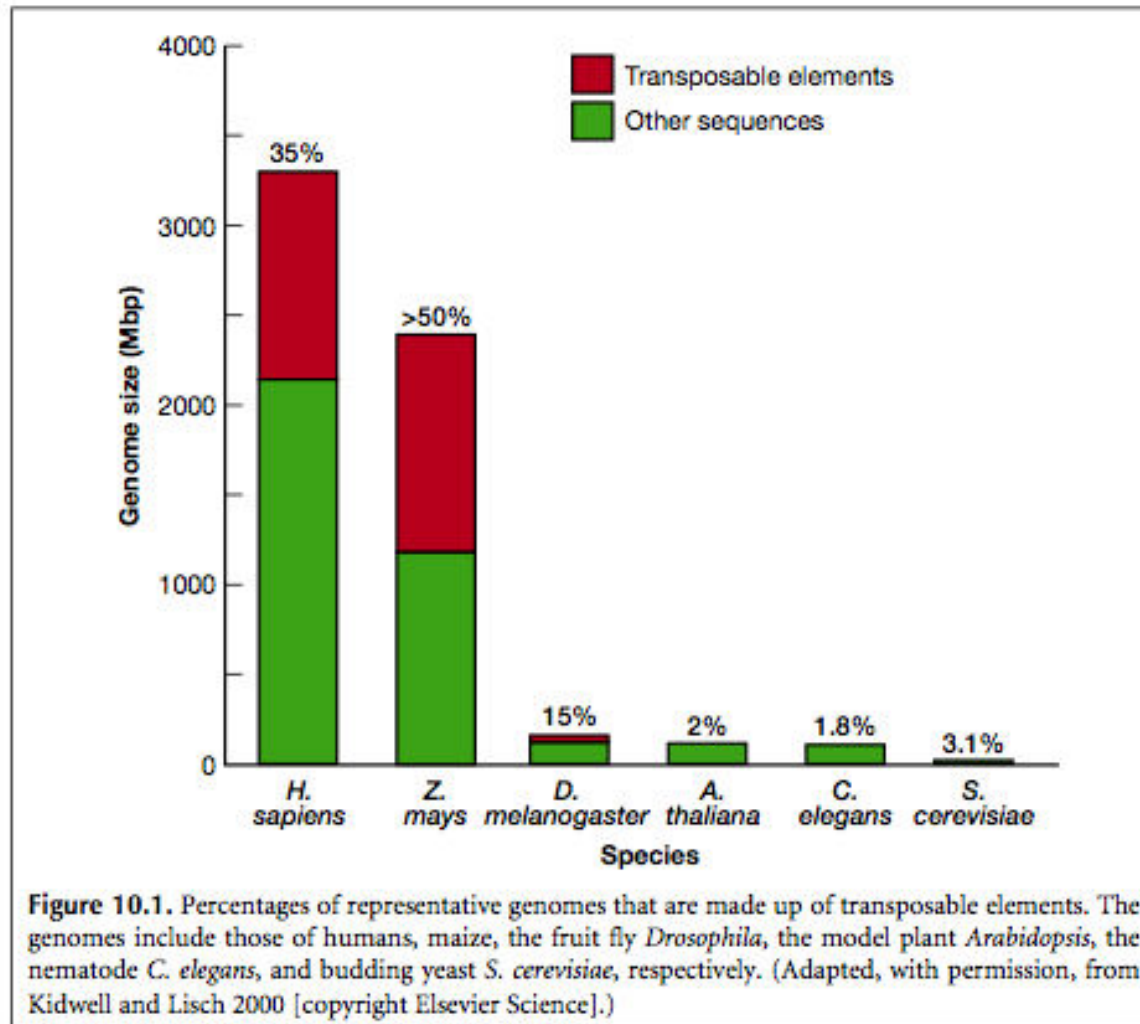
- Large portion of eukaryotic genome
- Thought to play important role in evolution of these genomes
- Move (Jump) from one location to another faster than chromosome replicate

Genome Repeats

Transposable Elements (TEs)

- Have a potential to increase in number
- Make up a large proportion of eukaryotic genome
- Detectable until blend into genome due to mutations

Genome Repeats



Genome Repeats

Conclusions

- Large proportion of eukaryotic genome is composed of repeats
- Different repeats act as markers to detect genetic variation and are also used to study evolution

Transposable Elements (TEs)

Introduction

Any segment of DNA,
able to transpose

- **Insertion Sequences (IS elements)**
 - The simplest transposable elements
 - Code only for the ability to transpose
 - prokaryotes

Transposable Elements (TEs)

IS elements

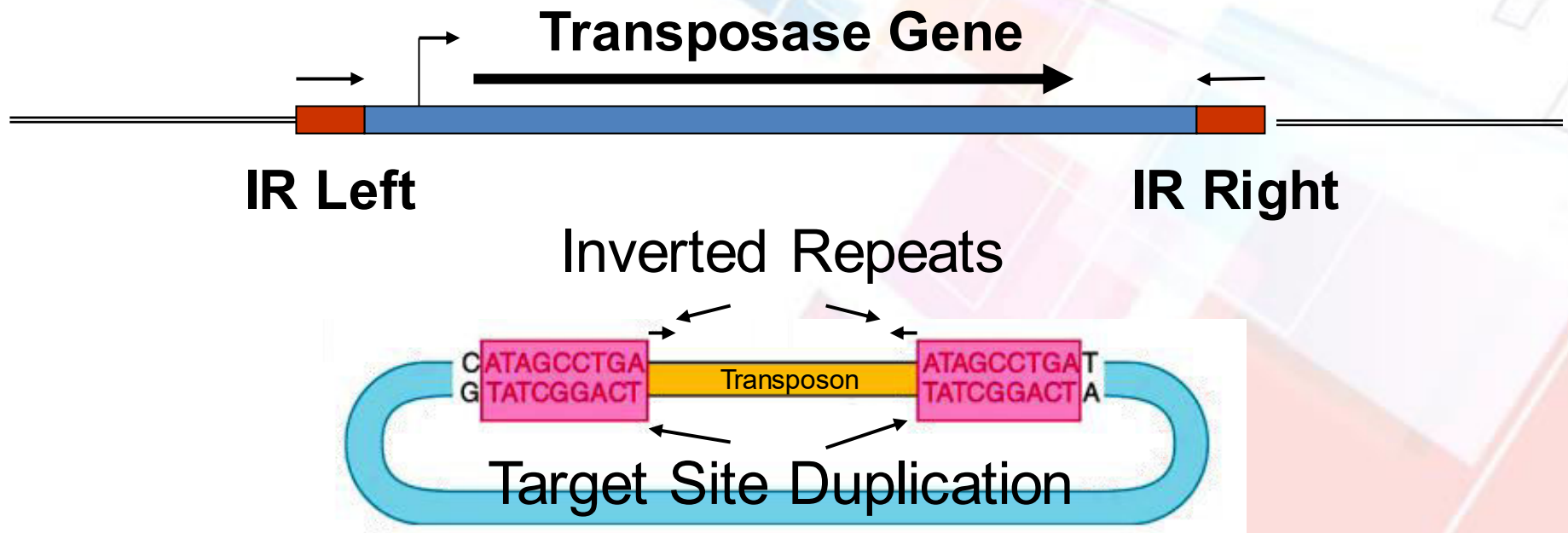
- usually very small (< 1 Kb to 2 Kb)
- Flanked by inverted repeat sequences (IRs)
- Encode at least one gene that provides their own transposition functions

Transposable Elements (TEs)

IS elements

- Do not code for noticeable (phenotypic) traits
- Can cause mutations by transposition into genes

Transposable Elements (TEs)



Transposable Elements (TEs)

- **Transposons**
 - More complex transposable elements
 - Code for the ability to transpose
 - Code for other detectable traits as well as ability to transpose

Transposable Elements (TEs)

Eukaryotic TEs

- 2 Classes; Class I and Class II
- Class I
 - Encode reverse transcriptase
 - Use RNA-mediated mechanisms of transcription

Transposable Elements (TEs)

Eukaryotic TEs

- **Class I**

Three classes

- LTR (Long Terminal Repeats)
retrotransposons
- Retrotransposons
- Retrovirus like
Elements

Transposable Elements (TEs)

Eukaryotic TEs

- **SINES** (Short Interspersed Nuclear Elements)
 - 80-300 bp
 - ***Alu* repeats**
- **LINES** (Long Interspersed Nuclear Elements)
 - 6-8 KB long

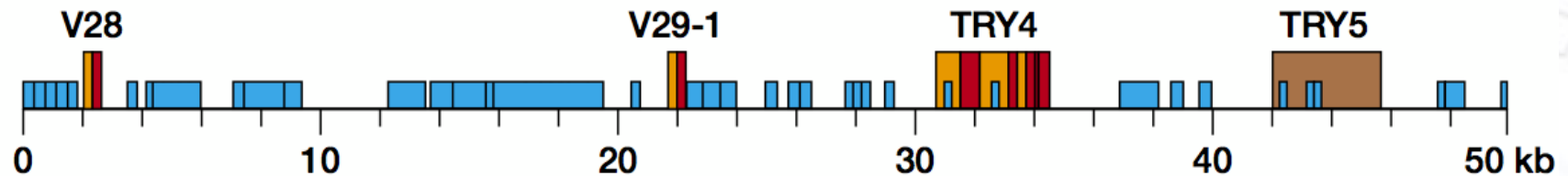
Transposable Elements (TEs)

Eukaryotic TEs

- **Class II TEs**
 - Ac-Ds in maize
 - P elements in Drosophila
- **MITES** (Miniature Inverted repeat TEs)
 - Features of both Class I and II
 - 400 bp

Transposable Elements (TEs)

A. Human



B. *Saccharomyces cerevisiae*

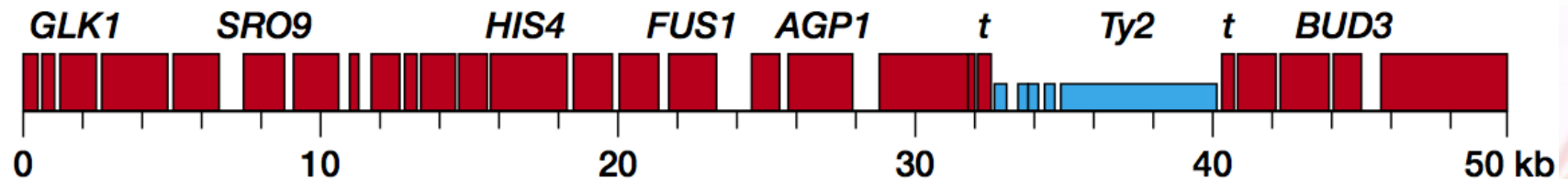
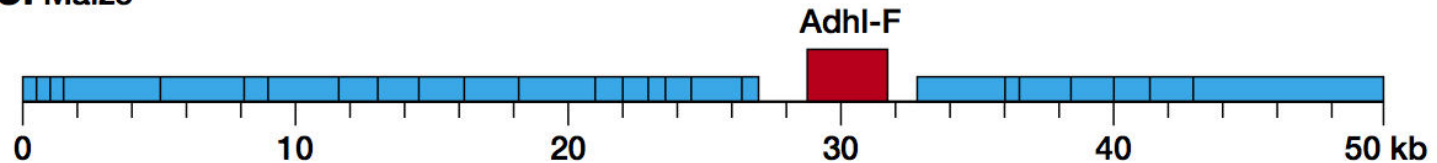


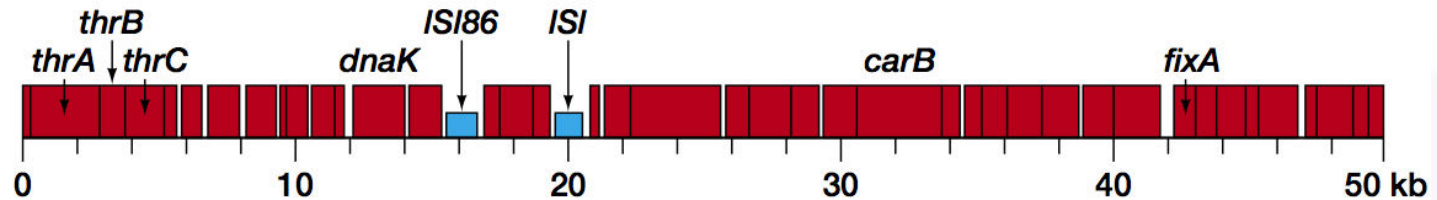
Figure 10.3. Comparison of genome composition in four genomes. (A) Human β T-cell receptor locus on chromosome 7. V28 and V29.1 encode parts of the β T-cell receptor proteins that are joined during development of the immune system (Rowen et al. 1996). TRY4, the gene for trypsinogen, and TRY5, a pseudogene related to the trypsinogen family, are not related to the receptor sequence. Why they are located here is not known. (B) Segment of yeast chromosome III (Oliver et al. 1992). (C, D) 50-kb fragments of the maize and *E. coli* chromosomes, respectively (SanMiguel et al. 1996; Blattner et al. 1997). The maize repeats are LTR retrovirus-like elements (Fig. 10.2) that have inserted within the last 3 million years (SanMiguel et al. 1998). (Redrawn, with permission, from Brown 1999 [BIOS Scientific].)

Transposable Elements (TEs)

C. Maize



D. *Escherichia coli*



KEY

Gene
 Intron
 Human pseudogene
 Genome-wide repeat
 t tRNA gene

Figure 10.3. Comparison of genome composition in four genomes. (A) Human β T-cell receptor locus on chromosome 7. V28 and V29.1 encode parts of the β T-cell receptor proteins that are joined during development of the immune system (Rowen et al. 1996). TRY4, the gene for trypsinogen, and TRY5, a pseudogene related to the trypsinogen family, are not related to the receptor sequence. Why they are located here is not known. (B) Segment of yeast chromosome III (Oliver et al. 1992). (C, D) 50-kb fragments of the maize and *E. coli* chromosomes, respectively (SanMiguel et al. 1996; Blattner et al. 1997). The maize repeats are LTR retrovirus-like elements (Fig. 10.2) that have inserted within the last 3 million years (SanMiguel et al. 1998). (Redrawn, with permission, from Brown 1999 [BIOS Scientific].)

Transposable Elements (TEs)

Conclusions

- Transposable elements make up a significant part of eukaryotic genome
- Move within and across genomes
- Cause genome expansion

Eukaryotic Gene Structure

Eukaryotic genes

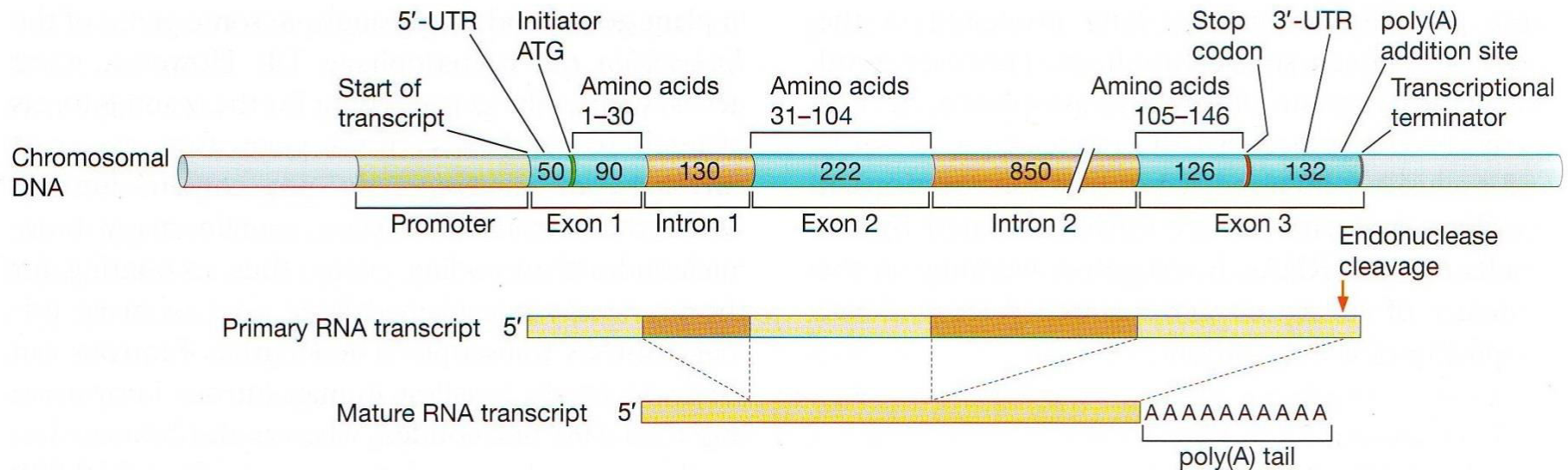
- Eukaryotic genes are complicated
- They possess exons which are protein coding regions and are interrupted with introns

Eukaryotic Gene Structure

Eukaryotic genes

- Whole gene is transcribed as a large transcript
- Introns are spliced leaving an ORF that is translated

Eukaryotic Gene Structure



Human β -globin gene

The “Universal Tree of Life” is based on comparisons of ribosomal small subunit RNAs

Eukaryotic Gene Structure

Eukaryotic genes

- Introns start on the 5' ends as GT and end as AG towards 3' end **GT-----
-AG**
- This trend is highly conserved

Eukaryotic Gene Structure

Introns proportion

- A small proportion in yeast, only 239 introns in genome
- Human genes have hundreds, often 95% of genes
- Might have embedded genes

Eukaryotic Gene Structure

Intron origin

- **Intron-early**
 - used to assemble first genes from ancient conserved introns
- **Intron-late**
 - Broke up previously continuous genes by inserting into them

Eukaryotic Gene Structure

Number of Genes

- Degree of compactness contributes to variation in gene density
- Compact genomes have higher genome density

Eukaryotic Gene Structure

Organism	Genome size (haploid MB)	Predicted genes
<i>A. thaliana</i> (plant)	130	~25,000
<i>C. elagans</i> (worm)	100	18,424
<i>Drosophila melangaster</i>	180	13,601
<i>Escherichia coli</i>	4.7	4,288
<i>Homo sapiens</i> (human)	3000	45,000 – 120,000
<i>S. cerevisiae</i> (yeast)	13.5	6,241

Mount: Table 10.3, 11.3 2nd edn

Eukaryotic Gene Structure

E.Coli 1 gene/4kb

AT & C. Elegans 1gene/6kb

D.M 1/14 kb

Human 1 gene/80 kb, half of it
are repeats so it's 1/40 kb

Pseudogenes

- Non functional genes
- Derived from functional genes through mutations following genome duplication
- Processed pseudogenes lack introns and promoters

Eukaryotic Gene Structure

Gene families

- Set of genes having similar sequences and functions
- Gene families arise from gene duplication and subsequent divergence

Eukaryotic Gene Structure

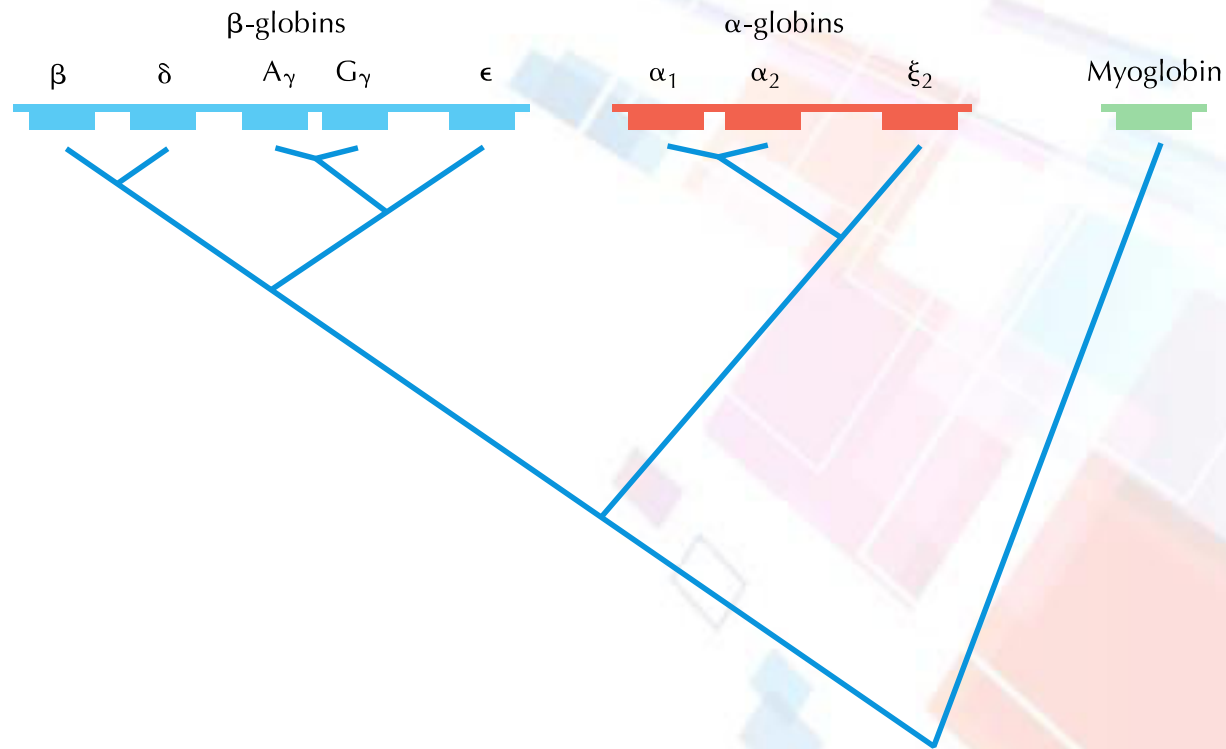


FIGURE 27.29. Gene duplications during the evolution of the human globin gene families. The initial split gave rise to two lineages, one leading to the modern gene for myoglobin and the other to the globin genes. Subsequently, the proto- α -globin and proto- β -globin lineages split following a duplication. Other duplications took place within the α and β lineages. (Modified from Strachan T. and Read A.P. *Human Molecular Genetics 2*, Fig. 14.16, © 1999 Garland Science.)

Eukaryotic Gene Structure

Conclusions

- Eukaryotic genes have exons and introns
- Introns make up a significant portion of Human genome
- Pseudogenes are non functional genes
- Similar genes make gene families

Comparative Genomics

Introduction

- Comparison of gene number, gene content and gene location in both prokaryotic and eukaryotic groups of organisms

Comparative Genomics

Introduction

- Availability of genome makes possible a comparison of all the proteins (proteome) encoded by one organism with those of another

Comparative Genomics

Orthologs

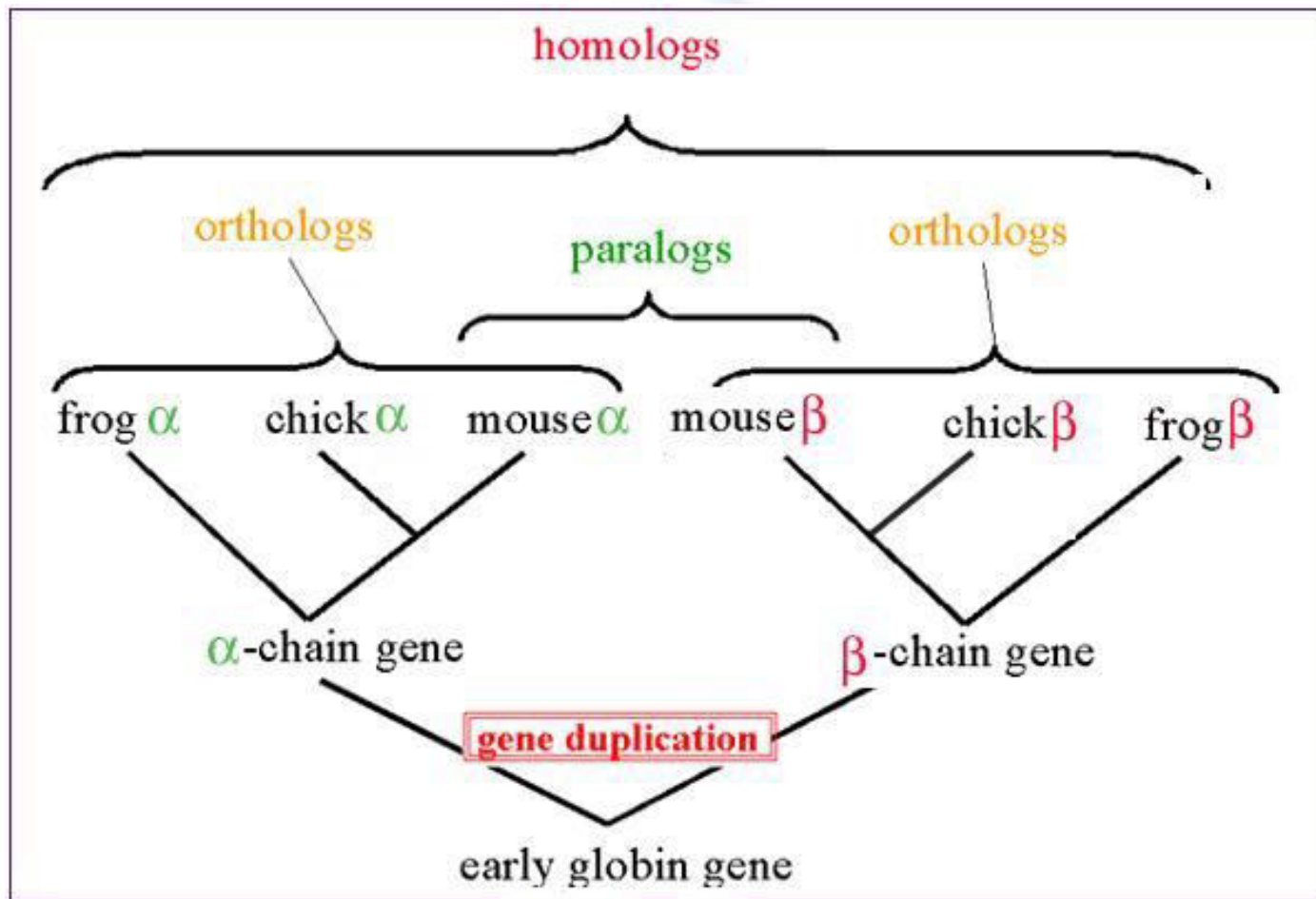
- Genes in two organisms that are so similar that they must have the same function and evolutionary history are called orthologs (Fitch, 1970)

Comparative Genomics

Paralogs

- Gene families originating from rare gene duplication events over the evolutionary time are called paralogs
- Unlike pseudogenes, 2nd copy of gene remains functional

Comparative Genomics



<http://www.ncbi.nlm.nih.gov/Education/BLASTinfo/Orthology.html>

Comparative Genomics

Drosophila and yeast

- *Drosophila* has core proteome only twice the size of that of yeast
- *Drosophila* proteome is more similar to mammalian proteomes than worm or yeast

Comparative Genomics

Drosophila and *C. elegans*

- Despite the large differences between fly and worm, they use a core proteome of similar size
- Nearly 30% of fly genes have putative orthologs in the worm

Comparative Genomics

Drosophila and Human

- Some human disease genes absent in *Drosophila*
- A number of previously unknown counterparts to human cancer and neurological disorders genes

Comparative Genomics

Conclusions

- Comparative genomics reveals the relationship among different organisms
- Fruit fly has more similarities with mammals

Genetics and Genomics

Functional Genomics

Functional Genomics

Functional Genomics

- Functions of genes, their regulation and end products.
- Functional genomics analyzes all genes in genomes to determine their functions and their gene control and expression.

Functional Genomics

Functional Genomics

- Classically, genetics analysis begins with a phenotype and moves for identification of genes.
- New approaches are needed to work in the opposite direction, from genes to phenotype.

Functional Genomics

**Functional Genomics
relies on Molecular
Biology, Biochemistry,
Genetics and
Bioinformatics tools**

- Functional genomics relies on molecular biology lab research and sophisticated computer analysis by bioinformatics tools.

Functional Genomics

Functional Genomics

- Fusion of biology with maths and computer science is used for many things. Examples:
- Finding genes within a genomic sequence.
- Aligning DNA/proteins sequences.

Functional Genomics

Functional Genomics – what is included

- Subtracted cDNA libraries
- Differential display
- Representational difference analysis
- Suppression subtractive hybridization

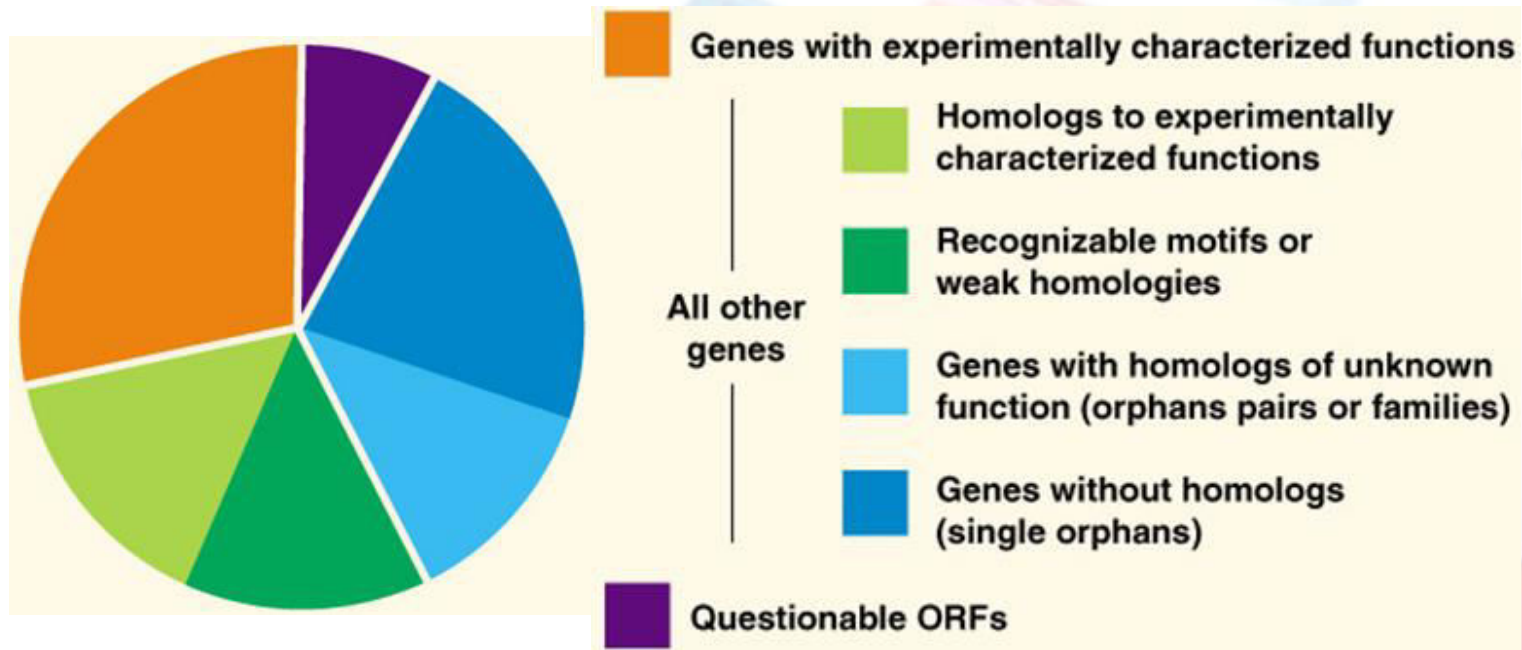
Functional Genomics

Functional Genomics – what is included

- cDNA Microarrays
- Serial analysis of
gene expression
- 2-D Gel
electrophoresis

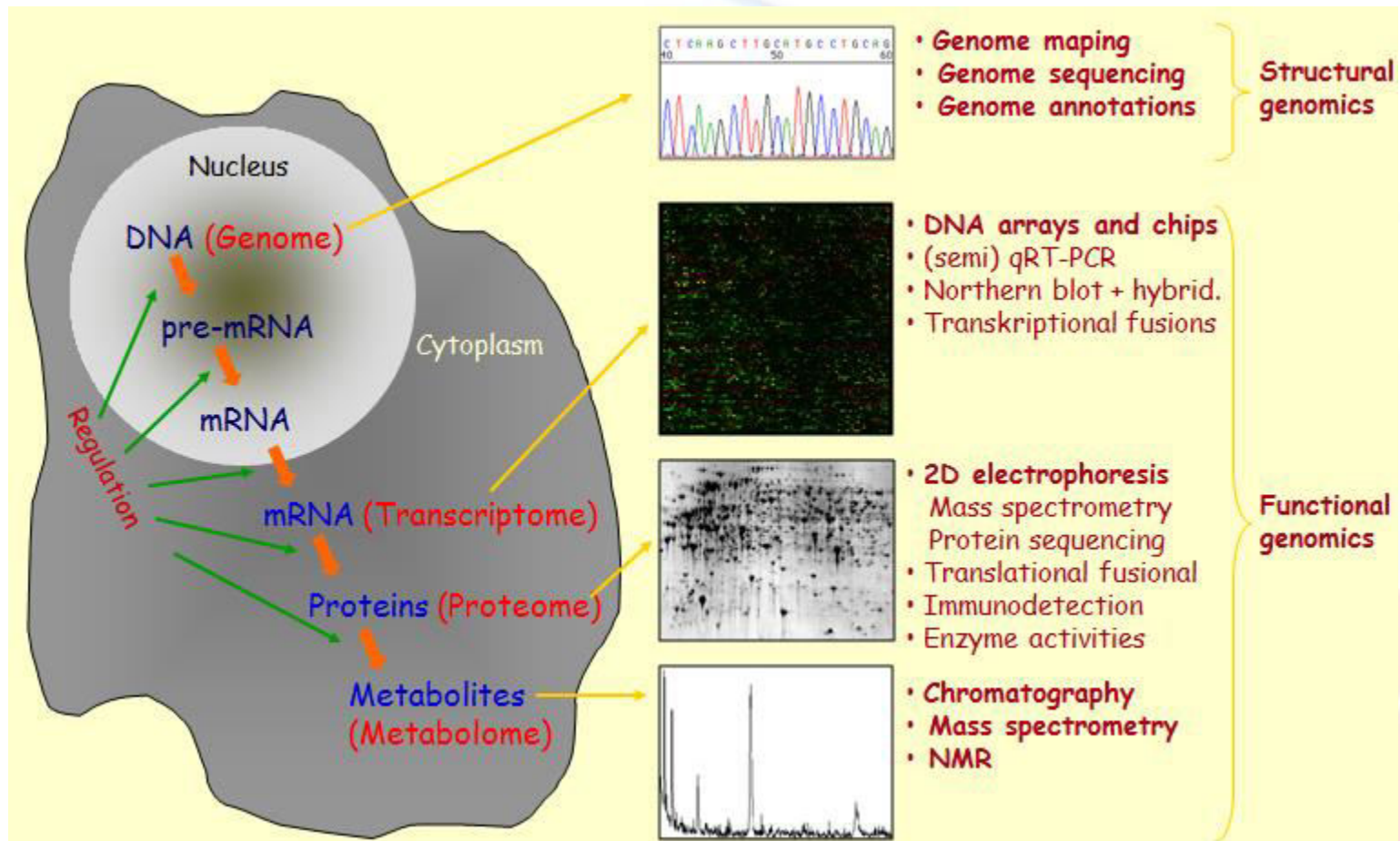
Functional Genomics

Yeast Genome with Experimentally Characterized Functions



Functional Genomics

Functional Genomics



Functional Genomics

Functional Genomics - Conclusion

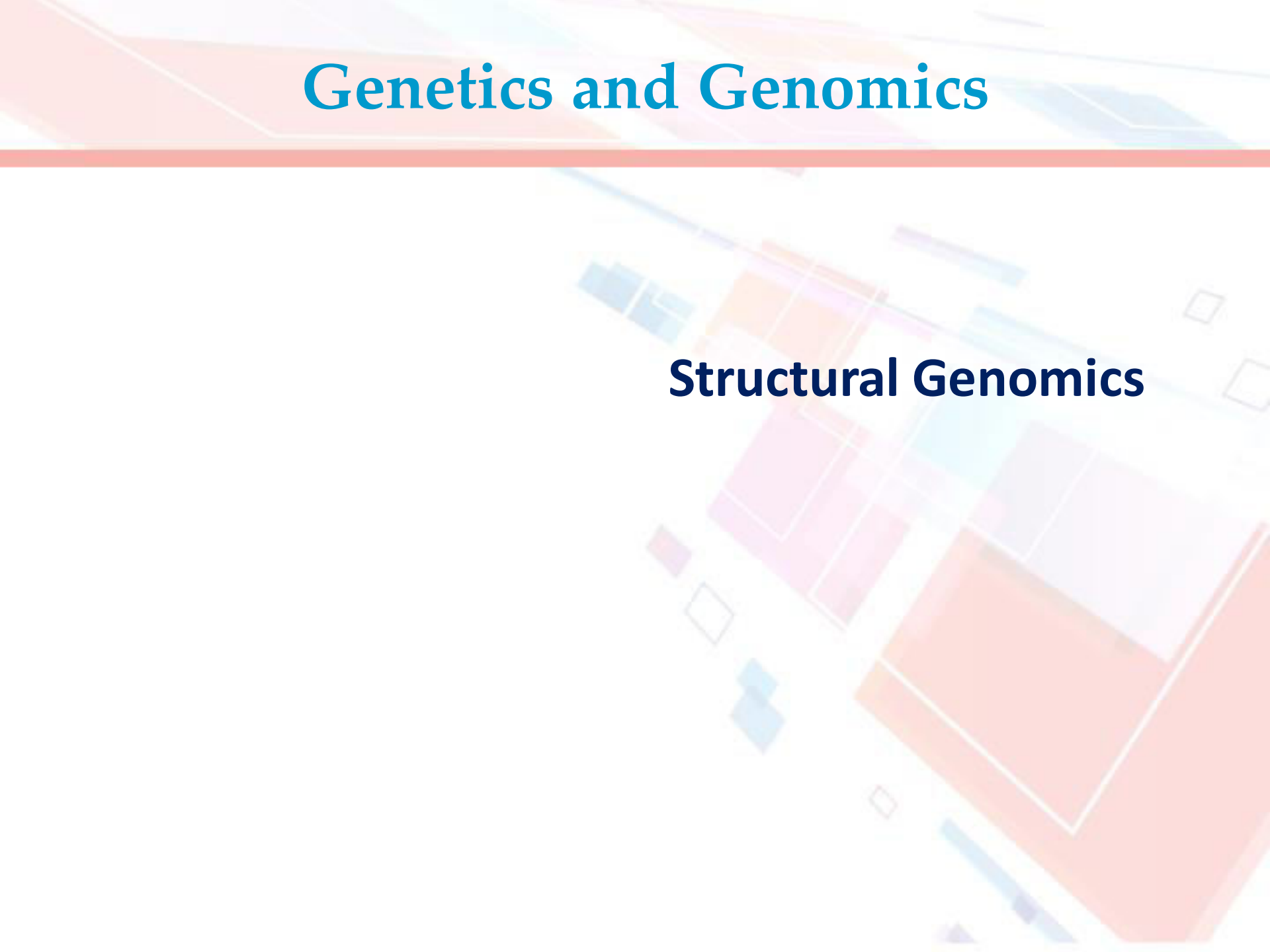
- Function of gene products.
- Describing interactions between genes and gene products in the cell, between cells and between organisms.

Functional Genomics

Functional Genomics - Conclusion

- Considering phylogenetic relationships.

Genetics and Genomics

The background of the slide features an abstract design of overlapping, semi-transparent rectangles in various colors including shades of red, orange, blue, and purple. These shapes are arranged in a way that creates a sense of depth and movement, with some lines intersecting to form a grid-like pattern in certain areas.

Structural Genomics

Structural Genomics

Structural Genomics

- The ultimate goal of genomic studies is to determine the nucleotide sequences of entire genomes of organisms.
- The genetic and physical mapping and sequencing of chromosomes.

Structural Genomics

Genetic Mapping

- Genetic mapping - approximate locations of genes, relative to the locations of other genes, based on the rates of recombinations.

Structural Genomics

Physical Mapping

- Physical mapping is based on the direct analysis of DNA.
- Physical mapping places genes on the genomes in relation to distances measured in bp, kbp, and mbp.

Structural Genomics

Structural Genomics includes

- Distinct components of genomes
- Abundance and complexity of mRNA
- Genome sequences
- Gene numbers
- Coding and non-coding DNA

Structural Genomics

Structural Genomics – Complex Genomes

- Complex genomes have roughly 10x to 30x more DNA than is required to encode all the RNAs or proteins in the organism

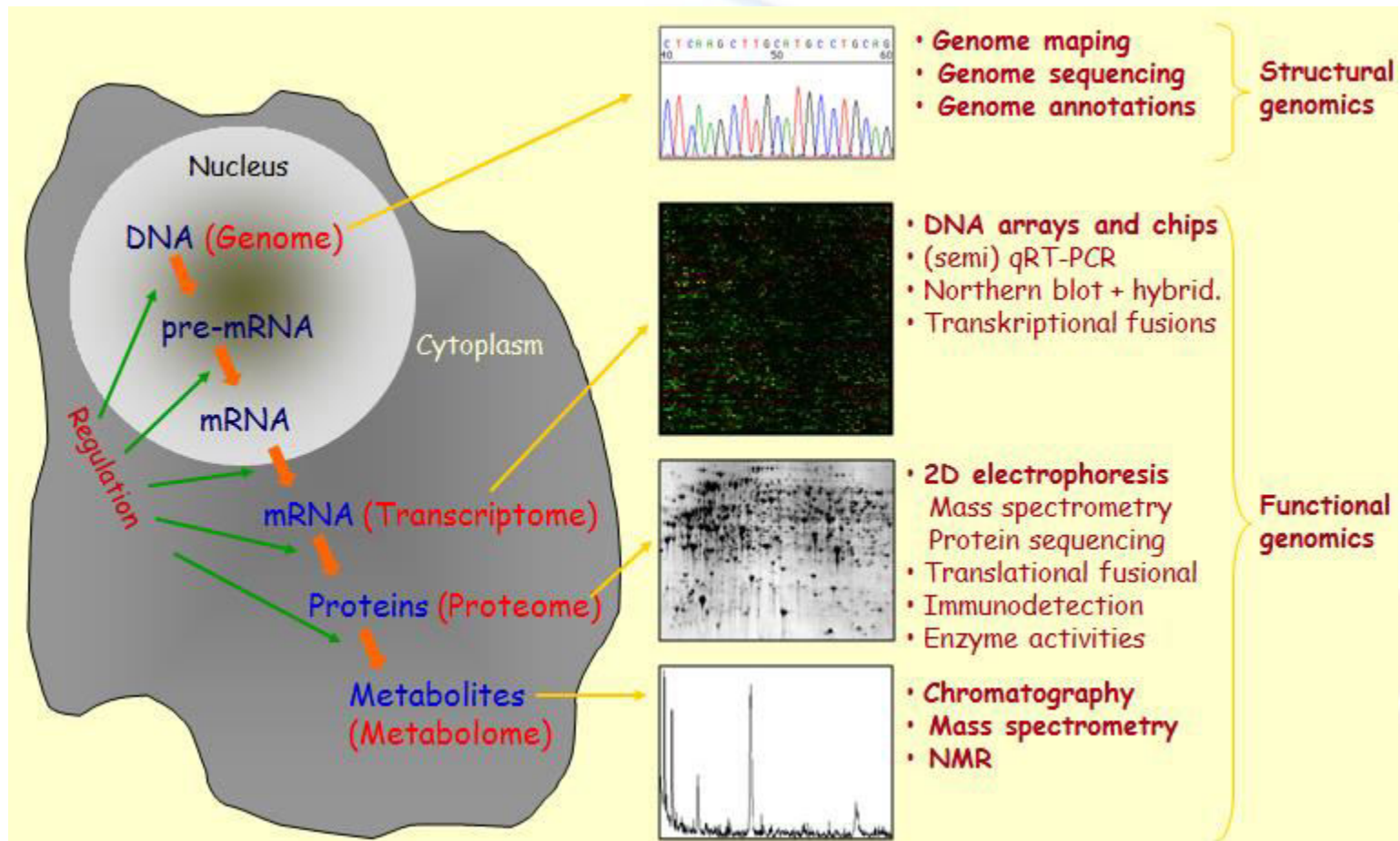
Structural Genomics

Structure of Complex Genomes

- Introns in genes
- Regulatory elements of genes
- Multiple copies of genes, including pseudogenes
- Intergenic sequences
- Interspersed repeats

Structural Genomics

Structural Genomics



Structural Genomics

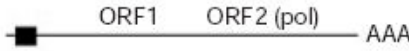
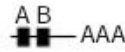
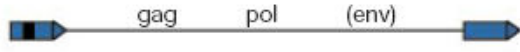
Structural Genomics – Transposable Elements

Vast majority of TEs can be classified into four families:

- **LINEs** (Long Interspersed Nuclear Elements, autonomous)
- **SINEs** (Short Interspersed Nuclear Elements, use LINE proteins for life cycle)
- **LTR elements** (Long Terminal Repeats; derived from retroviruses)
- **DNA transposons** (replicate without RNA intermediary)

Structural Genomics

Structural Genomics – Repetitive DNA in Humans

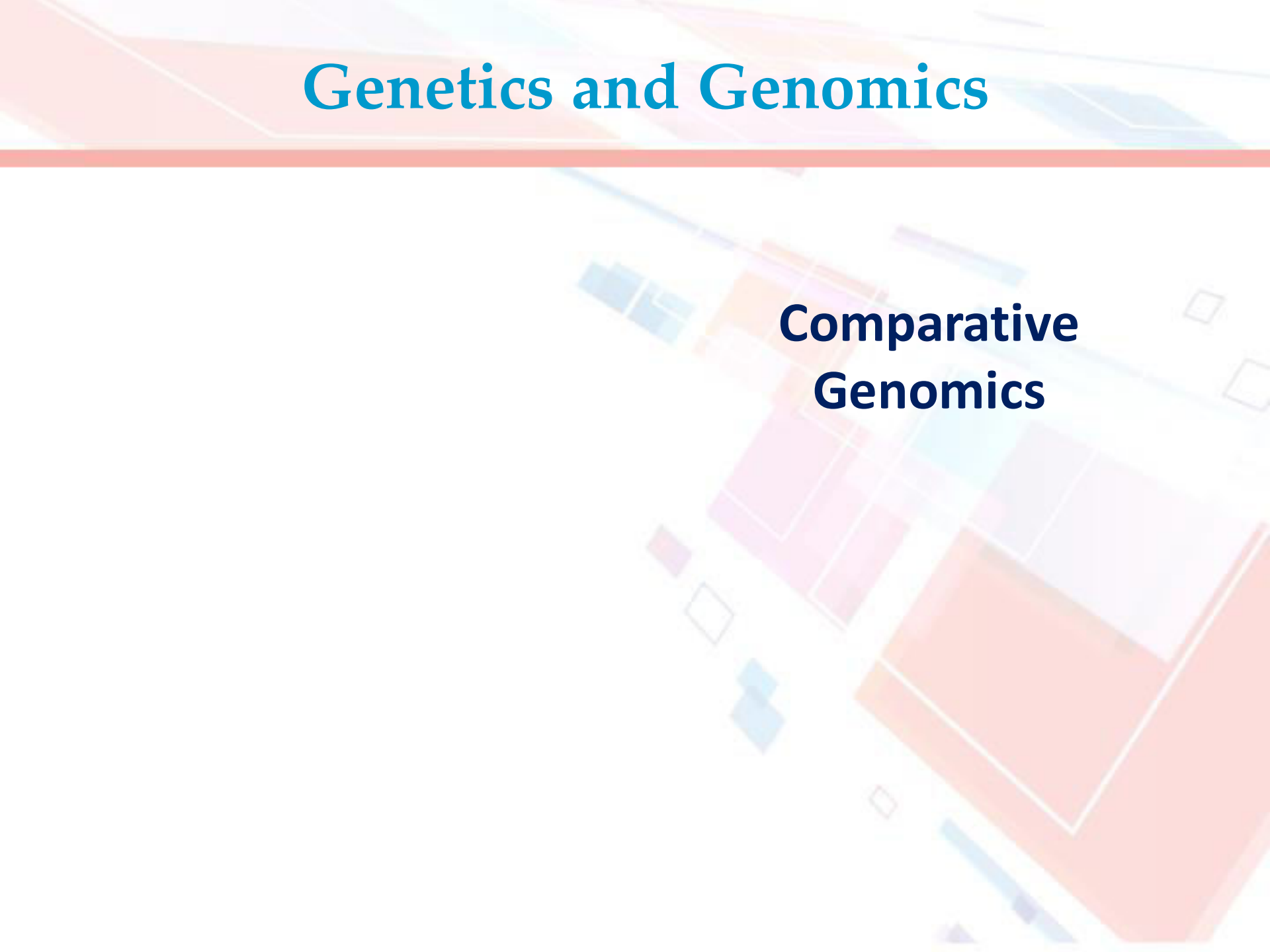
			Length	Copy number	Fraction of genome
LINES	Autonomous		6–8 kb	850,000	21%
SINEs	Non-autonomous		100–300 bp	1,500,000	13%
Retrovirus-like elements	Autonomous		6–11 kb	450,000	8%
	Non-autonomous		1.5–3 kb		
DNA transposon fossils	Autonomous		2–3 kb	300,000	3%
	Non-autonomous		80–3,000 bp		

Structural Genomics

Structural Genomics - Conclusion

- To determine the nucleotides sequences of genomes.
- To study genetic maps
- To study physical maps.

Genetics and Genomics



Comparative Genomics

Comparative Genomics

Comparative Genomics

- Comparison of gene numbers, gene locations and biological functions of genes, in the genomes of different organisms.

Comparative Genomics

Comparative Genomics

- To identify and compare groups of genes that play a unique biological role in different organisms.

Comparative Genomics

Homology

- Homology is the relationship of any two characters (genes or proteins) that have descended, usually through divergence, from a common ancestor/ ancestral character.

Comparative Genomics

Homologues

- Homologues are thus components or characters (such as genes/proteins with similar sequences) that can be attributed to a common ancestor of the two organisms during evolution.

Comparative Genomics

Homologues can be

- Orthologues
- Paralogues
- Xenologues
- Analogues

Comparative Genomics

Orthologues

- Orthologues are homologues that have evolved from a common ancestral gene by speciation.
- They usually have similar functions.

Comparative Genomics

Paralogues

- Paralogues are homologues that are related or produced by duplication within a genome followed by subsequent divergence.
- They often have different functions.

Comparative Genomics

Xenologues

- Xenologues are homologous that are related by an interspecies (horizontal transfer) of the genetic material for one of the homologues.
- The functions of the xenologues are quite often similar.

Comparative Genomics

Analogues

- Analogues are non-homologues genes/proteins that have descended convergently from an unrelated ancestor.
- Similar function, different sequence or structure.

Comparative Genomics

Comparative Genomics – Evolutionary Trends

- Comparison of genomes belonging to relatively similar group, like mammals, reveals some evolutionary trends.

Comparative Genomics

Comparative Genomics – Functional elements

- Provides a powerful and general approach to identify functional elements without previous knowledge of functions.

Comparative Genomics

Comparative Genomics – Ancestral Genome

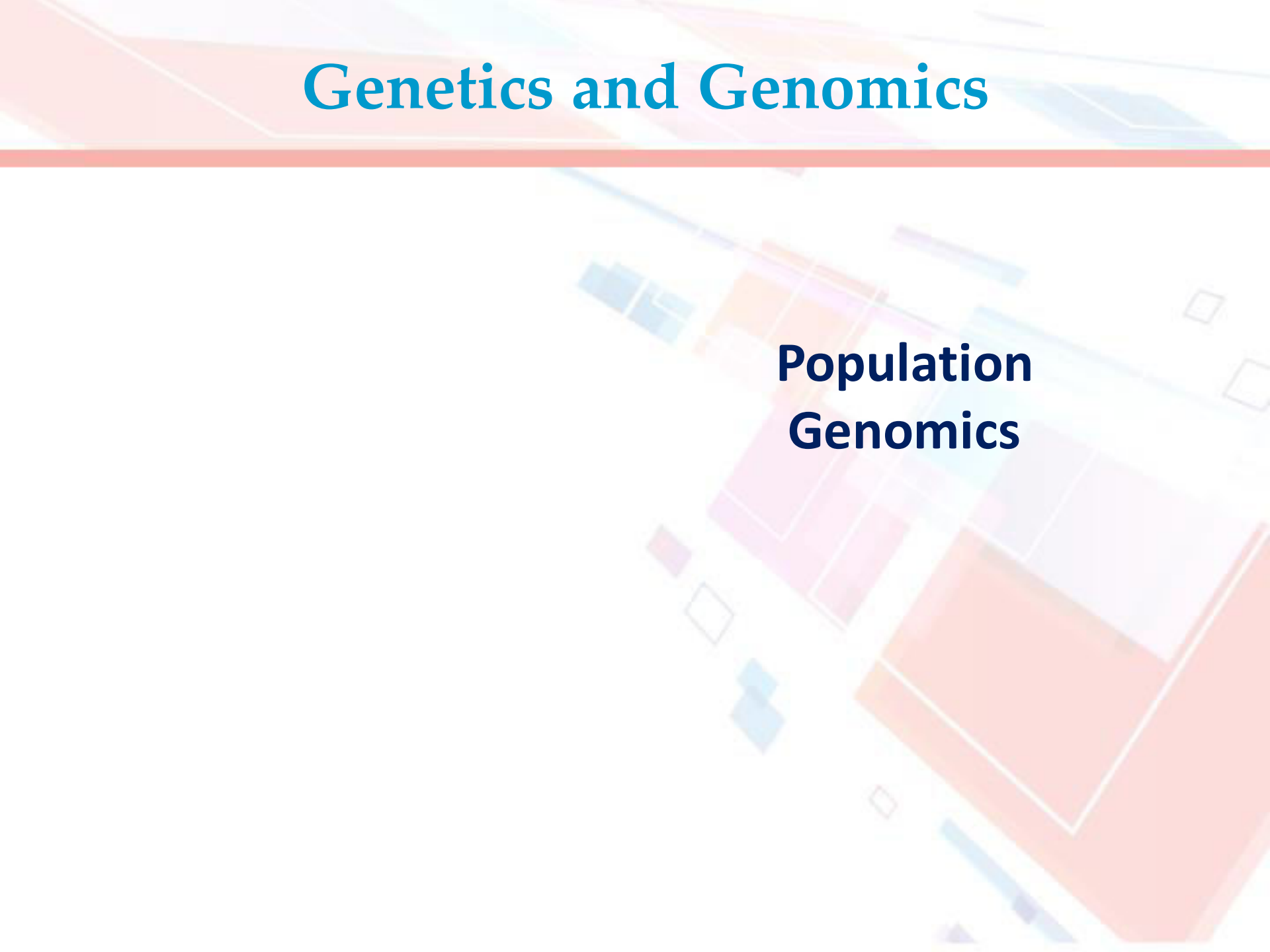
- Reconstruction of an ancestral genome for a group of organism.

Comparative Genomics

Conclusion

- Comparative genomics is the comparison of gene numbers, gene locations and biological functions of genes, in the genomes of different organisms.

Genetics and Genomics



Population Genomics

Population Genomics

Population Genomics

- Study of genomes of a specific population, strains, varieties or organisms.
- Study about the genetic diversity.
- Understanding new insights into disease and drug response.

Population Genomics

Population Genomics

- Deeper insights about genomic size.
- Further complexities in the genomes of the organisms.
- Variations between individuals or strains.

Population Genomics

Population Genomics

- Human genome is 200 times larger than yeast but 200 times smaller than Amoeba.
- Less than 2% of human genome is coding sequence

Population Genomics

1000 Genomes - Population Genomics

- International research consortium sequenced the genomes of at least 1000 people from around the world.
- Detailed and medically useful pictures of human genome variations.

Population Genomics

1000 Genomes - Population Genomics

- Any two humans are more than 99% identical at genetic level.
- Genetic variations may explain individual differences in susceptibility to diseases, responses to drugs.

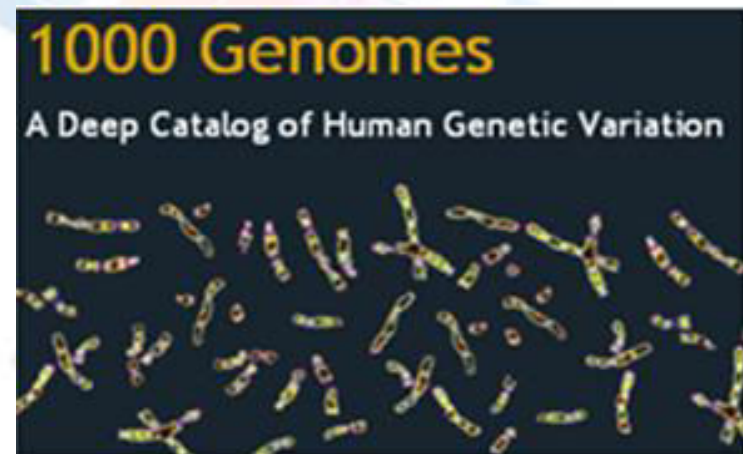
Population Genomics

HapMap Project - Population Genomics

- The HapMap project has already discovered many regions of the genome containing genetic variations associated with common human diseases.

Population Genomics

Population Genomics - HapMap Project and 1000 Genomes Project



Population Genomics

Goals of Population Genomics

- Produce a catalog of variants present at 1% or greater frequency in the human population.
- Down to 0.5 percent or lower within genes.
- Increase sensitivity of disease discovery.

Population Genomics

Goals of Population Genomics

- Provide better understanding of very rare genetic diseases.
- Understand contribution of common variants to common diseases like diabetes and heart diseases.

Population Genomics

Goals of Population Genomics

- Identify SNP but also large differences like rearrangements, deletions or duplications

Genetics and Genomics

Metagenomics

Metagenomics

Metagenomics

- Metagenome-environmental genome.
- Collection of genes sequenced from the environment could be analyzed in a way analogous to the study of a single genome.

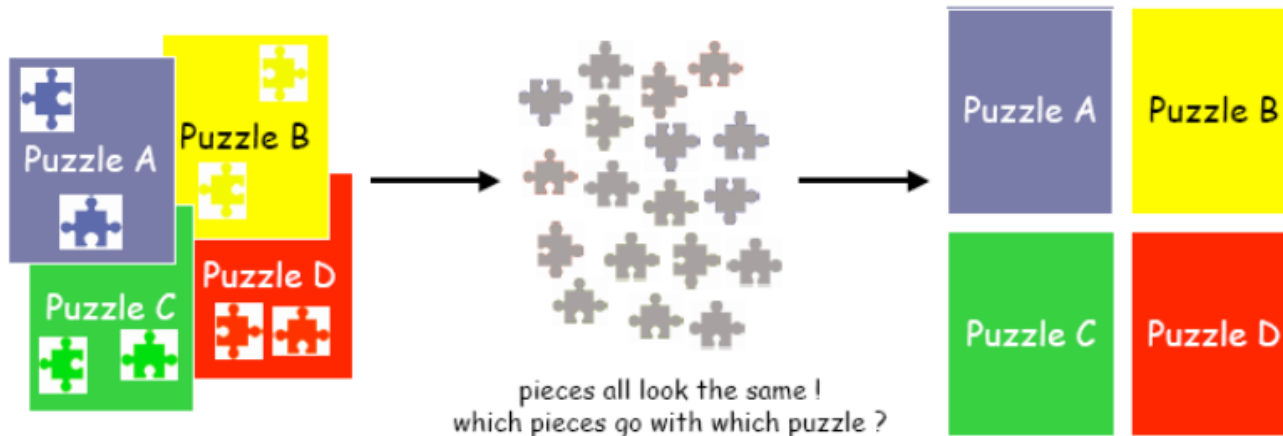
Metagenomics

Metagenomics

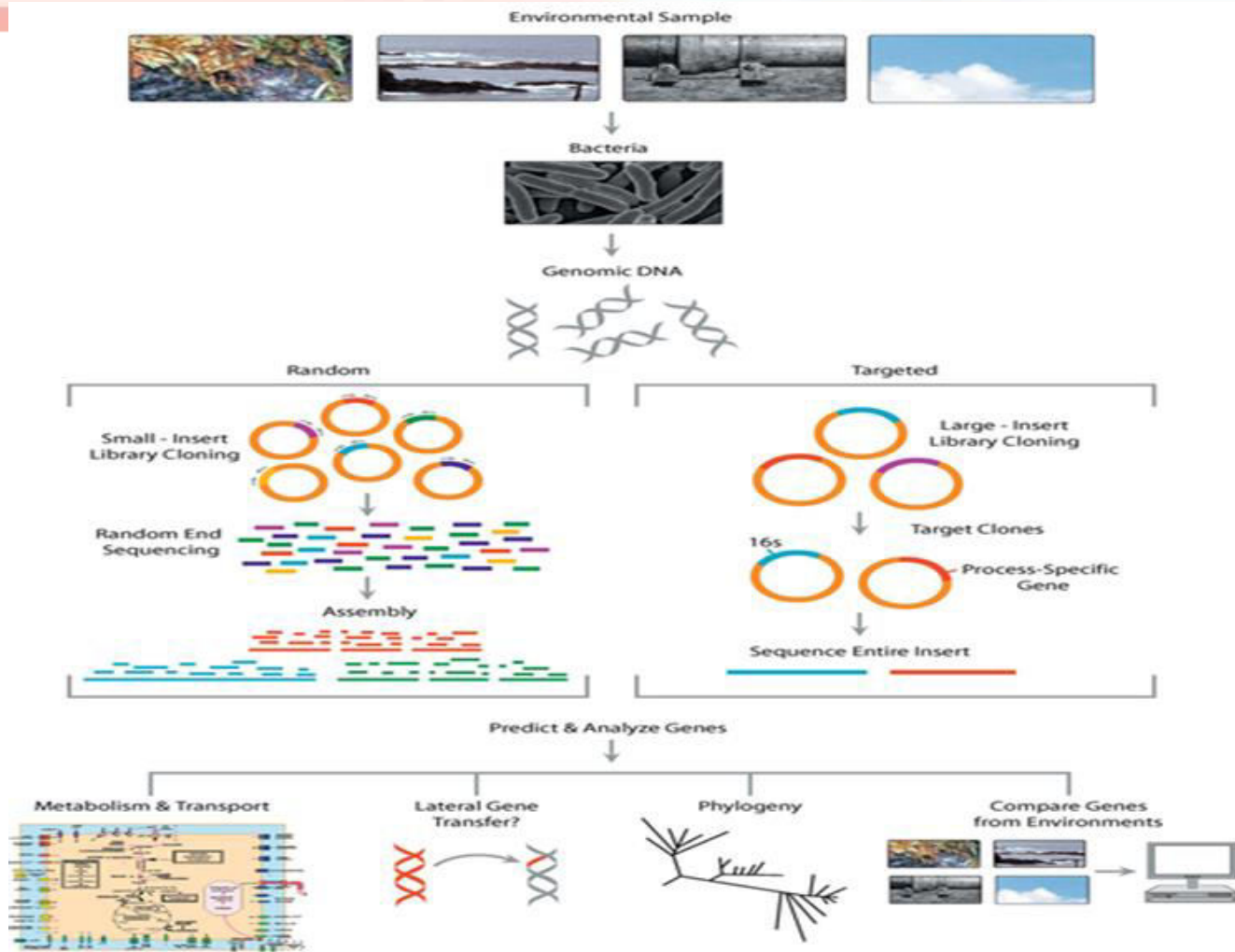
Isolate genome - single source of DNA



Environmental genome - multiple sources of DNA

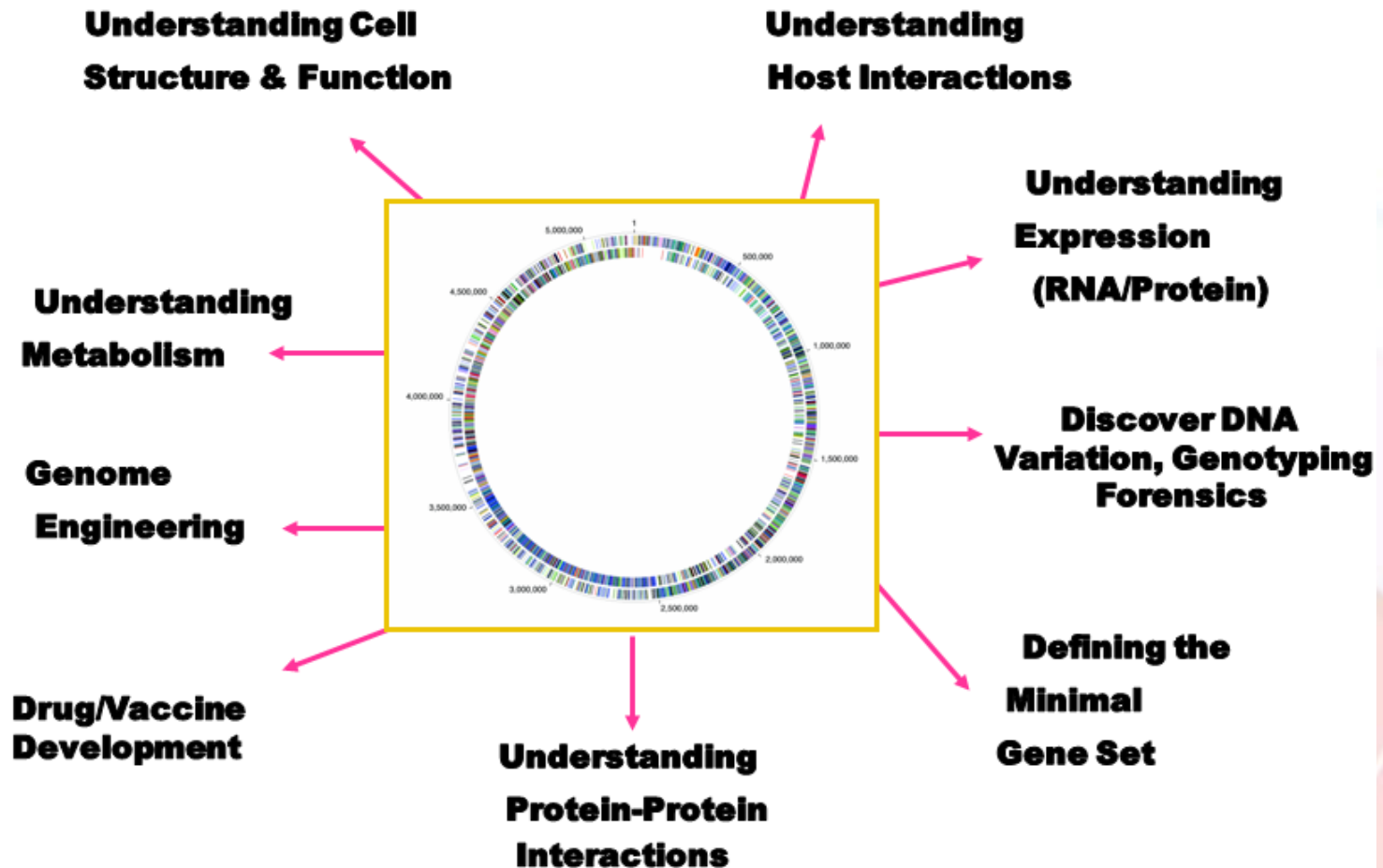


Metagenomics



Metagenomics

Why do Metagenomics



Metagenomics

Objectives of Metagenomics

- Examining phylogenetic diversity using 16s rRNA
- Diversity patterns of microorganisms for monitoring and predicting environmental conditions/change.

Metagenomics

Objectives of Metagenomics

- Examining genes/operons for desirable enzymes (cellulases, lipases, antibiotics, other natural products).
- Exploited for industrial or medical applications.

Metagenomics

Objectives of Metagenomics

- Examining secretory, regulatory, and signal transduction mechanisms associated with samples or genes of interest.

Metagenomics

Objectives of Metagenomics

- Examining bacteriophage or plasmid sequences. These potentially influence diversity and structure of microbial communities.

Metagenomics

Objectives of Metagenomics

- Examining potential lateral gene transfer events. Knowledge of genome plasticity may give us an idea of selective pressures for gene capture and evolution within a habitat.

Metagenomics

Objectives of Metagenomics

- Examining metabolic pathways.
- Directed approach towards designing culture media.
- Examining genes that predominate in a given environment compared to others.

Metagenomics

Metagenomics Data

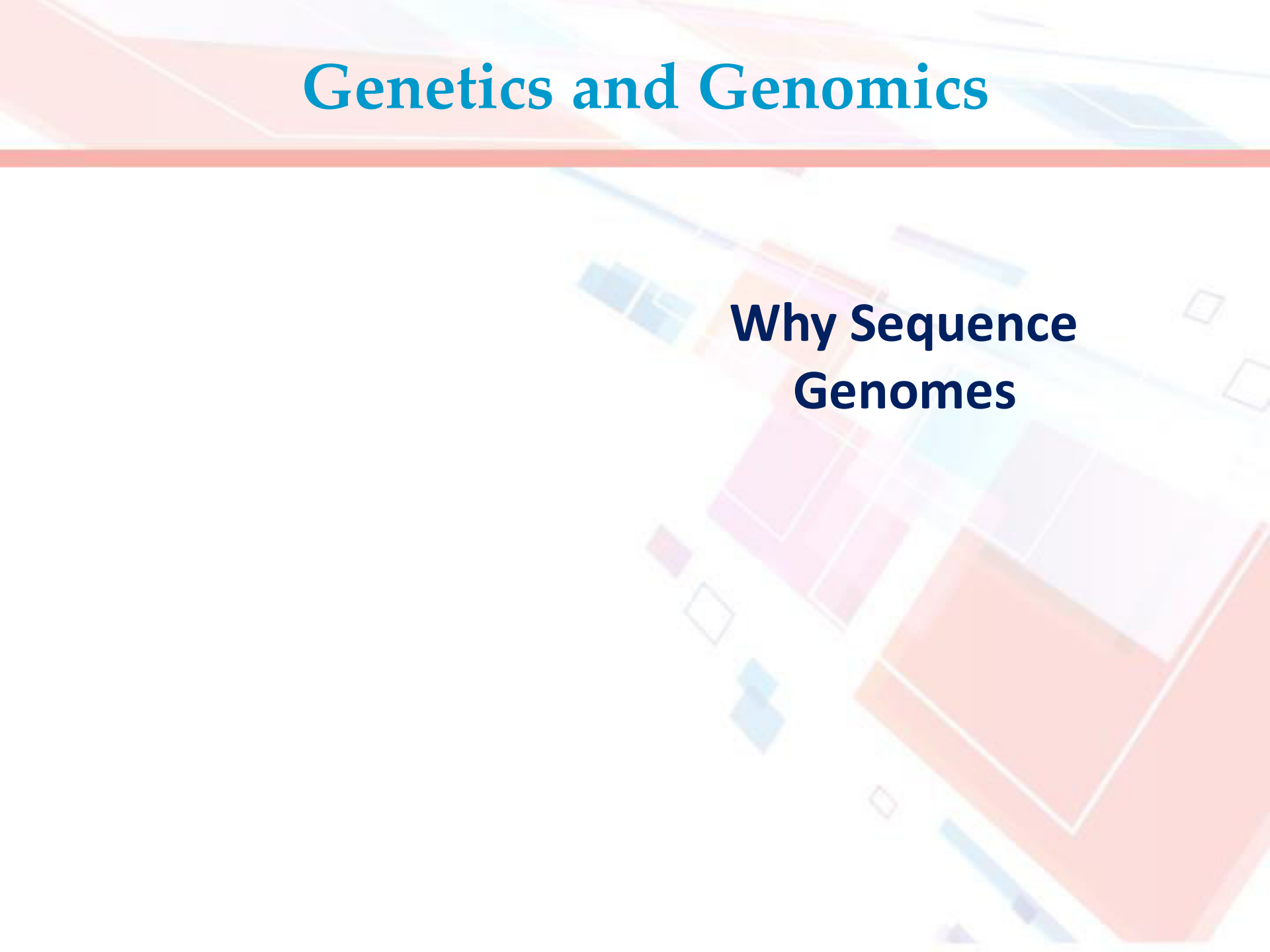
- Metagenomic data can be used towards designing low and high throughput experiments focused on defining the roles of genes and microorganisms in the establishment of dynamic microbial community.

Metagenomics

Conclusion

- Metagenome-environmental genome

Genetics and Genomics



Why Sequence Genomes

Why Sequence Genomes

Why Sequence Genomes

- To identify gene numbers, their locations on genomes, and to study their functions.
- Genes regulation

Why Sequence Genomes

Why Sequence Genomes

- DNA sequence
- Genome organization
- Chromosomal structure and organization

Why Sequence Genomes

Why Sequence Genomes

- Noncoding DNA types, amount, distribution and functions.
- Coordination of gene expression, protein synthesis, and post-translational events.

Why Sequence Genomes

Why Sequence Genomes

- Interaction of proteins in complex molecular machines
- Predicted vs experimentally determined gene function
- Evolutionary conservation

Why Sequence Genomes

Why Sequence Genomes

- Proteins structure and function.
- Proteomes (total protein content and function) in organisms.
- Correlation of SNPs with health and disease

Why Sequence Genomes

Why Sequence Genomes

- Disease-susceptibility prediction based on gene sequence variation
- Genes involved in complex traits and multigene diseases

Why Sequence Genomes

Novel Diagnostics

- Complex systems biology, developmental genetics.
- To provide platform for microchips and DNA microarrays.
- Gene expression - RNA

Why Sequence Genomes

Novel Diagnostics

- Complex systems biology, developmental genetics and genomics

Why Sequence Genomes

Novel Therapeutics

- Drug target discovery
- Rational drug design
- Molecular docking
- Gene therapy
- Stem cell therapy

Why Sequence Genomes

Understanding Metabolism

- To understand the metabolism of cells and tissues within different organisms.

Why Sequence Genomes

Understanding mechanism of diseases

- Inherited diseases
- Infectious diseases
- Pathogenic bacteria
- Viruses

Why Sequence Genomes

Conclusions

- Better understanding of the genomes would be possible by sequencing of the genomes.

Genetics and Genomics

Major Techniques used for Genomes Characterization

Genomes Characterization -Techniques

Major Techniques used for Genomes Characterization

- Cloning
- Hybridization
- PCR amplification
- Sequencing
- Computational tool

Genomes Characterization - Techniques

Genomes Characterization Techniques - Cloning

- Genomes digested with restriction enzymes and inserted in vectors to produce genomic libraries.
- BACs
- YACs

Genomes Characterization - Techniques

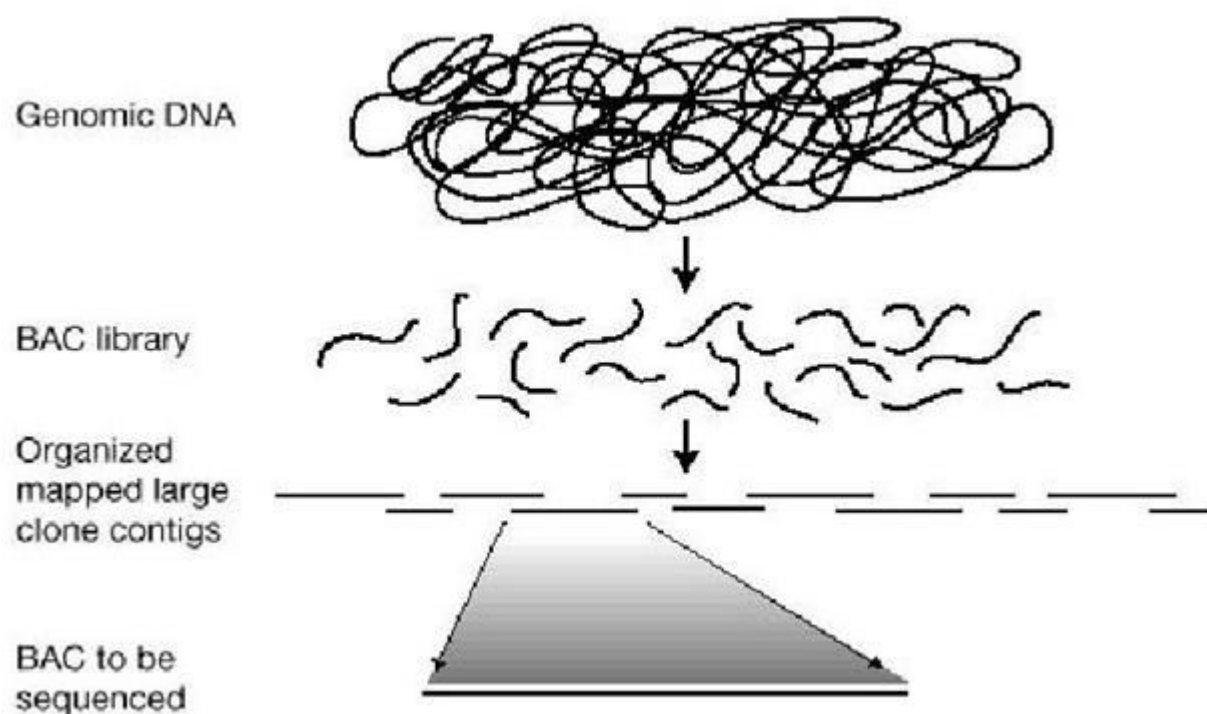
Genomes Characterization Techniques - Hybridization

- To arrange large contigs of genomes to produce genetic maps and physical maps of genomes.

Genomes Characterization - Techniques

Genomes Characterization Techniques –

Hybridization: To arrange large contigs of genomes to produce genetic and physical maps



Genomes Characterization - Techniques

Genomes Characterization Techniques – PCR

- Technique to amplify the DNA. Different variants of the technique used

Genomes Characterization - Techniques

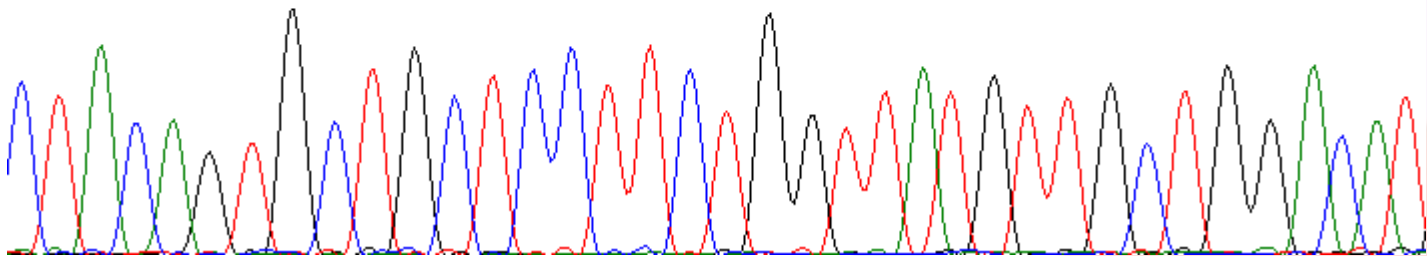
Genomes Characterization Techniques – DNA Sequencing

- One of the important technique used to characterize the genomes
- To study structure and function of genomes.

Genomes Characterization - Techniques

Genomes Characterization Techniques – DNA Sequencing: DNAs are amplified and sequenced

C T A C A G T G C T G C T C C T T C T G G T T A T G T T G C T G G A C A T



Genomes Characterization - Techniques

Genomes Characterization Techniques – Computational Tools

- Used to align the sequenced DNA to produce physical maps of the genomes.

Genomes Characterization - Techniques

Genomes Characterization Techniques – Conclusion

- Different techniques used for genomes characterizations.

Genetics and Genomics

Steps of Genomes Analysis

Genomes Analysis - Steps

Steps of Genomes Analysis

- Genome sequence assembled
- Identify repetitive sequences – mask out
- Gene prediction – train a model for each genome

Genomes Analysis - Steps

Steps of Genomes Analysis

- Look for EST and cDNA sequences
- Genome annotation
- Microarray analysis
- Metabolic pathways and regulation
- Protein 2D gel electrophoresis

Genomes Analysis - Steps

Steps of Genomes Analysis

- Functional genomics
- Gene location/gene map
- Self-comparison of proteome
- Comparative genomics

Genomes Analysis - Steps

Steps of Genomes Analysis

- Identify clusters of functionally related genes
- Evolutionary modeling

Genetics and Genomics

Steps of Genomes Analysis

Genomes Analysis - Steps

Steps of Genomes Analysis

- Genome sequence assembled
- Identify repetitive sequences – mask out
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Genomes Analysis - Steps

Steps of Genomes Analysis

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Genomes Analysis - Steps

Steps of Genomes Analysis

- Functional genomics
- Gene location/gene map
- Self-comparison of proteome
- Comparative genomics

Genomes Analysis - Steps

Steps of Genomes Analysis

- Identify clusters of functionally related genes
- Evolutionary modeling

Genetics and Genomics

Genes and Size of Genomes

Genes and Size of Genomes

Size of Genomes

- Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb).
- Genomes of eukaryotes are usually larger

Genes and Size of Genomes

Size of Genomes

- Most plants and animals have genomes greater than 100 Mb.
- Humans have genome size of 3,000 Mb

Genes and Size of Genomes

Size of Genomes

- Within each domain there is no systematic relationship between genome size and phenotype

Genes and Size of Genomes

Size of Genomes

Organism	Haploid Genome Size (Mb)	Number of Genes	Genes per Mb
Bacteria			
<i>Haemophilus influenzae</i>	1.8	1,700	940
<i>Escherichia coli</i>	4.6	4,400	950
Archaea			
<i>Archaeoglobus fulgidus</i>	2.2	2,500	1,130
<i>Methanosarcina barkeri</i>	4.8	3,600	750

Genes and Size of Genomes

Size of Genomes

Organism	Haploid Genome Size (Mb)	Number of Genes	Genes per Mb
Eukaryotes			
<i>Saccharomyces cerevisiae</i> (yeast, a fungus)	12	6,300	525
<i>Caenorhabditis elegans</i> (nematode)	100	20,100	200
<i>Arabidopsis thaliana</i> (mustard family plant)	120	27,000	225
<i>Drosophila melanogaster</i> (fruit fly)	165	13,700	83
<i>Oryza sativa</i> (rice)	430	42,000	98
<i>Zea mays</i> (corn)	2,300	32,000	14
<i>Mus musculus</i> (house mouse)	2,600	22,000	11
<i>Ailuropoda melanoleuca</i> (giant panda)	2,400	21,000	9
<i>Homo sapiens</i> (human)	3,000	<21,000	7

Genes and Size of Genomes

Conclusion

- Although most eukaryotes have large size of genomes.
- Within each domain there is no systematic relationship between genome size and phenotype

Genetics and Genomics

Viral Genomes

Viral Genomes

Genomes of Viruses

- Viral genomes can be
- ssRNA
- dsRNA
- ssDNA
- dsDNA
- Linear
- Circular

Viral Genomes

Viruses Genomes

- A viral genome is the genetic material of the virus.
- Also termed the viral chromosome.
- Viral genomes vary in size -few thousand to more than a hundred thousand nucleotides.

Viral Genomes

Viruses with RNA Genomes

- Almost all plants viruses and some bacterial and animal viruses
- Genomes are rather small (a few thousands nucleotides)

Viral Genomes

Viruses with DNA Genomes

- Often a circular genome
- lambda = 48,502 bp

Viral Genomes

Replicative form of Viral Genomes

- All ssRNA viruses produce dsRNA molecules
- Many linear DNA molecules become circular

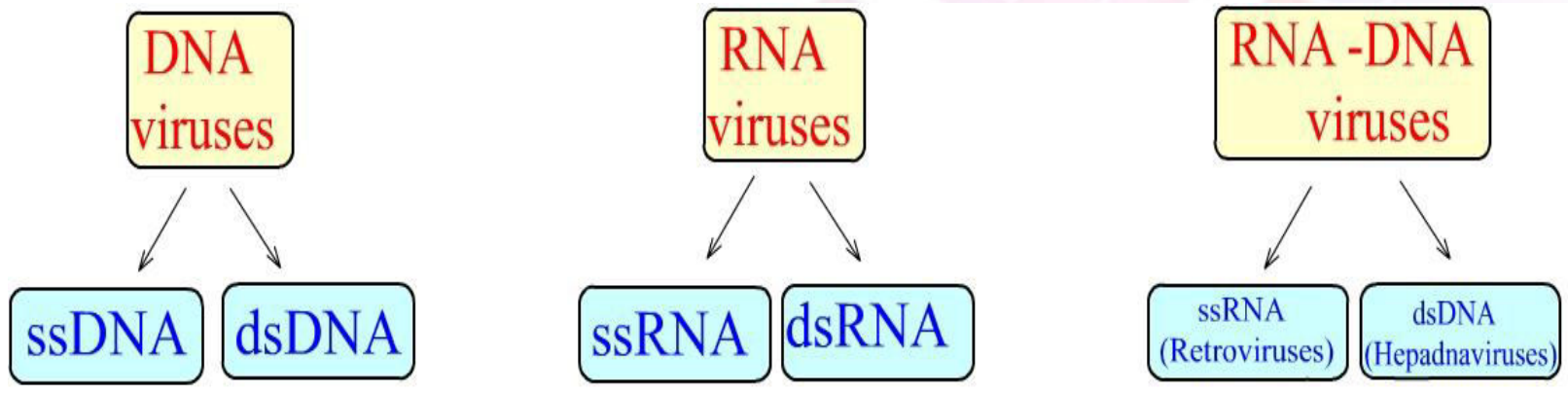
Viral Genomes

Viruses and Kingdoms

- Many plants viruses contain ssRNA genomes.
- Many fungal viruses contain dsRNA genomes.
- Many bacterial viruses contain dsDNA genomes.

Viral Genomes

Genomes in Virions: The genomes of viruses can be composed of either DNA or RNA, and some use both as their genomic material at different stages in their life cycle. However, only one type of nucleic acid is found in the virion of any particular type of virus.



Viral Genomes

Viruses and Number of Genes

Virus	Host	Type of Nucleic Acid	Number of Genes
Parvovirus	Mammals	ssDNA	5
Phage fd	<i>E. coli</i>	ssDNA	10
Lambda	<i>E. coli</i>	dsDNA	36
T4	<i>E. coli</i>	dsDNA	>190
Q β	<i>E. coli</i>	ssRNA	4
TMV	Many plants	ssRNA	6
Influenza virus	Mammals	ssRNA	12

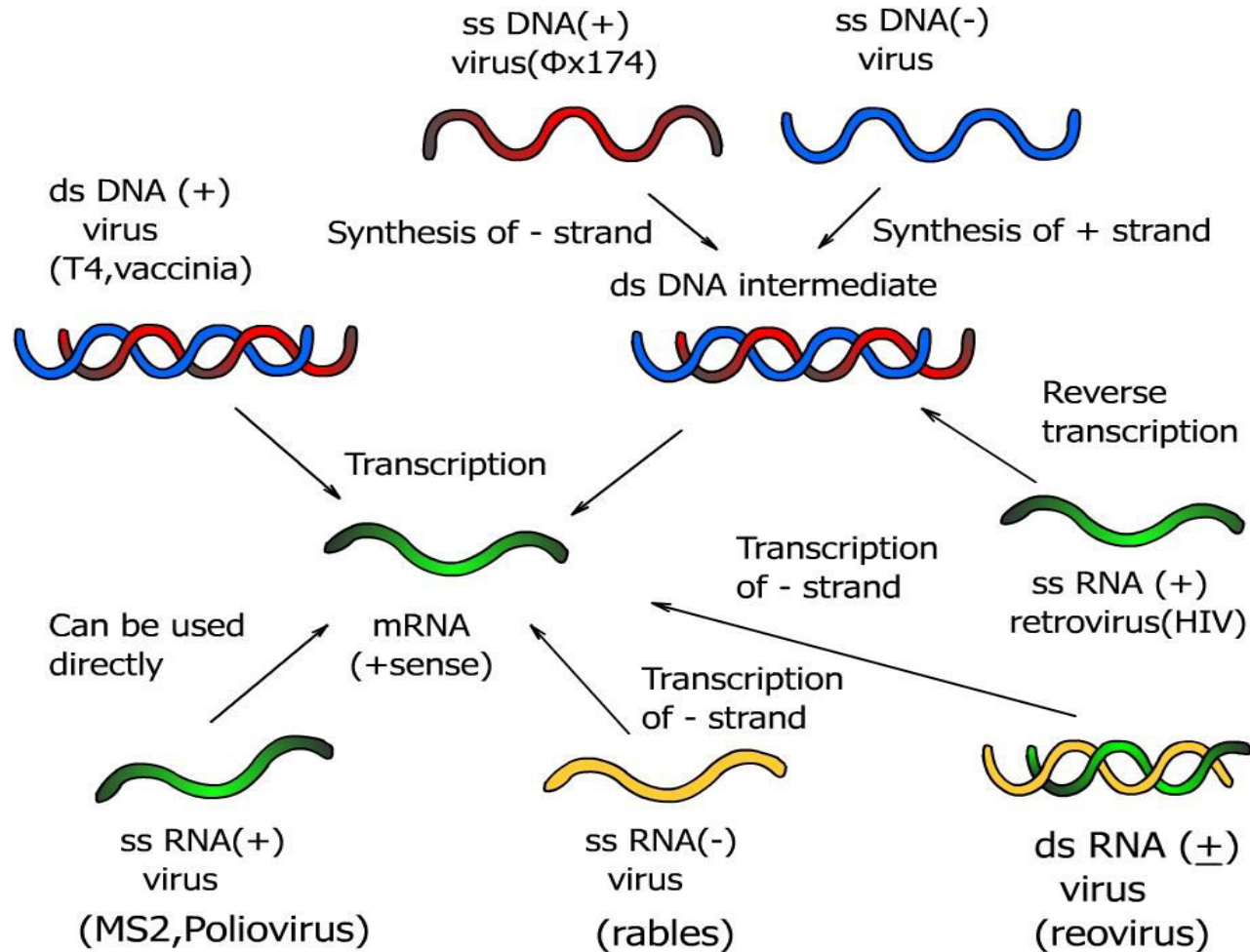
Viral Genomes

Viruses and Number of Genes

Virus	Genome structure	Genome size (kb)	Number of genes
Adenovirus	Double-stranded linear DNA	36.0	30
Hepatitis B	Partly double-stranded circular DNA	3.2	4
Influenza virus	Single-stranded segmented linear RNA	22.0	12
Parvovirus	Single-stranded linear DNA	1.6	5
Poliovirus	Single-stranded linear RNA	7.6	8
Reovirus	Double-stranded segmented linear RNA	22.5	22
Retroviruses	Single-stranded linear RNA	6.0–9.0	3
SV40	Double-stranded circular DNA	5.0	5
Tobacco mosaic virus	Single-stranded linear RNA	6.4	6
Vaccinia virus	Double-stranded circular DNA	240	240

Viral Genomes

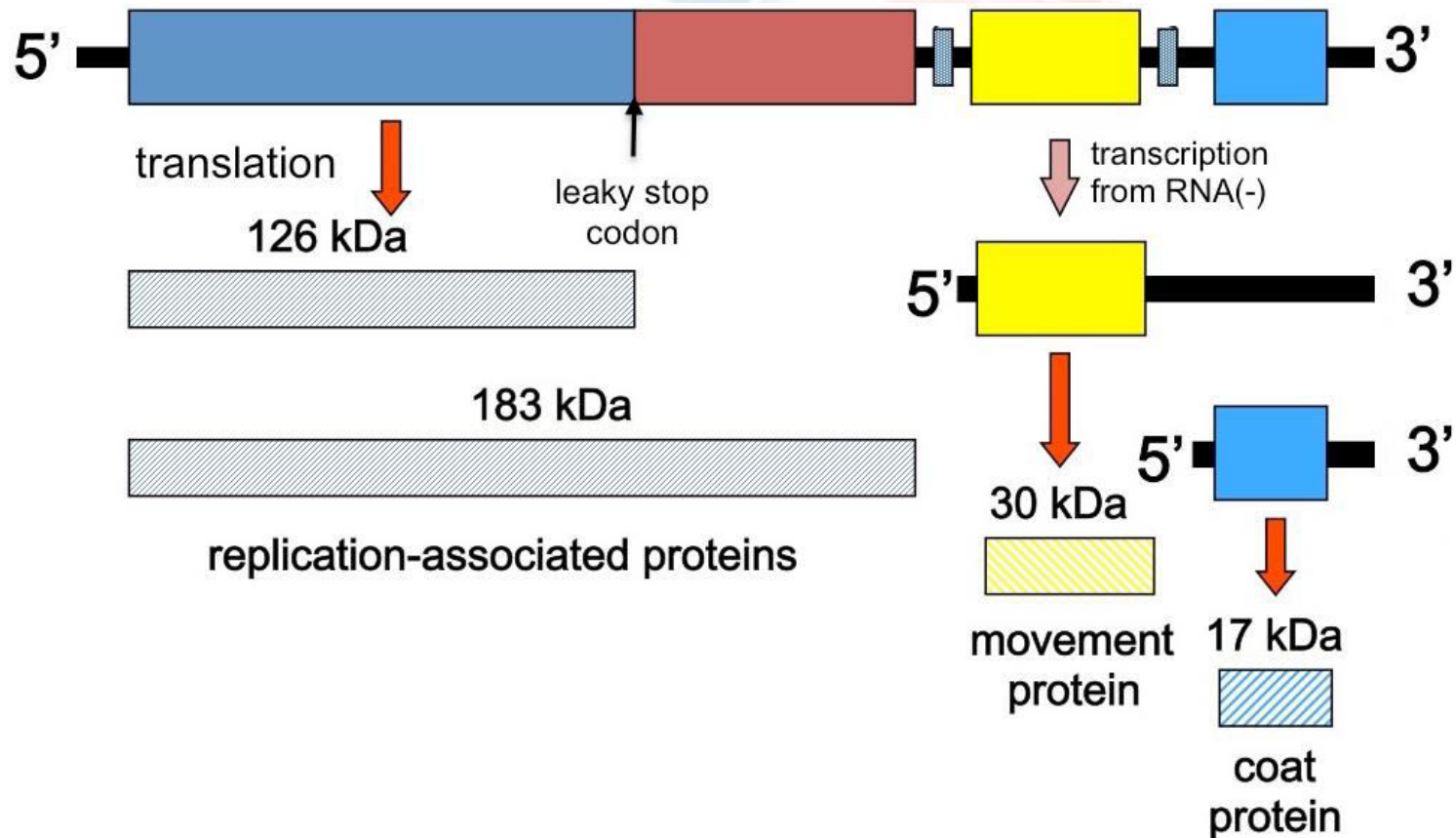
Genomes in Virions



Viral Genomes

Genome of Tobacco Mosaic Virus

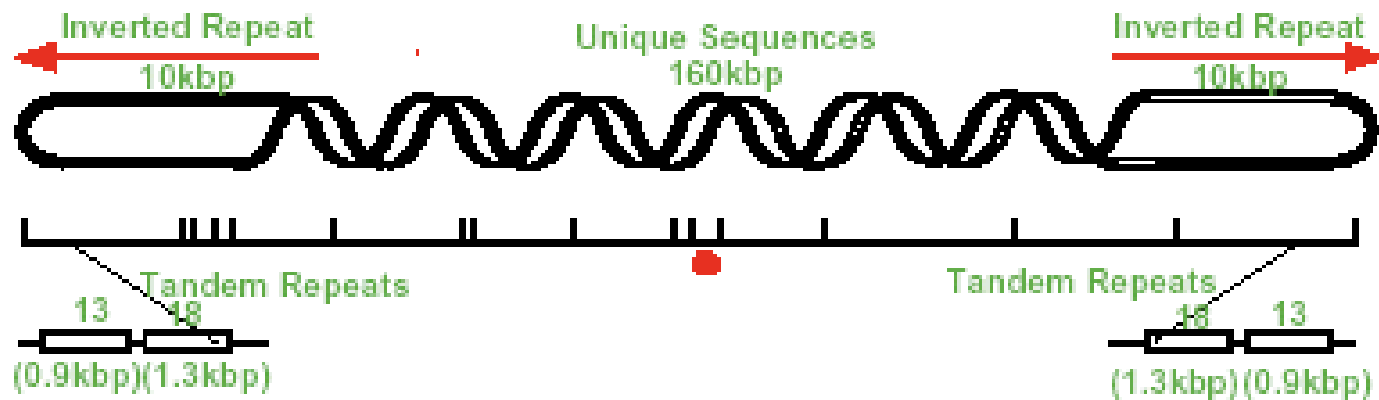
- Single, 6400 nucleotides RNA, 3 Essential Genes



Viral Genomes

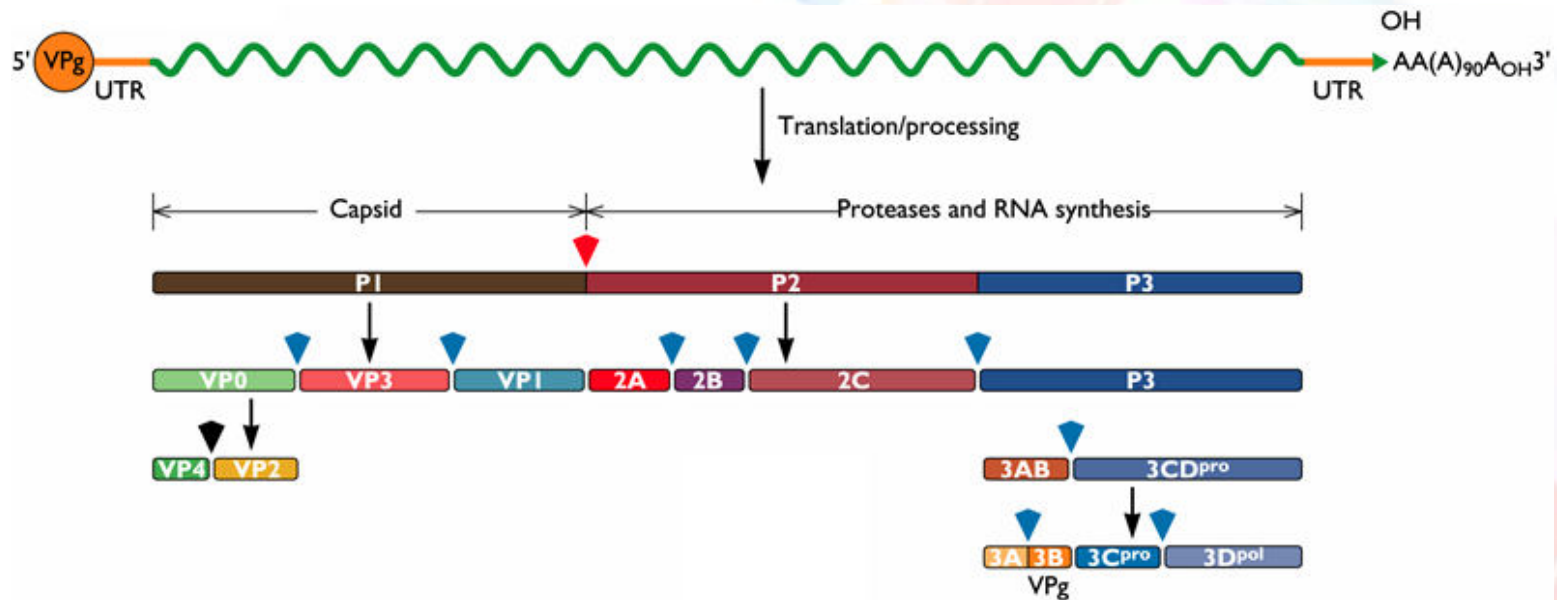
Genome of Poxvirus –A typical large dsDNA Virus

- 180 kb DNA, >100 Essential Genes



Viral Genomes

Genome of Polio Virus: Single-stranded positive-sense RNA genome that is about 7500 nucleotides long



Viral Genomes

Genome of Pox Virus

- Linear dsDNA 130-375 kbp; covalently closed termini.
- Large hairpin structure at each terminus - up to 10 kb total at each end is repeat sequence.
- Encode 150-300 proteins.
- Coding regions are closely spaced, no introns.
- Coding regions are on both strands of genome, and are not tightly clustered with respect to time of expression or function.

Viral Genomes

Viral Genomes

- Viral genomes can be
- ssRNA
- dsRNA
- ssDNA
- dsDNA
- Linear
- Circular

Genetics and Genomics

Bacterial Genomes

Bacterial Genomes

Genomes of Bacteria

- Small organisms carry high coding density (85-90%)
- 1 gene per 1000 bases in prokaryotes
- Large variation in genome size between bacteria

Bacterial Genomes

Genomes of Bacteria – Large Variation

- *Tremblaya princeps*
140kb, 121 coding sequences
- *Sorangium cellulosum*
- 14000kb
- 11599 coding sequences

Bacterial Genomes

Genomes of Bacteria

Comparison of regulatory genes in bacterial genomes

Microorganism	# Genes in the Genome	# Regulatory Proteins	% of Total
<i>Pseudomonas aeruginosa</i>	5570	468	8.4
<i>Escherichia coli</i>	4289	250	5.8
<i>Bacillus subtilis</i>	4100	217	5.3
<i>Mycobacterium tuberculosis</i>	3918	117	3.0
<i>Helicobacter pylori</i>	1566	18	1.1

Bacterial Genomes

Distribution of genes among selected bacterial genomes and their sizes

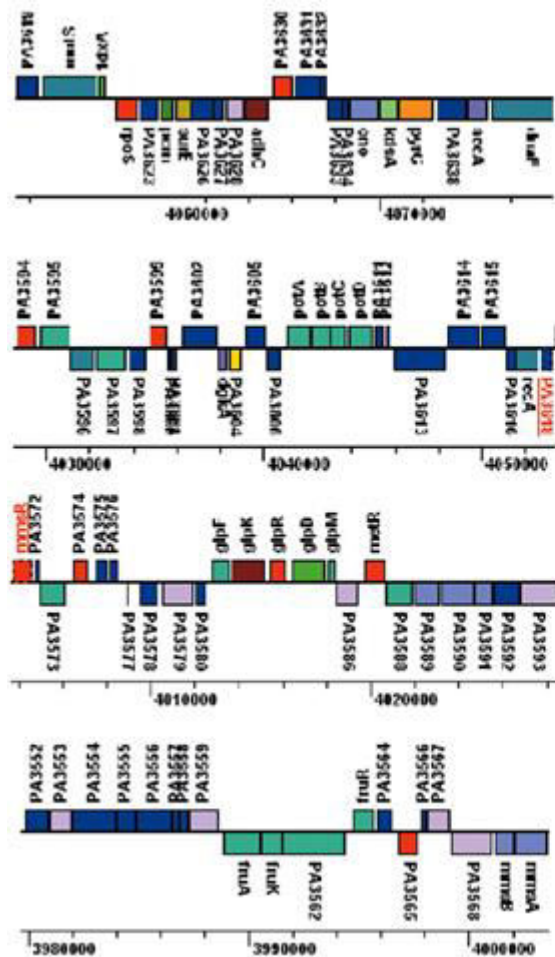
Organism	Genome Size (Mbp)	No. of ORFs (% coding)		Unknown Function		Unique ORFs	
<i>Aeropyrum pernix</i> K1	1.67	1,885	(89%)				
<i>A. aeolicus</i> VF5	1.50	1,749	(93%)	663	(44%)	407	(27%)
<i>A. fulgidus</i>	2.18	2,437	(92%)	1,315	(54%)	641	(26%)
<i>B. subtilis</i>	4.20	4,779	(87%)	1,722	(42%)	1,053	(26%)
<i>B. burgdorferi</i>	1.44	1,738	(88%)	1,132	(65%)	682	(39%)
<i>Chlamydia pneumoniae</i> AR39	1.23	1,134	(90%)	543	(48%)	262	(23%)
<i>Chlamydia trachomatis</i> MoP _n	1.07	936	(91%)	353	(38%)	77	(8%)
<i>C. trachomatis</i> serovar D	1.04	928	(92%)	290	(32%)	255	(29%)
<i>Deinococcus radiodurans</i>	3.28	3,187	(91%)	1,715	(54%)	1,001	(31%)
<i>E. coli</i> K-12-MG1655	4.60	5,295	(88%)	1,632	(38%)	1,114	(26%)
<i>H. influenzae</i>	1.83	1,738	(88%)	595	(35%)	237	(14%)
<i>H. pylori</i> 26695	1.66	1,589	(91%)	744	(45%)	539	(33%)
<i>Methanobacterium thermotautotrophicum</i>	1.75	2,008	(90%)	1,010	(54%)	496	(27%)

Bacterial Genomes

Distribution of genes among selected bacterial genomes and their sizes

Organism	Genome Size (Mbp)	No. of ORFs (% coding)		Unknown Function		Unique ORFs	
<i>Methanococcus jannaschii</i>	1.66	1,783	(87%)	1,076	(62%)	525	(30%)
<i>M. tuberculosis</i> CSU#93	4.41	4,275	(92%)	1,521	(39%)	606	(15%)
<i>M. genitalium</i>	0.58	483	(91%)	173	(37%)	7	(2%)
<i>M. pneumoniae</i>	0.81	680	(89%)	248	(37%)	67	(10%)
<i>N. meningitidis</i> MC58	2.24	2,155	(83%)	856	(40%)	517	(24%)
<i>Pyrococcus horikoshii</i> OT3	1.74	1,994	(91%)	589	(42%)	453	(22%)
<i>Rickettsia prowazekii</i> Madrid E	1.11	878	(75%)	311	(37%)	209	(25%)
<i>Synechocystis</i> sp.	3.57	4,003	(87%)	2,384	(75%)	1,426	(45%)
<i>T. maritima</i> MSB8	1.86	1,879	(95%)	863	(46%)	373	(26%)
<i>T. pallidum</i>	1.14	1,039	(93%)	461	(44%)	280	(27%)
<i>Vibrio cholerae</i> El Tor N1696	4.03	3,890	(88%)	1,806	(46%)	934	(24%)
	50.60	52,462	(89%)	22,358	(43%)	12,161	(23%)

Genes in a portion of a bacterial genome

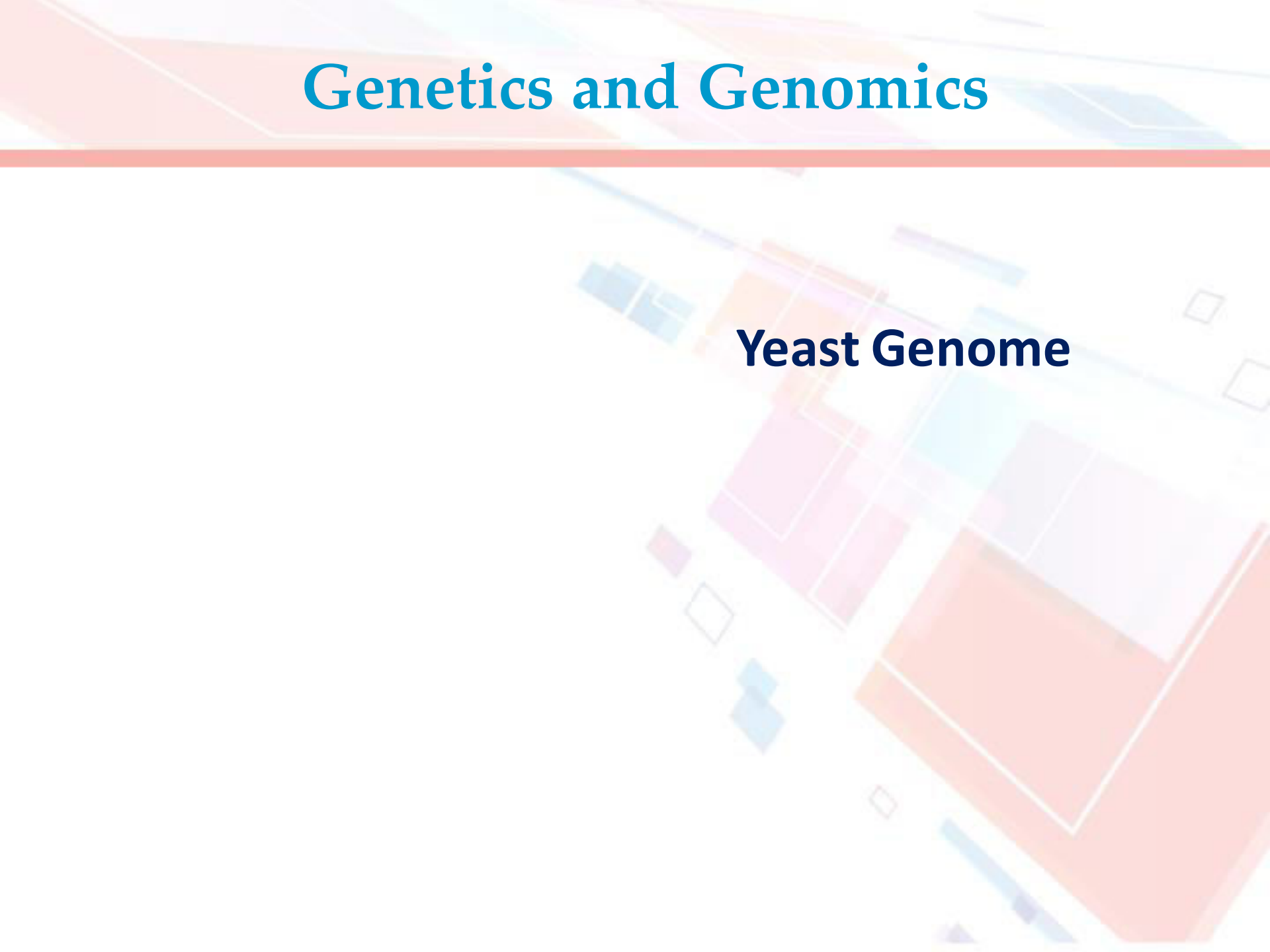


Bacterial Genomes

Bacterial Genomes - Conclusion

- Small organisms carry high coding density.
- Large variation in genome size between bacteria.

Genetics and Genomics



Yeast Genome

Yeast Genome

Yeast Genome

- The nuclear genome consists of 16 chromosomes.
- In addition, there is a mitochondrial genome and a plasmid, 2 micron circle.

Yeast Genome

Yeast Genome

- The haploid yeast genome consists of ~ 12.1 Mb
- Yeast genome was completely sequenced by 1996

Yeast Genome

Yeast Genome - Characteristics

- Small and compact
- Small intergenic sequences
- Few transposable elements
- Few introns
- Limited RNA interference

Yeast Genome

Yeast Genome

- The yeast genome is predicted to contain about 6,200 genes
- 274 tRNA
- 287 introns
- Small percentage of yeast genes have introns
- The intergenic space between genes is only between 200bp - 1,000bp

Yeast Genome

Yeast Genome: Genome of Yeast Cell

Characteristic	Chromosomes	Plasmid	Mitochondria
Relative amount (%)	85	5	10
Number of copies	2 x 16	60-100	~50 (8-130)
Size (kbp)	~ 12,100	6.318	70-76

Yeast Genome

Yeast Genome

- The largest known regulatory sequences are spread over about 2,800bp
- MUC1/FLO11

Yeast Genome

Yeast Genome

- Yeast genes have names consisting of three letters and up to three numbers
- GPD1, HSP12, PDC6
- Usually they are meaningful

Yeast Genome

Yeast Genome – Genes Nomenclature

- Wild type genes are written with capital letters in italics: *TPS1, RHO1, CDC28*
- Recessive mutant genes are written with small letters in italics: *tps1, rho1, cdc28*

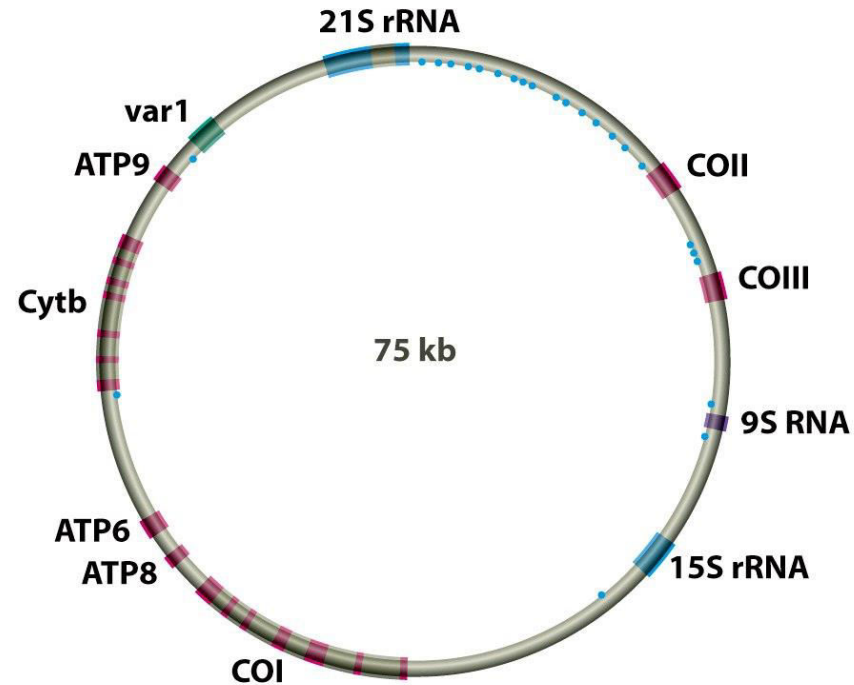
Yeast Genome

Yeast Genome – Genes Nomenclature

- Three letters provides information about a function, mutant phenotype, or process related to that gene.
- CDC - Cell Division Cycle ; ADE-ADEnine biosynthesis

Yeast Genome

Yeast – Mitochondrial DNA



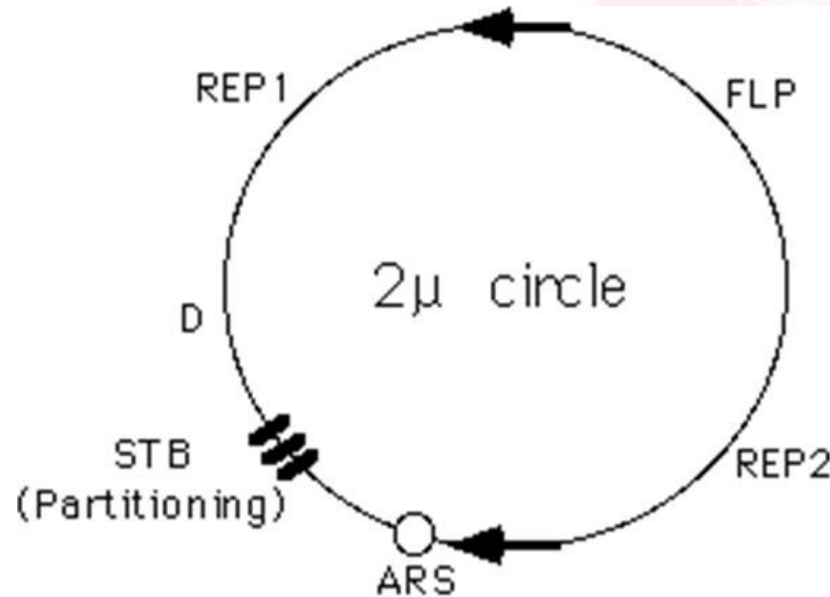
KEY

- | | | |
|--------------------------|------------------------|----------------|
| Respiratory complex gene | Ribosomal protein gene | Intron |
| Ribosomal RNA gene | Transfer RNA gene | Other RNA gene |

Yeast Genome

Yeast – Plasmid DNA

- The 2 μ circle is a 6.3 kb
- 50 to 100 copies per haploid genome of the yeast cells
- ARS, the FLP gene, the three genes which encode proteins required for regulation of FLP expression (REP2, REP1, and D)
- Set of small direct repeats (called "STB") required for partitioning into daughter cells during mitosis and meiosis.



Yeast Genome

Yeast Genome

- Yeast nuclear genome has 16 chromosomes.
- A mitochondrial genome.
- A plasmid.

Genetics and Genomics

Mitochondrial Genome

Mitochondrial Genome

Mitochondrial Genome

- Multiple identical circular chromosomes
- ~15-16 Kb in animals
- ~ 200 kb to 2,500 kb in plants

Mitochondrial Genome

Mitochondrial Genome

- Over 95% of mitochondrial proteins are encoded in the nuclear genome.
- Often A+T rich genomes.

Mitochondrial Genome

Human Mitochondrial Genome

- Circular, double stranded, 16.6 kb
- The two strands are notably different in base composition, leading to one strand being heavy (H strand) and the other light (L strand).

Mitochondrial Genome

Human Mitochondrial Genome

- Both strands encode genes, although more are on the H strand.
- A short region (1121 bp), the D loop is a DNA triple helix: two overlapping copies of the H strand.

Mitochondrial Genome

Human Mitochondrial Genome

- The D loop is also the site where most of replication and transcription is controlled.
- Genes are tightly packed, with almost no non-coding DNA outside of D loop.

Mitochondrial Genome

Human Mitochondrial Genome

- Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants)

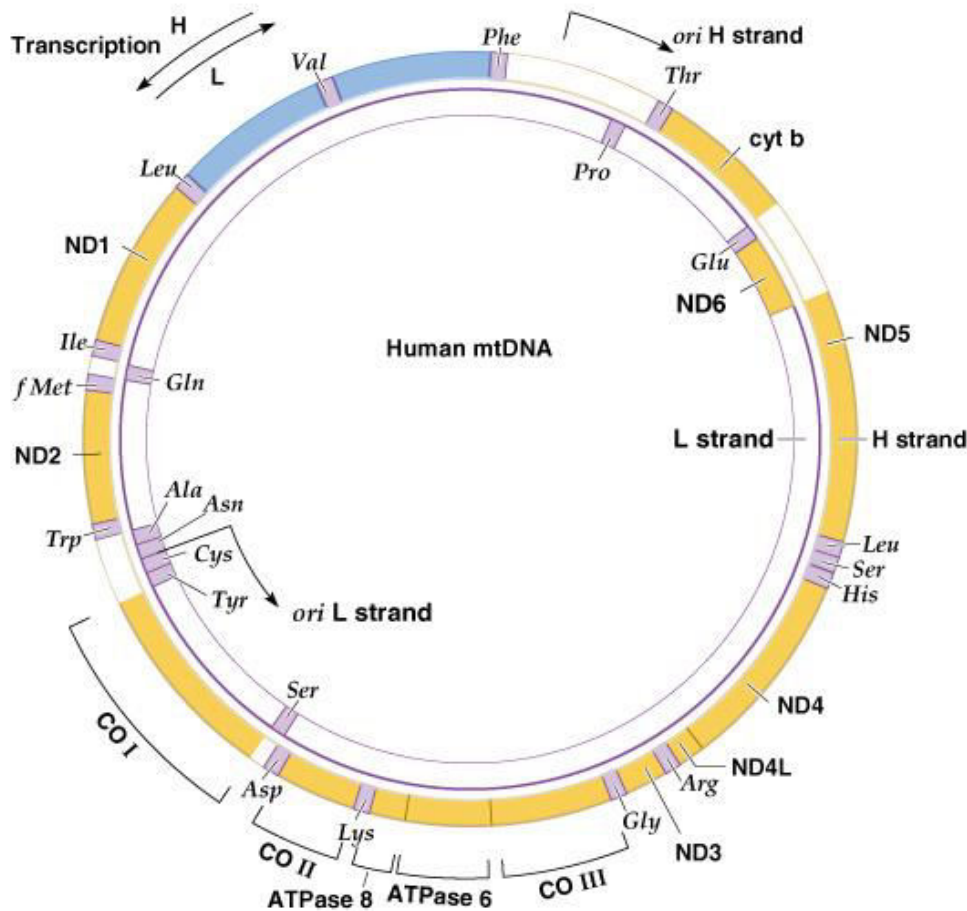
Mitochondrial Genome

Human Mitochondrial Genome

- 37 Genes
- 22 tRNAs
- 2 rRNAs
- 13 polypeptides
- tRNA: only 60 of the 64 codons code for amino acids.

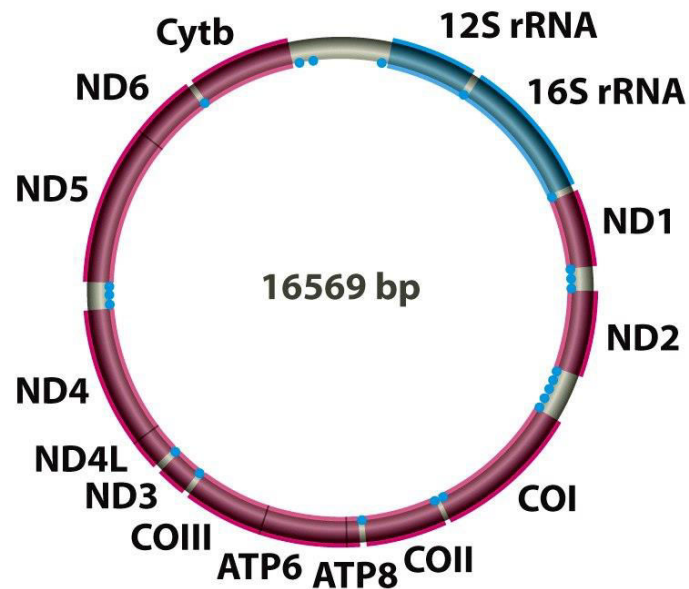
Mitochondrial Genome

Mitochondrial Genome - Humans



Mitochondrial Genome

Mitochondrial Genome - Humans

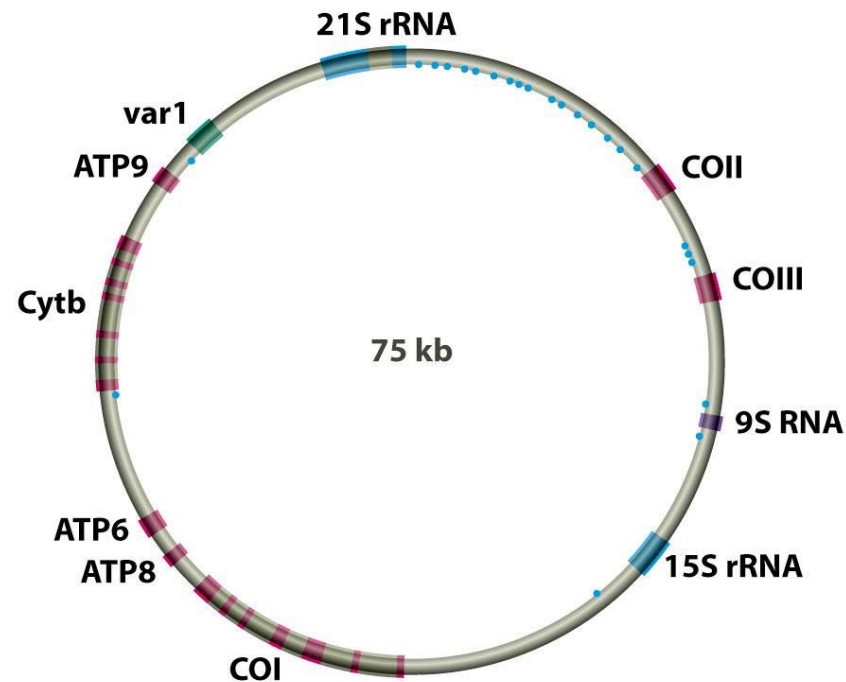


KEY

-  Respiratory complex gene
-  Ribosomal RNA gene
-  Transfer RNA gene

Mitochondrial Genome

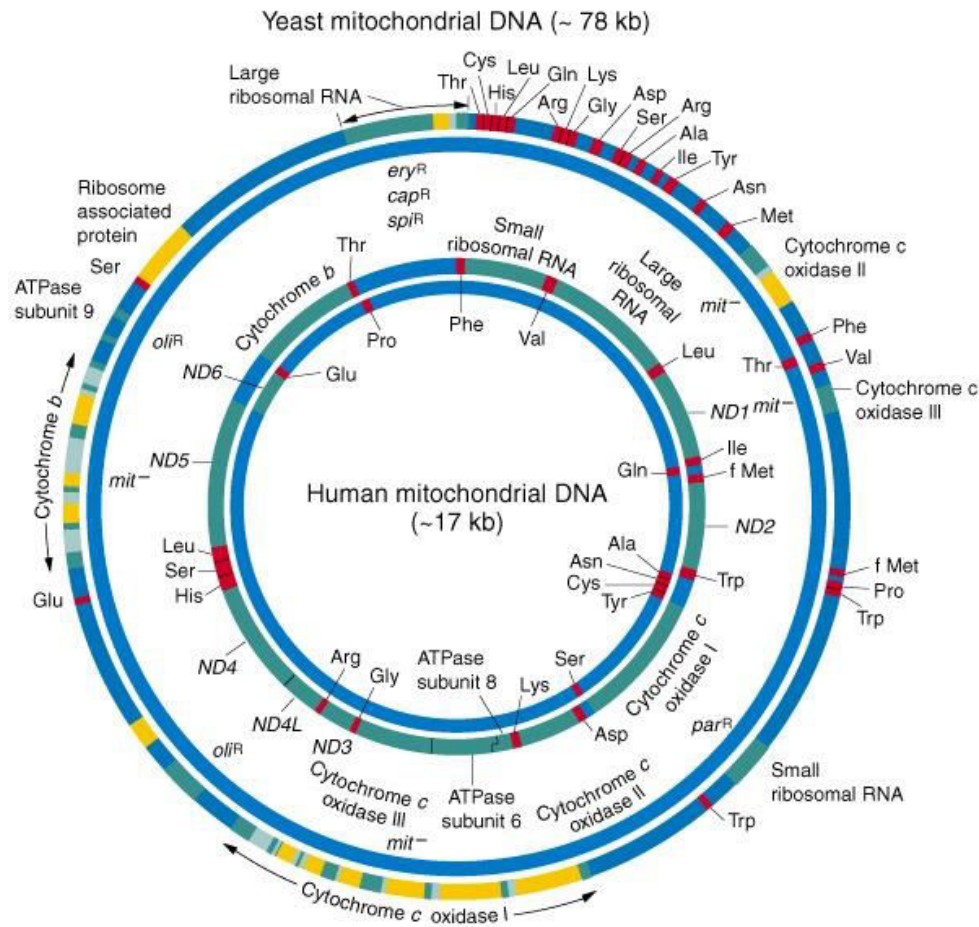
Mitochondrial Genome – Yeast



KEY

- | | | |
|--------------------------|------------------------|----------------|
| Respiratory complex gene | Ribosomal protein gene | Intron |
| Ribosomal RNA gene | Transfer RNA gene | Other RNA gene |

Mitochondrial Genome – Human and Yeast



Mitochondrial Genome

Mitochondrial Genomes

Species	Type of organism	Genome size (kb)
Mitochondrial genomes		
<i>Plasmodium falciparum</i>	Protozoan (malaria parasite)	6
<i>Chlamydomonas reinhardtii</i>	Green alga	16
<i>Mus musculus</i>	Vertebrate (mouse)	16
<i>Homo sapiens</i>	Vertebrate (human)	17
<i>Metridium senile</i>	Invertebrate (sea anemone)	17
<i>Drosophila melanogaster</i>	Invertebrate (fruit fly)	19
<i>Chondrus crispus</i>	Red alga	26
<i>Aspergillus nidulans</i>	Ascomycete fungus	33
<i>Reclinomonas americana</i>	Protozoa	69
<i>Saccharomyces cerevisiae</i>	Yeast	75
<i>Suillus grisellus</i>	Basidiomycete fungus	121
<i>Brassica oleracea</i>	Flowering plant (cabbage)	160
<i>Arabidopsis thaliana</i>	Flowering plant (vetch)	367
<i>Zea mays</i>	Flowering plant (maize)	570
<i>Cucumis melo</i>	Flowering plant (melon)	2500

Mitochondrial Genome

Mitochondrial Genomes

Feature	<i>Plasmodium falciparum</i>	<i>Chlamydomonas reinhardtii</i>	<i>Homo sapiens</i>	<i>Saccharomyces cerevisiae</i>	<i>Arabidopsis thaliana</i>	<i>Reclinomonas americana</i>
Total number of genes	5	12	37	35	52	92
Types of genes						
Protein-coding genes	3	7	13	8	27	62
Respiratory complex	3	7	13	7	17	24
Ribosomal proteins	0	0	0	1	7	27
Transport proteins	0	0	0	0	3	6
RNA polymerase	0	0	0	0	0	4
Translation factor	0	0	0	0	0	1
Functional RNA genes	2	5	24	27	25	30
Ribosomal RNA genes	2	2	2	2	3	3
Transfer RNA genes	0	3	22	24	22	26
Other RNA genes	0	0	0	1	0	1
Number of introns	0	1	0	8	23	1
Genome size (kb)	6	16	17	75	367	69

Mitochondrial Genome

Universal Code and Mitochondrial Code

Mitochondrial code

		Second letter			
		U	C	A	G
First letter	U	Phe Phe Leu Leu	Ser Ser Ser Ser	Tyr Tyr Stop Stop	Cys Cys (Stop) Trp Trp
	C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gin Gin	Arg Arg Arg Arg
	A	Ile (Met) Ile (Ile) Met Ile	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser (Arg) Stop (Arg) Stop
	G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly
		U	C	A	G

Universal code

		Second letter			
		U	C	A	G
First letter	U	UUU } Phe UUC } UUA } UUG } Leu	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG } Stop	UGU } Cys UGC } UGA } Stop UGG } Trp
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gin CAG }	CGU } CGC } Arg CGA } CGG }
	A	AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }
		U	C	A	G

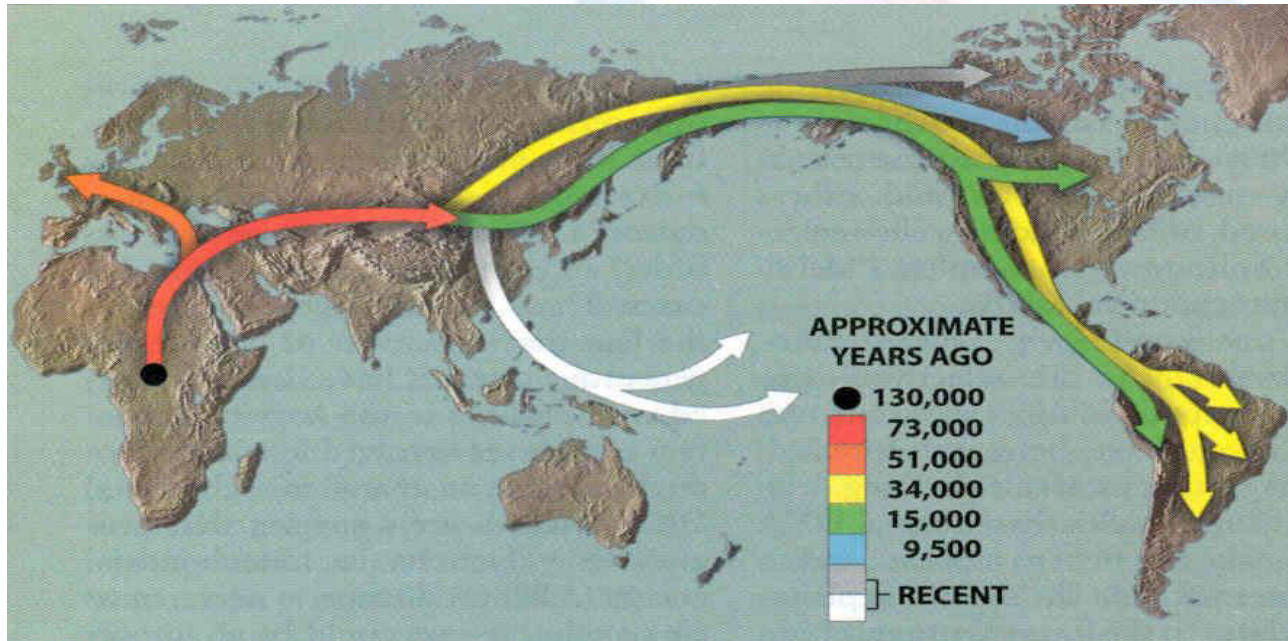
Mitochondrial Genome

Mitochondrial Genome – Tumors due to mutations



Mitochondrial Genome

Mitochondrial DNA polymorphisms track human migrations



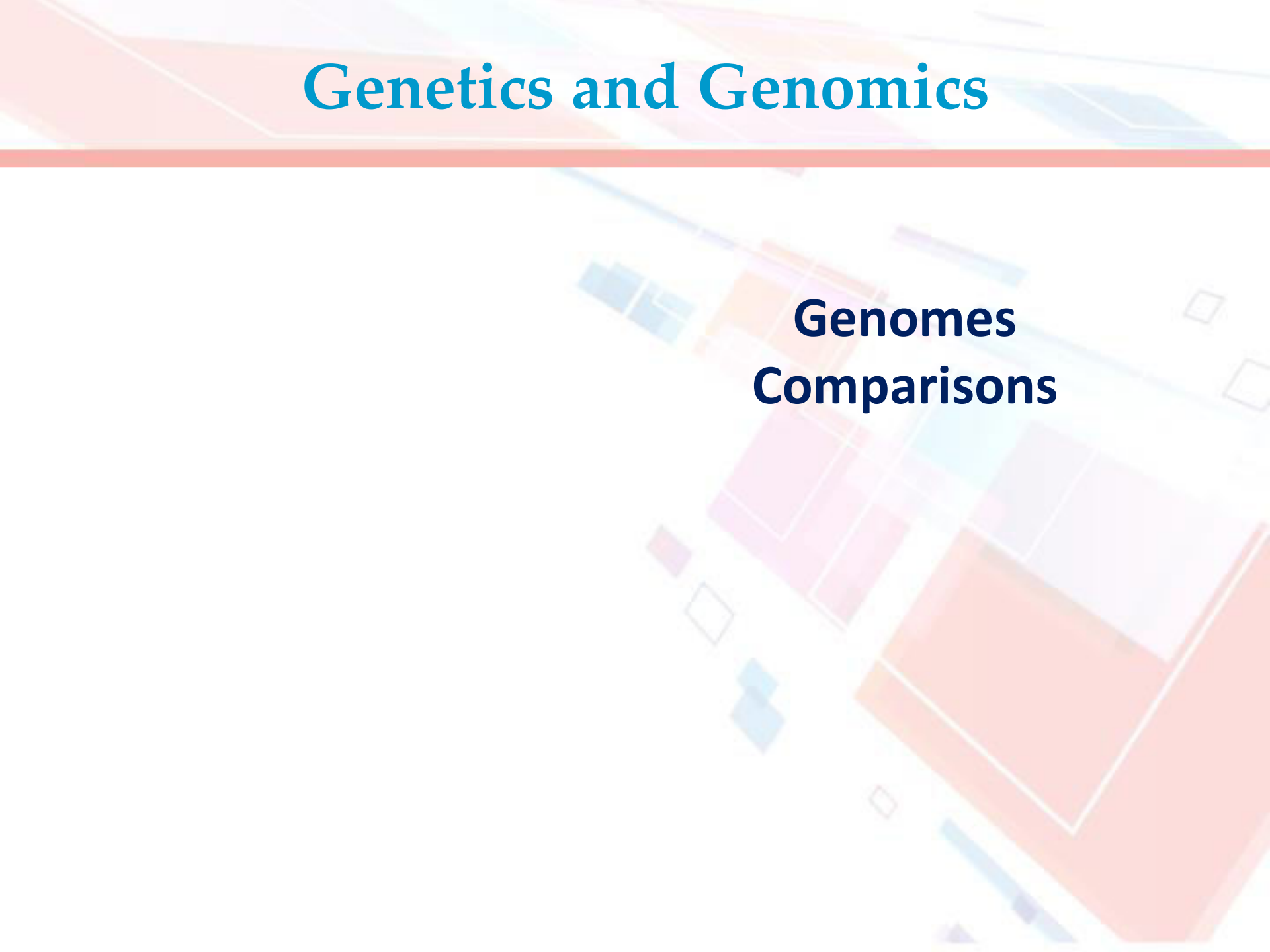
All humans descend from a small group of Africans
This group originated in central Africa ~200,000 years ago
The founding group was small (10^2 - 10^4 people)

Mitochondrial Genome

Conclusion

- Multiple identical circular chromosomes
- 15-16 Kb in animals
- 200 kb to 2,500 kb in plants

Genetics and Genomics



Genomes Comparisons

Genomes Comparisons

Genomes vary in size

- Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb)
- Most plants and animals have genomes greater than 100 Mb; humans have 3,000 Mb

Genomes Comparisons

Genomes vary in genes numbers

- Free-living bacteria and archaea have 1,500 to 7,500 genes
- Fungi have about 5,000 genes and multicellular eukaryotes upto 40,000 genes

Genomes Comparisons

Genomes vary in genes numbers

- Number of genes is not correlated to genome size
- Nematode
C. elegans has 100 Mb and 20,000 genes, while
Drosophila has 165 Mb and 13,700 genes

Genomes Comparisons

Genomes vary in genes numbers

- Vertebrate genomes can produce more than one polypeptide per gene because of alternative splicing of RNA transcripts

Genomes Comparisons

Humans and Mammals have low gene density

- Humans and other mammals have the lowest gene density, or number of genes, in a given length of DNA. Multicellular eukaryotes have many introns within genes and noncoding DNA between genes

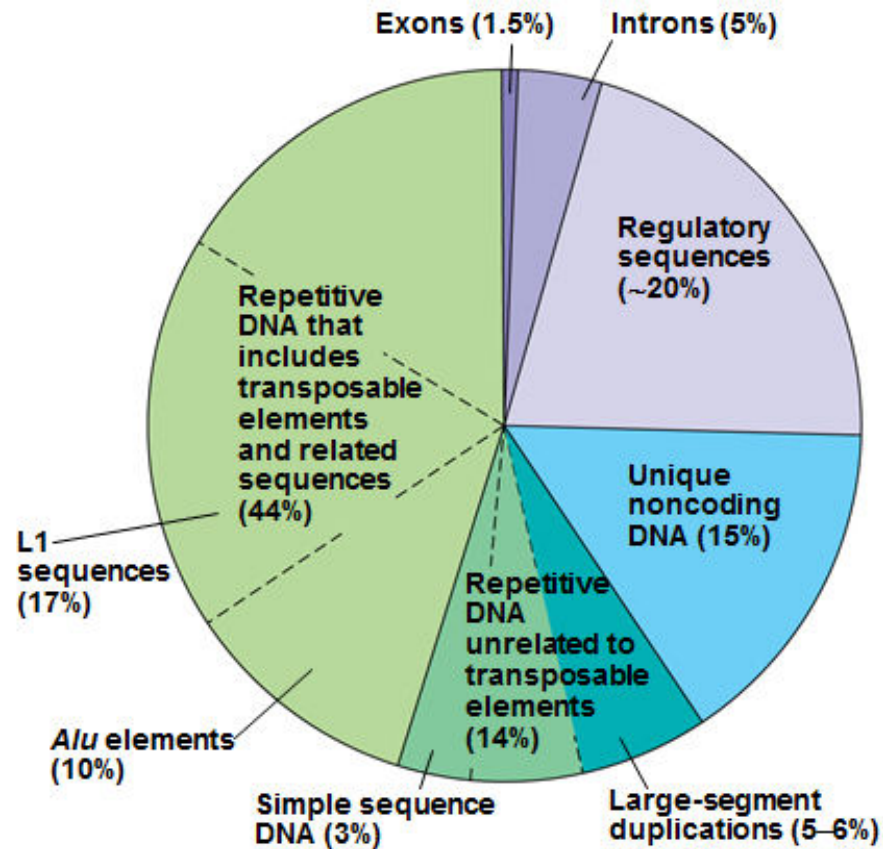
Genomes Comparisons

Multicellular eukaryotes have much noncoding DNA and multigene families

- Most of eukaryotic genomes neither encodes proteins nor functional RNAs
- Evidence indicates that noncoding DNA plays important roles in the cell

Genomes Comparisons

Human Genome: Distribution of coding and non-coding DNA



Genomes Comparisons

Comparing Genomes

- Significant similarity between genomes of "distant" species (Man – Yeast 23%)
- Similarity increases for taxonomically close species.

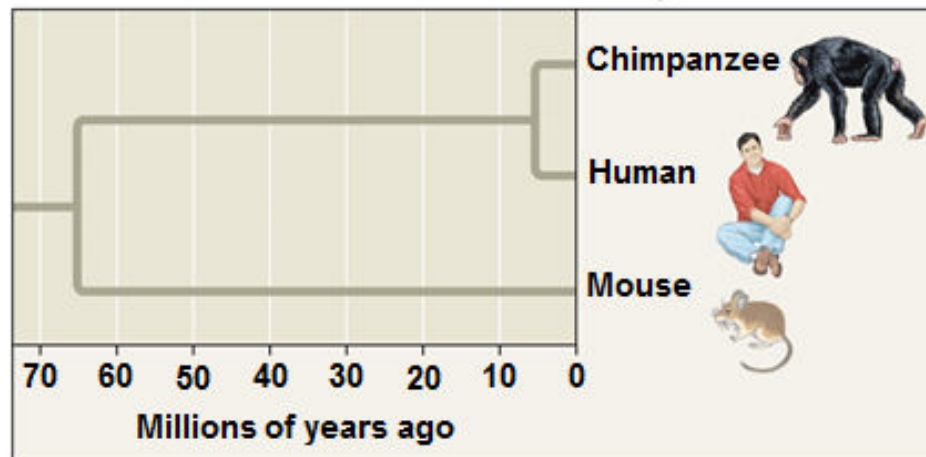
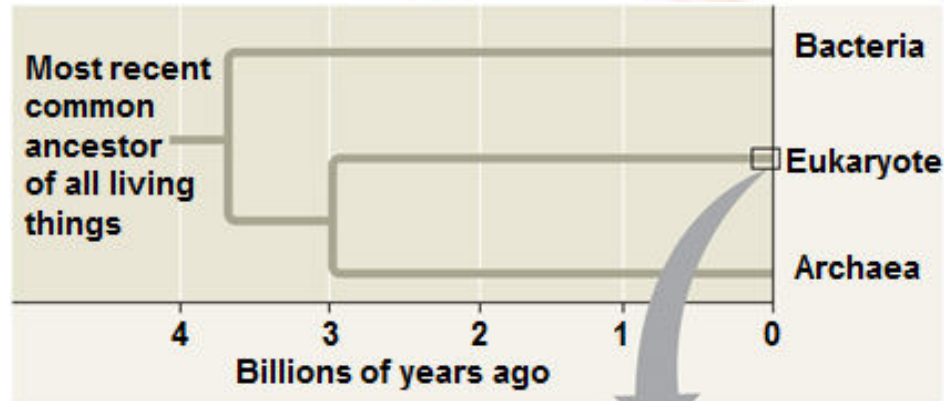
Genomes Comparisons

Comparing Genomes

- Closely related species help us understand recent evolutionary events
- Distantly related species help us understand ancient evolutionary events

Genomes Comparisons

Comparing Genomes: Bacteria, archaea, and eukaryotes diverged from each other between 2 and 4 billion years ago



Genomes Comparisons

Comparing Genomes

	Bacteria	Archaea	Eukaryotes
Genome size	Most are 1–6 Mb		Most are 10–4,000 Mb, but a few are much larger
Number of genes	1,500–7,500		5,000–40,000
Gene density	Higher than in eukaryotes		Lower than in prokaryotes (Within eukaryotes, lower density is correlated with larger genomes.)
Introns	None in protein-coding genes	Present in some genes	Unicellular eukaryotes: present, but prevalent only in some species Multicellular eukaryotes: present in most genes
Other noncoding DNA	Very little		Can be large amounts; generally more repetitive noncoding DNA in multicellular eukaryotes

Genomes Comparisons

Comparing Genomes

- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions

Genetics and Genomics

**Comparing
distantly/closely
related species**

Distantly/Closely Related Species

Comparing distantly related species

- Highly conserved genes have changed very little over time
- These help to clarify relationships among species that diverged from each other long ago

Distantly/Closely Related Species

Comparing distantly related species

- Bacteria, archaea, and eukaryotes diverged from each other 2 and 4 billion years ago
- Highly conserved genes can be studied in one model organism.

Distantly/Closely Related Species

Comparing closely related species

- Genetic differences between closely related species can be correlated with phenotypic differences

Distantly/Closely Related Species

Comparing closely related species

- Genetic comparison of several mammals with non-mammals helps to identify what make mammals

Distantly/Closely Related Species

Comparing closely related species

- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions

Distantly/Closely Related Species

Comparing closely related species

- Several genes are evolving faster in humans than chimpanzees

Distantly/Closely Related Species

Comparing closely related species

- Genes involved in defense against malaria and tuberculosis and in regulation of brain size, genes code for transcription factors

Distantly/Closely Related Species

Comparing closely related species

- Humans and chimpanzees differ in the expression of the *FOXP2* gene, whose product turns on genes involved in vocalization

Distantly/Closely Related Species

Comparing closely related species

- Differences in the *FOXP2* gene may explain why humans but not chimpanzees communicate by speech

Distantly/Closely Related Species

Conclusion

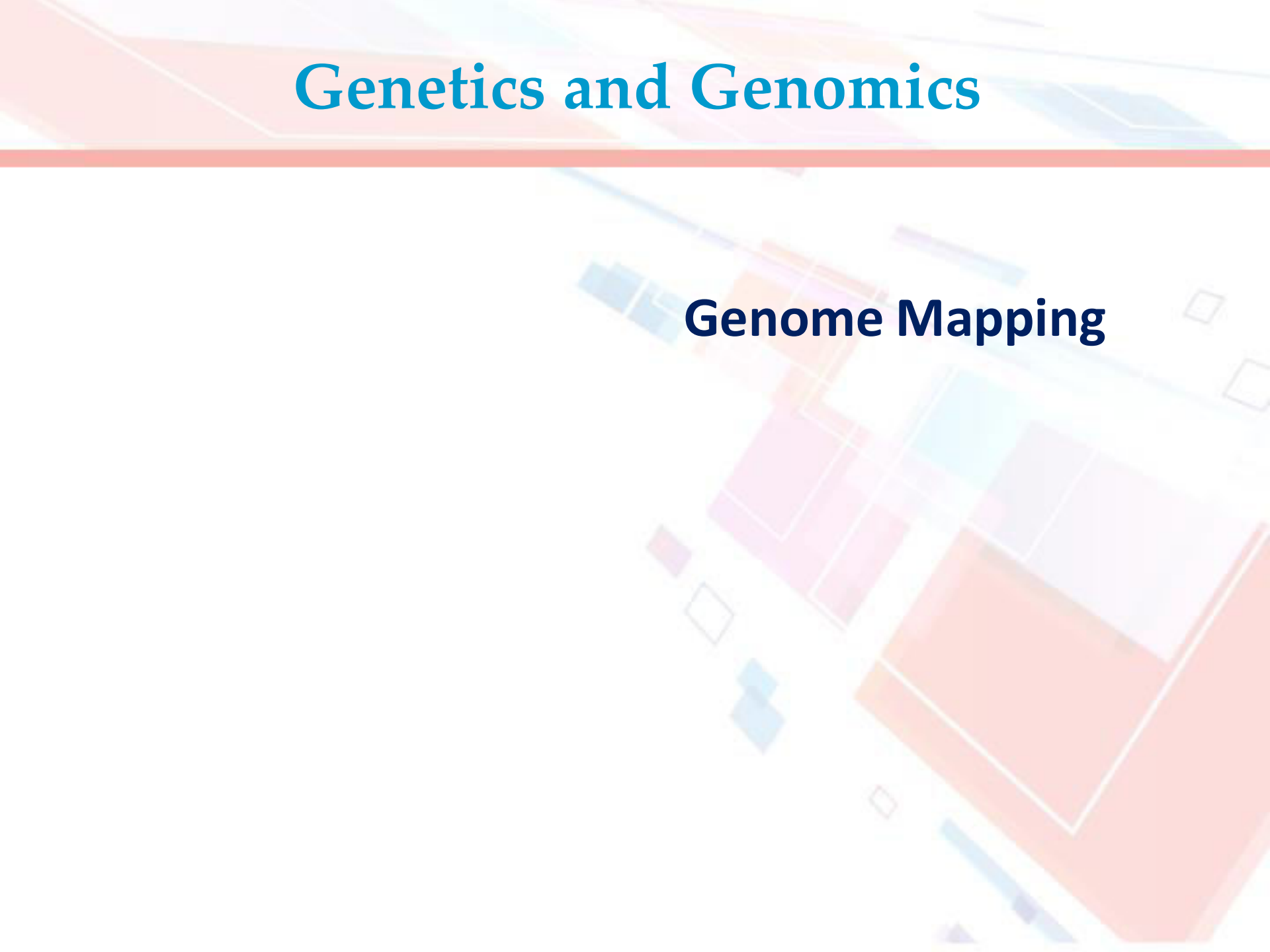
- Highly conserved genes have changed very little over the time
- These help to clarify relationships among species that diverged from each other long ago

Distantly/Closely Related Species

Conclusion

- Highly conserved genes have changed very little over the time
- These help to clarify relationships among species that diverged from each other long ago

Genetics and Genomics



Genome Mapping

Genome Mapping

Genome mapping

- Different types
- Genetic mapping
- Physical mapping

Genome Mapping

Genetic Mapping of Genomes

- Genetic mapping is based on the use of genetic techniques to construct maps showing the positions of genes and other sequence features on a genome.

Genome Mapping

Genetic Mapping

- Genetic techniques include cross-breeding experiments
- Case of humans, the examination of family histories (pedigrees).

Genome Mapping

Physical Mapping of Genomes

- Physical mapping uses molecular biology techniques to examine DNA molecules directly in order to construct maps showing the positions of sequence features, including genes.

Genome Mapping

Mapping of Genomes

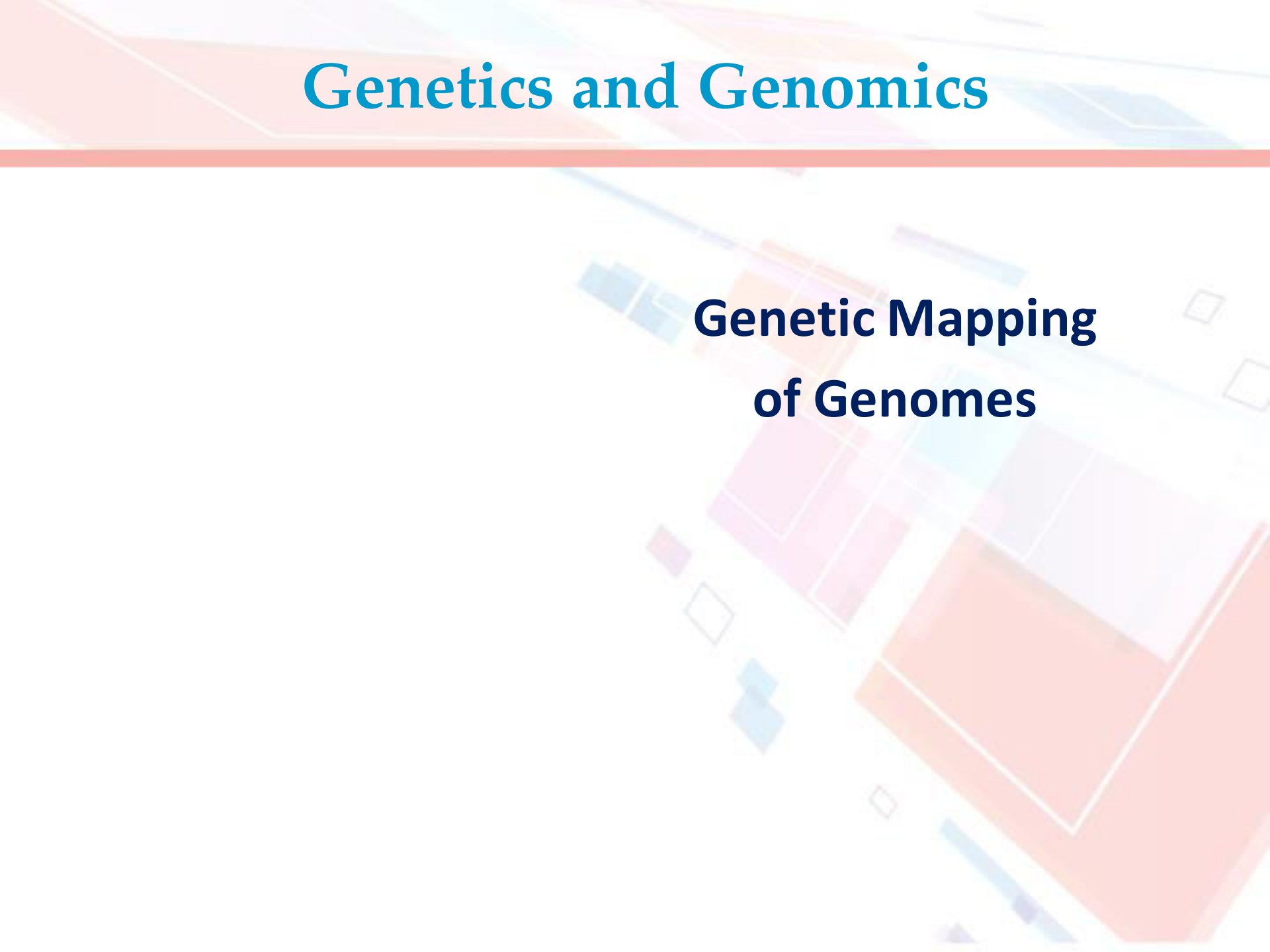
Mapping Strategy	Requires	Resolution	How to increase resolution
Genetic	Polymorphic Markers, and Pedigrees	Medium to High	Increase number of markers or people
Restriction	Restriction Enzymes	High	Increase number of enzymes
Somatic Cell Hybrid	Somatic Hybrid Panel, and STSs	Low to Medium	Increase number of deletion hybrids
Radiation Hybrid	Radiation Hybrid Panel, and STSs	High	Use additional hybrids, or make a new panel

Genome Mapping

Genome Mapping

- Different types
- Genetic mapping
- Physical mapping

Genetics and Genomics



Genetic Mapping of Genomes

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- Linkage analysis is the basis of genetic mapping
- Recombination fraction is a measure of the distance between two loci.

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- Two loci that show 1% recombination are defined as being 1 centimorgan (cM) apart on a genetic map.
- 1 map unit = 1 cM centimorgan

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- Genetic mapping involves determining, the location of genes on a chromosome relative to other genes, using genetic crosses and pedigree analysis.

Genetic Mapping of Genomes

Genetic Mapping

- Genetic map of the human genome has 24 different maps; one for each autosome, X and Y.
- Marker alleles in genetic crosses help to determine crossover rate between linked genes

Genetic Mapping of Genomes

Genetic Mapping of Genomes - Procedure

- Individuals with different alleles at two or more loci are crossed, and their offspring examined.
- Most of the offspring will have phenotypes corresponding to the linked alleles. A few progeny will be recombinant.
- The frequency of the recombinant phenotype is calculated as a percentage of the total offspring, giving the recombination frequency or genetic distance.

Genetic Mapping of Genomes

Genetic Mapping in Humans

- Experimental crosses are not done in humans and so genetic mapping relies on pedigree analysis, and is limited by rarity of large, multigenerational pedigrees showing segregation of defined linked traits.
- Usually, the lod (logarithm of odds) score method is used for statistical analysis of pedigree data.

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- A lod score compares the expected distributions of traits if they are linked or not linked.
- The lod score is the $\log_{10}$ of the ratio of the two probabilities. The higher the lod score, the closer the two genes.
- The map distance for linked markers is computed from the recombination frequency

Genetic Mapping of Genomes

Genetic Mapping

- Mapped features that are not genes are called as DNA markers.
- A DNA marker must have at least two alleles to be useful.
- Mostly, three types RFLP, SSLP, SNP

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- To test linkage between the genes for two traits, certain types of matings are examined whether or not the pattern of the combinations of traits exhibited by the offspring follows the law of independent assortment.
- If not, the gene pairs for those traits must be linked, that is they must be on the same chromosome pair.

Genetic Mapping of Genomes

Recombinations Frequencies (RF)

- Two genes that undergo independent assortment have RF of 50 % and are located on non-homologous chromosomes
- They are unlinked

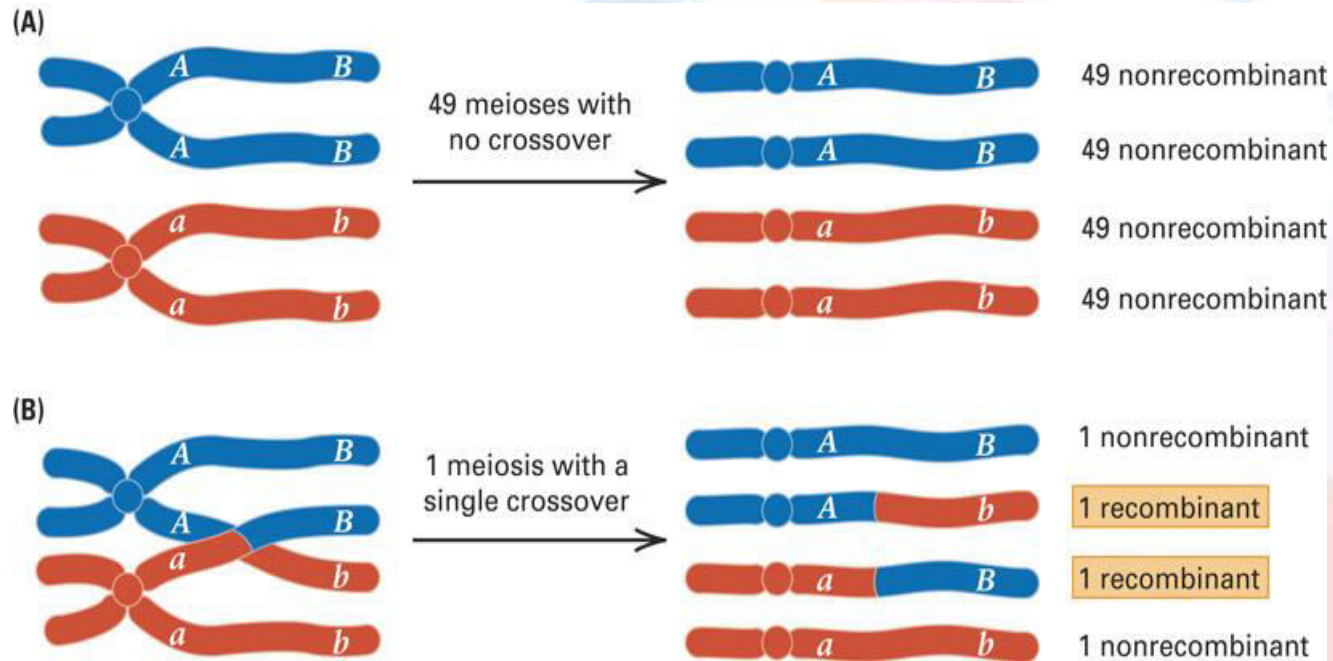
Genetic Mapping of Genomes

Recombinations Frequencies

- Genes with recombination frequencies less than 50 percent are on the same chromosome
- They are linked

Genetic Mapping of Genomes

Genetic Mapping of Genomes – Recombinations Frequency Calculated as:



(C) Frequency of recombination:

$$r = \frac{1 + 1}{49 + 49 + 49 + 49 + 1 + 1 + 1 + 1} = \frac{2}{200}$$

= 1 percent = 1 map unit = 1 cM

Genetic Mapping of Genomes

Genetic mapping of Genomes – Recombinations Frequency

$$\text{recombinant frequency} = \frac{\text{number of recombinant progeny}}{\text{total number of progeny}} \times 100\%$$

Genetic Mapping of Genomes

Genetic Mapping of Genomes – Recombinations frequency which is calculated as follows:

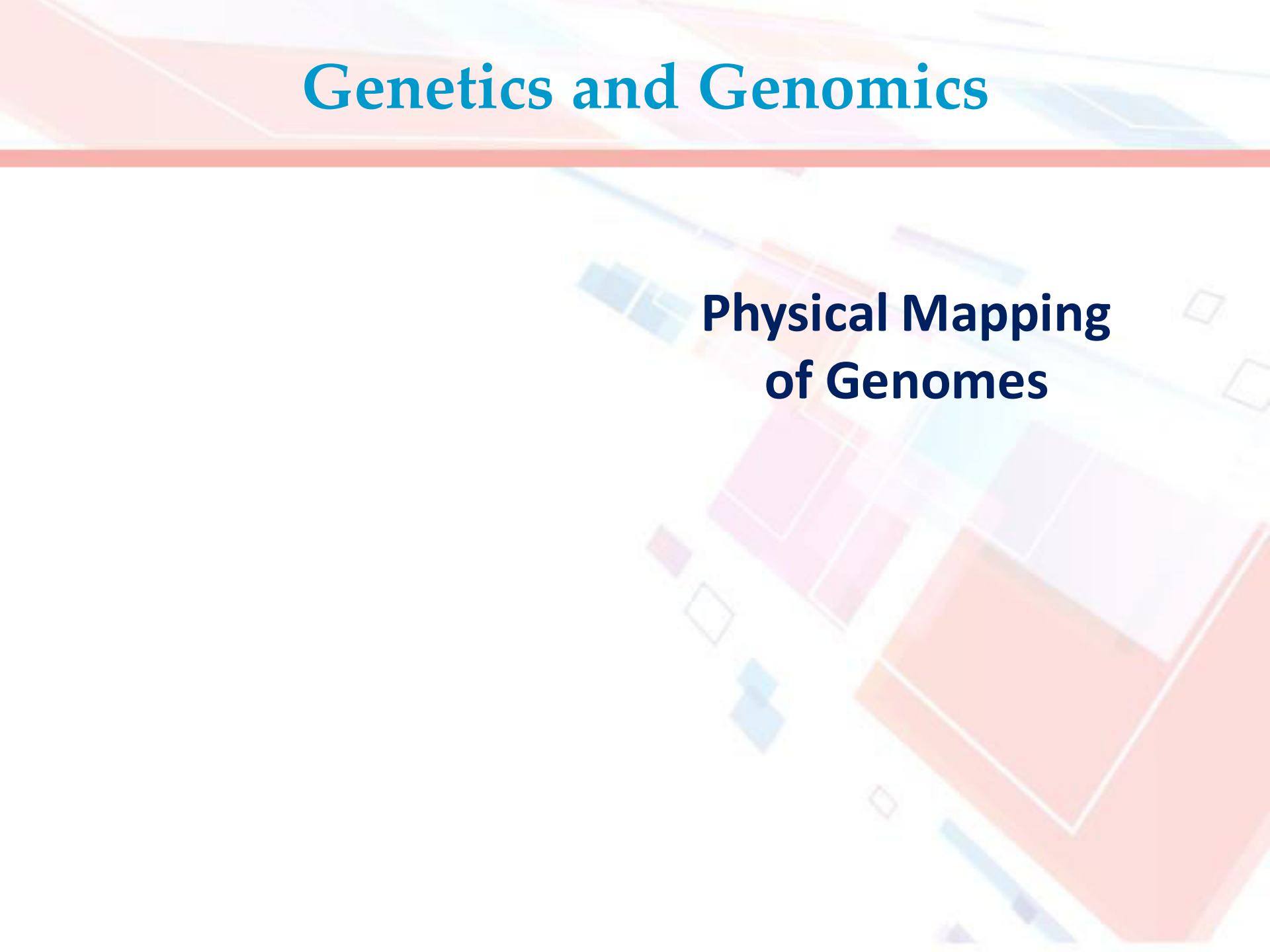
- The LOD score is calculated as follows:
- $LOD = Z = \log_{10} \frac{\text{probability of birth sequence with a given linkage}}{\text{probability of birth sequence with no linkage}}$
- A LOD score greater than 3.0 is considered evidence for linkage.
- A LOD score less than -2.0 is considered evidence to exclude linkage.

Genetic Mapping of Genomes

Genetic Mapping of Genomes

- Linkage analysis is the basis of genetic mapping
- Recombination fraction is a measure of the distance between two loci.

Genetics and Genomics



Physical Mapping of Genomes

Physical Mapping of Genomes

Physical Map of Genomes

- A physical map of a chromosome or genome that shows the physical locations of genes and other DNA sequences

Physical Mapping of Genomes

**Genetic Mapping
provides in-sufficient
information about
exact locations of
genes**

- Genetic Information by genetic map rarely sufficient for directing the sequencing phase of a genome project.

Physical Mapping of Genomes

Genetic Mapping – limited accuracy

- Two reasons
- The resolution of a genetic map depends on the number of crossovers that have been scored.
- Genetic maps have limited accuracy

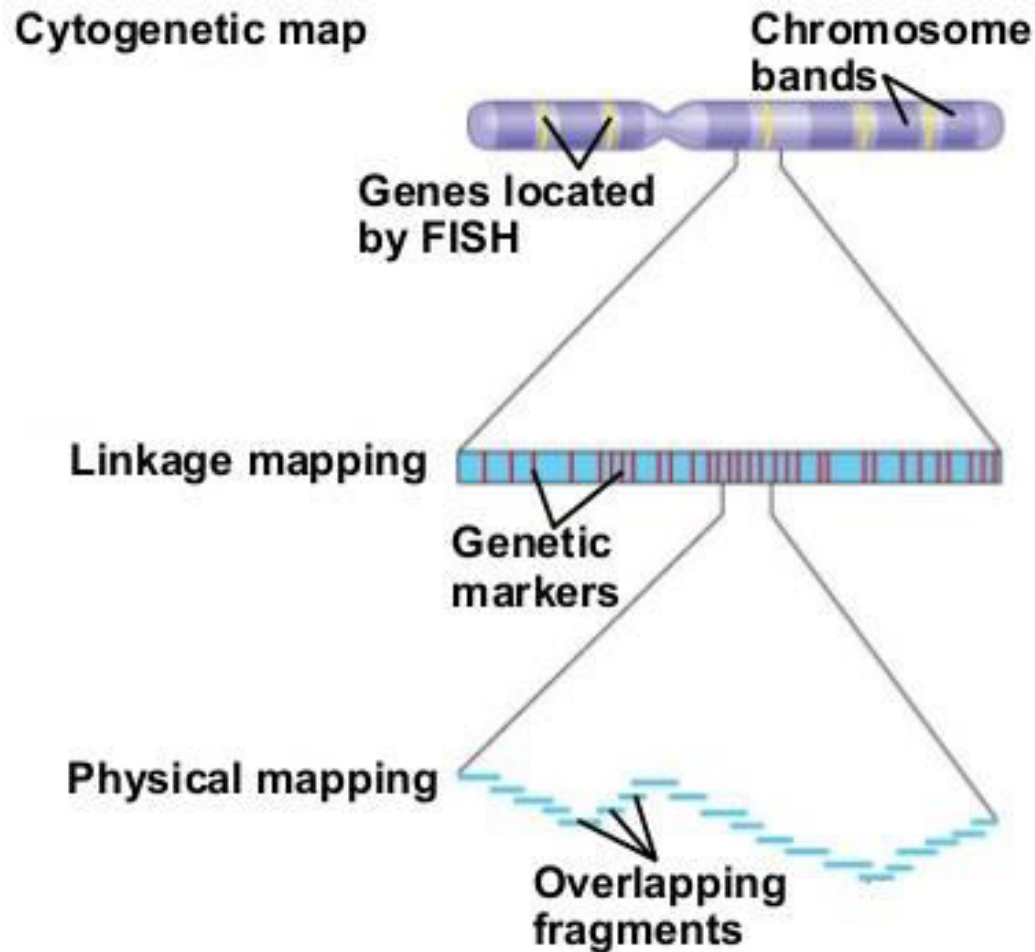
Physical Mapping of Genomes

Physical Mapping

- In humans, even the detailed genetic map lacks the required resolution.
- Therefore, a physical map derived directly from genomic DNA rather than analysis of recombinants has been generated.

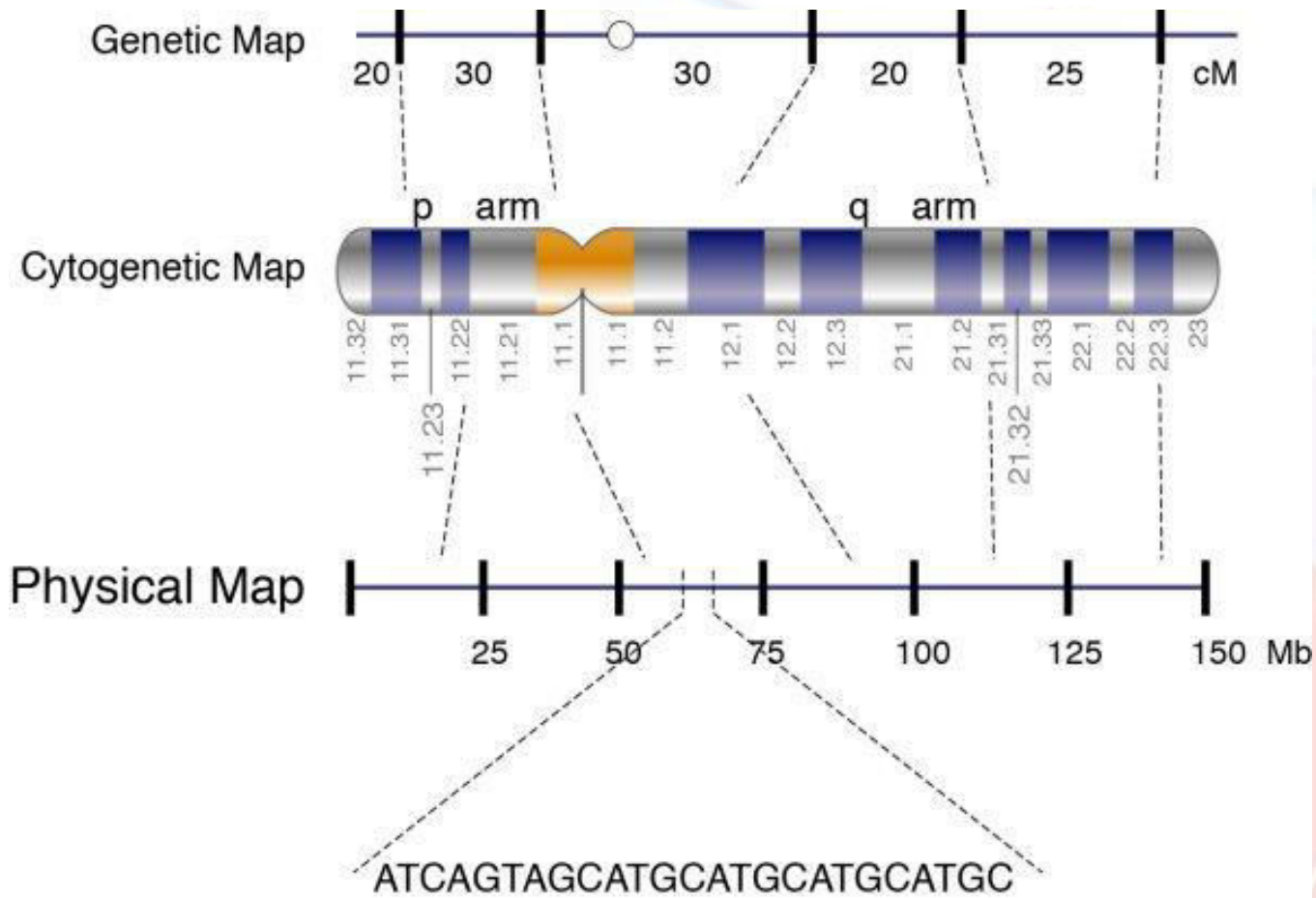
Physical Mapping of Genomes

Cytogenetic Map, Genetic Map, Physical Map



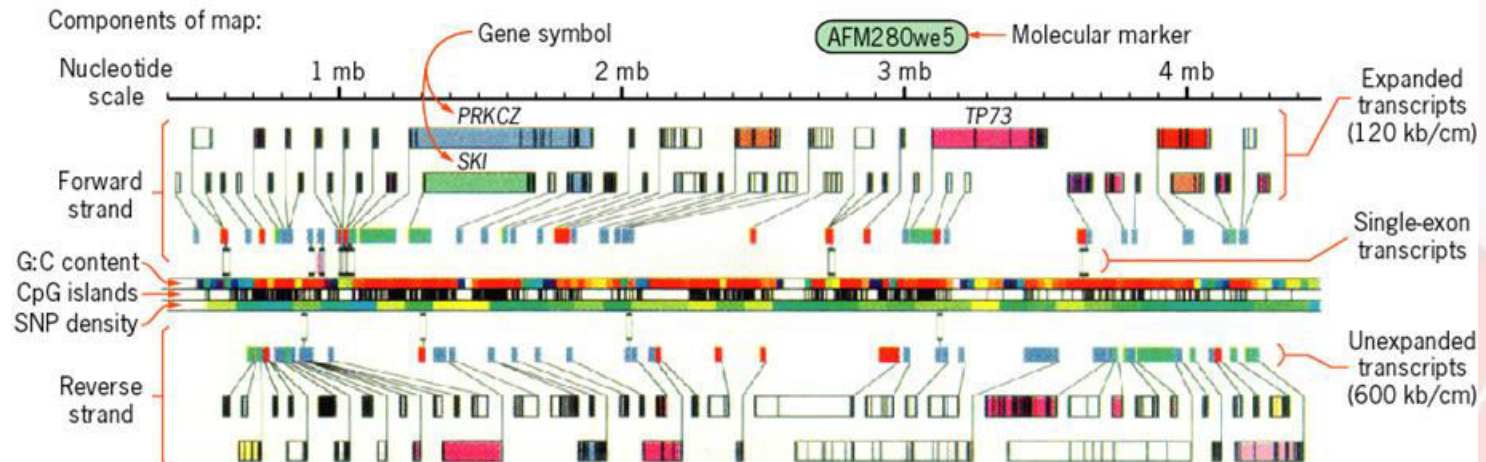
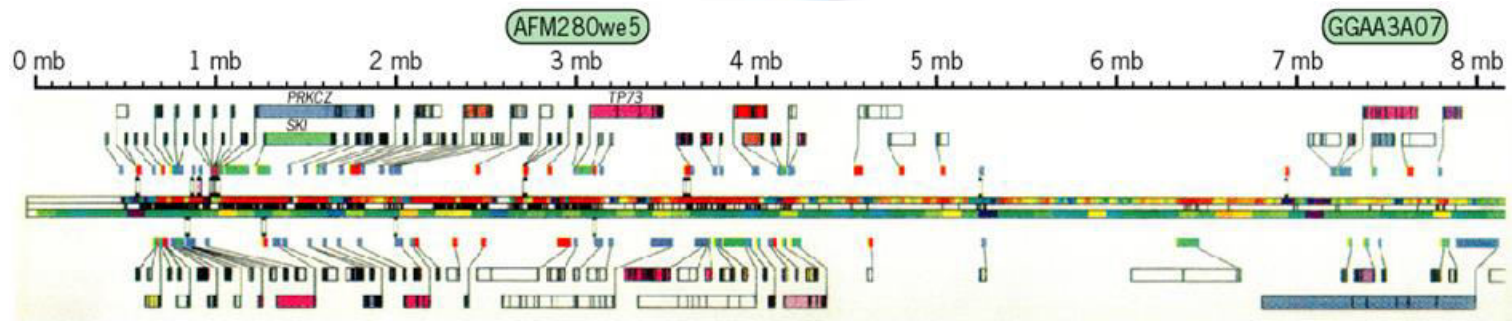
Physical Mapping of Genomes

Genetic Map and Physical Map



Physical Mapping of Genomes

Physical Map



Physical Mapping of Genomes

Physical Mapping of Genomes

- As in human, there are 24 physical maps, 22 autosomes plus X and Y.
- Types of physical maps are presented in order of increasing resolution:

Physical Mapping of Genomes

Physical Mapping Techniques

- Cytotogenetic map
- A restriction map
- Fluorescent in situ hybridization
- Sequence tagged site (STS) map
- Nucleotide sequence map

Physical Mapping of Genomes

Physical Mapping Techniques

- Plasmid
- Phagemid
- Cosmid
- YACs
- BACs

Physical Mapping of Genomes

Physical Map of Genomes

- A physical map of a chromosome or genome that shows the physical location of genes/DNA
- Low resolution - banding patterns; highest-resolution nucleotide sequence

Genetics and Genomics

Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes

Cytogenetic Mapping

- Microscopic examination of stained chromosome reveals a banding patterns
- Regions - designated based on their chromosomal position relative to the centromere

Cytogenetic Mapping of Genomes

Cytogenetic Mapping

- Chromosomal banding pattern.
- These methods allow a rough determination of locations, but not to yield a direct measure of distance.

Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes

p13.3

PLM
SCK
MAdCAM-1
CDC34
GZMM
BSG
POLRMT
PALM
AZU1
PRTN3
ELA2
Adipsin
POLR2E
GPX4
STK11
CIRP
EFNA2
PSST
RPS15
PCSK4
TCF3
AMH
OAZ

p13.2

SCYA25
RPS28
ZNF20
ZNF121
DNMT
ICAM1
ICAM3
TYK2
PDE4A
CDKN2D
LDLR
EPOR
PRKCSH
CNN1
AC P5
ZNF136
ZNF58
ZNF55
MANB
DHPS
JUNB

p13.1

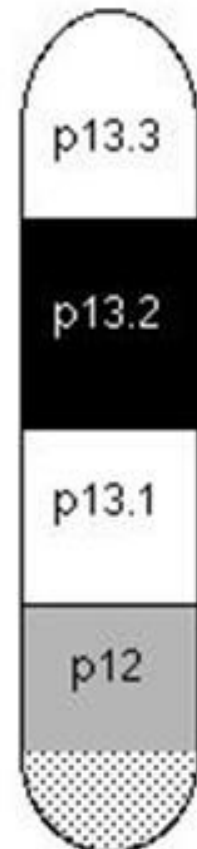
RFX1
CD97
SLC1A6R
NOTCH3
CYP4F3
MEL
MYO9B
MELL1
ERBAL2
INSL3
JAK3

q13.2

AKT2
SNRPA
CYP2F1
CYP2A6
CYP2A7P

p12

RAB3A
PDE4C
JUND
FKBP38
ELL
UBA52
RPL40
COMP
pNORF1
GDF-1
UOG-1
MEF2B
NEUC
ZNF91
ZNF95
ZNF98
ZNF97



Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes

- Cytogenetic
 - Visual study of chromosomes at microscopic level
- Karyotype
 - Chromosome complement
 - also applied to picture of chromosomes
- Idiogram
 - Stylised form of karyotype

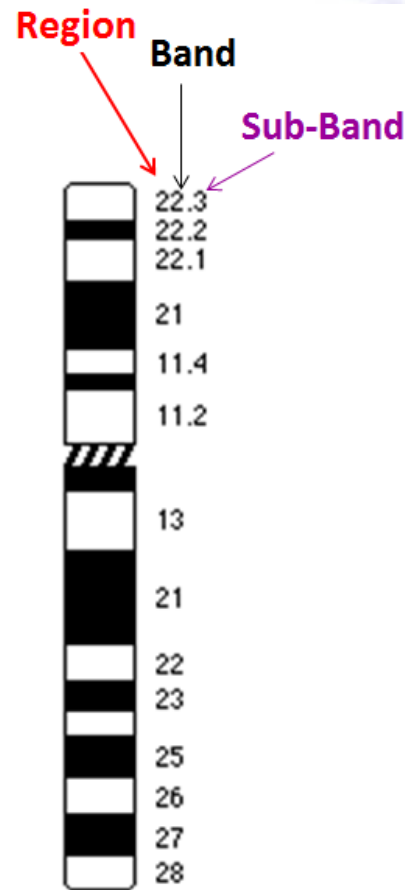
Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes

- Classified according to position of centromere
- Central centromere - metacentric
- Sub-terminal centromere - acrocentric
 - have satellites which contain multiple copies of genes for ribosomal RNA on short arm
- Intermediate centromere – submetacentric
- Terminal centromere - Telocentric

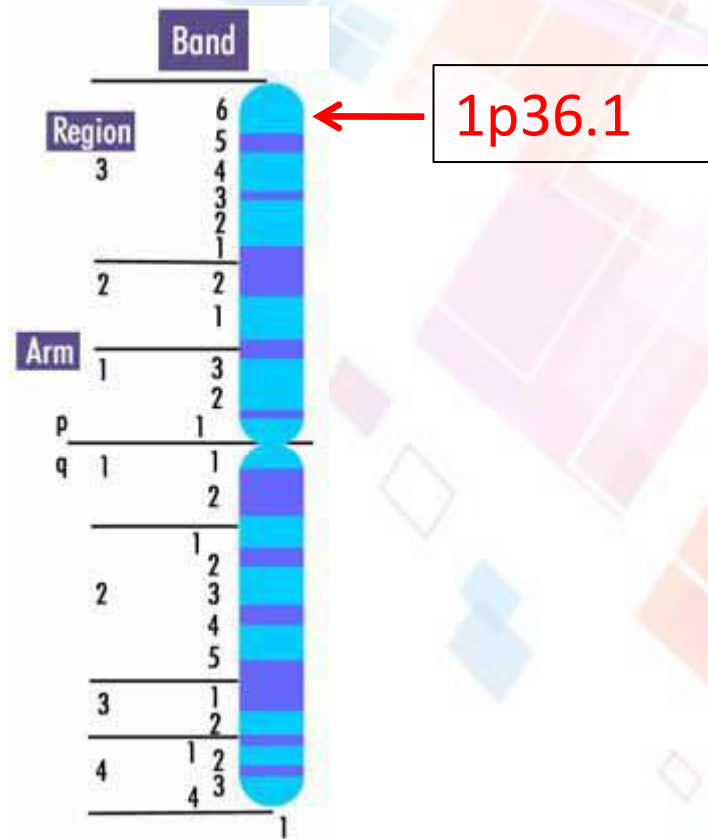
Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes



Cytogenetic Mapping of Genomes

Cytogenetic Mapping of Genomes: Gene is located at Chromosome 1p36.1



Cytogenetic Mapping of Genomes

Cytogenetic Mapping

- Microscopic examination of stained chromosome reveals a pattern of bands.
- Regions are designated based on their chromosomal position relative to the centromere

Genetics and Genomics

FISH Mapping of Genomes

FISH Mapping of Genomes

FISH Mapping

- Fluorescence in situ hybridization (FISH) is a type of mapping that uses fluorescent probes that bind to only those parts of the chromosome with a high degree of sequence complementarity.

FISH Mapping of Genomes

FISH Mapping

- It is used to detect and localize the presence or absence of specific DNA sequences on chromosomes.

FISH Mapping of Genomes

FISH Mapping

- Fluorescence microscopy used to find out where fluorescent probe is bound to the chromosomes.
- Used in genetic counseling, medicine and species identification.

FISH Mapping of Genomes

FISH Mapping

- Individual metaphase chromosomes are probed in situ with specific fluorescently labeled DNA sequences, identifying homologous sequences in the chromosome

FISH Mapping of Genomes

FISH Mapping

- Different probes labeled with different fluorescent dyes may be used in same experiment.
- Fluorescence microscopy provides data for computer imaging analysis to determine binding site for each probe.

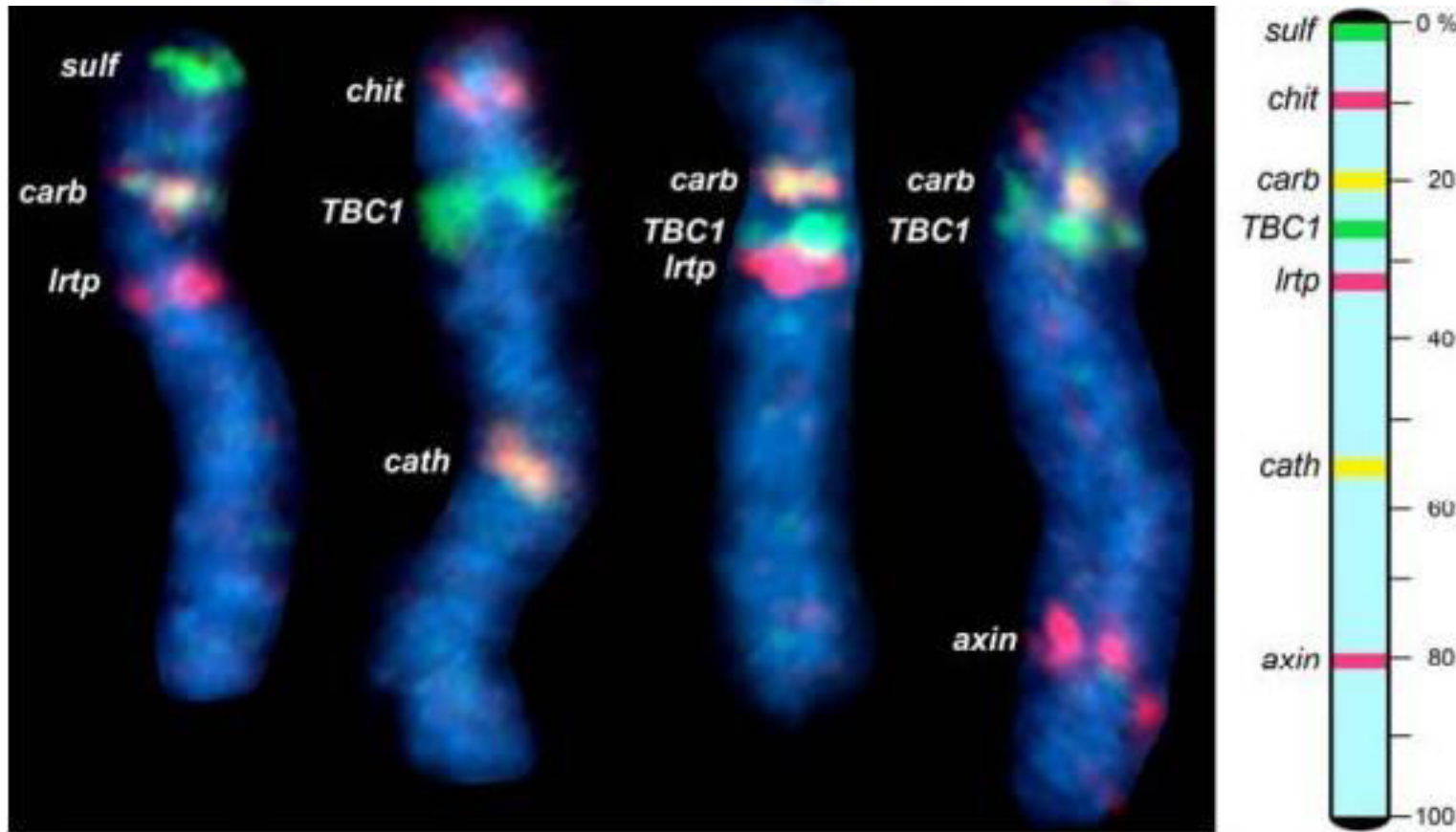
FISH Mapping of Genomes

FISH Mapping

- With a resolution of 2-5 Mb in metaphase chromosomes, FISH can localize markers to sub-regions of chromosomal bands.
- Less condensed chromosomes may be resolved in the 5-700 kb range.

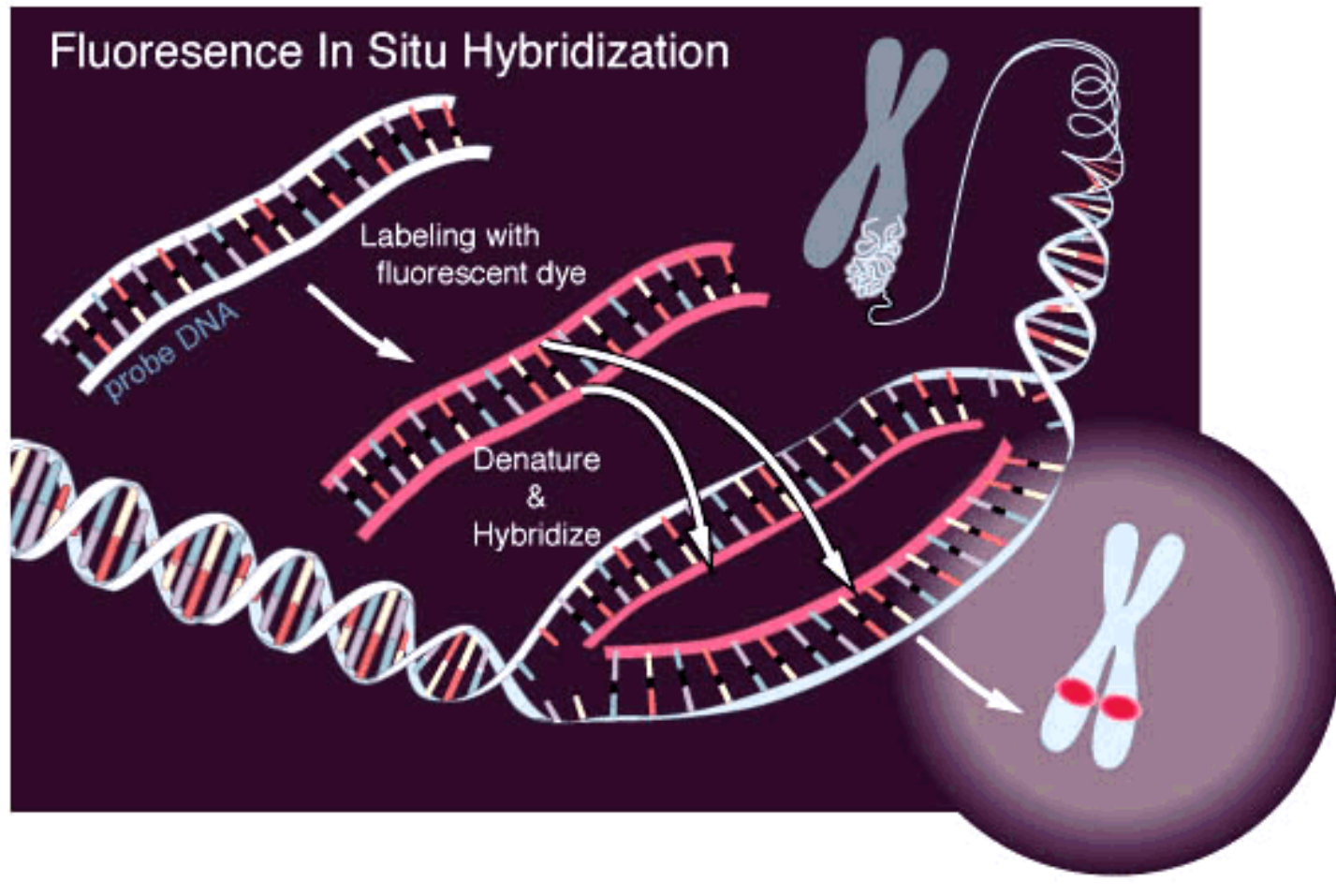
FISH Mapping of Genomes

FISH Mapping of Genomes



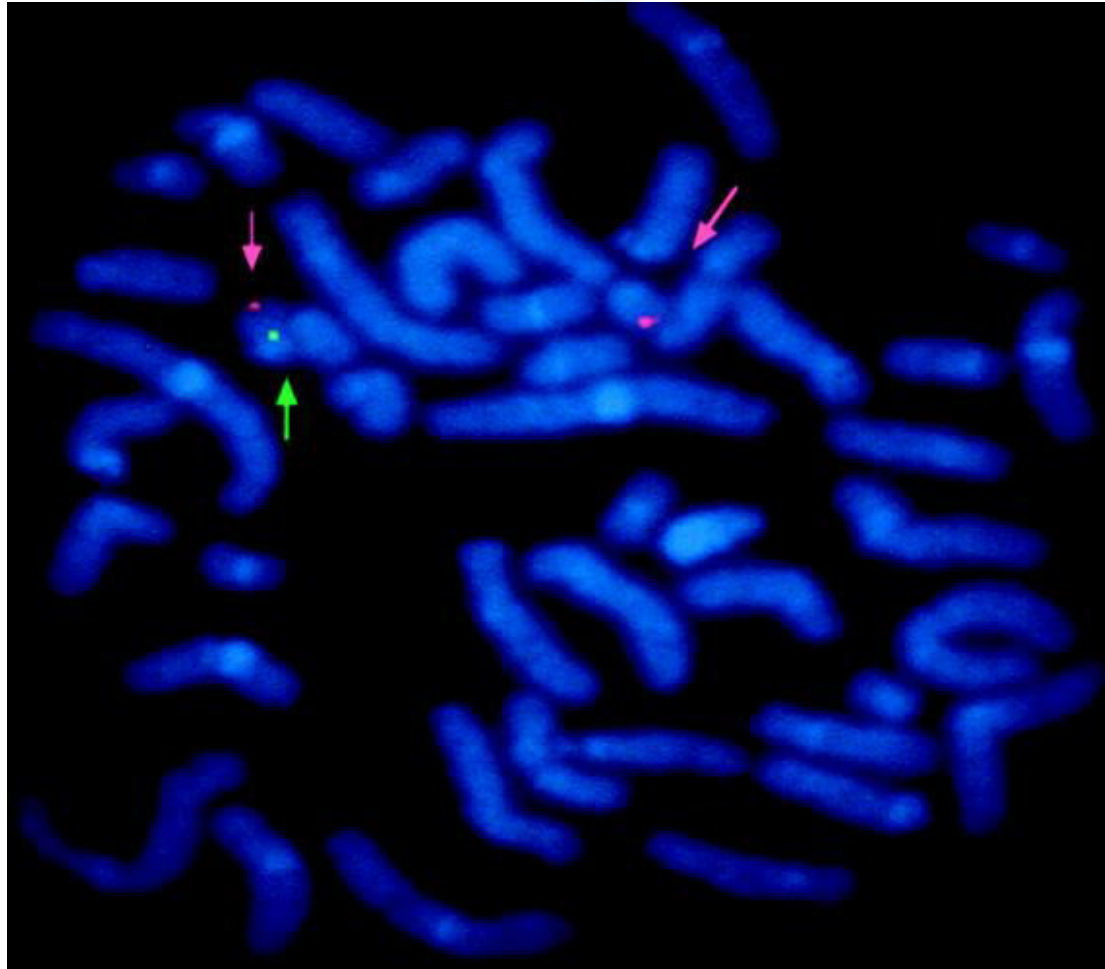
FISH Mapping of Genomes

FISH Mapping of Genomes



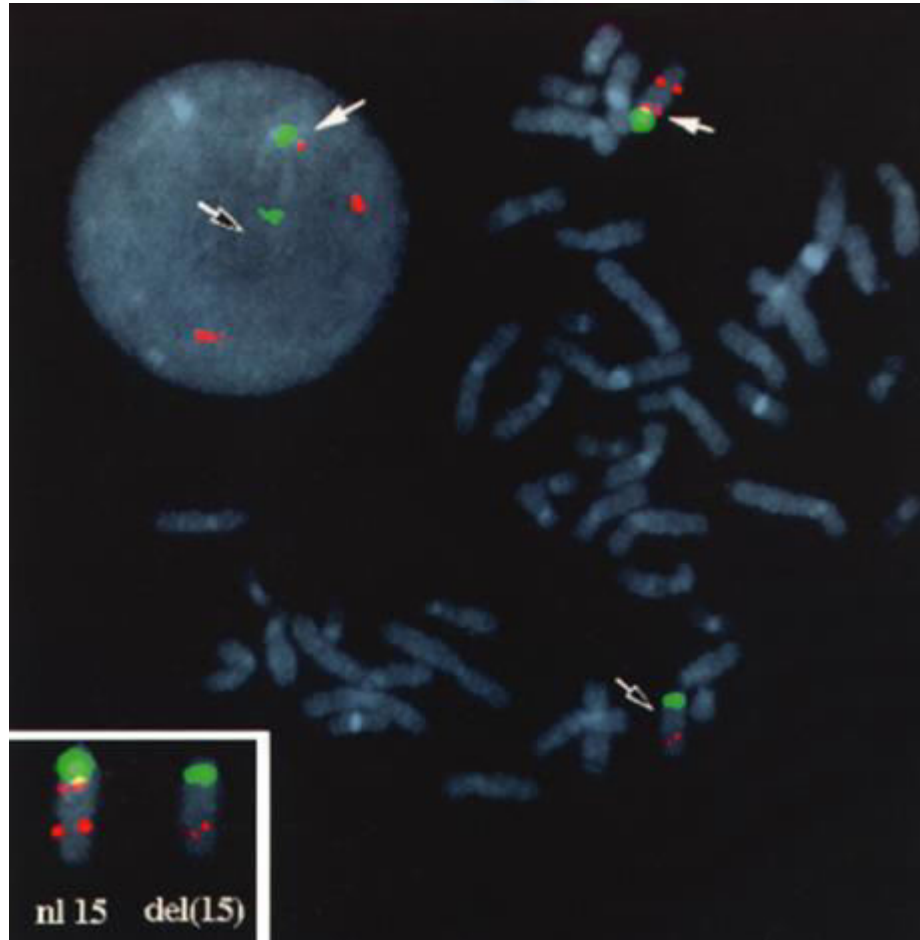
FISH Mapping of Genomes

FISH Mapping of Genomes - Microdeletions



FISH Mapping of Genomes

FISH Mapping of Genomes - Microdeletions



FISH Mapping of Genomes

FISH Mapping of Genomes

- Advantages
Highly specific,
Microdeletions
/Microduplications
- Disadvantages
500-600 probes
needed to match
the power of
karyotyping

Genetics and Genomics

Restriction Mapping of Genomes

Restriction Mapping of Genomes

Restriction Mapping

- A restriction map is a map of known restriction sites within a sequence of DNA.
- Restriction mapping requires the use of restriction enzymes.

Restriction Mapping of Genomes

Restriction Mapping

- Restriction enzymes are used that cut DNA due to the recognition sequence in the DNA/genome under study

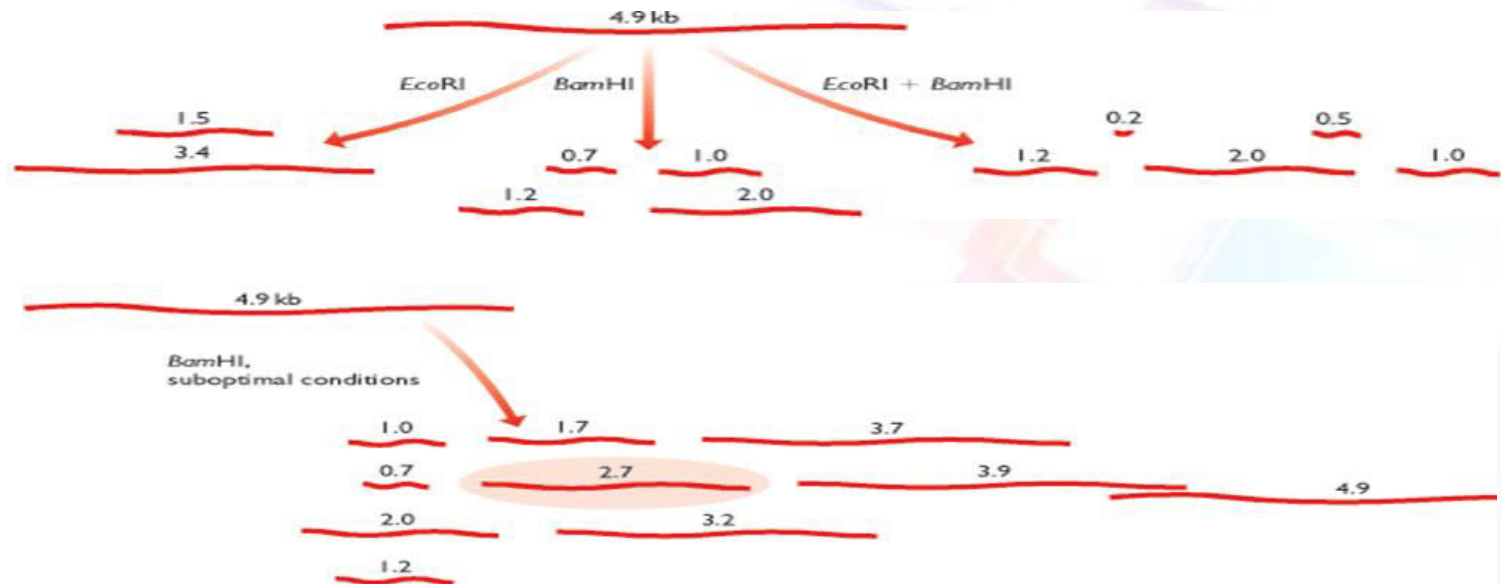
Restriction Mapping of Genomes

Restriction Mapping

- To construct a restriction map is to compare the fragment sizes produced when a DNA molecule is digested with different restriction enzymes that recognize different target sequences.

Restriction Mapping of Genomes

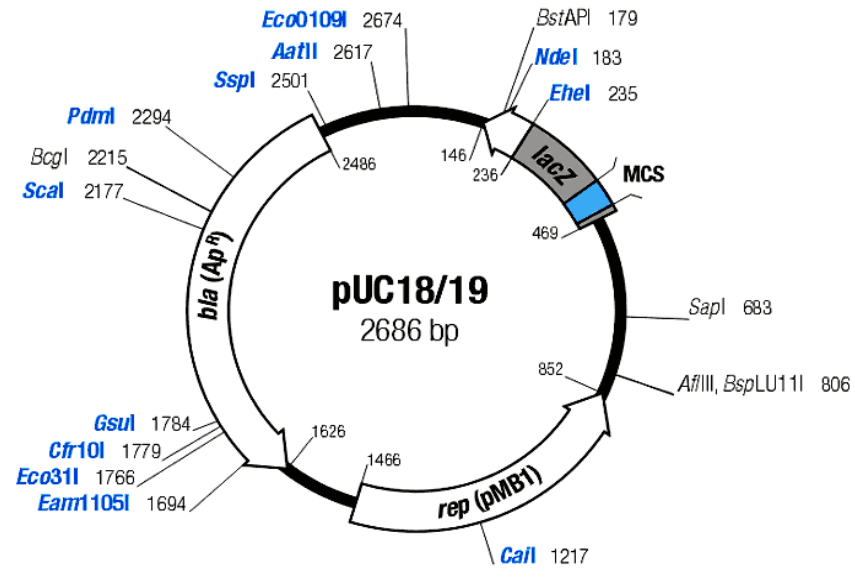
Restriction Mapping of Genomes



Restriction Mapping of Genomes

Restriction Mapping of Genomes

- A map showing positions of restriction sites in a DNA sequence
- If DNA sequence is known then construction of restriction map is a trivial exercise
- In early days of molecular biology DNA sequences were often unknown



Restriction Mapping of Genomes

Restriction Mapping of Genomes: Example

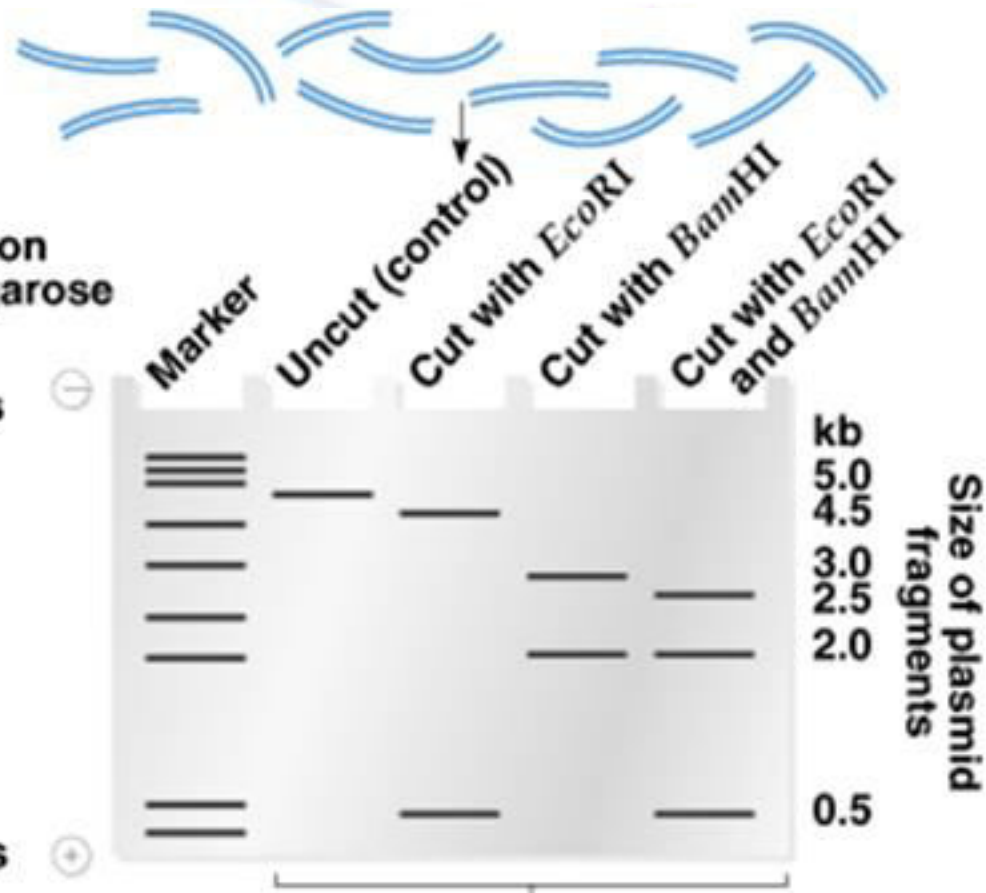
Many copies of a cloned 5.0 kb linear DNA fragment

Analyze DNA by restriction enzyme cleavage and agarose gel electrophoresis.

Longer DNAs

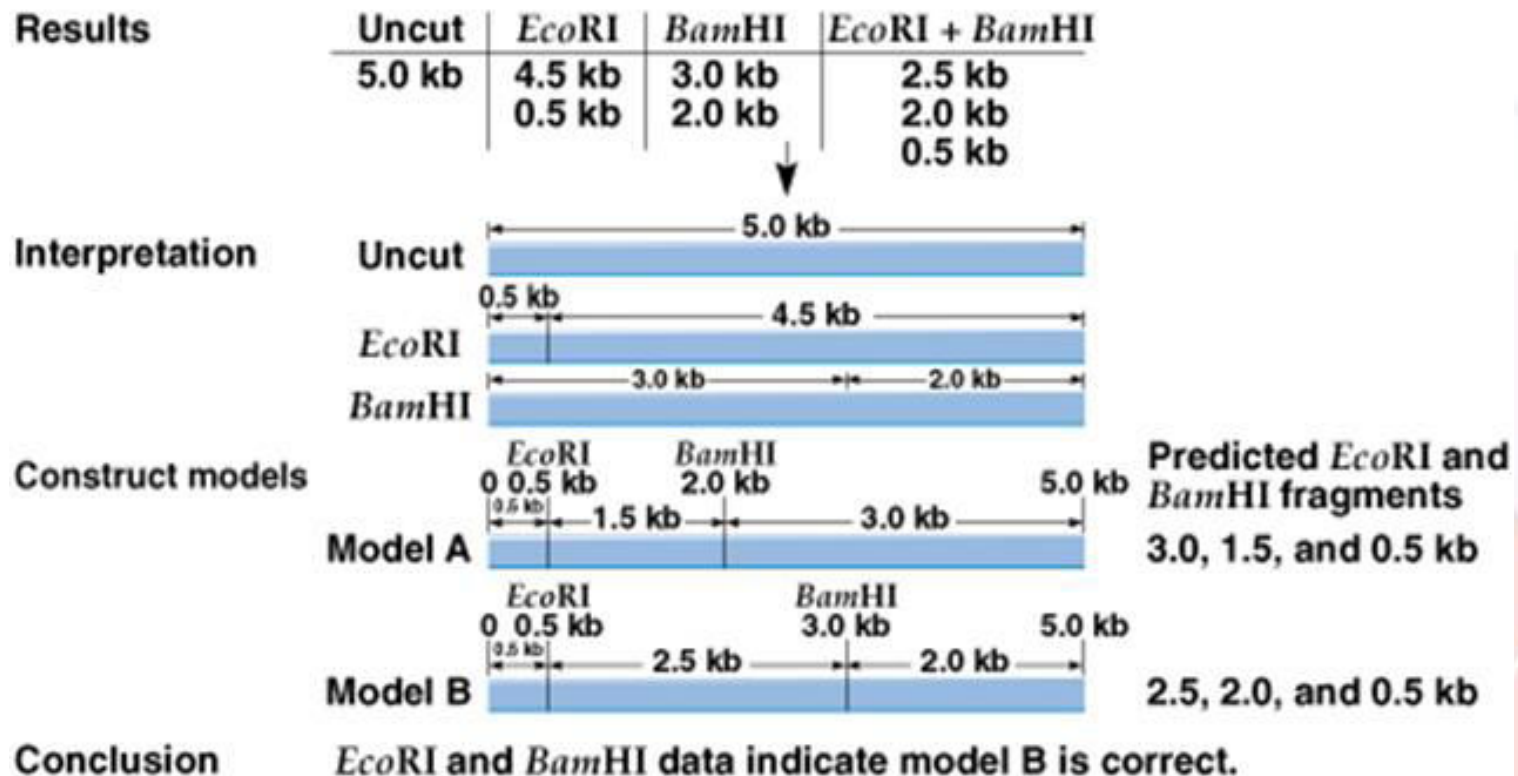
Migration

Shorter DNAs



Restriction Mapping of Genomes

Restriction Mapping of Genomes: How to Construct Restriction Enzyme Map



Restriction Mapping of Genomes

Limitations of Restriction Mapping

- Restriction mapping is more applicable to small rather than large molecules.

Restriction Mapping of Genomes

Limitations of Restriction Mapping

- In practice, if a DNA molecule is less than 50 kb in length it is usually possible to construct a restriction map.
- To construct map for large genomic size is difficult

Restriction Mapping of Genomes

Limitations of Restriction Mapping

- The limitations of restriction mapping can be eased slightly by choosing enzymes expected to have infrequent cut sites(rare cutter) in the target DNA molecule.

Genetics and Genomics

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping of Genomes

- A radiation hybrid is a rodent cell line carrying a small genomic DNA molecule from another organism e.g. a human.

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping of Genomes

- Exposure to X rays breaks the DNA in human cells.
- The fragments become smaller with more X ray exposure
- Fragments length determines the map resolution

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping of Genomes

- Irradiation kills the human cells, which are then fused with rodent cells, rescuing chromosomal fragments that are typically a few Mb in length.

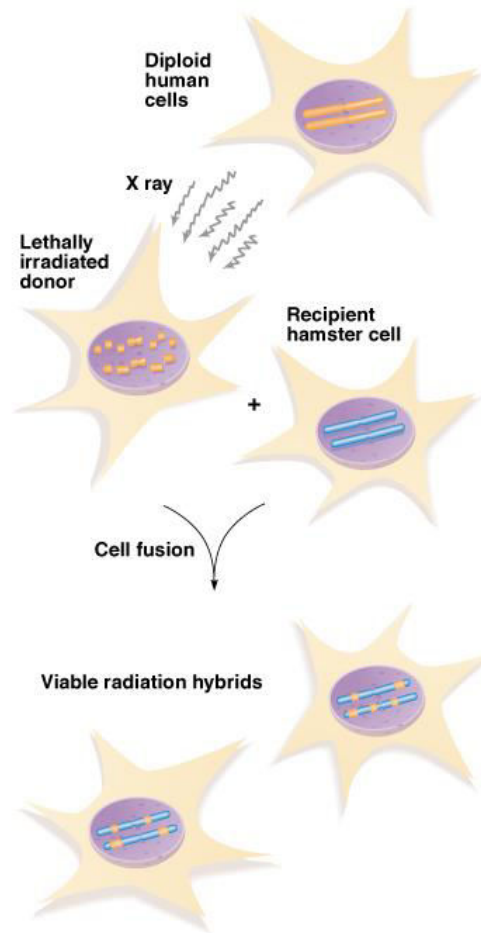
Radiation Hybrid Mapping of Genomes

Radiation Hybrid

- Human DNA in the RH is analyzed for gene and/or DNA markers.
- Closer the two markers to each other – chromosome; the more likely they are to be found together in an RH.

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping of Genomes



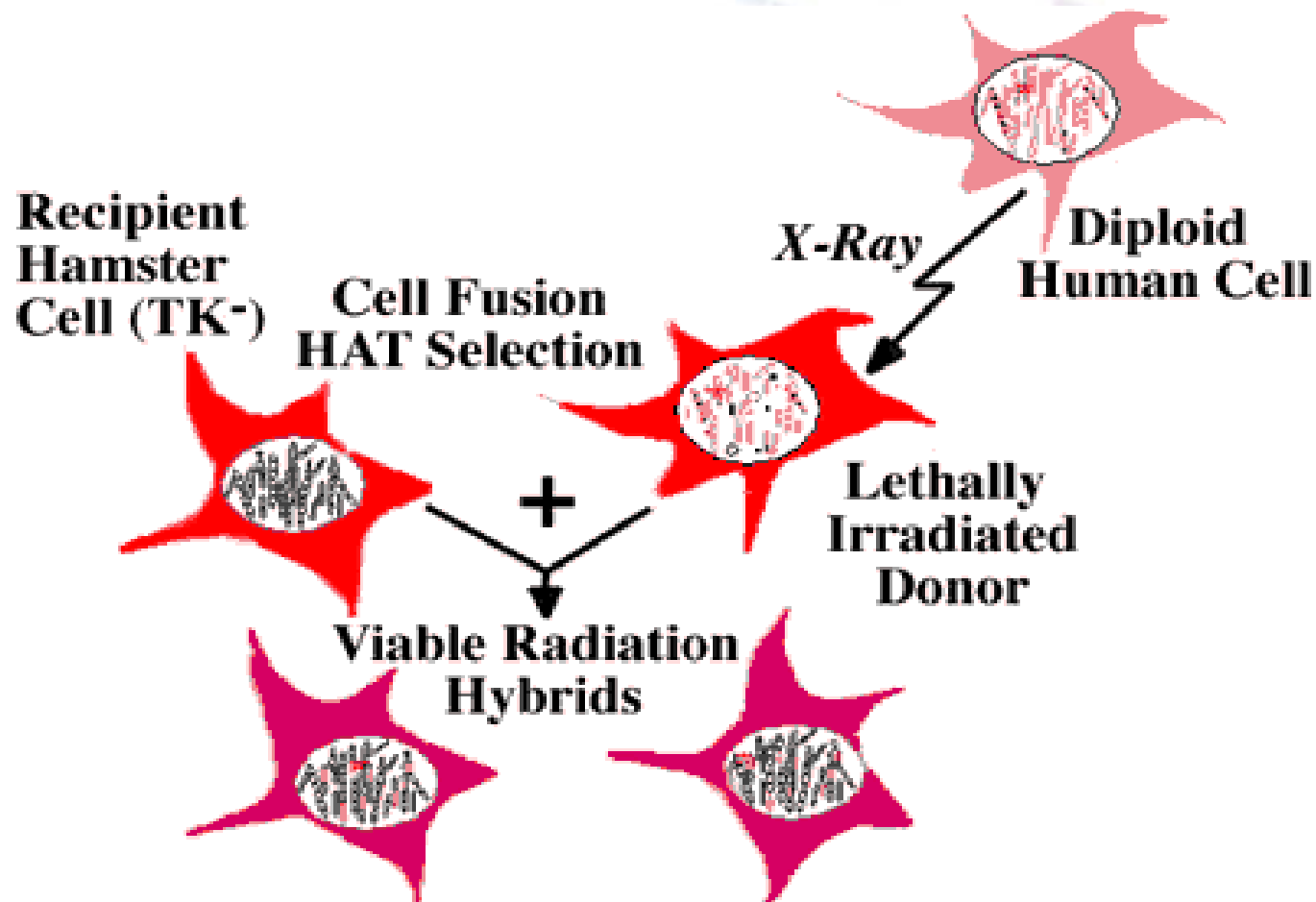
Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping: Methodology

- Standard somatic cell fusions contain entire human chromosomes. To locate a gene more closely, you need to use chromosome fragments.
- Start by irradiating human cells with a controlled dose of X-rays: chromosomes break up. Then, fuse the cells to rodent cells.
- The human chromosome fragments get integrated into mouse/rodent chromosomes
- Hybrid cell lines contain random human chromosome fragments.

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping



Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping

- Radiation hybrid mapping is a method for high-resolution mapping.
- Exploits the ability of rodent cells (hamster cells) to stably incorporate genetic material from fused cells.

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping

- Resolution is tunable
- Relatively cheap
- Difficult to compare results from different groups

Genetics and Genomics

Clone Contigs Mapping of Genomes

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- A partial restriction digest produces a set of large, overlapping DNAs, which are cloned into YAC/BAC vector

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- Shearing may also be used to make high-molecular-weight DNA that is blunt-end cloned into a YAC/BAC.

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- An entire genome or single chromosome may be represented in a YAC clone library
- YAC clones are then assembled into a map.

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- Maps can be generated either by matching with a FISH-generated chromosome map or by DNA fingerprinting
- Assembly of clone contigs is based on clone overlaps.

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- Non-polymorphic short tandem sequence are especially useful for YAC contig mapping

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- A complete library should yield a complete contig map that indicates the order in which the cloned fragments occur in the chromosome

Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- Problems arise when some of the YAC inserts contain DNA from more than one chromosomal location.
- Complicated the efforts at generating a YAC contig map of human chromosome

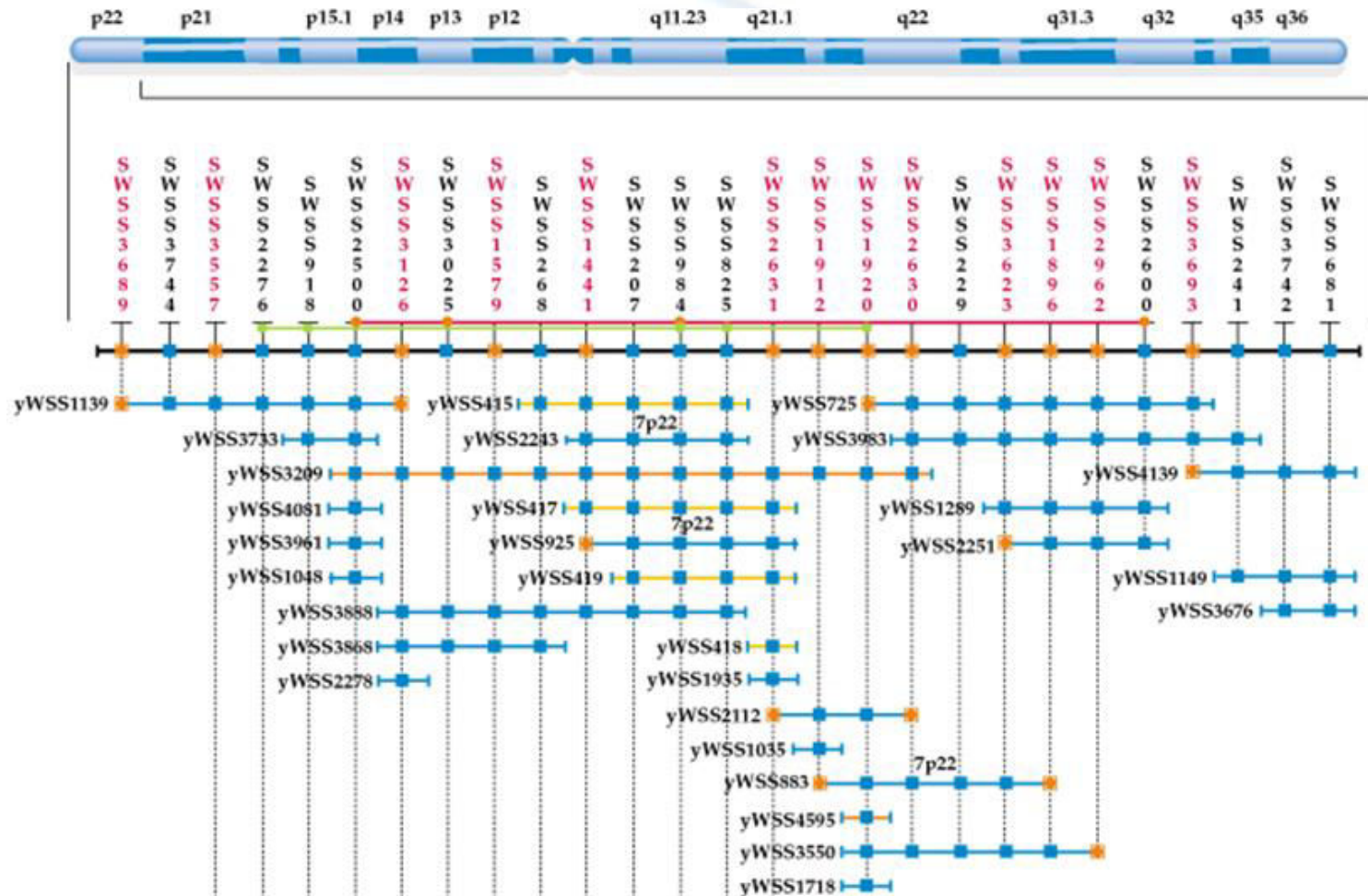
Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- Many labs have switched to BAC vectors with a capacity of 300 kb and the ability to replicate in *E. coli* as a resource for their sequencing projects.

Clone Contigs Mapping of Genomes

YAC Contigs Map of Chromosome 7



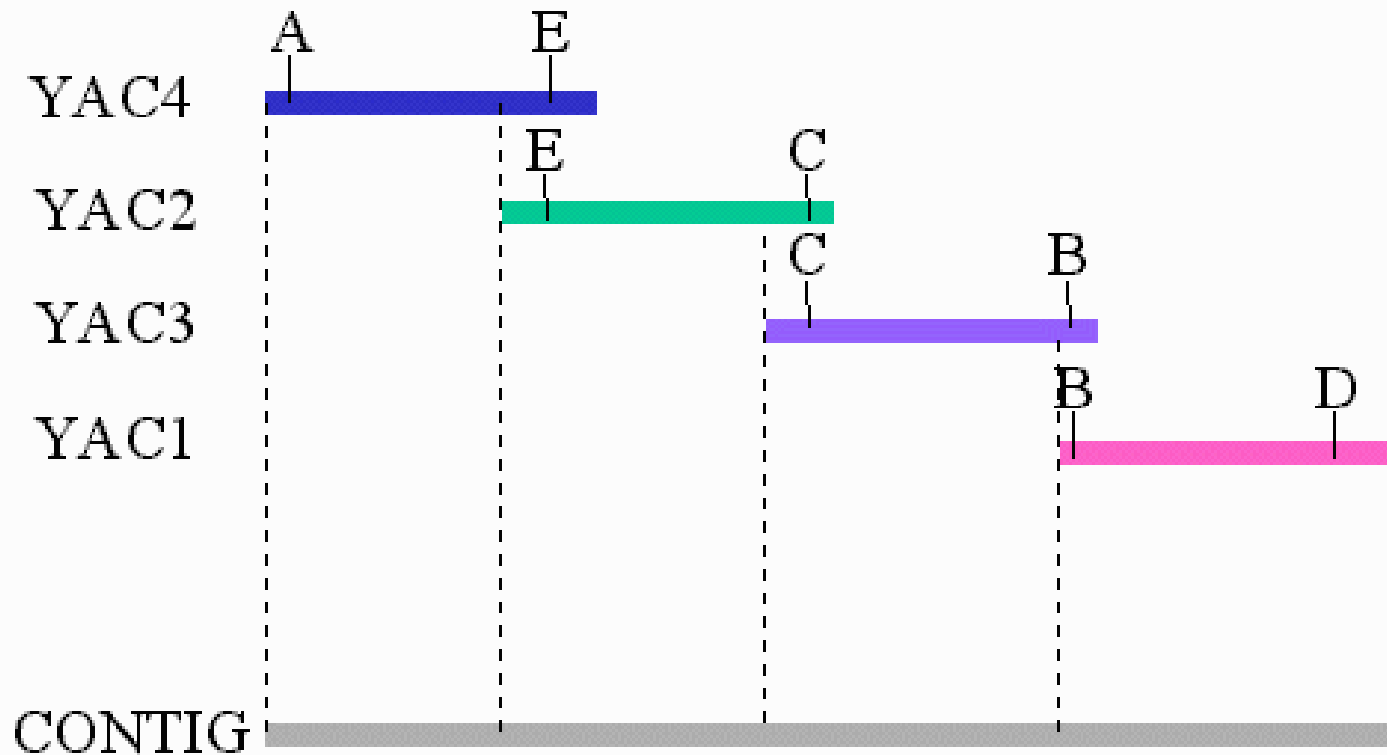
Clone Contigs Mapping of Genomes

Clone Contigs Mapping of Genomes

- A contig is a set of partially overlapping clones, a contiguous set of clones. No gaps between them.
- Contigs allow you to build up the sequence of the chromosome over much larger regions than any single clone.
- The first reasonably complete physical map of the human genome involved contigs generated by YACs.

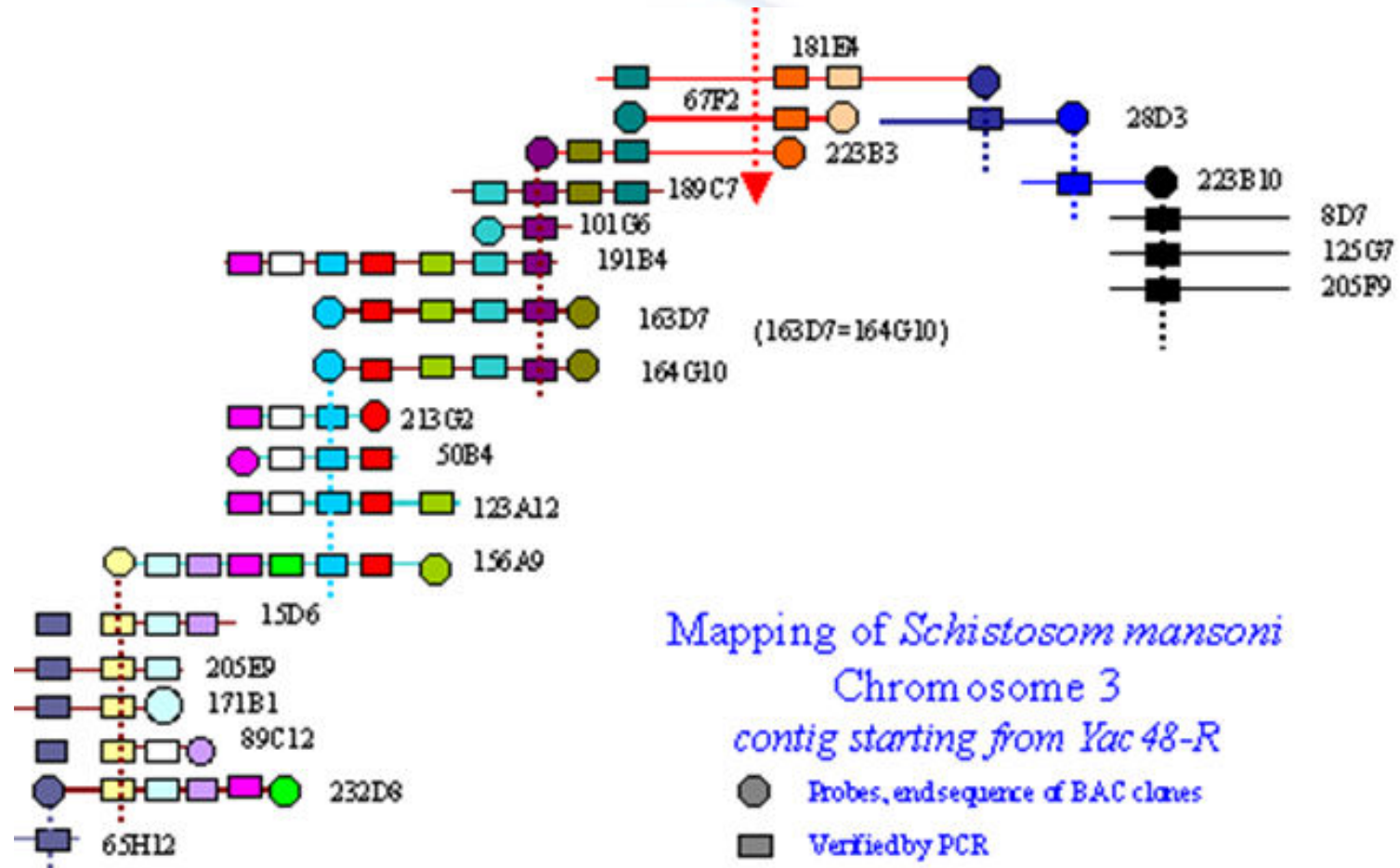
Clone Contigs Mapping of Genomes

Clone Contigs Mapping of Genomes



Clone Contigs Mapping of Genomes

Clone Contigs Mapping of Genomes: Yac48-R

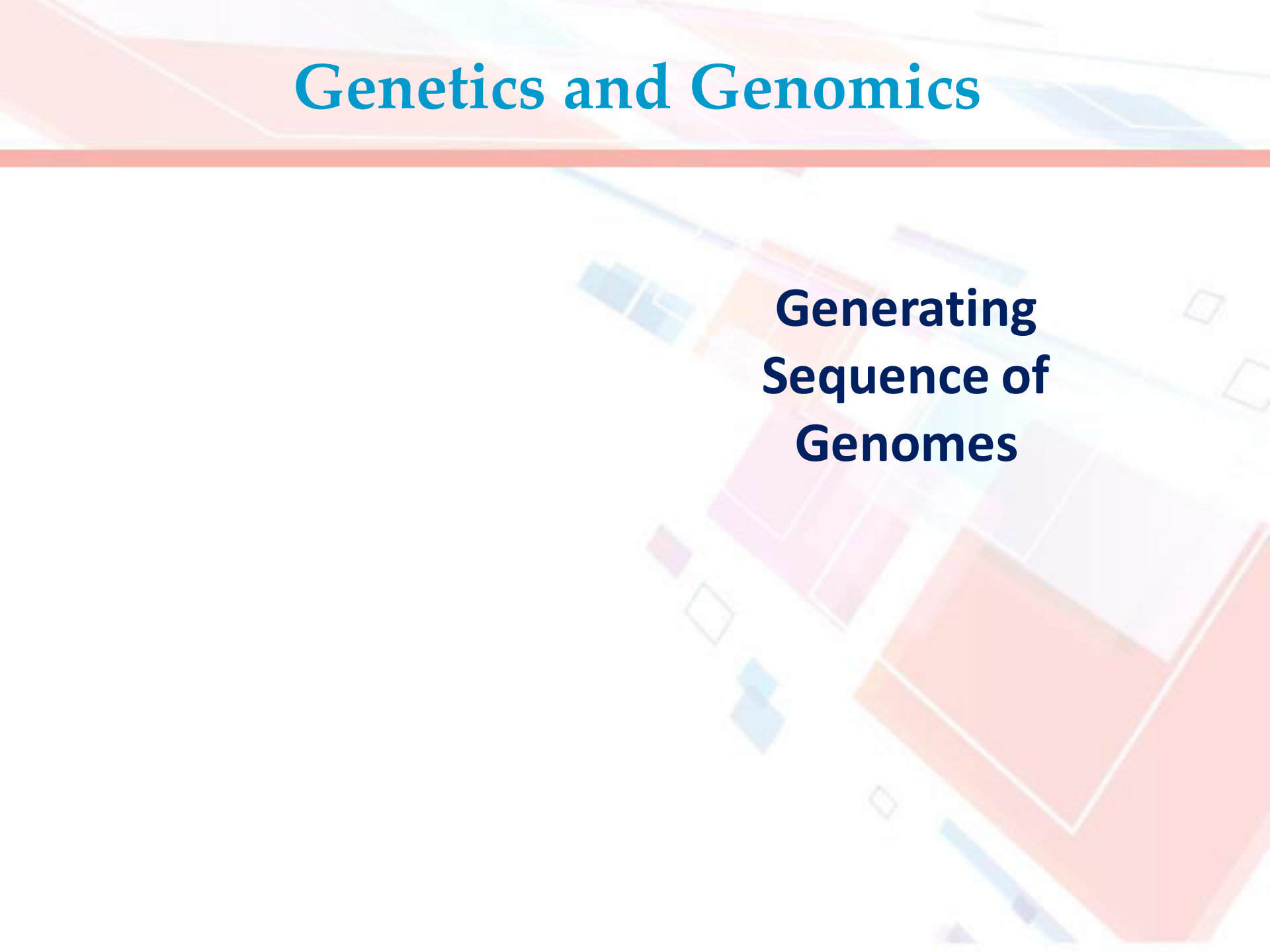


Clone Contigs Mapping of Genomes

Clone Contigs Mapping

- An entire genome or single chromosome may be represented in a YAC/BAC clone library

Genetics and Genomics



Generating Sequence of Genomes

Generating Sequence of Genomes

Generating Sequence of Genomes

- Dideoxy sequencing used to sequence the genomes

Generating Sequence of Genomes

Generating Sequence of Genomes

- One sequencing reaction is limited to on average 500 nucleotides, and for accurate sequences both strands were sequenced.

Generating Sequence of Genomes

Generating Sequence of Genomes

- Two competing sequencing strategies used
- Clone-by-clone
- Whole-genome shotgun

Generating Sequence of Genomes

Generating Sequence of Genomes: Clone-by-Clone Shotgun Sequencing

- Map construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

Generating Sequence of Genomes

Generating Sequence

- Human genome sequencing by the mapping approach used BACs
- BAC insert too large to sequence in one reaction.
- BAC inserts were sequenced using a shotgun approach

Generating Sequence of Genomes

Generating Sequence of Genomes

- Each insert is cut from the vector, sheared into fragments that will be partially overlapping and cloned into a plasmid vector.

Generating Sequence of Genomes

Generating Sequence of Genomes

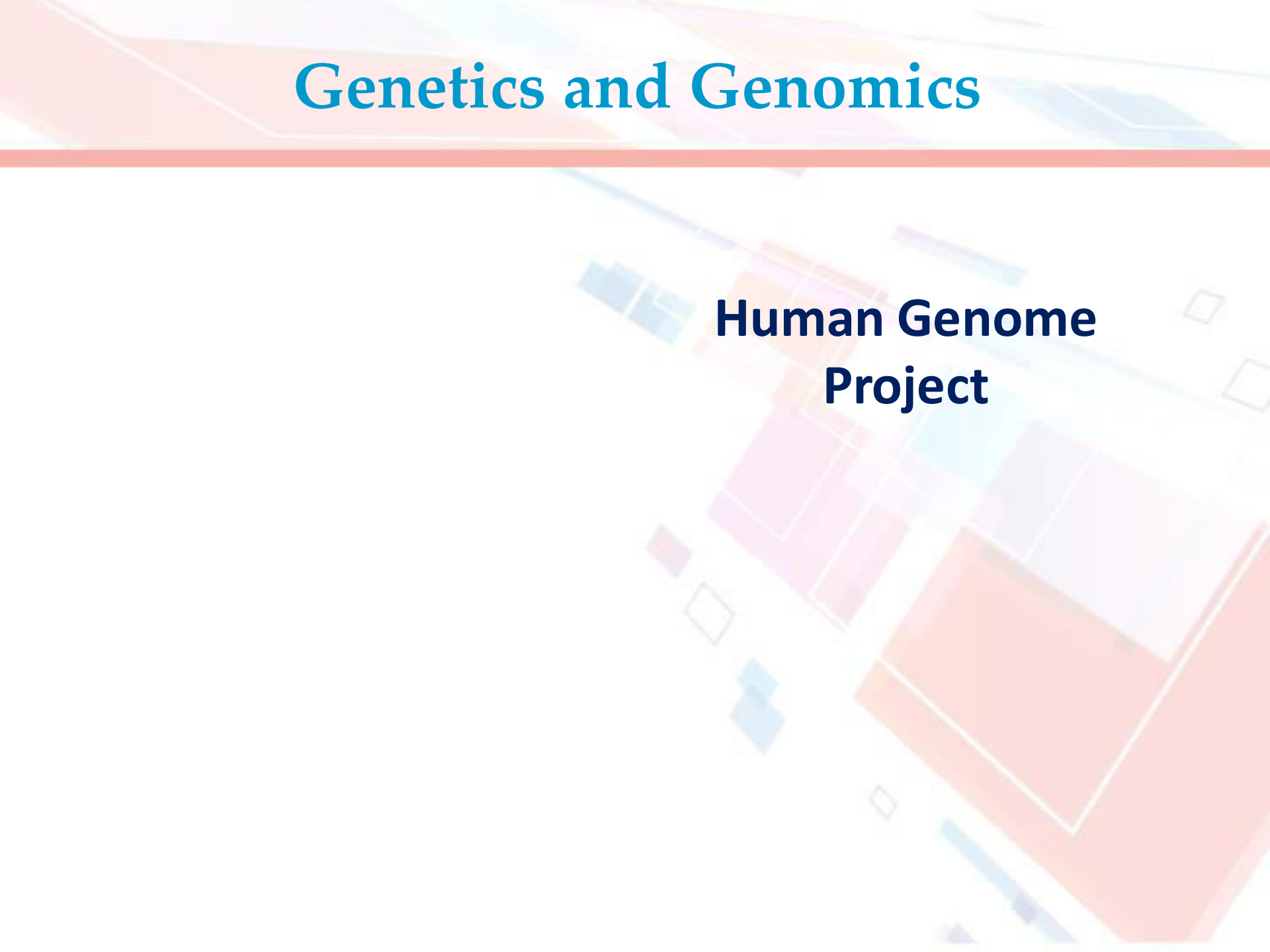
- Each subclone is sequenced, and overlaps are used by a computer to assemble the data into one contiguous sequence representing the BAC insert.

Generating Sequence of Genomes

Generating Sequence of Genomes

- Using the chromosomal map for BAC clones, the BAC insert sequences are put in order to yield the complete chromosome sequence

Genetics and Genomics



Human Genome Project

Human Genome Project

HGP

- 1990 – Human Genome Project was started (NHGRI)
- Later many institutes of UK, France, Japan, Germany, China involved.
- In 1998, Celera announced a 3-year plan to sequence human genome

Human Genome Project

Challenges to Sequence Human Genome

- Size
- Polymorphism
- Repeats - smaller repeats are technically difficult to sequence
- Some DNA sequence are repeated all over the genome

Human Genome Project

Challenges to Sequence Human Genome

- Relies on cloning (Some regions are difficult to clone - Heterochromatin).
- Some sequences rearrange or are deleted when cloned.

Human Genome Project

Human Genome Project - Objectives

- Create a genetic and physical map of the 24 human chromosomes (22 autosomes, X & Y)
- Identify the entire set of genes & map them all to their chromosomes

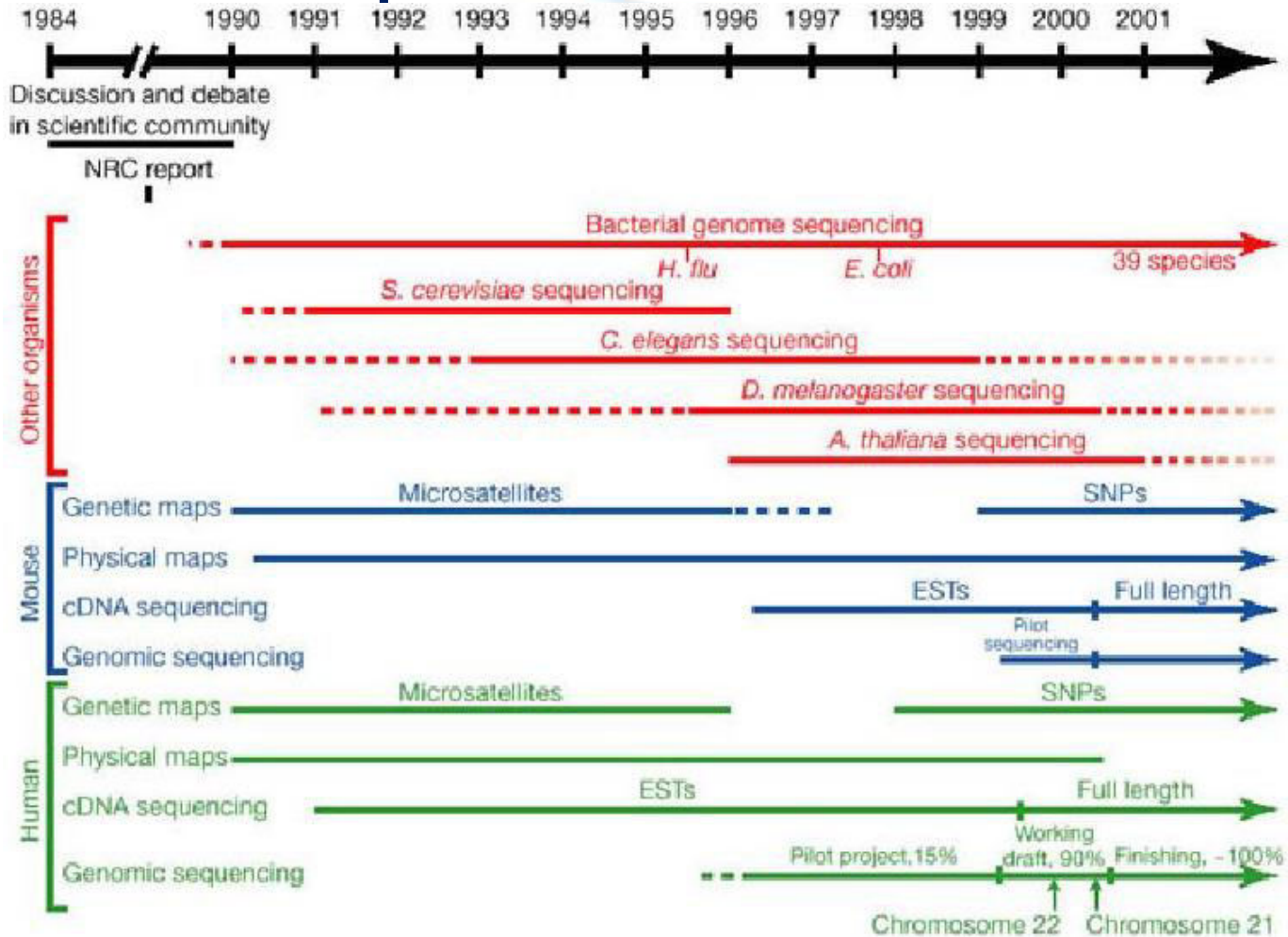
Human Genome Project

Human Genome Project - Objectives

- Determine the nucleotide sequence of the estimated 3 billion base pairs
- Analyze genetic variations in humans
- Map and sequence the genomes of model organisms

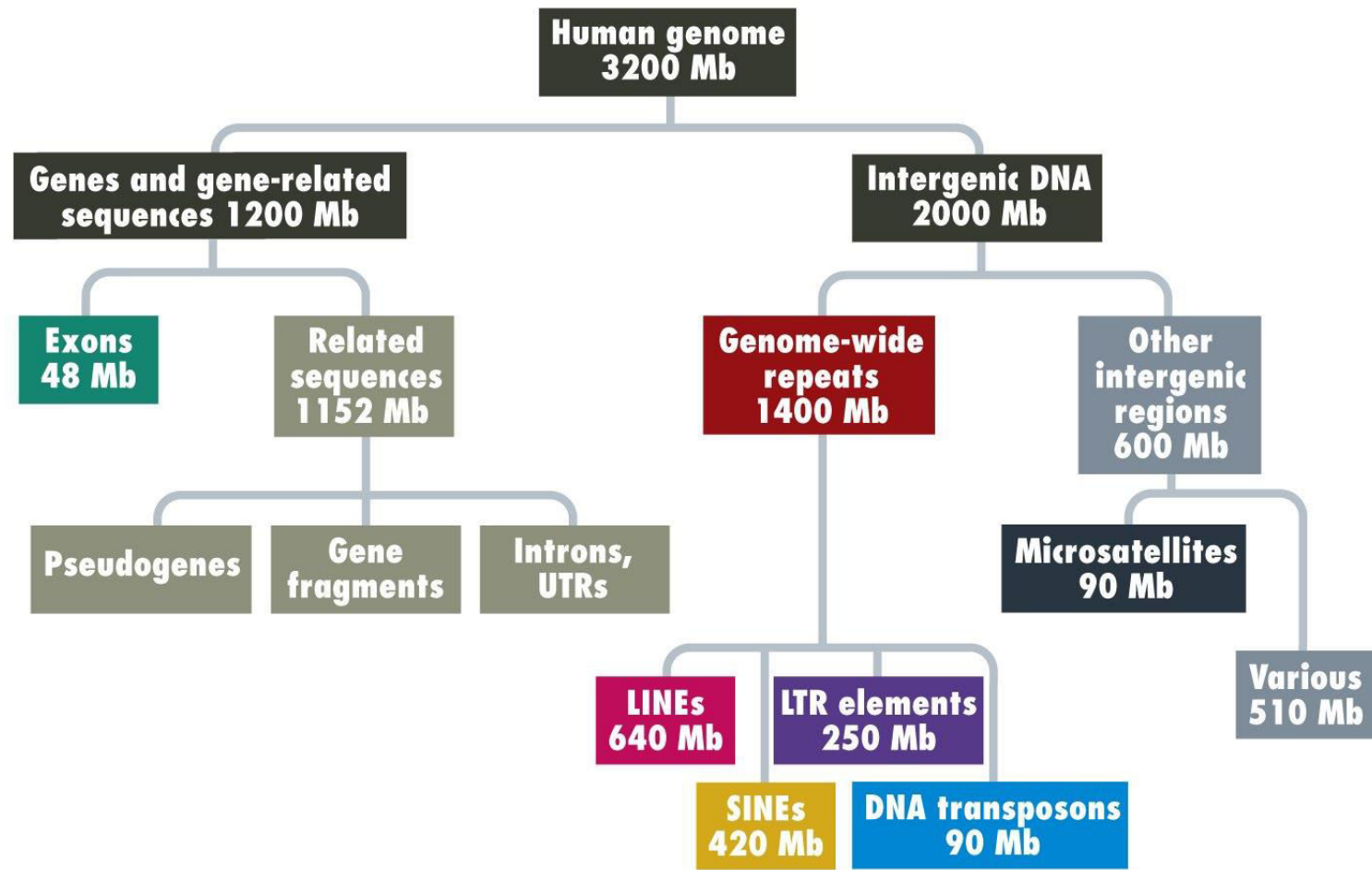
Human Genome Project

Time line to sequence Human Genome



Human Genome Project

Human Genome



Human Genome Project

Human Genome Project

- Two strategies used to sequence human genome
- Hierarchical Shotgun Sequencing
- Whole Genome Shotgun Sequencing

Human Genome Project

Human Genome Project - conclusion

- Determine the nucleotide sequence of the estimated 3 billion base pairs

Genetics and Genomics

**HGP –
Hierarchical
Shotgun
Sequencing**

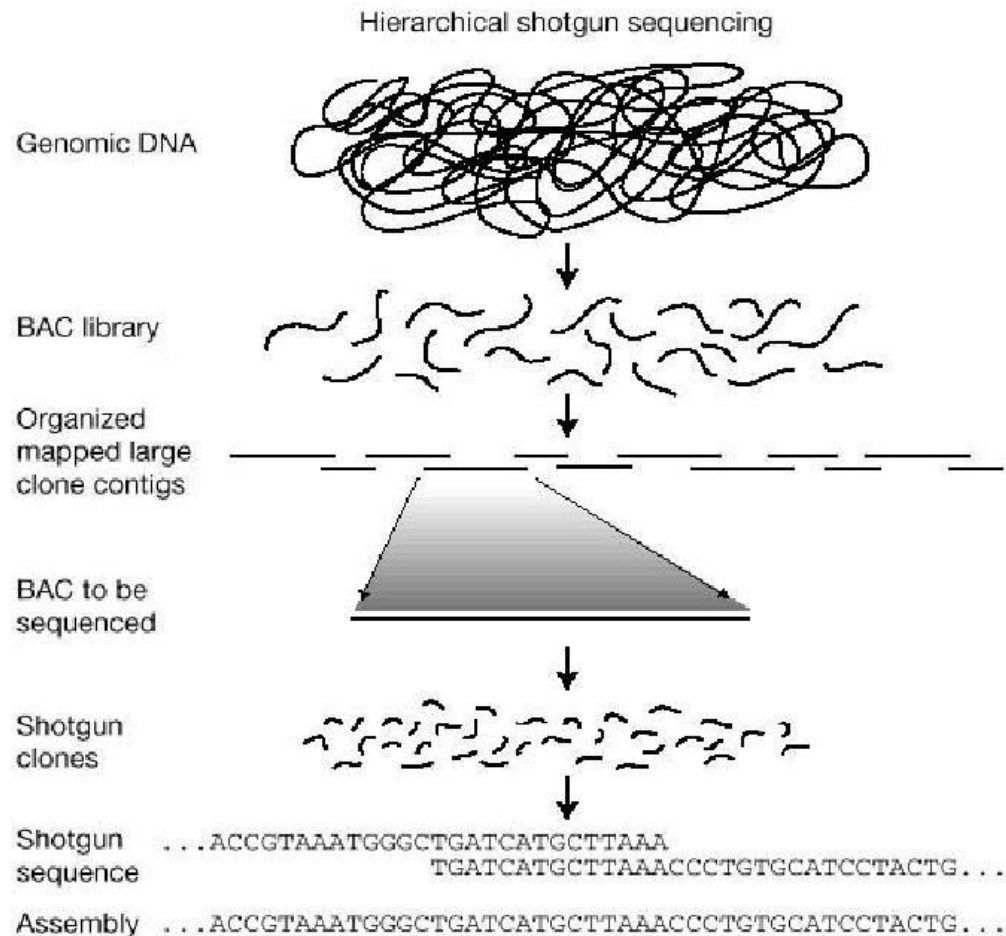
HGP–Hierarchical Shotgun Sequencing

HGP – Hierarchical Shotgun Sequencing

- Map Construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

HGP–Hierarchical Shotgun Sequencing

HGP – Hierarchical Shotgun Sequencing



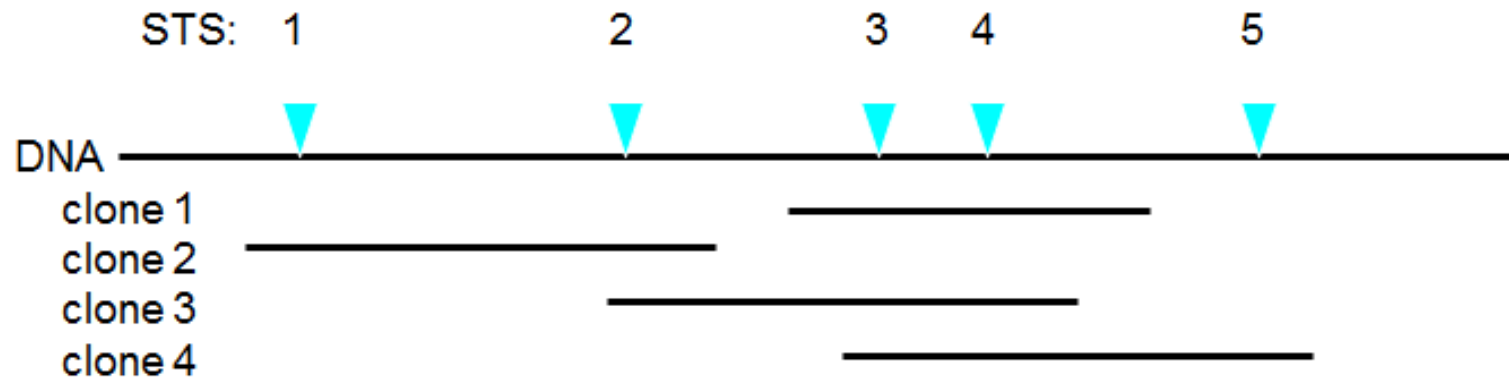
HGP–Hierarchical Shotgun Sequencing

Hierarchical Shotgun Sequencing: Map Construction

- Clone genomic DNA in YACs (~1MB) or BACs (~200-300Kb)
- Map the relative location of clones
 - Sequenced-tagged sites (STS, e.g. EST) mapping
 - PCR or probe hybridization to screen STS
 - Restriction site fingerprint
- Most time consuming
 - 1990-98 to generate physical maps for human

HGP–Hierarchical Shotgun Sequencing

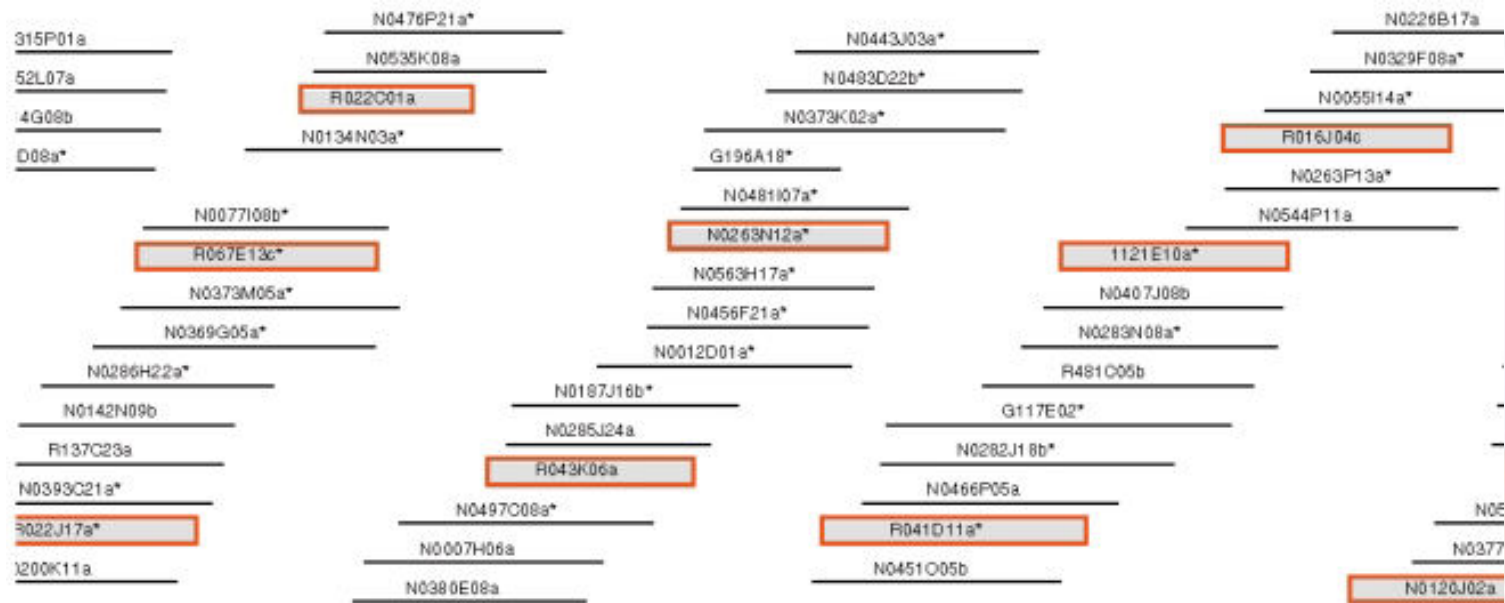
Hierarchical Shotgun Sequencing: Resolve Clone Relative Location



HGP–Hierarchical Shotgun Sequencing

Hierarchical Shotgun Sequencing: Clone Selection:

Based on clone map, select authentic clones to generate a minimum tiling path

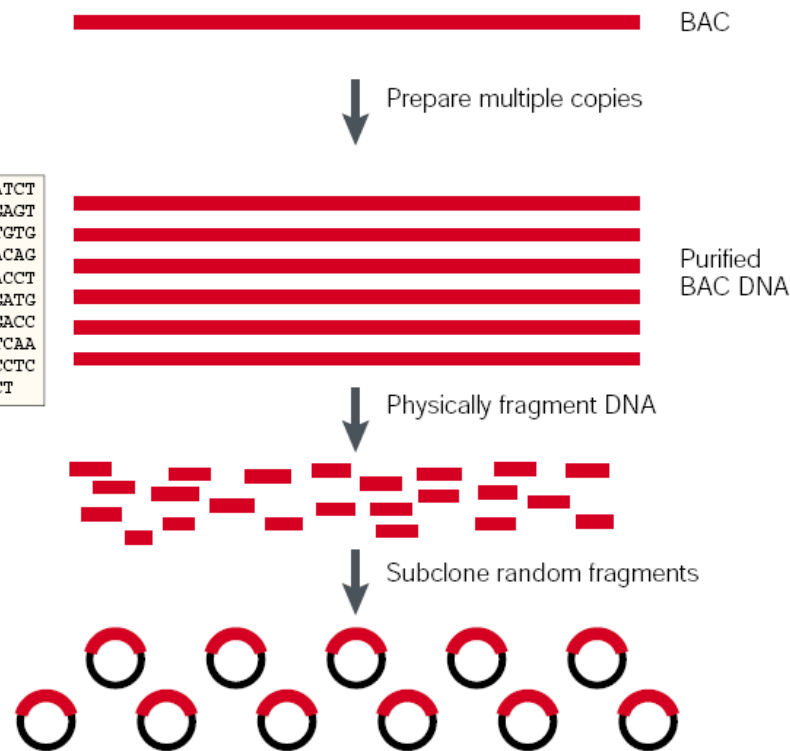


HGP–Hierarchical Shotgun Sequencing

Subclone Library Construction: DNA fragmented by sonication or RE cut; Fragment size ~ 2-5 kb

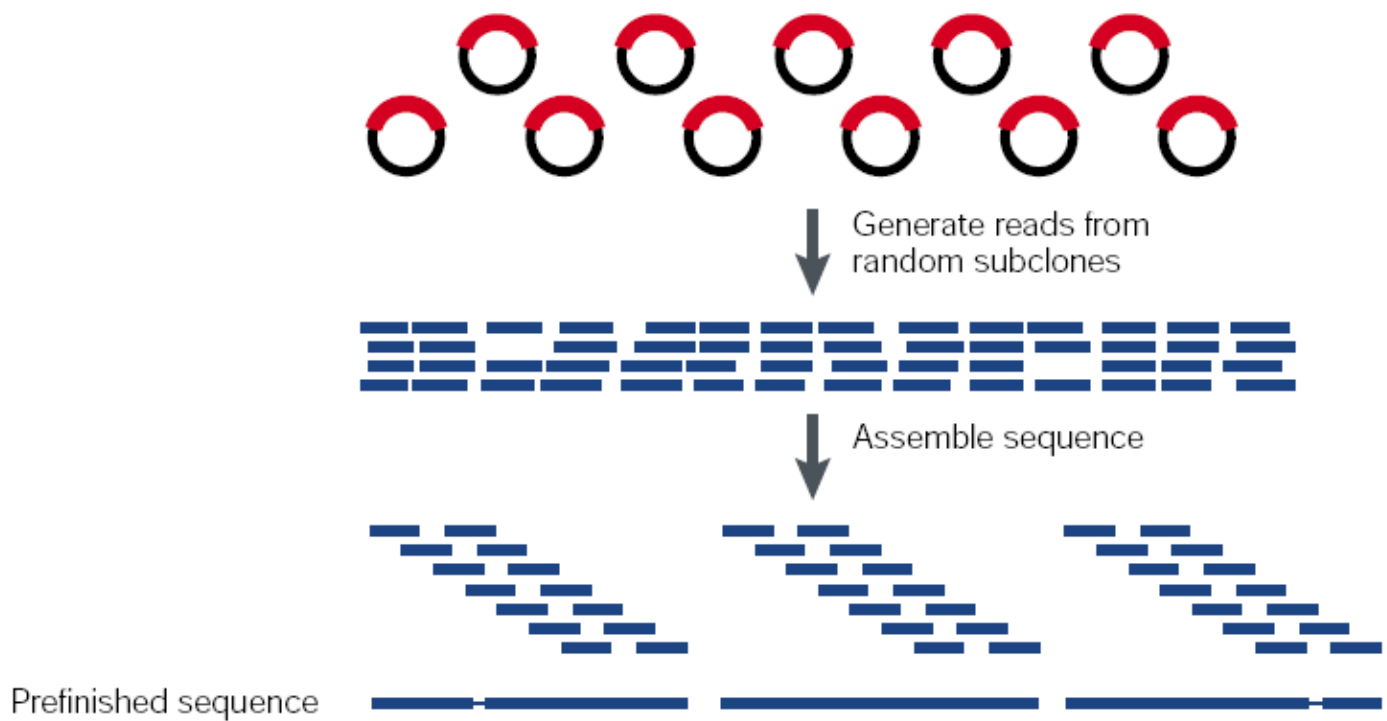
GATCGTCTAGAATCT
GAGATCTCTGAGAGT
GTGGGAAACTGTGTG
TGTGACTAGCCACAG
ACTGGGATCCTACCT
TACGTGTGAGAGATG
ATGATGCACCTGACC
GGGTTTCACTCTCAA
GACTCACTCCACCTC
GATCCCCAACCT

GF	GF	GF	GF	GF	GF	GATCGTCTAGAATCT
GF	GF	GF	GF	GF	GF	GAGATCTCTGAGAGT
GT	GT	GT	GT	GT	GT	GTGGGAAACTGTGTG
TG	TG	TG	TG	TG	TG	TGTGACTAGCCACAG
AC	AC	AC	AC	AC	AC	ACTGGGATCCTACCT
TA	TA	TA	TA	TA	TA	TACGTGTGAGAGATG
AT	AT	AT	AT	AT	AT	ATGATGCACCTGACC
GC	GC	GC	GC	GC	GC	GGGTTTCACTCTCAA
GA	GA	GA	GA	GA	GA	GACTCACTCCACCTC
GA	GA	GA	GA	GA	GA	GATCCCCAACCT



HGP–Hierarchical Shotgun Sequencing

Random Shotgun Phase: Dideoxy termination reaction; Informatics programs; Coverage and contigs



HGP–Hierarchical Shotgun Sequencing

HGP – Hierarchical Shotgun Sequencing

- Map Construction
- Clone selection
- Subclone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

Genetics and Genomics

**HGP – Whole
Genome Shotgun
Sequencing**

HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- Whole genome randomly digested three times
- Plasmid library constructed with ~ 2kb inserts and ~10 kb inserts
- BAC library with ~ 200 kb inserts

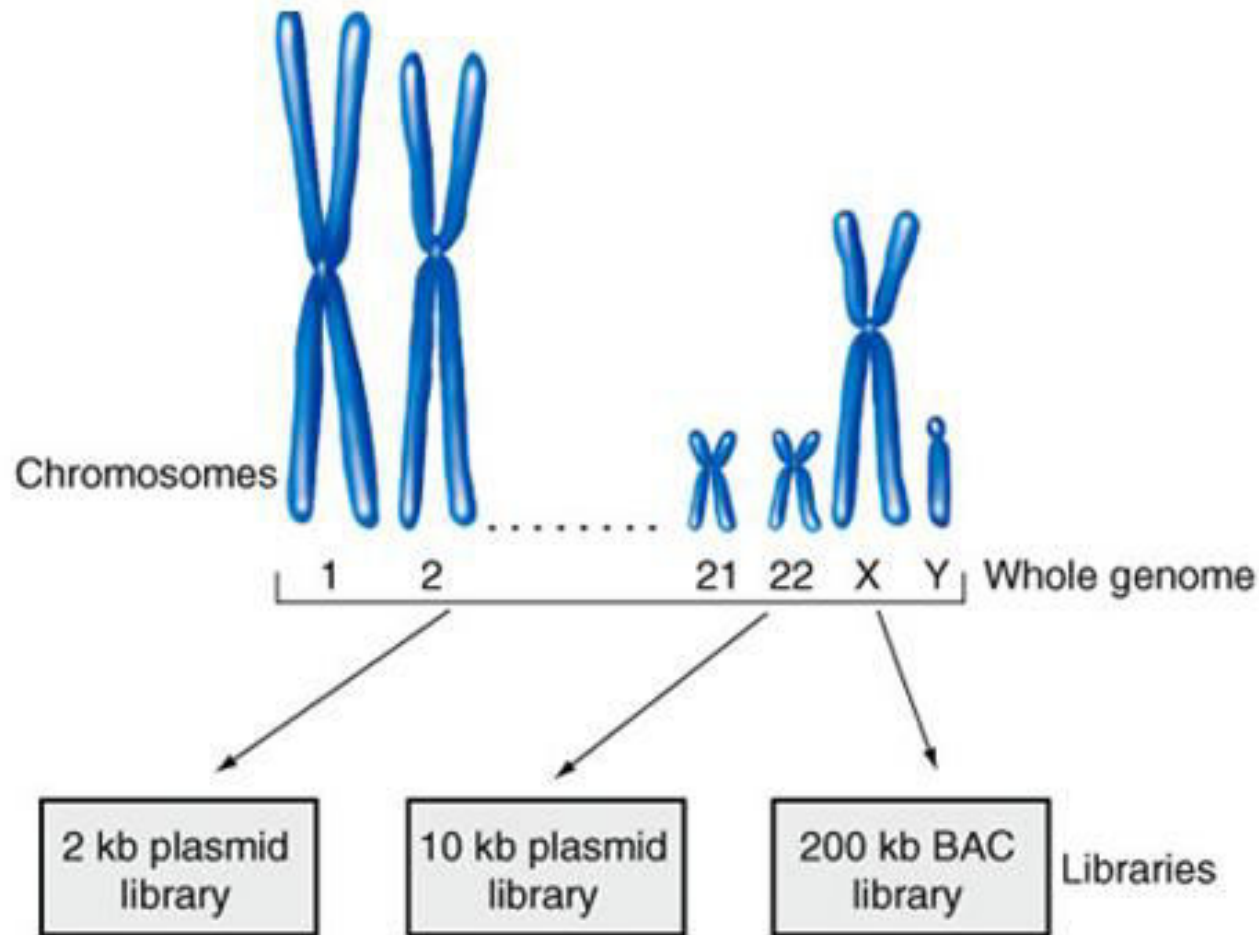
HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- Computer program assembles sequences into chromosomes
- No physical map construction
- Reduces problems of repeat sequences

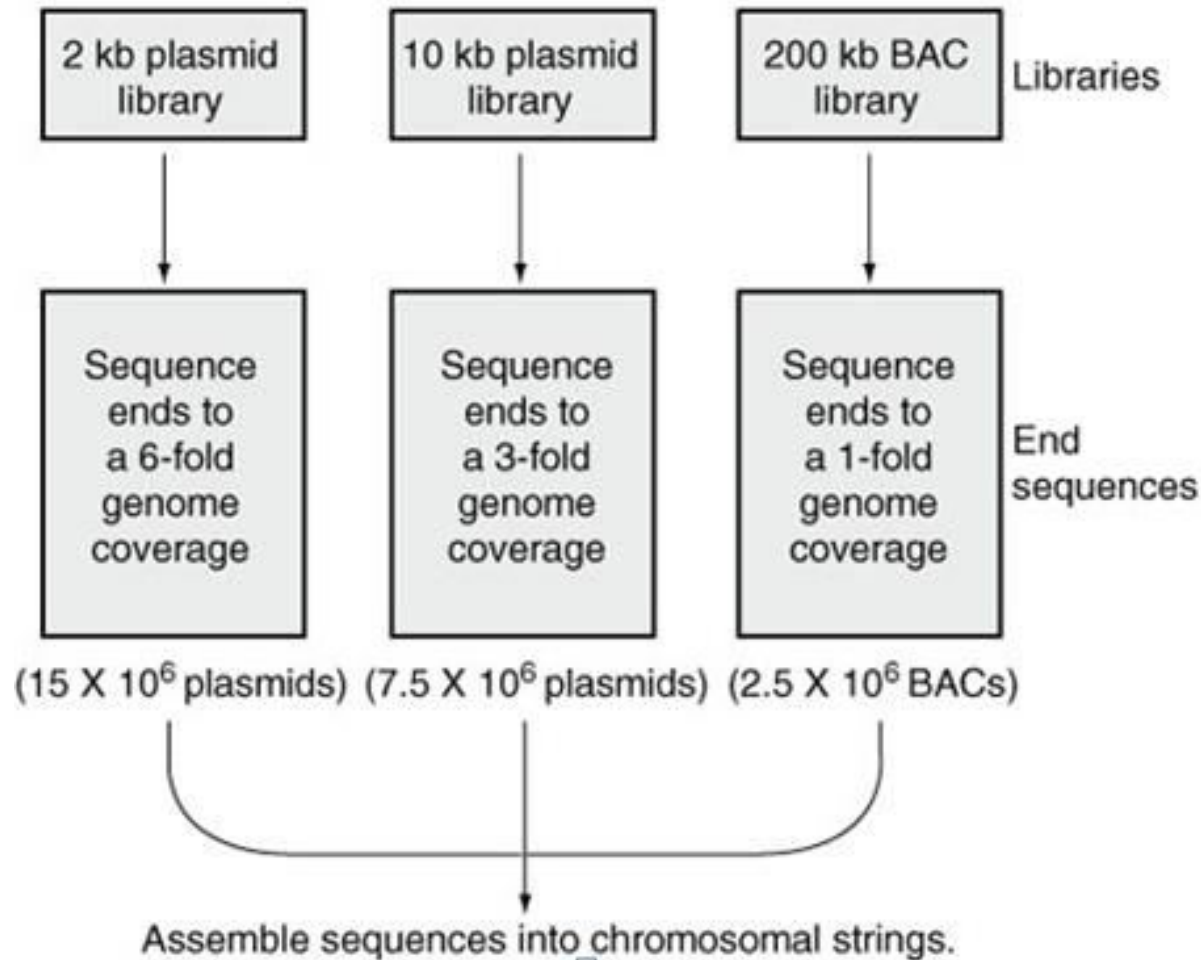
HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing



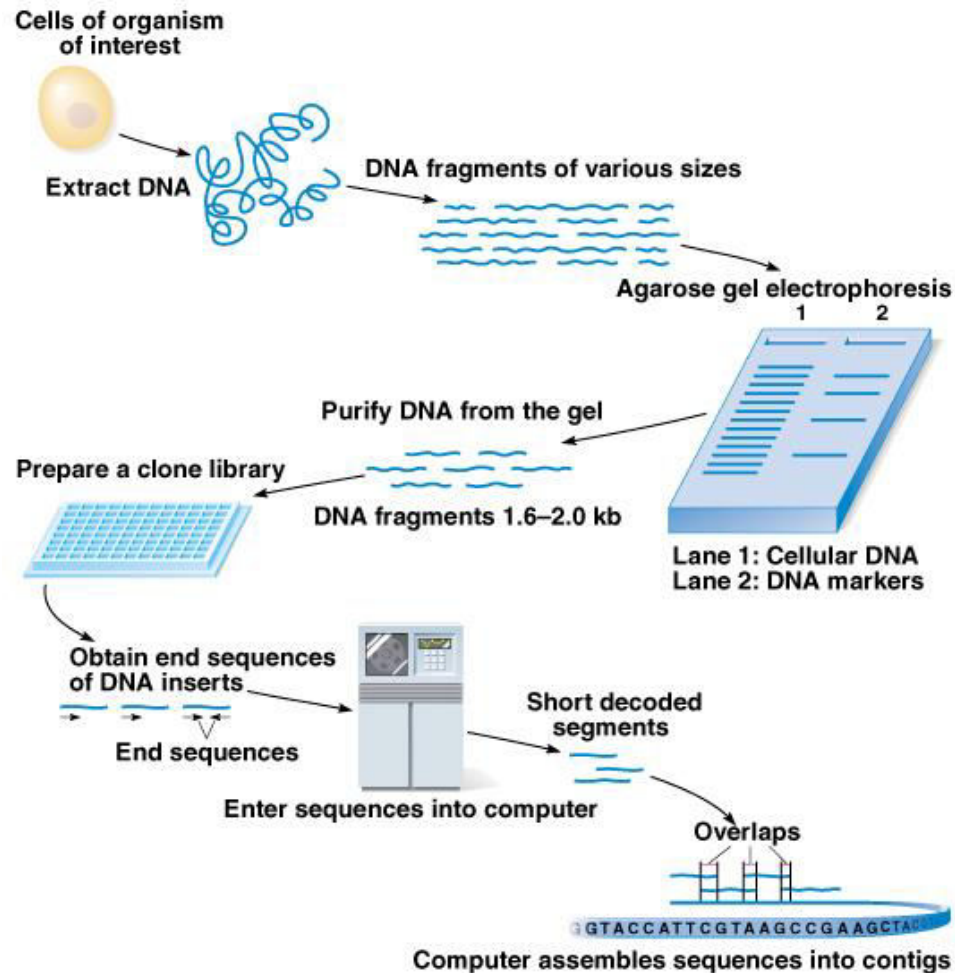
HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing



HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing



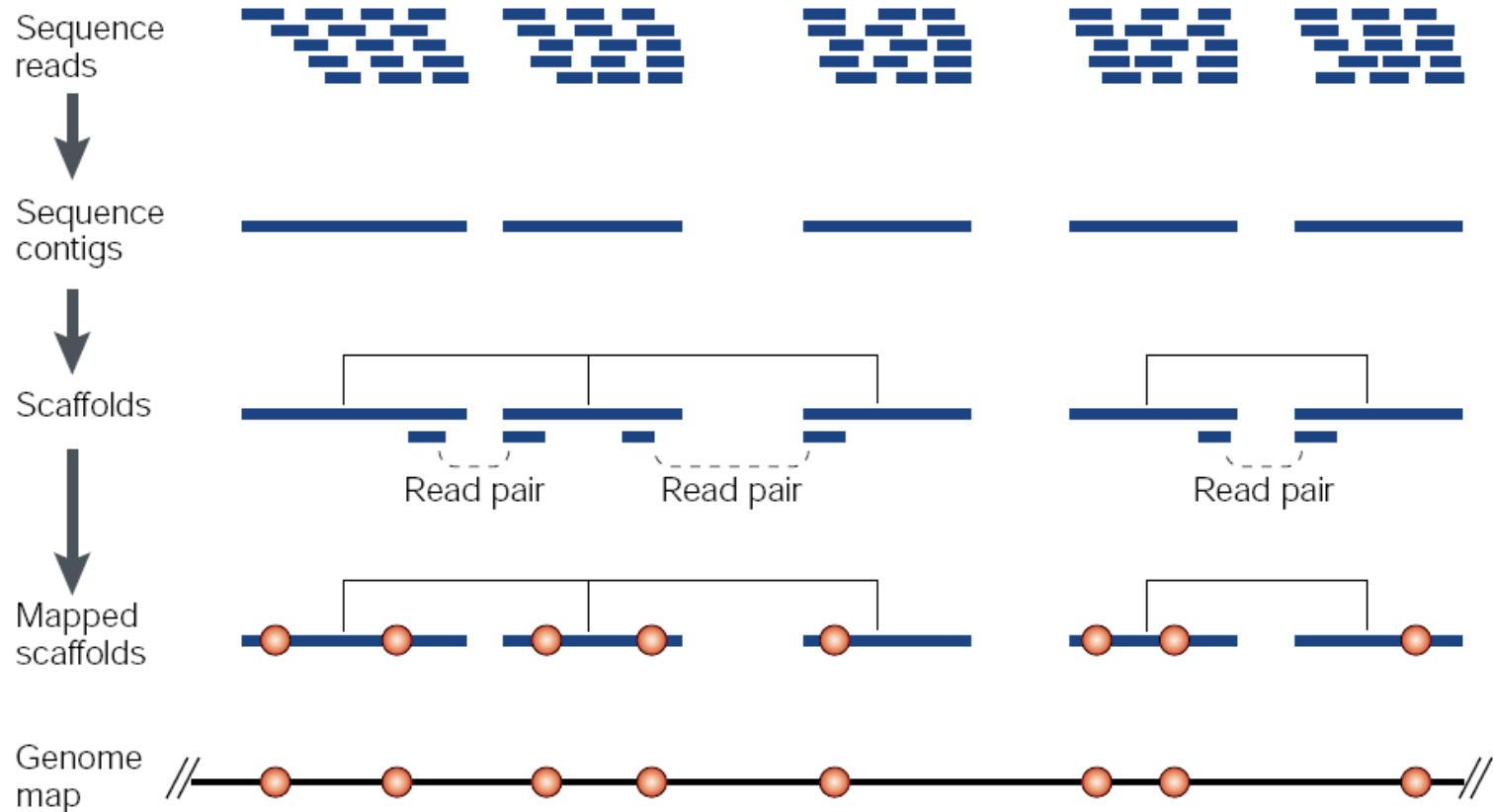
HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- After the sequence is shotgunned, 10 million fragments of the genomic sequences need to be recompiled into the readable base pairs in the proper order.

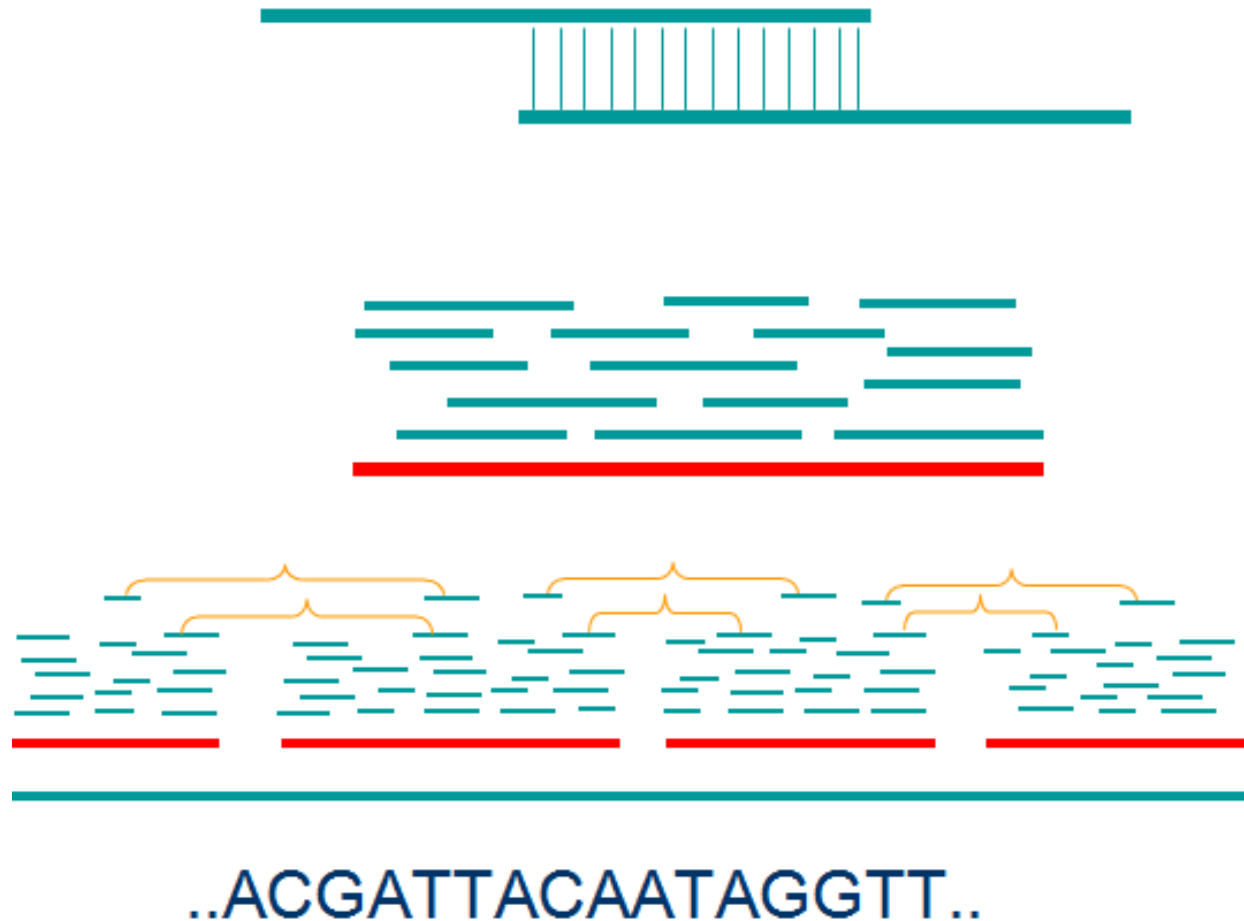
HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing



HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing



HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- The Celera Assembler was one of the core competencies and makes the task possible.

HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- Shotgunned fragments are compared against each other and equivalent sequences greater than 40 base pairs long are identified.
- These 40 base pairs matches are statistically impossible to occur by chance. These matches are then determined to be true or repeat induced.
- True matches are overlapping sections and are the desired fragments

HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- The assembler then searches for overlapping fragments that have a common sequence and are not contested elsewhere in the dataset.
- The uncontested data is assembled into unitigs containing approximately 30 fragments
- These assembled unitigs are 99 % accurate
- Unitigs passing this filter are ready for ordering.

HGP–Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- By looking at these contigs and orientation the scaffold become complete
- Mapped scaffolds were arranged to prepare chromosome maps

Genetics and Genomics

Human Genome - Characteristics (A)

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Human Genome contain ~3 billion bp
- The average gene consists of 3000 bases, but sizes vary
- Largest known human gene dystrophin at 2.4 million bases

Human Genome-Characteristics (A)

Human Genome - Characteristics

- The total number of genes is estimated at around 23,000-much lower than previous estimates of 30,000.
- Almost all (99.9%) nucleotide bases are exactly the same in all people.

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Human genome's gene-dense areas are predominantly composed of the DNA building blocks G and C.

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Gene poor areas are rich in the DNA building blocks A and T.
- GC- and AT-rich regions can be seen through microscope as light and dark bands-chromosomes

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Genes appear to be concentrated in random areas along the genome
- Large areas of noncoding DNA between the genes

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Stretches of up to 30,000 C-G bases repeating over and over often occur adjacent to gene-rich areas.
- CpG islands are believed to regulate gene activity

Human Genome-Characteristics (A)

Human Genome - Characteristics

- Chromosome 1 has the most genes ~ (2100), and the Y chromosome has the fewest ~ (458).

Human Genome-Characteristics (A)

Human Genome - Characteristics

- 3,000 Mb
- ~ 23,000 genes
- Exons ~ 1.5%
- Introns ~ 3.5%
- Repeats ~ 45%

Genetics and Genomics

Human Genome - Characteristics (B)

Human Genome-Characteristics (B)

Human Genome - Characteristics

- ~ 1.5% of the genome codes for proteins.
- Repeated sequences make up at least 45 % of the human genome

Human Genome-Characteristics (B)

Human Genome - Characteristics

- Average size of an exon is 120 -145bp.
- Average number of exons is 7-9.
- Average coding sequence encodes proteins of 370 - 440 amino acids

Human Genome-Characteristics (B)

Human Genome - Characteristics

Chromosome	Length (mm)	Base pairs	Variations	Confirmed proteins	Putative proteins	Pseudogenes	miRNA	rRNA	snRNA	snoRNA	Misc ncRNA
1	85	249,250,621	4,401,091	2,012	31	1,130	134	66	221	145	106
2	83	243,199,373	4,607,702	1,203	50	948	115	40	161	117	93
3	67	198,022,430	3,894,345	1,040	25	719	99	29	138	87	77
4	65	191,154,276	3,673,892	718	39	698	92	24	120	56	71
5	62	180,915,260	3,436,667	849	24	676	83	25	106	61	68
6	58	171,115,067	3,360,890	1,002	39	731	81	26	111	73	67
7	54	159,138,663	3,045,992	866	34	803	90	24	90	76	70
8	50	146,364,022	2,890,692	659	39	568	80	28	86	52	42
9	48	141,213,431	2,581,827	785	15	714	69	19	66	51	55
10	46	135,534,747	2,609,802	745	18	500	64	32	87	56	56
11	46	135,006,516	2,607,254	1,258	48	775	63	24	74	76	53
12	45	133,851,895	2,482,194	1,003	47	582	72	27	106	62	69
13	39	115,169,878	1,814,242	318	8	323	42	16	45	34	36
14	36	107,349,540	1,712,799	601	50	472	92	10	65	97	46
15	35	102,531,392	1,577,346	562	43	473	78	13	63	136	39

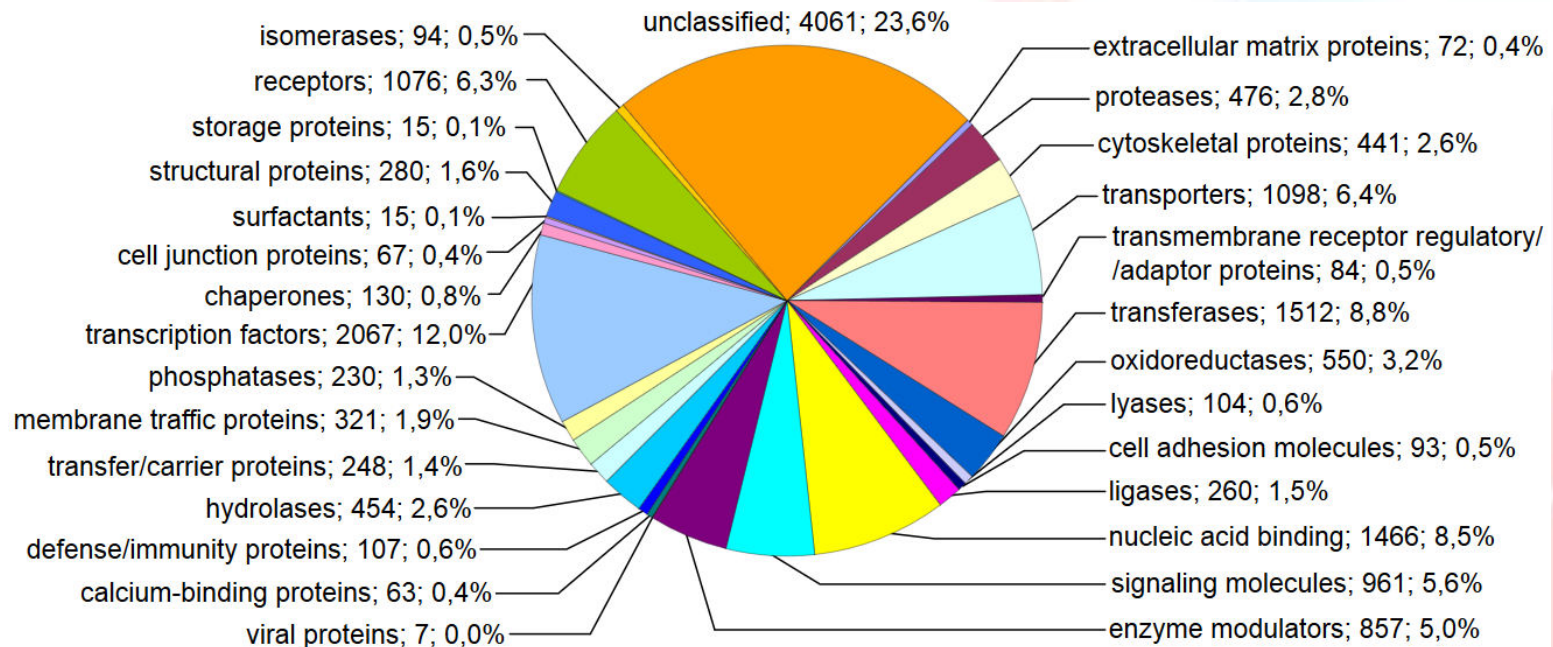
Human Genome-Characteristics (B)

Human Genome - Characteristics

Chromosome	Length (mm)	Base pairs	Variations	Confirmed proteins	Putative proteins	Pseudogenes	miRNA	rRNA	snRNA	snoRNA	Misc ncRNA
16	31	90,354,753	1,747,136	805	65	429	52	32	53	58	34
17	28	81,195,210	1,491,841	1,158	44	300	61	15	80	71	46
18	27	78,077,248	1,448,602	268	20	59	32	13	51	36	25
19	20	59,128,983	1,171,356	1,399	26	181	110	13	29	31	15
20	21	63,025,520	1,206,753	533	13	213	57	15	46	37	34
21	16	48,129,895	787,784	225	8	150	16	5	21	19	8
22	17	51,304,566	745,778	431	21	308	31	5	23	23	23
X	53	155,270,560	2,174,952	815	23	780	128	22	85	64	52
Y	20	59,373,566	286,812	45	8	327	15	7	17	3	2

Human Genome-Characteristics (B)

Human Genome - Human genes categorized by function of the transcribed proteins, given both as number of encoding genes and percentage of all identified genes



Human Genome-Characteristics (B)

Human Genome - Characteristics

- The human genome has many different regulatory sequences which are crucial to controlling gene expression.
- Conservative estimates indicate that these sequences make up 8% of the genome.

Human Genome-Characteristics (B)

Human Genome - Characteristics

- About 8% of the human genome consists of tandem DNA arrays or tandem repeats

Human Genome-Characteristics (B)

Human Genome - Characteristics

- Mobile elements:
- LTR retrotransposons (8.3% of total genome)
- SINEs (13.1% of total genome) including Alu elements
- LINEs (20.4% of total genome)

Human Genome-Characteristics (B)

Human Genome - Characteristics

- Pericentromeric and sub-telomeric regions of chromosomes filled with large transposable elements

Human Genome-Characteristics (B)

Human Genome - Characteristics

- Chimpanzee genome differs from that of the human genome by 1.23% in direct sequence comparisons

Human Genome-Characteristics (B)

Human Genome - Characteristics

- Male mutation rate about twice female
- Most mutations occurs in males
- Recombination rates much higher in distal regions of chromosomes

Genetics and Genomics

Genome Browser - UCSC

Genome Browser - UCSC

Genome Browsers

- The genomes are so large that useful information is hard to find.
- Researchers at UCSC decided to make a computational microscope to help scientists search the genomes.

Genome Browser - UCSC

Genome Browsers

- Researchers can use the UCSC Genome Browser to find information in the human genome and other genomes that have been sequenced

Genome Browser - UCSC

Genome Browsers

- UCSC Genome Browser

<http://genome.ucsc.edu>

Genome Browser - UCSC

The UCSC Genome Browser - Homepage

UCSC Genome Bioinformatics

 Genomes Genome Browser Tools Mirrors Downloads My Data Help About Us

Genome Browser

Blat

Table Browser

Gene Sorter

In Silico PCR

Genome Graphs

Galaxy

VisiGene

Utilities

Downloads

Release Log

About the UCSC Genome Bioinformatics Site

Welcome to the UCSC Genome Browser website. This site contains the reference sequence and working draft assemblies for a large collection of genomes. It also provides portals to [ENCODE](#) data at UCSC (2003 to 2012) and to the [Neandertal](#) project. Download or purchase the Genome Browser source code, or the Genome Browser in a Box ([GBiB](#)) at our [online store](#).

We encourage you to explore these sequences with our tools. The [Genome Browser](#) zooms and scrolls over chromosomes, showing the work of annotators worldwide. The [Gene Sorter](#) shows expression, homology and other information on groups of genes that can be related in many ways. [Blat](#) quickly maps your sequence to the genome. The [Table Browser](#) provides convenient access to the underlying database. [VisiGene](#) lets you browse through a large collection of *in situ* mouse and frog images to examine expression patterns. [Genome Graphs](#) allows you to upload and display genome-wide data sets.

The UCSC Genome Browser is developed and maintained by the Genome Bioinformatics Group, a cross-departmental team within the [UC Santa Cruz Genomics Institute](#) at the University of California Santa Cruz ([UCSC](#)). If you have feedback or questions concerning the tools or data on this website, feel free to contact us on our [public mailing list](#).

The Genome Browser project team relies on public funding to support our work. Donations are welcome -- we have many more ideas than our funding supports! If you have ideas, drop a comment in our [suggestion box](#).

News  

News Archives ►

To receive announcements of new genome assembly releases, new software features, updates and training seminars by email, subscribe to the [genome-announce](#) mailing list. Please see our [blog](#) for posts about Genome Browser tools, features, projects and more.

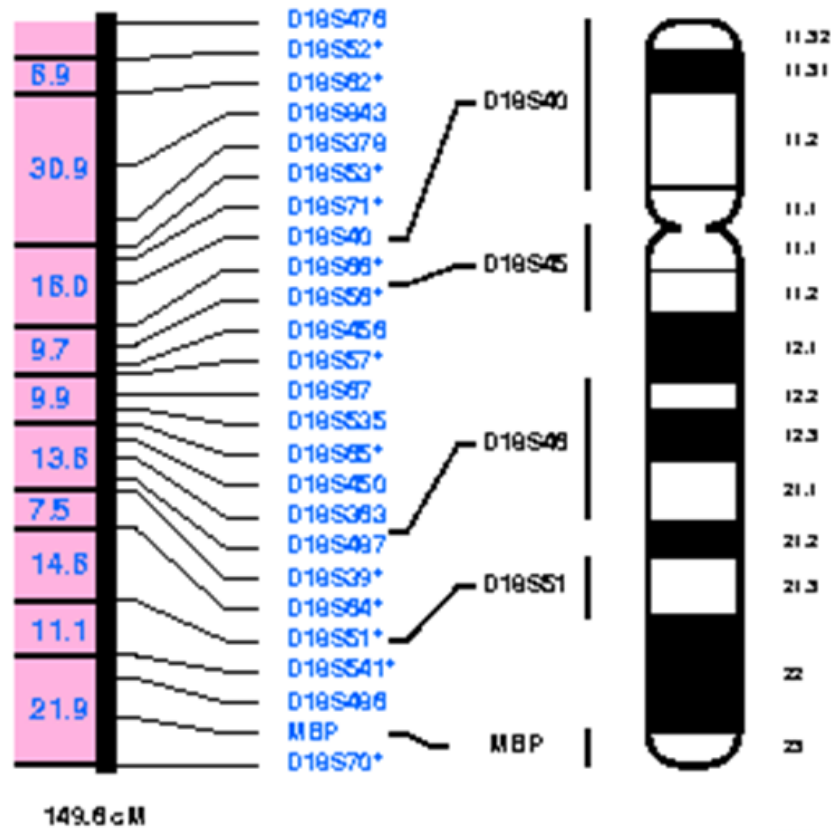
Genome Browser - UCSC

The UCSC Genome Browser

The screenshot displays the UCSC Genome Browser interface for Human July 2003 Freeze. The top navigation bar includes links for Home, BLAT, DNA, Tables, Convert, PDF/PS, and Guide. The main title is "UCSC Genome Browser on Human July 2003 Freeze". Below the title, there are controls for moving and zooming: "move" with left and right arrow buttons, "zoom in" with 1.5x, 3x, and 10x buttons, and "zoom out" with 1.5x, 3x, and 10x buttons. The "position" field shows "chr22:20000000-30000000" and the "size" is "10,000,001" with an "image width" of "610" and a "jump" button. The central track area shows various genomic features: Base Position (25000000), Chromosome Bands (22q11.22, 22q11.23, 22q12.1, 22q12.2), STS Markers, Gap, Known Genes, Genscan Genes, Human mRNAs, Spliced ESTs, NonHuman mRNAs, Mouse Net, and RepeatMasker. Below the tracks, there are controls for moving the start and end of the view, with "2.0" in the input fields. A "reset all" button and a "hide all" button are also present. A "Guidelines" section has checkboxes for "Labels: left" (checked), "center" (checked), and "refresh". A "Chromosome Color Key" shows a row of colored boxes for chromosomes 1 through 22, X, and Y. A note at the bottom states: "Note: Tracks with lots of items will automatically be displayed in more compact modes."

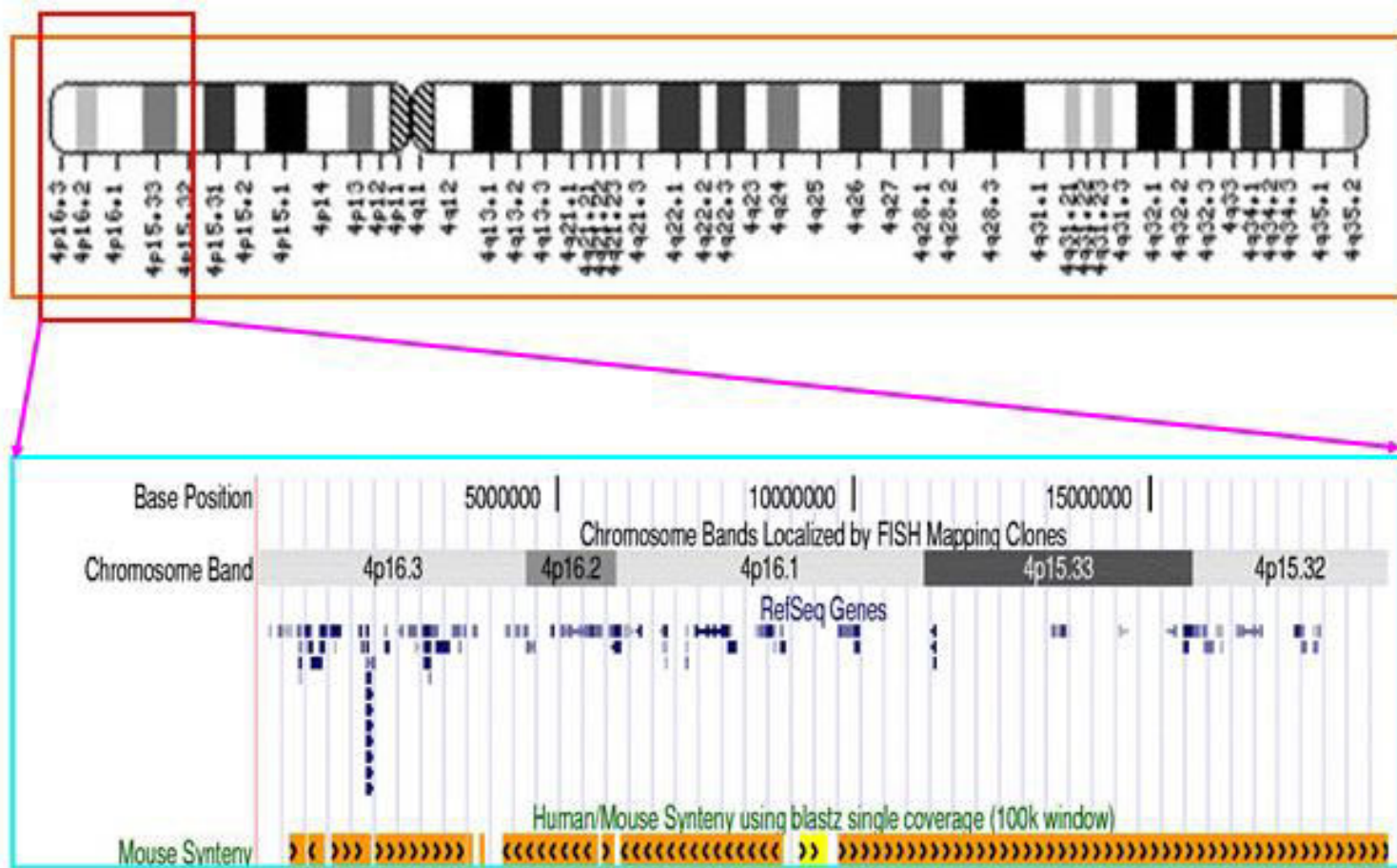
Genome Browser - UCSC

The browser takes you from early maps of the genome . . .



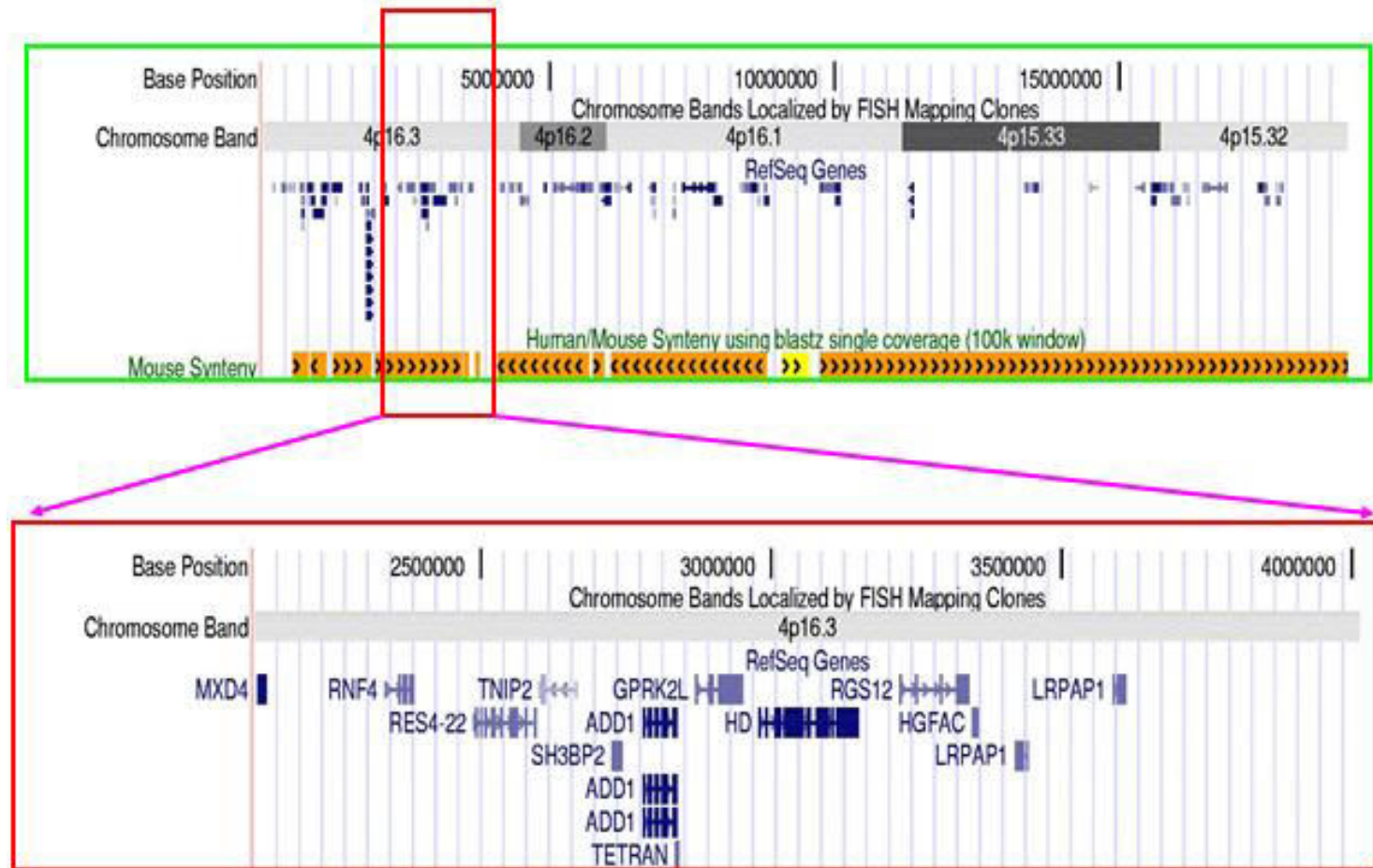
Genome Browser - UCSC

Multi Resolution View



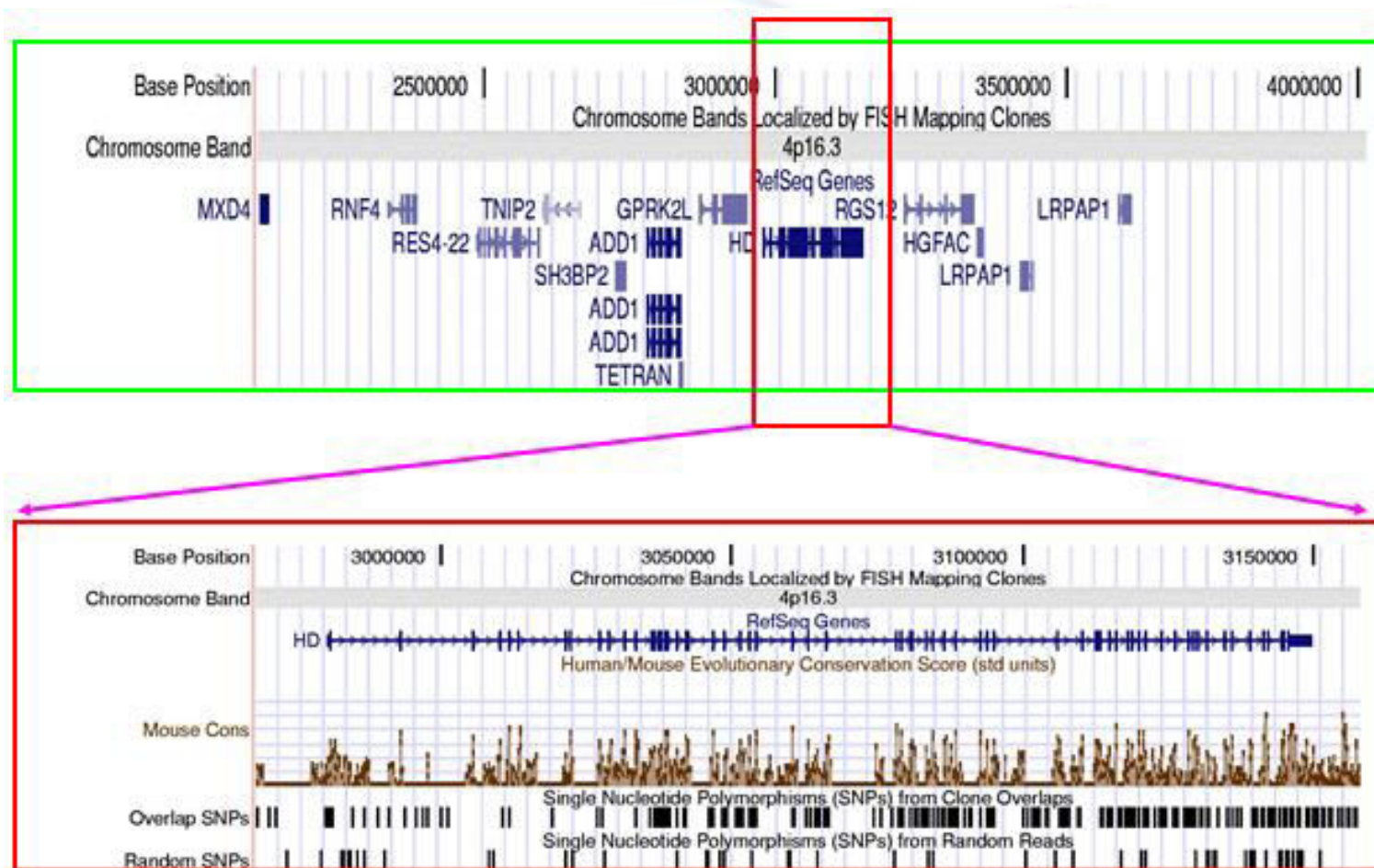
Genome Browser - UCSC

Gene Cluster View



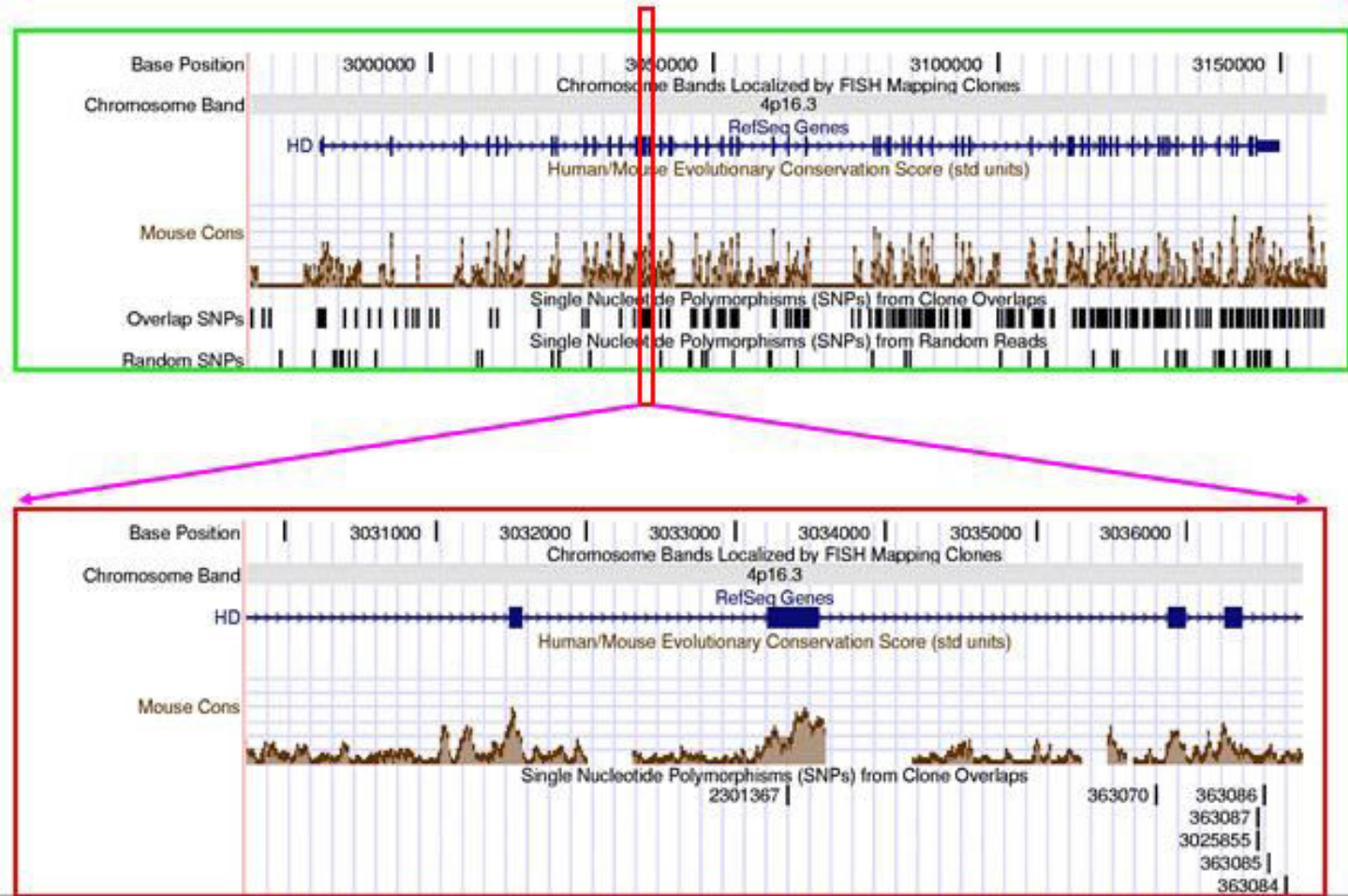
Genome Browser - UCSC

Single Gene View



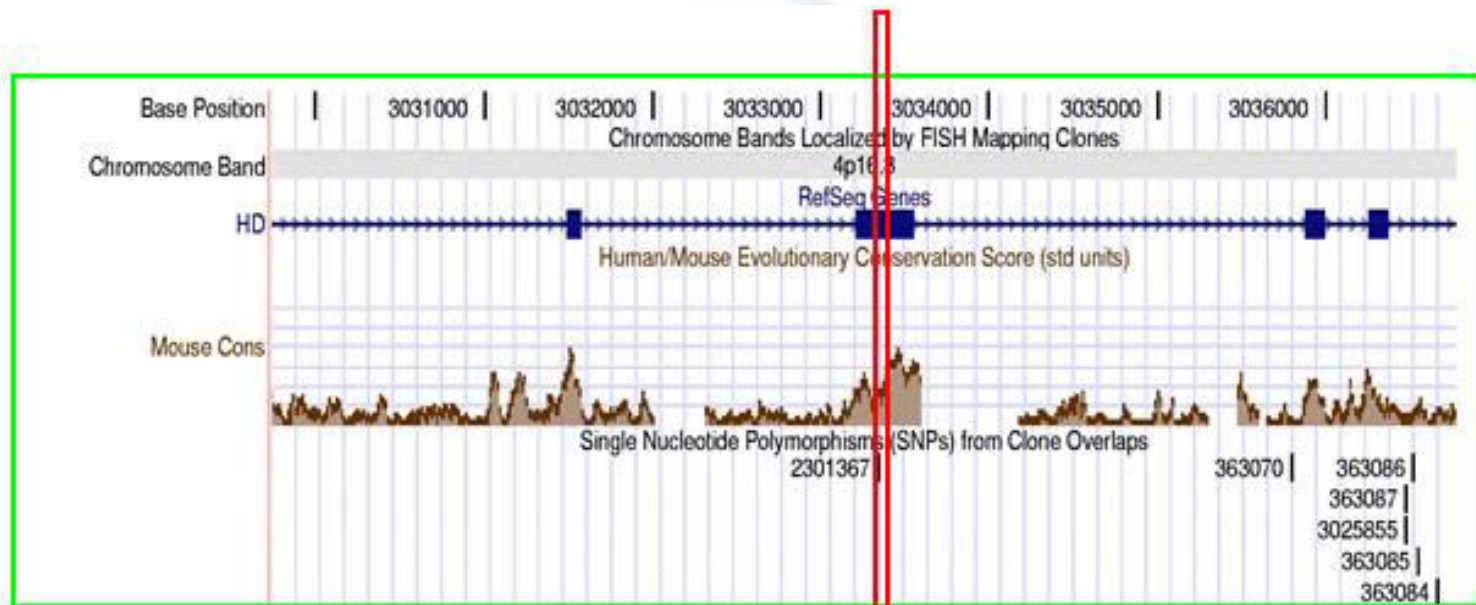
Genome Browser - UCSC

Single Exon View



Genome Browser - UCSC

Single Base/Nucleotide View



caggcggactcagtggatctggccagctgtgacttgacaag
caggcggactcagtggatctagccagctgtgacttgacaag

Genome Browser - UCSC

Genome Browser - UCSC

- Researchers can use the UCSC genome browser to find information in the human genome and other genomes that have been sequenced

Genetics and Genomics

Genome Browser - Ensembl

Genome Browser - Ensembl

Ensembl Genome Browser

- Ensembl is a joint project between 3 organizations
 - EMBL
 - EBI
 - WTSI

Genome Browser - Ensembl

Ensembl Genome Browser

- Ensembl does not gather any genome project directly
- Works in relation with the sequencing centers that generate the genome assembly

Genome Browser - Ensembl

Ensembl Genome Browser

- 4 Main Databases
 - Ensembl Core Database
 - Ensembl EST Database
 - Ensembl Compara Database
 - Ensembl Variation Database

Genome Browser - Ensembl

Ensembl

- Ensembl core databases store genome sequence and annotation information
- Gene, transcript, and protein models
- Databases stores information about cDNA and protein alignments

Genome Browser - Ensembl

Ensembl Genome Browser

- Within one genome: regulatory elements, gene order etc
- Comparative studies: Evolution, conserved regions rearrangements
- Gene quality and prediction

Genome Browser - Ensembl

Ensembl Genome Browser

- Ensembl Genome browser

<http://www.ensembl.org>

Genome Browser - Ensembl

Ensembl Genome Browser – Home page

e!Ensembl

Ensembl release 42 - Dec 2006 (ecs3.3307 - homo_sapiens_core_42_36d) HOME · BLAST · BIOMART · SITEMAP · **HELP**

Your Ensembl

- Show account · Log out
- Save bookmark

Healthchecks

- Health checks
- Old Health checks

Help & Documentation

- Table of Contents
- Helpdesk
- About Ensembl
- Downloading data
- Displaying your own data
- Ensembl software

Select a species

- Mammals
- Other chordates
- Other eukaryotes

Ensembl Archive

- View previous release of page in Archive!
- Stable Archive! link for this page


Now in Ensembl **Pre!**

Search Ensembl

Search: for **Go**

e.g. mouse chromosome 2 or X10000-20000 or human gene BRCA2

Ensembl tools

- Start a sequence search → Search Ensembl for nucleotide and peptide sequences with BLAST and SRAHA.
- Mine Ensembl with BioMart → Cross-reference Ensembl datasets with BioMart, a powerful data-mining tool.
- Customise Your Ensembl → Register with Ensembl to bookmark your favourite pages, customise your home page and much more!
- Fetch data with the Ensembl API → Learn how to extract data from the public Ensembl database with this tutorial.

You are logged in as **Güldensta.** Go to your account home page or log out.

Ensembl headlines: Release 42 (December 2006)

-  **New - User accounts** (all species)
-  **New species - Duck-billed Platypus** (*Ornithorhynchus anatinus*)
-  **New Dog assembly and genebuild** (*Canis familiaris*)
-  **New Chicken assembly and genebuild** (*Gallus gallus*)
-  **New Human Ensembl Vega** (*Homo sapiens*)

More news...

Go to your account to customise this news panel

About Ensembl

Ensembl is a joint project between EMBL, EBI and the Sanger Institute to develop a software system which produces and maintains automatic annotation on selected eukaryotic genomes. Ensembl is primarily funded by the Wellcome Trust.

This site provides free access to all the data and software from the Ensembl project. Click on a species name to browse the data.

Access to all the data produced by the project, and to the software used to analyse and present it, is provided free and without constraints. Some data and software may be subject to third-party constraints.

For all enquiries, please contact the Ensembl HelpDesk (helpdesk@ensembl.org).

Other Ensembl websites

- archive - past releases of Ensembl
- VEGA - Vertebrate Genome Annotation
- EBI Genome Reviews database - mainly archaea and bacteria.
- Trace server

Other sites using Ensembl software...

Ensembl 42

Popular genomes · [Reorder](#)

-  **Homo sapiens**
NCBI 36 | Vega
-  **Ciona intestinalis**
JGI 2
-  **Mus musculus**
NCBI m38 | Vega
-  **Danio rerio**
GDB | Vega

More genomes · [Reorder](#)

-  **Aedes aegypti** AaegL1
-  **Anopheles gambiae** AgamP3
-  **Bos taurus** Btau 2.0
-  **Caenorhabditis elegans** WS100
-  **Canis familiaris** Canfam 2.0 **UPDATED!**
-  **Ciona savignyi** Csav 2.0
-  **Dasyatis novemcinctus** ARMA
-  **Drosophila melanogaster** BDGP 4.3
-  **Echinops telfairi** TENREC
-  **Gallus gallus** WASHUC2 **UPDATED!**
-  **Gasterosteus aculeatus** BROAD S1
-  **Loxodonta africana** BROAD E1
-  **Macaca mulatta** MMUL 1.0
-  **Monodelphis domestica** ManDom 4.0
-  **Ornithorhynchus anatinus** Qana 5.0 **NEW!**
-  **Oryctolagus cuniculus** RABBIT
-  **Oryzias latipes** HsR
-  **Pan troglodytes** PanTro 2.1
-  **Rattus norvegicus** RNOSE 3.4
-  **Saccharomyces cerevisiae** S001.01 **UPDATED!**
-  **Takifugu rubripes** FUGU 4.0
-  **Tetraodon nigroviridis** TETRAODON 7
-  **Xenopus tropicalis** JGI 4.1

You are using the web team's integration server. [More →](#)

© 2006 WTSI / EBI. Ensembl is available to [download for public use](#) - please see the [code licence](#) for details.

Genome Browser - Ensembl

Ensembl Genome Browser – Map View

 **Human MapView**

Search e/Human:

e.g. [5, 22](#)

[Help](#)

Chromosome X

- View Chromosome X
- View Chr X Synteny
- Map your data onto this chromosome
- Browse OMIM diseases

Use Ensembl to...

- Run a BLAST search
- Search Ensembl
- Data mining [BioMart]
- Upload your own data
- Download data
- Export data

Docs and downloads

- Information
- About Ensembl
- Using Ensembl data
- Software

Other links

- Home
- Sitemap
- What's New
- Stable (archive) link for this page

Known Genes
Genes

% GC
Repeats

SNPs

Chromosome X

Click on the image above to zoom into that point

Chromosome X

Length: 154,824,264 bps

Gene Count: 931

Known Gene Count: 766

PseudoGene Count: 380

SNP Count: 320,997

Change Chromosome

Chromosome

Fields marked with * are required

Jump to ContigView

Choose two features from this chromosome as anchor points and display the region between them. Both features must be mapped to the current Ensembl golden tiling path. If you select "None" for the second feature, the display will be based around the first feature.

Please note that if you select widely spaced features there may be an significant delay while Ensembl builds the DNA display.

Region

From (type):

To (type):

Context

Bp downstream

Bp upstream

Fields marked with * are required

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Genome Browser - Ensembl

Ensembl Genome Browser – Blast

[new](#) [SETUP](#) [CONFIG](#) [RESULTS](#) [DISPLAY](#)

[refresh](#) [Online Help](#)

Enter the Query Sequence

Either Paste sequences (max 30) in FASTA or plain text:

```
ACCACCCCTACCCAGGCTACAGAACTACATGGCAGCCAGAAGCATGCGGAAGC
CTGCCTCTCCTTGAACAAACACATCCGGATACGTTCTAAAGAATCTGAGAAAAC
ACGACTCGATTGGCTGTACGCAAACCGTAATAAATTCAGCAGTTTATCGTATC
ATTACACCATAT
```

Or Upload a file containing one or more FASTA sequences
[Choose File](#) no file selected

Or Enter a sequence ID or accession (EMBL, UniProt, RefSeq)
 [Retrieve](#)

Or Enter an existing ticket ID:
 [Retrieve](#)

☒ dna queries
☐ peptide queries

Select the databases to search against

Select species:
Use 'ctrl' key to select multiple species

Anopheles_gambiae
Apis_mellifera
Caenorhabditis_elegans
Canis_familiaris

☒ dna database ☐ peptide database

Genomic sequence
Ab-initio Peptides (Genscan/SNAP)

Select the Search Tool

BLASTN
SSAHA
TBLASTX

[configure](#) [RUN](#)

Search sensitivity:
Optimise search parameters to find the following alignments

Near-exact matches

Genome Browser - Ensembl

Ensembl Genome Browser – Contig View

e!Ensembl Human ContigView

Search e!Human:

Q: [AC007037.4](#) [1.168084](#) [AL359894.9](#) [1.161040](#)

Ensembl v32 - May 2005

Chromosome X
138,229,875 - 138,686,223

- View of Chromosome X
- Graphical view
- Graphical overview
- Export sequence as FASTA
- Export EMBL file
- Export Gene info in region
- Export SNP info in region
- Export Vega info in region
- View alongside ...
- View Syntenic regions ...
- View region in NCBI browser
- View region in UCSC browser

Use Ensembl to...

- Run a BLAST search
- Search Ensembl
- Data mining [BioMart]
- Upload your own data
- Download data
- Export data

Chromosome X

Overview

Chr. X band

Rat. synteny

Chimp. synteny

Mouse synteny

Chicken synteny

Dog synteny

DNA(contigs)

Markers

Ensembl Genes

ncRNA Genes

e! ncRNA Genes

Vega Genes

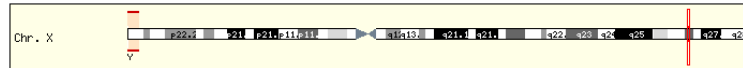
Gene Legend

Curated Known Gene
Curated Novel CDS
Ensembl Gene (GENE)
Ensembl Predicted Genes (NOVEL)
Ensembl Predicted Genes (PREDICT)
Ensembl Predicted Genes (KNOWN)
Curated Processed Pseudogene
Curated Pseudogene

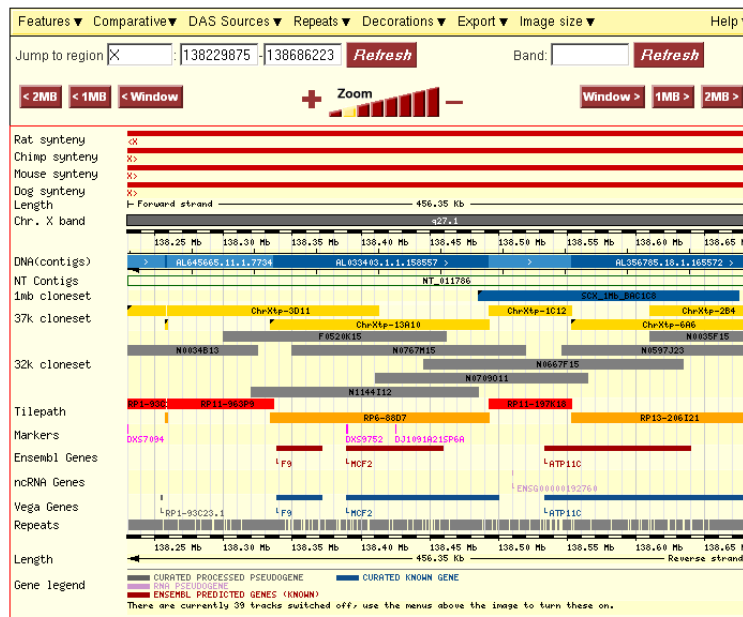
Genome Browser - Ensembl

Ensembl Genome Browser – Cyto View

☐ Chromosome X



☐ Detailed view



☐ Export data

Select Set of features to render

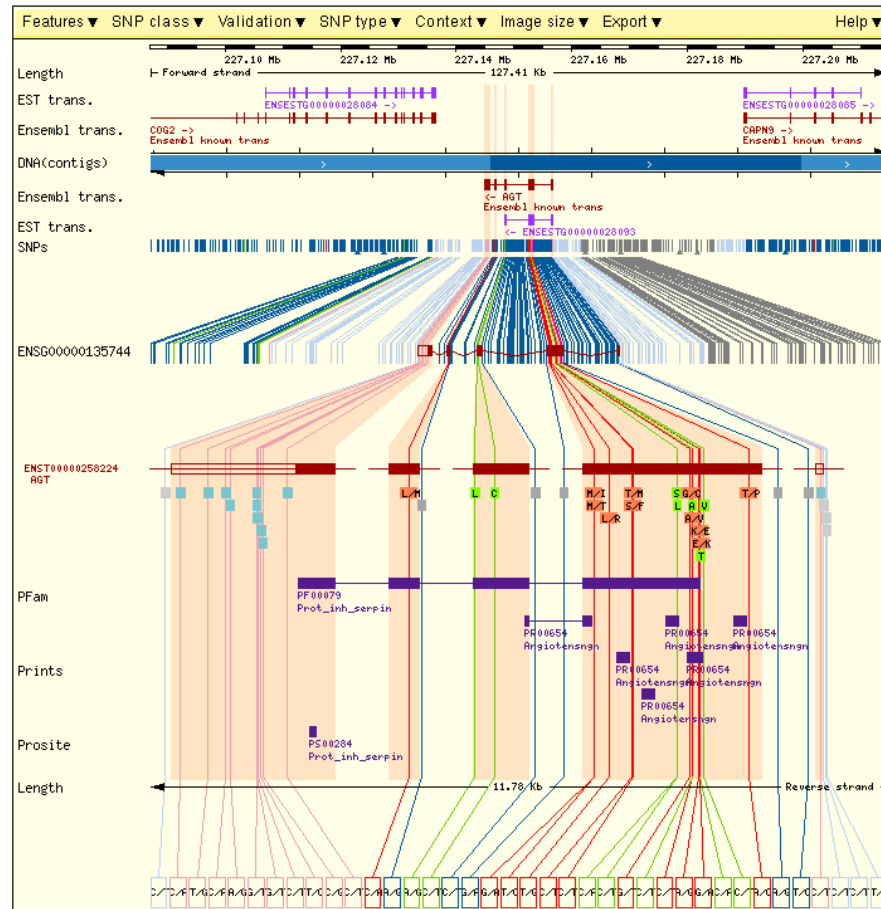
Output format

Select type to export

Genome Browser - Ensembl

Ensembl Genome Browser – SNP View

Variations in region of gene ENSG00000135744



Genome Browser - Ensembl

Ensembl Genome Browser – Marker View

Chromosome Map Marker DXS9752

Marker Source	82913 (database: unists)	
Marker Location	Basepairs 138389786 - 138390044 on chromosome X [Export data]	
Marker Synonyms	Gdb : GDB:737728 GDB:738733 Genbank : G13636 Other : SHGC-11927 DXS9752 RH8108	
Marker Primers	Expected Product Size Left Primer 259 TTTTCAGGTTAATGGACACGC	Right Primer CCATTTTGCAGCCGTAATTT

Marker DXS9752 map locations

Map Name	Synonym	Chromosome	Position	LOD Score
gm99g3	RH8108	X	4259	3.5

Genome Browser - Ensembl

Ensembl Genome Browsers – Gene View

e!Ensembl Human GeneView

Search #Human:
 e.g. [ENSG00000139616](#), [BRCA2](#)

Ensembl v92 May 2005 [Help](#)

ENSG00000101981

- Gene information
- Gene splice site image
- Gene variation info.
- Genomic sequence
- Export data
- Transcript information
- Exon information
- Peptide information

Chromosome X
138,338,415 - 138,371,137

- View of Chromosome X
- Graphical view
- Graphical overview
- Export information about region
- Export sequence as FASTA
- Export EMBL file
- Export Gene info in region
- Export SNP info in region
- Export Vega info in region

Ensembl Gene Report for ENSG00000101981

Gene	F9 (HGUGO ID) (to view all Ensembl genes linked to the name click here) This gene is a member of the human CCDS set: CCDS14666
Ensembl Gene ID	ENSG00000101981
Genomic Location	This gene can be found on Chromosome X at location: 138,338,415-138,371,137 This start of this gene is located in Contig AL033403.1.1.158557 .
Description	Coagulation factor IX precursor (EC 3.4.21.27) (Christmas factor) (Plasma thromboplastin component) (PTC). Source: UniProtKB/SwissProt P00740
Prediction Method	Genes were annotated by the Ensembl automatic analysis pipeline using either a GeneWise model from a human/vertebrate protein, a set of aligned human cDNAs followed by GenomeWise for ORF prediction or from Genscan exons supported by protein, cDNA and EST evidence. GeneWise models are further combined with available aligned cDNAs to annotate UTRs.
Transcripts	ENST00000218099 F9 [Transcript info] [Exon info] [Protein info] Features ▾ Chr. X Length Forward strand Ensembl trans. [NW(contigs)] Length Reverse strand

Genome Browser - Ensembl

Ensembl Genome Browsers – Exons View

Ensembl Exon Report

Transcript	F9 (HUGO ID) (to view all Ensembl genes linked to the name click here) This transcript is a member of the human CCDS set: CCDS14666
Ensembl Transcript ID	ENST00000218099
Transcript information	Exons: 8 Transcript length: 2,802 bps Translation length: 461 residues This transcript is a product of gene: F9
Genomic Location	This transcript can be found on Chromosome X at location: 138,338,415-138,371,137 This start of this transcript is located in Contig AL033403.1.1.58557 .
Description	Coagulation factor IX precursor (EC 3.4.21.22) (Christmas factor) (Plasma thromboplastin component) (PTC). Source: UniProt/SwissProt P00740

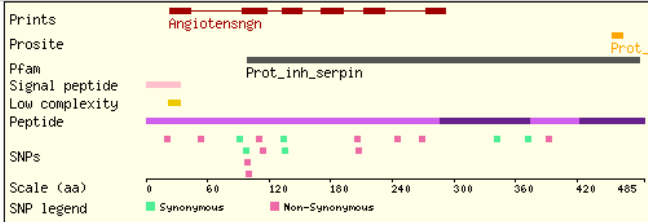
Exon Information

No.	Exon / Intron	Chr	Strand	Start	End	Length	Sequence
	5' upstream sequence						ccagaagtaaatcacagctcagcttgactttgtacacactaactgacctt
1	ENSE00000677288	X	1	138,338,415	138,338,531	117 bp	ACCACCTTTCACAATCTGCTAGCAAAGGTTATGCGAGCGCGTGAACATGATCATGCGAGAAT CACCAGGCCCTCATCACCATCTGCGCTTTTAGGATATCTACTCACTGCTGAATGTACAG
	Intron 1-2	X	1	138,338,532	138,344,688	6,157 bp	gtttgttctcttttttaaatcat.....aaagaattattctttacatttcag
2	ENSE00000677287	X	1	138,344,689	138,344,852	164 bp	TTTTTCTTGATCATGAAAACGCCAACAAATTTGAAATCGGCCAAAGAGGTATAATTTCAG GTAAATTGGAAGAGTTTGTTCAGGGGAACCTTGAGAGAGAAATGTATGGAAGAAAAGTGTAT GTTTTGAAGAAGCAGGAGAAGTTTTTGAAGCACTGAAAGAACAA
	Intron 2-3	X	1	138,344,853	138,345,040	188 bp	gtgagtatctccacataataaccctt.....atttgttctcttttattctttatag
3	ENSE00001173335	X	1	138,345,041	138,345,065	25 bp	ACTGAATTTTGAAGCAGTATGTTG
	Intron 3-4	X	1	138,345,066	138,348,754	3,689 bp	gtaagcaattcattttatctcttag.....ttcaatttcttaacctatctcaag
4	ENSE00001173327	X	1	138,348,755	138,348,868	114 bp	ATGGAGATCAGTGTGAGTCCAATCCATGTTTAAATGCGCGCAGTTGCAAGGATGACATTA ATTCTATGAATGTGTGTGCTTGTGGATTGGAAGGAAAGAACTGTGAATTAG
	Intron 4-5	X	1	138,348,869	138,356,041	7,173 bp	gtaagtaactattttttgaatactc.....tactgtctattttgtctcttttag
5	ENSE00001173320	X	1	138,356,042	138,356,170	129 bp	ATGTAACATGTAAACATTAAGAATGGCAGATGCGAGCACTTTTGTAAAAATAGTGCTGATA ACAAGGTGTTTGTCTCTGTACTGAGGGATATGCACTTGCAGAAAACAGAAAGTCTGTG AACCAGCAG
	Intron 5-6	X	1	138,356,171	138,358,740	2,570 bp	gtcataactcgaataagatttttta.....actaatttttctctatttttttag
6	ENSE00001173315	X	1	138,358,741	138,358,943	203 bp	TGCCAATTCATGTGGAAGAGTTTCTGTTTACAAACTTCTAAAGCTCACCCGTGCTGAGA CTGTTTTCTCTGATGTGCACTATGTAAATCTACTGAACTGAAACCATTTTGTGATAACA

Genome Browser - Ensembl

Ensembl Genome Browser – Proteins View

Ensembl Protein Report

Peptide	AGT (HUGO ID) (to view all Ensembl genes linked to the name click here) This peptide is a member of the human CCDS set: CCDS1585
Ensembl Peptide ID	ENSP00000258224
Translation information	This peptide is a translation of transcript: ENST00000258224 , which is a product of gene: ENSG00000135744 .
Genomic Location	This peptide can be found on Chromosome 1 at location: 227,145,622-227,153,331 This start of this peptide is located in Contig AL158214.33.1.181175 .
Description	Angiotensinogen precursor [Contains: Angiotensin I (Ang I); Angiotensin II (Ang II); Angiotensin III (Ang III) (Des-Asp1- angiotensin II)]. Source: UniProt/SWISSPROT P01019
Prediction Method	Genes were annotated by the Ensembl automatic analysis pipeline using either a GeneWise model from a human/vertebrate protein, a set of aligned human cDNAs followed by GenomeWise for ORF prediction or from Genscan exons supported by protein, cDNA and EST evidence. GeneWise models are further combined with available aligned cDNAs to annotate UTRs.
InterPro	IPR000227 Angiotensinogen - [View other genes with this domain] IPR000215 Proteinase inhibitor I4, serpin - [View other genes with this domain]
Protein Family	ENSF00000005553 : ANGIOTENSINOGEN PRECURSOR [CONTAINS: ANGIOTENSIN I ANG I ; ANGIOTENSIN II ANG II ; ANGIOTENSIN III ANG III DES ASP[1] ANGIOTENSIN II] This cluster contains 1 Ensembl gene member(s)
Protein Features	
Peptide Sequence	<pre>MRKRAPQSEMAPAGVSLRATILCLLAWAGLAAGDRVYIHPHLVIHNHSTCEQLAKANAG KPKDPTFIPAPIQAKTSPVDEKALQDQLVLVAAKLDTEKLRPAAHVGHANFLGFRYCH HSELWGVVHGATVLSPTAVFGTLASLYLGALDHTADRLQAILGVFWKDKNCTSRDLAHKV LSALQAVQCLLVACQCRADSSQAQLLLSTVVGVPFTAPGLHLKQPFVQCLALYTPVVLPRSLD FTELDVAEEKIDRFMQAVTGWKTCGLMGASVDSFLAFNTYVHFQCKMKGFSLLAEPQEF WVDNSTSVSVPMLSGCGTFFQHWSDIQDNFVTVQVPFTESACLLIQPHYASDLDKVECLT FQNSLNMHGKLSPTIHLTHPQLVLCQSYDLQDLLAQAEPLAILHTELNLQKLSNDRIR VCEVLNIFFELEADEREPTSTQQLNRPVLEVLNRPFLPAVYDQSATALHFLGRVAN PLSTA</pre>

Genome Browser - Ensembl

Ensembl Genome Browser

- Homo Sapiens - Human
- Pan troglodytes- Chimpanzee
- Macaca mulatta - Rhesus monkey
- Mus musculus - Mouse
- Rattus norvegicus - Rat
- Canis familiaris - Dog

Genome Browser - Ensembl

Ensembl

- All data generated is free for download
- Includes genes sequence, transcript, protein predictions
- Ensembl provides a dedicated Export View page, can be exported into HTML, text or zipped format

Genetics and Genomics

HGP – Ethical, Legal and Social Issues

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- The ability to identify human genes raises complex ethical issues involving
- The right to information about one's own genome

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Right to genetic information by employers, insurance companies and government agencies, and concerns about the ability to diagnose but not treat genetic disorders

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Privacy and confidentiality of genetic information
- Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Psychological impact and discrimination due to an individual's genetic differences.

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Reproductive issues including adequate and informed consent and use of genetic information in reproductive decision making.

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Clinical issues including the education of doctors and other health-service providers

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- People identified with genetic conditions, and the general public about capabilities, limitations, and social risks.

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease, diabetes, and alzheimer's disease).

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Fairness in access to advanced genomic technologies.

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Health and environmental issues concerning genetically modified (GM) foods and microbes.

HGP – Ethical, Legal, Social Issues

HGP- Ethical, Legal and Social Issues

- Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials.

Genetics and Genomics

**Genome
Annotations –
Proteins Coding
Genes**

Genome Annotations-Proteins Coding Genes

Genome Annotations

- To make use of the genome sequences, all components of genomes need to understand.
- Assigning identities and functions to sequences within the genome is called genome annotation.

Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Genes transcribed, which means that we can identify them
- Traditionally cDNA /EST sequencing
- More recently by microarray

Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Structural Annotation – Finding genes and other biologically relevant sites

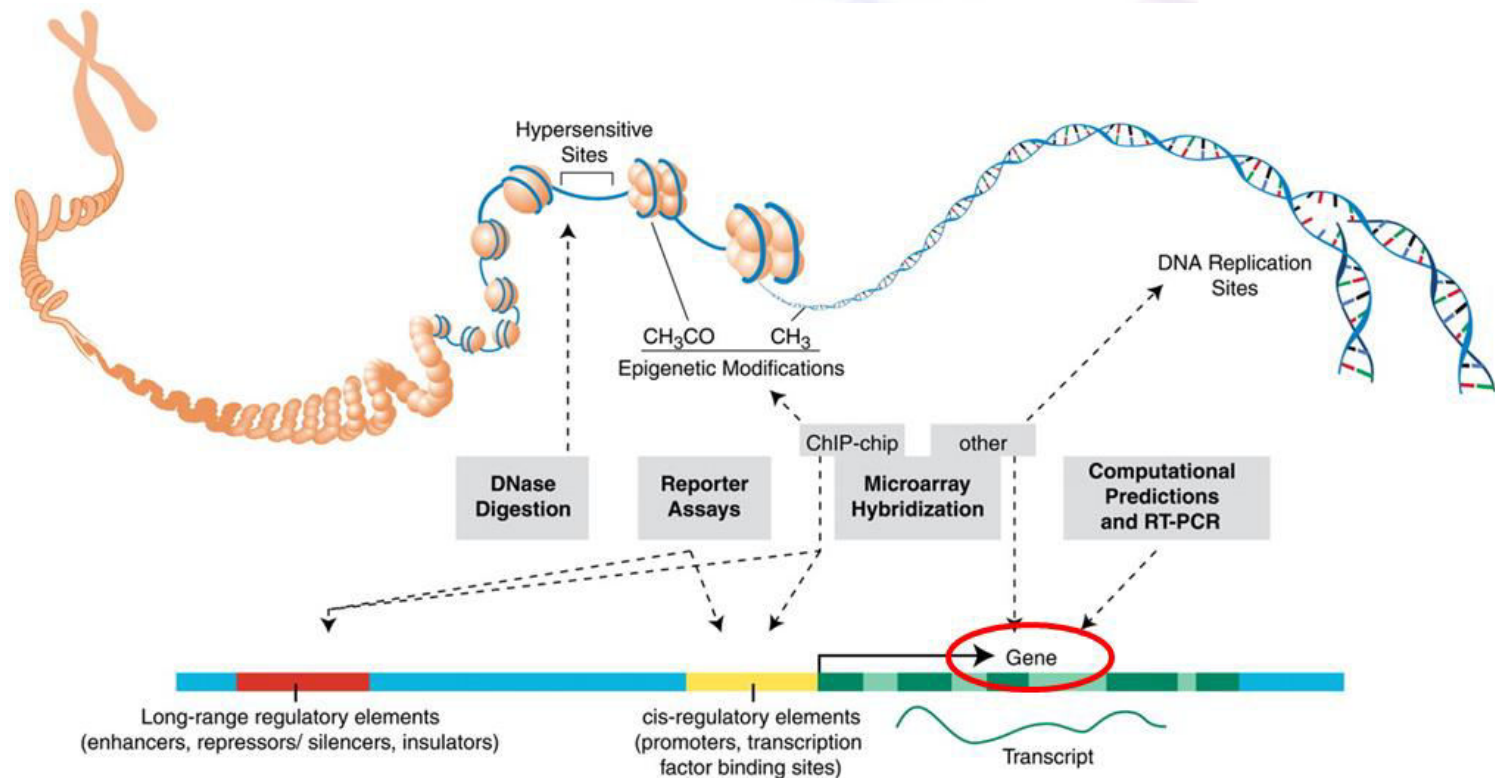
Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Annotation/function can be mapped to different levels:
- Organism, cellular, molecular level

Genome Annotations-Proteins Coding Genes

Genome Annotations



Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Genes can be identified *in vitro* using computational methods
- Protein-coding genes have recognizable features

Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Open reading frames
- Codon bias
- Known transcription and translational start and stop motifs
- Splice consensus sequences at intron-exon boundaries

Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Protein-coding genes recognizable feature
- Software to scan the genome and identify these features
- Some of these programs work quite well, in bacteria and simple eukaryotes

Genome Annotations-Proteins Coding Genes

Genome Annotations-Protein Coding Genes

- Harder for the higher eukaryotes where there are a lot of long introns, genes can be found within introns of other genes

Genome Annotations-Proteins Coding Genes

Validating Predictions & Refining Gene Models

- Validation of predictions include;
- Match to previously annotated cDNA
- Match to EST from same organism
- Similarity of nucleotide or translated protein sequence to GenBank
- Protein structure prediction match - PFAM domain
- Associated with recognized promoter sequences, i.e TATA box, CpG island

Genome Annotations-Proteins Coding Genes

Finding Non-Protein-Coding Genes

- Harder to find
- No poly-A tailed—
- No ORF
- Rely on sequence divergence at nucleotide not protein level, so homology is harder to detect

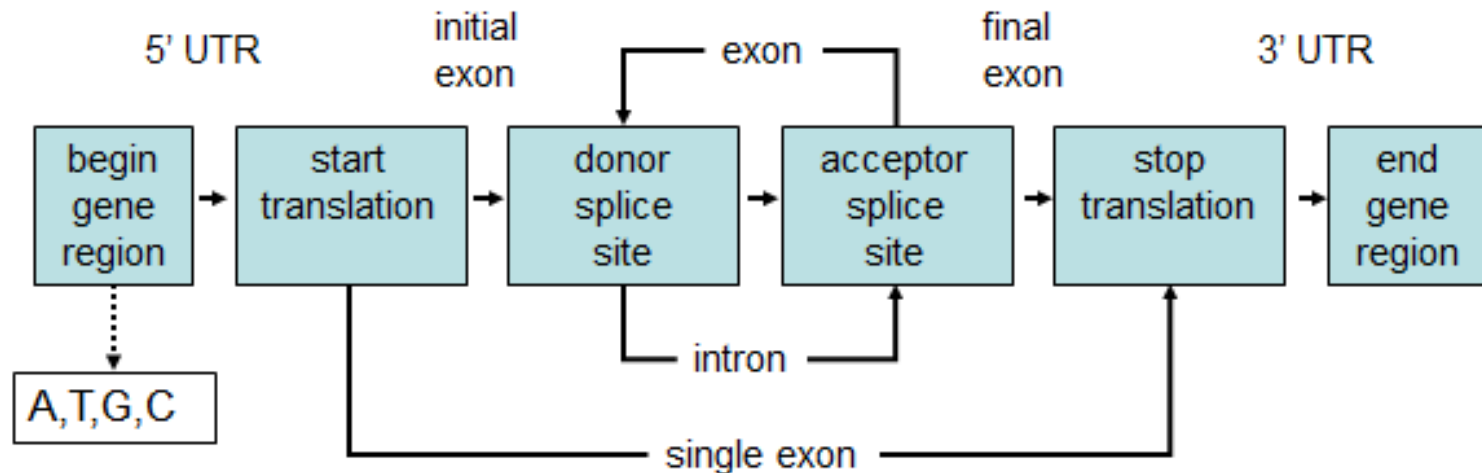
Genome Annotations-Proteins Coding Genes

Gene Discovery Programs

- Most gene-discovery programs makes use of machine learning algorithm.
- Two approaches;
- Artificial Neural Networks and Hidden Markov Models.

Genome Annotations-Proteins Coding Genes

Gene Discovery Programs- Diagram for an HMM for Gene Discovery



Examples of known gene models; provide the probability that a stretch of sequence is a gene.

Genome Annotations-Proteins Coding Genes

Genome Annotations

- Assigning identities and functions to sequences within the genome is called genome annotation.

Genetics and Genomics

Genome Annotations – Transcription Factors

Genome Annotations-Transcription Factors

Genome Annotations

- Other than protein coding genes
- Structural sequences
- Regulatory sequences
- Non-functional junk
- Annotate regulatory sequences such as transcription factor binding sites (Cis-)

Genome Annotations-Transcription Factors

Genome Annotations

- Transcription factors (TFs) are proteins that bind to the DNA and help to control gene expression.
- Sequences to which TF bind called transcription factor binding sites (Cis-acting elements)

Genome Annotations-Transcription Factors

Genome Annotations

- Transcription factors bind to specific DNA sequences
- Most transcription factors can bind to a range of similar sequences.

Genome Annotations-Transcription Factors

Genome Annotations

- Once we know the binding site, we can search the genome to find all of the (predicted) binding sites.

Genome Annotations-Transcription Factors

Genome Annotations

7 characterized binding sites for certain transcription factor

Consensus Sequence

Frequency matrix and its graphical depiction, a sequence logo

TCCGGAAGC
TCCGGATGC
TCCGGATCT
CATGGATGC
CCAGGAAGT
GGTGGATGC
ACCGGATGC

$T_C C^C_T GGAAGC$

A	111007200
T	302000502
G	110770060
C	254000015



Genome Annotations-Transcription Factors

Genome Annotations - Conclusion

- Genome annotations can also be performed by searching transcription factor binding sites

Genetics and Genomics

**Arabidopsis
Thaliana Genome –
A Model Organism
in Plant Biology and
Genetics**

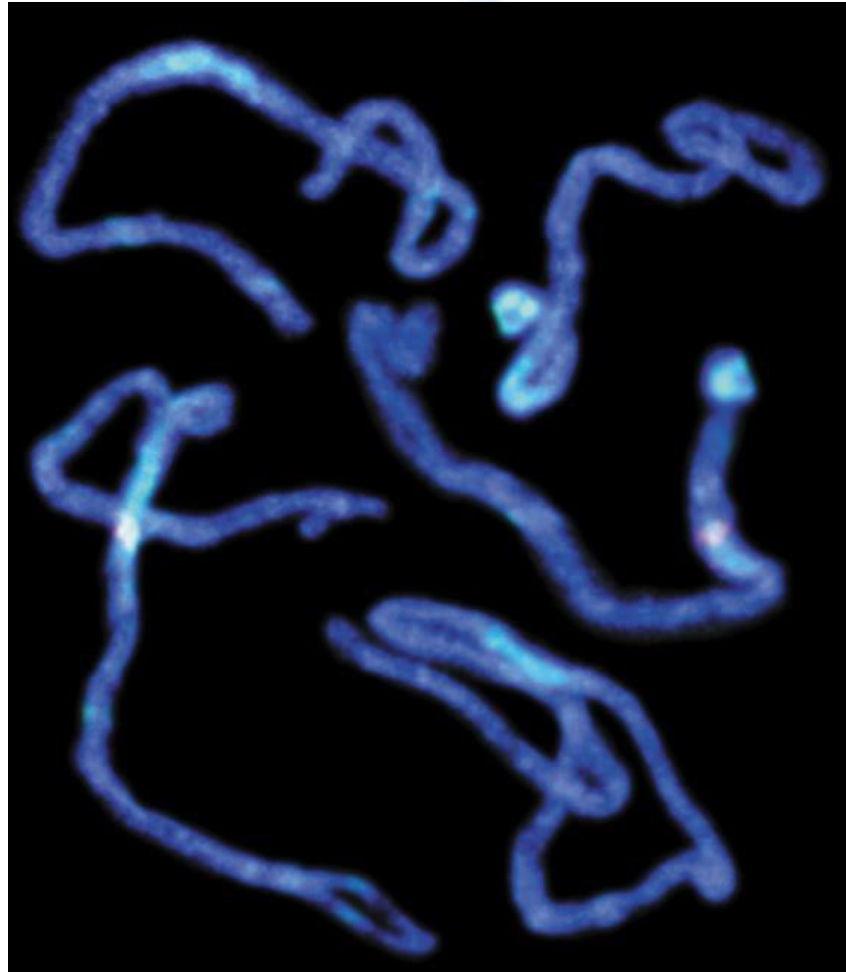
Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- Arabidopsis Thaliana
 - A model Plant
- Arabidopsis has 5 chromosomes
($2n=10$)

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Chromosomes



Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- Contains about 135 megabases of sequence
- Encodes approximately 27,000 genes and 35,000 proteins.

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- Has 35% unique genes
- Has 37% genes that exist as members of large gene families (families of 5 or more members)

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- An estimated 58-60% of the Arabidopsis genome exists as large segmental duplications

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- Arabidopsis genome contains genes encoding RNA polymerase subunits not seen in other eukaryotic organisms

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

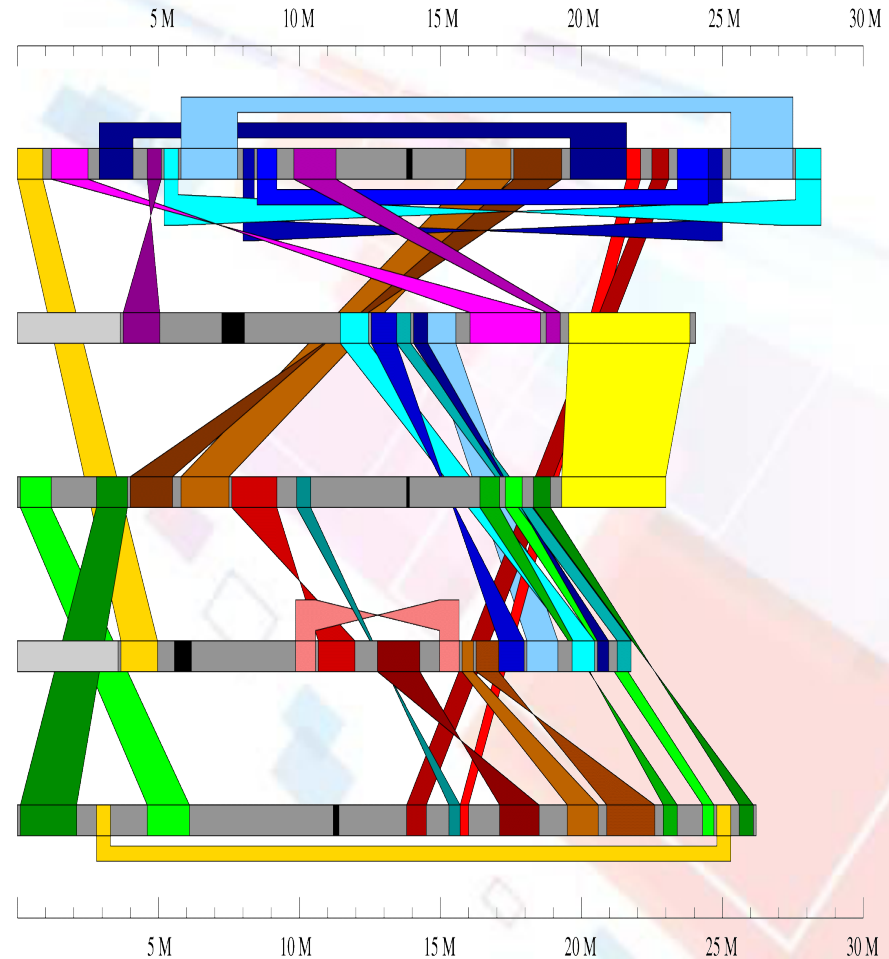
- Arabidopsis has genes unique to plants – approximately 150 unique protein families were found, including 16 unique families of transcription factors

Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome:

**Segmental
duplications in
Arabidopsis
thaliana.**

**Colored bands
serve as
representations
of duplicated
segments.**



Arabidopsis Thaliana Genome

Arabidopsis Thaliana Genome

- Arabidopsis has 5 chromosomes ($2n=10$)
- Contains about 135 Mb of sequence
- Encodes ~ 27,000 genes and 35,000 proteins.

Genetics and Genomics

Mouse Genome – A Model Organism in Animal Biology and Genetics

Mouse Genome

Mouse Genome

- Number of chromosome $2n=40$
- Human and mouse genomes have conserved blocks of genetic material
- Humans and mice suffer from similar diseases

Mouse Genome

Mouse genome

- Estimated genome Size 2.6 billion bp
- Number of estimated coding genes 22,500
- At the nucleotide level, approximately 40% of the human genome aligned to the mouse genome.

Mouse Genome

Mouse Genome: Chromosomes and their Sizes

Chromosome	cM	Mbp	Chromosome	cM	Mbp
1	98.5	195	11	88.0	122
2	103.9	182	12	63.9	120
3	82.7	160	13	67.3	120
4	88.6	157	14	66.4	125
5	90.2	152	15	59.0	104
6	79.0	150	16	57.8	91
7	89.1	145	17	61.3	95
8	76.2	129	18	59.4	91
9	75.1	125	19	56.9	61
10	77.9	131			

Mouse Genome

Mouse Genome: Chromosomes and their Sizes

Type	Name	Size (Mb)	GC%	Protein	rRNA	tRNA	Other RNA	Gene	Pseudogene
Chr	1	195.47	41.3	4,613	-	37	2,010	2,709	557
Chr	2	182.11	42.2	6,093	-	8	2,712	3,529	596
Chr	3	160.04	40.7	3,476	-	40	1,643	2,268	469
Chr	4	156.51	42.5	4,637	-	8	1,925	2,621	484
Chr	5	151.84	42.7	4,509	-	10	1,884	2,543	391
Chr	6	149.74	41.6	3,835	1	53	1,578	2,616	537
Chr	7	145.44	43.2	6,198	-	11	2,011	3,769	906
Chr	8	129.4	42.6	3,609	32	7	1,682	2,222	366
Chr	9	124.6	42.9	4,239	-	7	1,598	2,323	368
Chr	10	130.7	41.6	3,455	-	11	1,609	2,119	381
Chr	11	122.08	44.0	5,511	-	47	1,987	2,851	366
Chr	12	120.13	42.0	2,509	-	3	1,542	1,997	501
Chr	13	120.42	41.9	2,494	-	106	1,553	2,161	470
Chr	14	124.9	41.4	2,840	-	12	1,492	2,145	459
Chr	15	104.04	42.2	2,802	-	4	1,356	1,644	272
Chr	16	98.21	41.2	2,385	-	2	1,035	1,380	255
Chr	17	94.99	42.9	3,581	-	12	1,406	2,058	422
Chr	18	90.7	41.7	1,803	-	-	923	1,244	260
Chr	19	61.43	43.1	2,309	-	10	841	1,312	210
Chr	X	171.03	39.7	3,005	-	17	942	2,282	873
Chr	Y	91.74	39.3	238	-	-	84	372	140

Mouse Genome

Mouse Genome

- Mouse genome contains fewer CpG islands (15,500)
- The (G+C) content for each of the mouse chromosome is relatively similar

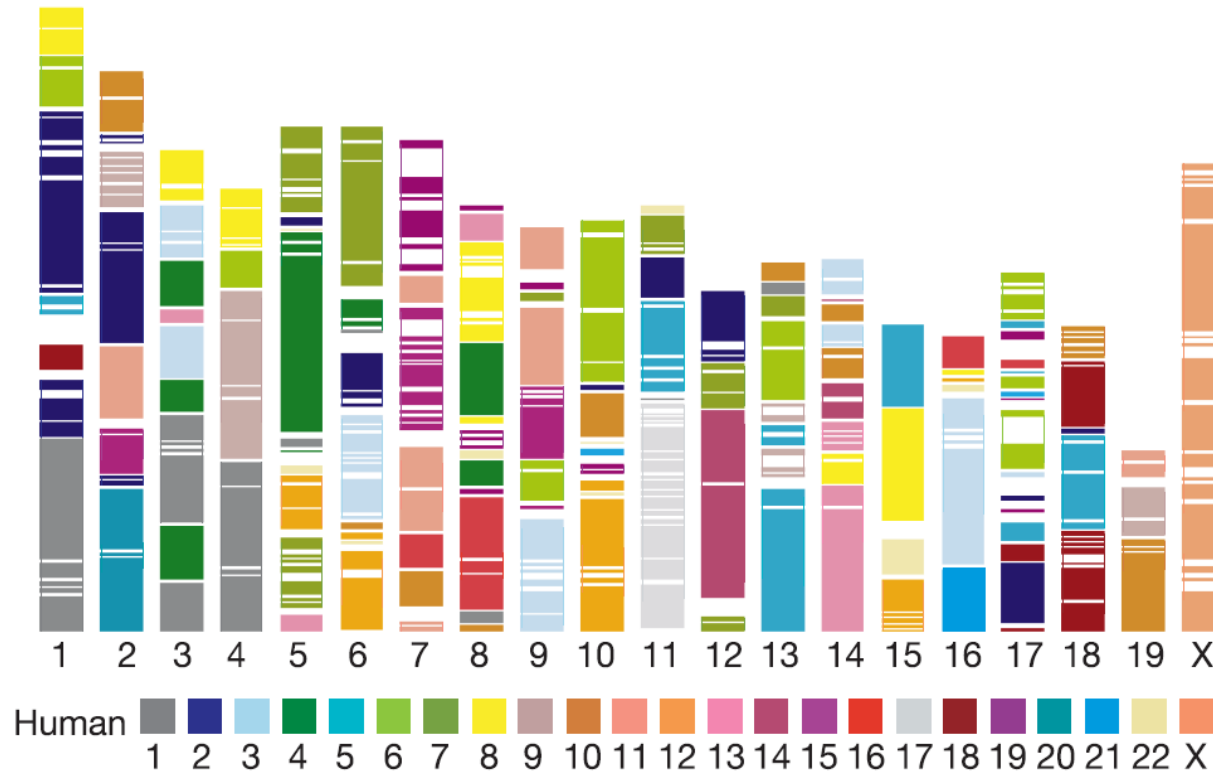
Mouse Genome

Mouse Genome

- Approximately 99% of mouse genes have a homologue in the human genome

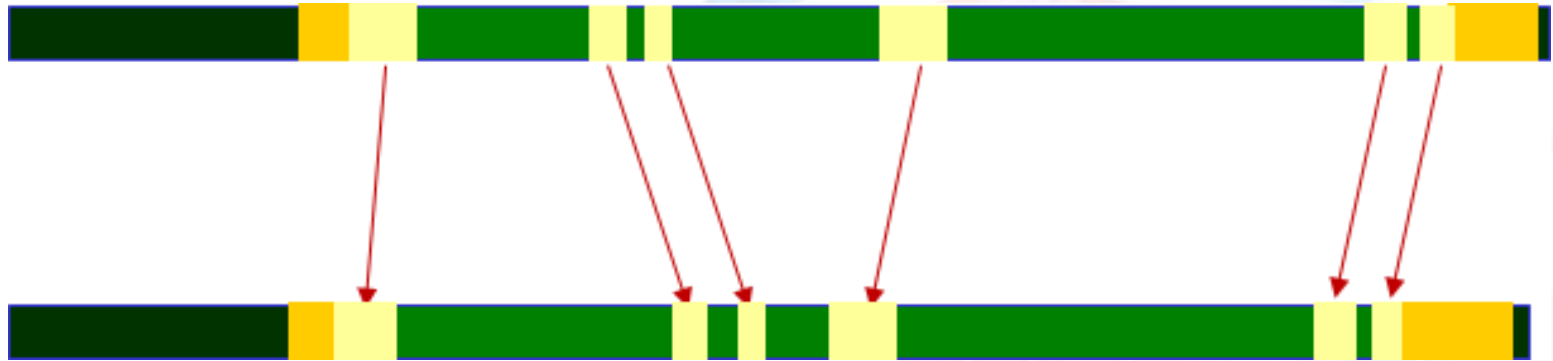
Mouse Genome

Human and Mouse Chromosomes



Mouse Genome

Human and Mouse: Chromosomes Orthologs



- Most human genes have mouse orthologs
- Coding exons usually correspond 1-1
- Coding sequence similarity ~ 85%

Mouse Genome

Mouse Genome Contigs

- Most mouse BAC contigs contained multiple mouse markers.
- Coverage of the mouse genome (2.6 billion bp) in mapped BACs is virtually complete: 296 contigs of average size 9.3 Mb.

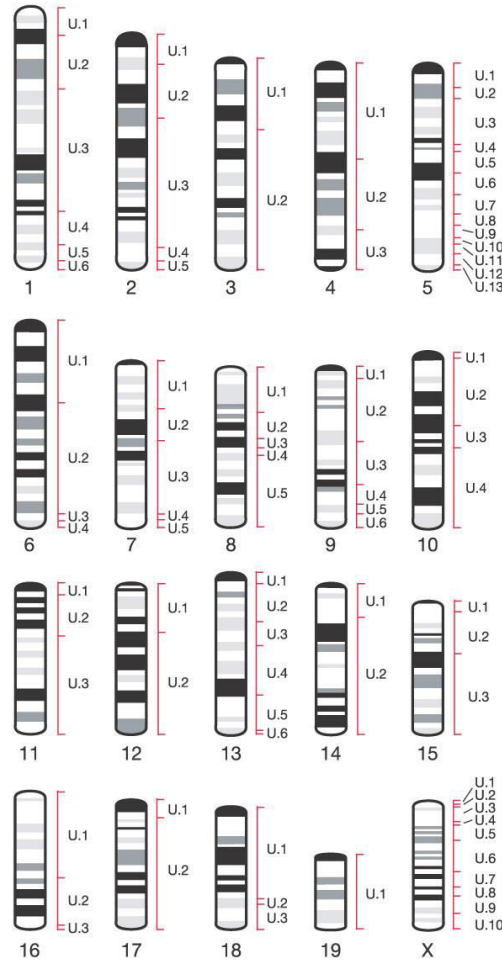
Mouse Genome

Mouse Genome

- All mouse chromosomes are acrocentric, with the centromeric end at the top of each chromosome.

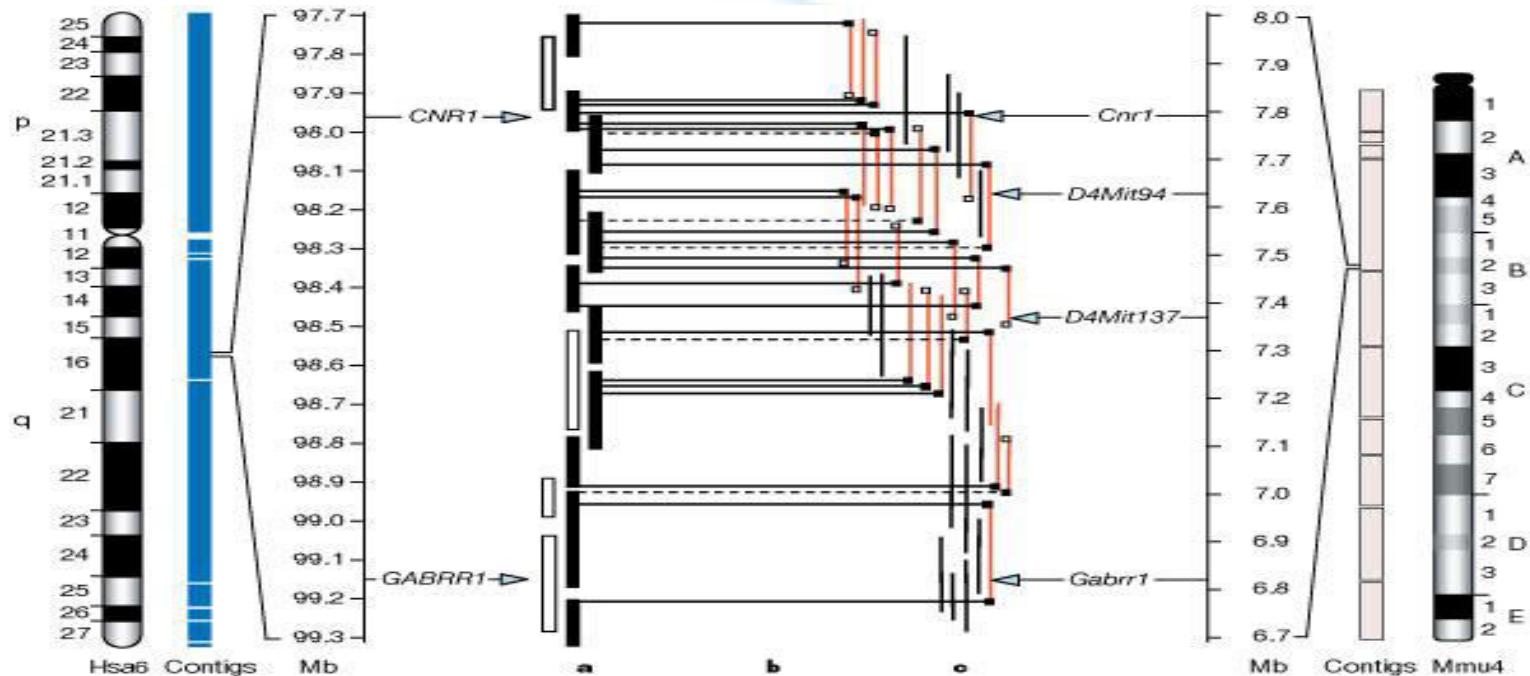
Mouse Genome

Cytogenetic View of Chromosomes



Mouse Genome

Human–Mouse Homology Clone Map



Alignment between part of human chromosome 6 and mouse chromosome 4 . 1.6-Mb interval is enlarged, showing part of 16.1 aligned to a 1.3-Mb mouse BAC contig. 11 of the 15 segments of human sequence match to 29 of segments of mouse BAC contig

Mouse Genome

Mouse Gene and Human Gene

Human Disease and Mouse Model Detail

Human Disease	Term: Von Hippel-Lindau Syndrome; VHL OMIM ID: 193300			
Associated Genes	Orthologous mouse and human markers where mutations in one or both species have been associated with phenotypes characteristic of this disease.			
	Mouse Gene	Human Gene	Characteristics of this human disease are associated with mutations in...	
	Vhl	VHL	 ...both mouse and human orthologous genes.	
	Hif1a	HIF1A	 ...the mouse gene. <i>OMIM data currently do not associate this disease with the orthologous human gene.</i>	
	Ccnd1	CCND1	 ...the human gene. <i>MGI data currently do not associate this disease with a mouse model.</i>	
Mouse Models	Genotype			Ref(s)
	Allelic Composition	Note	Genetic Background	
	Models with phenotypic similarity to human disease where etiologies involve orthologs. ¹			
	Hif1a^{tm3Rsio}/Hif1a^{tm3Rsio} Hprt1^{tm1(Pck1-cre)Vhh}/Y Vhl^{tm1Jae}/Vhl^{tm1Jae}	2	involves: 129 * BALB/c * C57BL/6	1:97652 , 1:106705
	Hif1a^{tm3Rsio}/Hif1a^{tm3Rsio} Vhl^{tm1Jae}/Vhl^{tm1Jae} Tg(Alb-cre)21Mqn/0	2	involves: 129 * BALB/c * C57BL/6 * DBA	1:97652

Mouse Genome

Mouse Genome

- Number of chromosome $2n=40$
- Humans and mice suffer from similar diseases
- Estimated genome Size 2.6 billion bp
- Number of estimated coding genes 22,500

Genetics and Genomics

Genome Comparison - Human and Mouse

Genome Comparison-Human and Mouse

Comparison

- The mouse genome is about 14% smaller than the human genome
- Over 90% of mouse and human genomes can be partitioned into corresponding regions of conserved synteny

Genome Comparison-Human and Mouse

Comparison

- At the nucleotide level, 40% of the human genome can be aligned to the mouse genome

Genome Comparison-Human and Mouse

Comparison

- Mouse and human genomes each seem to contain ~ 23,000 protein-coding genes.

Genome Comparison-Human and Mouse

Comparison

- The proportion of mouse genes without any homologue currently detectable in the human genome (and vice versa) is $< 1\%$.

Genome Comparison-Human and Mouse

Comparison

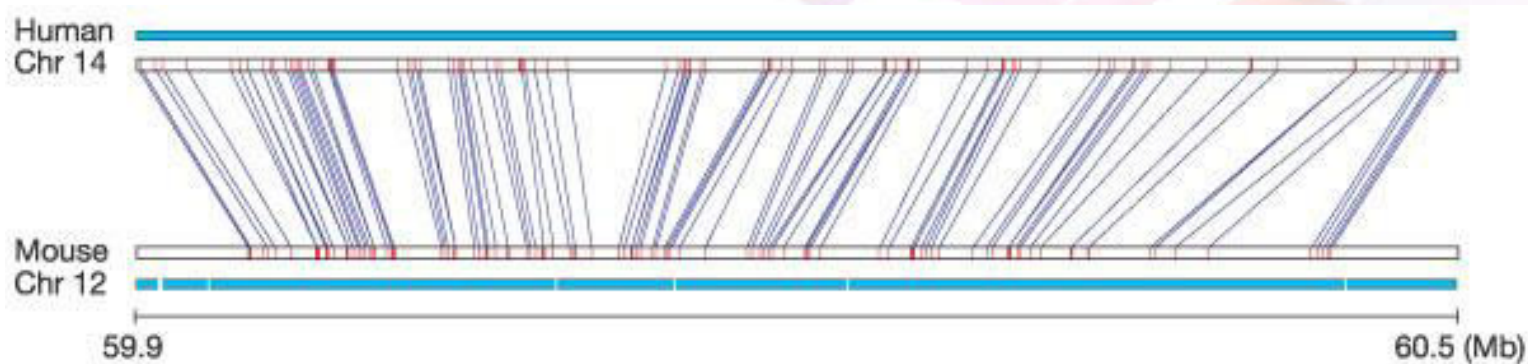
- Despite marked differences in activity of transposable elements, similar types of repeat sequences have accumulated in the corresponding genomic regions

Genome Comparison-Human and Mouse

Comparison - Human and Mouse Genome

A typical 510-kb segment of mouse chromosome 12 that shares common ancestry with a 600-kb section of human chromosome 14.

Blue lines connect the reciprocal unique matches in the two genomes.



Genome Comparison-Human and Mouse

Comparison – Human and Mouse Genome

- Mouse genome contains fewer CpG islands than human.
- (G+C) content and density of CpG islands shows more variability in human than mouse

Genome Comparison-Human and Mouse

Comparison

- Human chromosomes show more variation (G+C) on chromosomes 16, 17, 19 and 22 have higher (G+C) content, and chromosome 13 lower (G+C) content.

Genome Comparison-Human and Mouse

Comparison

- The density of CpG islands is relatively homogenous for all mouse chromosomes and more variable in human.

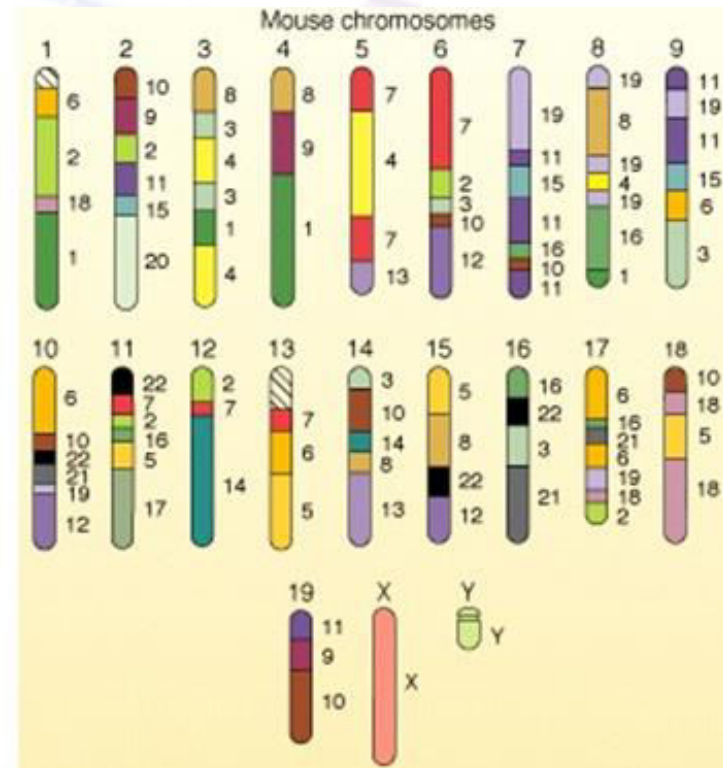
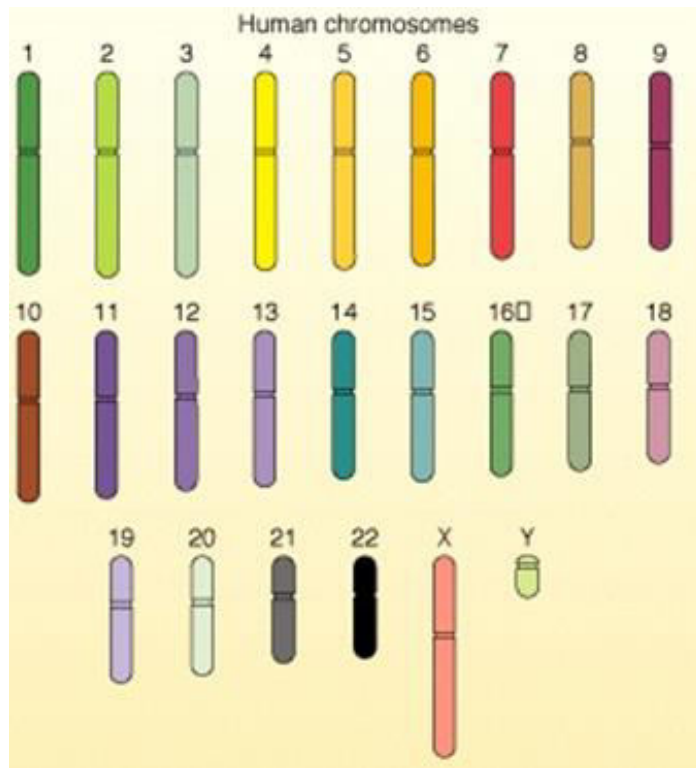
Genome Comparison-Human and Mouse

Comparison

- Approximately 99% of mouse genes have a homologue in the human genome

Genome Comparison-Human and Mouse

Comparison – Human and Mouse Genome



Genome Comparison-Human and Mouse

Comparison - Conclusion

- The mouse genome is about 14% smaller than the human genome
- Mouse and human genomes each seem to contain ~ 23,000 protein-coding genes.

Genetics and Genomics

Mapping of Disease Genes

Mapping of Disease Genes

Mapping of Disease Genes

- Disease gene mapping is one of the main objectives of genotyping
- Different approaches for identification of disease genes

Mapping of Disease Genes

Position Independent Method

- No information about the location of the gene.
- Starting at the phenotype, determining which protein was involved and getting to gene through the protein.

Mapping of Disease Genes

Position Dependent Method

- Starting from the approximate location of gene, to finding the gene itself, then translating it to learn about the protein and its function.
- Positional cloning or reverse genetics

Mapping of Disease Genes

Position Independent Method

- Mostly identifying and isolating the protein product of the gene.
- Such genes usually produce large amounts of well-known and well studied proteins.

Mapping of Disease Genes

Position Independent Method

- Two common methods used
- Gene-specific oligonucleotides
- Antibody Methods

Mapping of Disease Genes

Position Dependent Method

- Recombination mapping and/or somatic cell mapping to defined the region of interest as tightly as possible.

Mapping of Disease Genes

Position Dependent Method

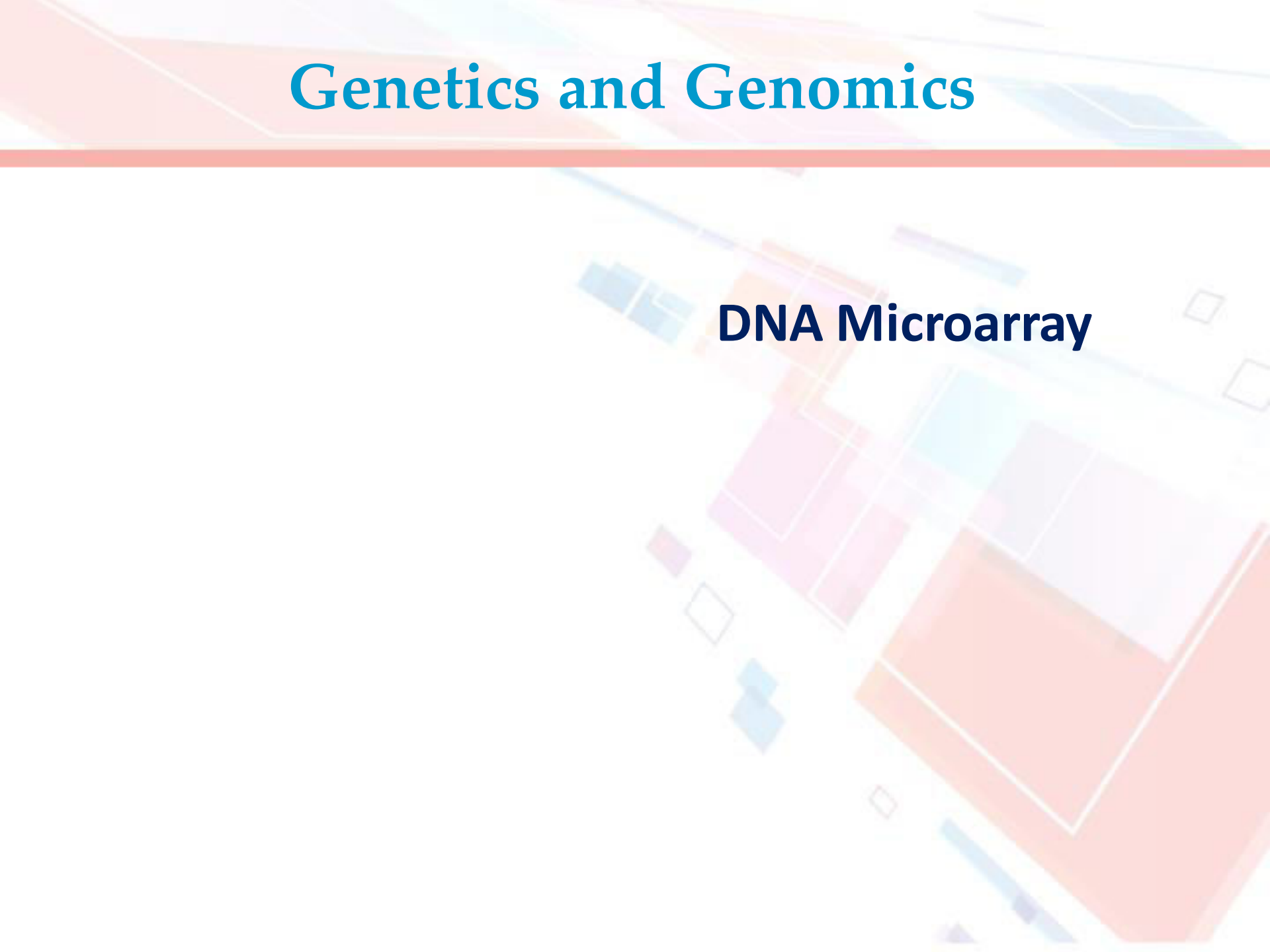
- Chromosomes are inherited as intact units, so it is reasoned that the alleles of some pairs of genes will inherit together because they are on the same chromosome.

Mapping of Disease Genes

Mapping of Disease Genes

- Different approaches are used for identification of disease genes

Genetics and Genomics



DNA Microarray

DNA Microarrays

DNA Microarrays

- Microarray technology evolved from Southern blotting, where fragmented DNA is attached to a substrate and then probed with a known DNA sequence.

DNA Microarrays

DNA Microarrays

- There are several synonyms of DNA microarrays such as
- DNA chips
- Gene chips
- DNA arrays
- Gene arrays
- Biochips

DNA Microarrays

Microarray - Principle

- The principle of DNA microarrays lies on the hybridization between the nucleotides.

DNA Microarrays

Microarray

- Presence of one genomic or cDNA sequence in 100,000 or more sequences can be screened in a single hybridization.

DNA Microarrays

DNA Microarray

- DNA microarray allows us to analyze thousands of genes in one experiment
- DNA microarrays are solid supports; glass or silicon, upon which DNA is attached

DNA Microarrays

Simultaneous Studies of Many Genes

- Patterns or clusters of genes are more informative regarding total cellular function than looking at one or two genes – can figure out new pathways

DNA Microarrays

Simultaneous Studies of Many Genes

- Just sequencing genomes is not sufficient
- Thousands of genes remain without an assigned function
- Each spot of DNA, (probe), represents a single gene

DNA Microarrays

Microarray Use

- Determine which genes are active in a cell and at what levels
- Compare the gene expression profiles of a control vs treated
- Assessing gene expression levels

DNA Microarrays

Microarray Use

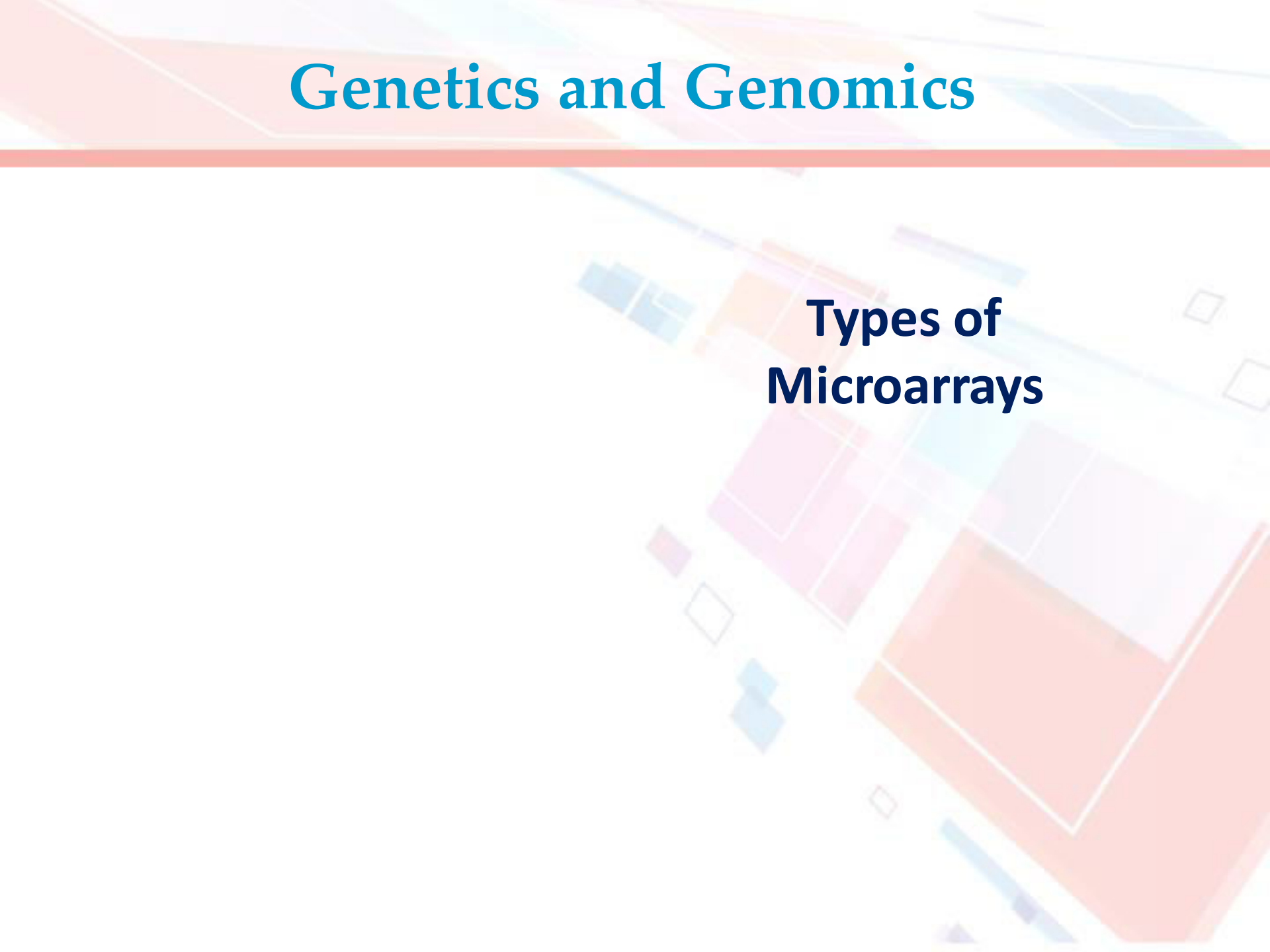
- Genome-wide studies and genotyping
- Evaluating microRNA levels
- Determine which genes have biological significance in a system

DNA Microarrays

Microarray Use

- Discovery of new genes, pathways, and cellular trafficking

Genetics and Genomics



Types of Microarrays

Types of Microarrays

Types of Microarrays

- DNA microarrays
- Exon arrays
- Comparative Genomic Hybridization array

Types of Microarrays

Types of Microarrays

- DNA microarrays-
such as cDNA
microarrays and
oligonucleotide
microarrays, SNPs
- Exon arrays-
Alternative splice
variant detection

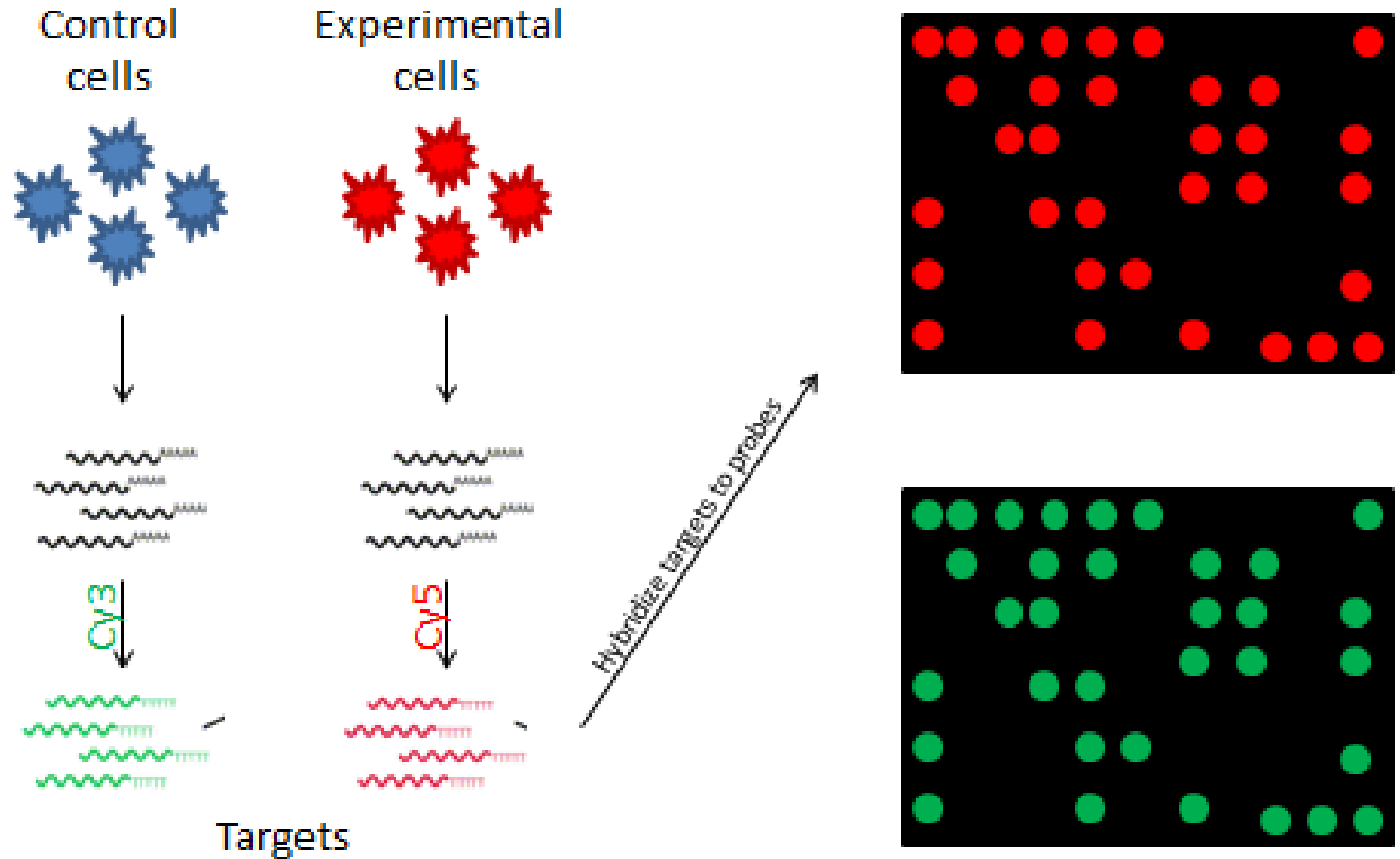
Types of Microarrays

Types of Microarrays

- CGH arrays-
Comparative
genomic
hybridization

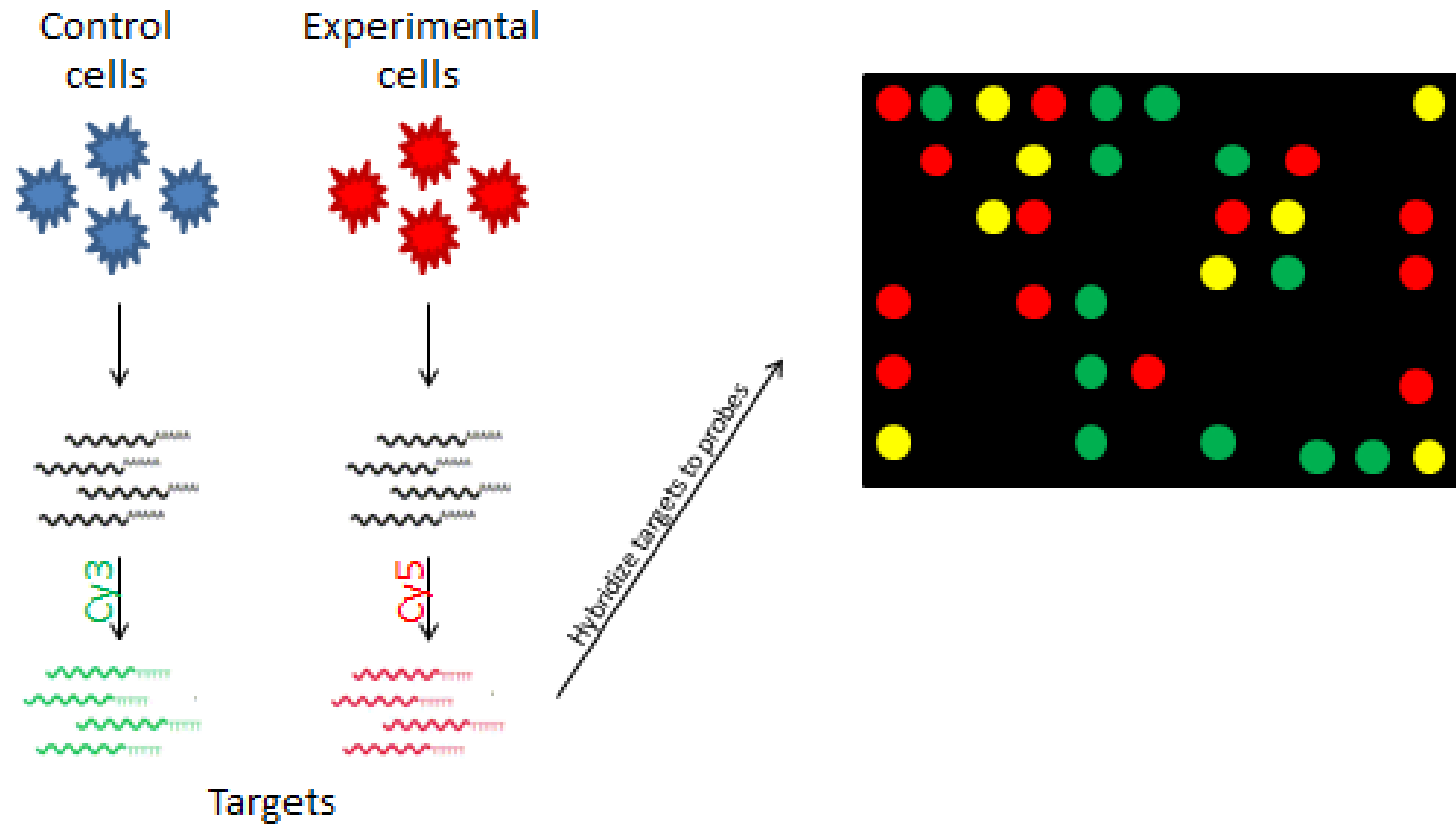
Types of Microarrays

Oligonucleotide Arrays



Types of Microarrays

cDNA Arrays



Types of Microarrays

Types of Microarray

- Protein microarrays (protein-protein interactions)
- Tissue microarrays
- Cellular microarrays (also called transfection microarrays)
- Expression Arrays

Types of Microarrays

Types of Microarray

- Chemical compound microarrays
- Antibody microarrays (proteomics)
- Carbohydrate arrays (glycoarrays)
- Re-sequencing arrays

Types of Microarrays

Microarray Require

- Microarrays combine genomics, silicon chip manufacturing, DNA and Protein chemistry, signal and image processing, statistics, software skills and traditional molecular biology experiments.

Types of Microarrays

Microarray and Genes Expression Profiling

- Gene expression profiling can be monitored for thousands of genes simultaneously
- CGH arrays assesses genome content in different cells or related organisms

Types of Microarrays

Microarray and SNPs

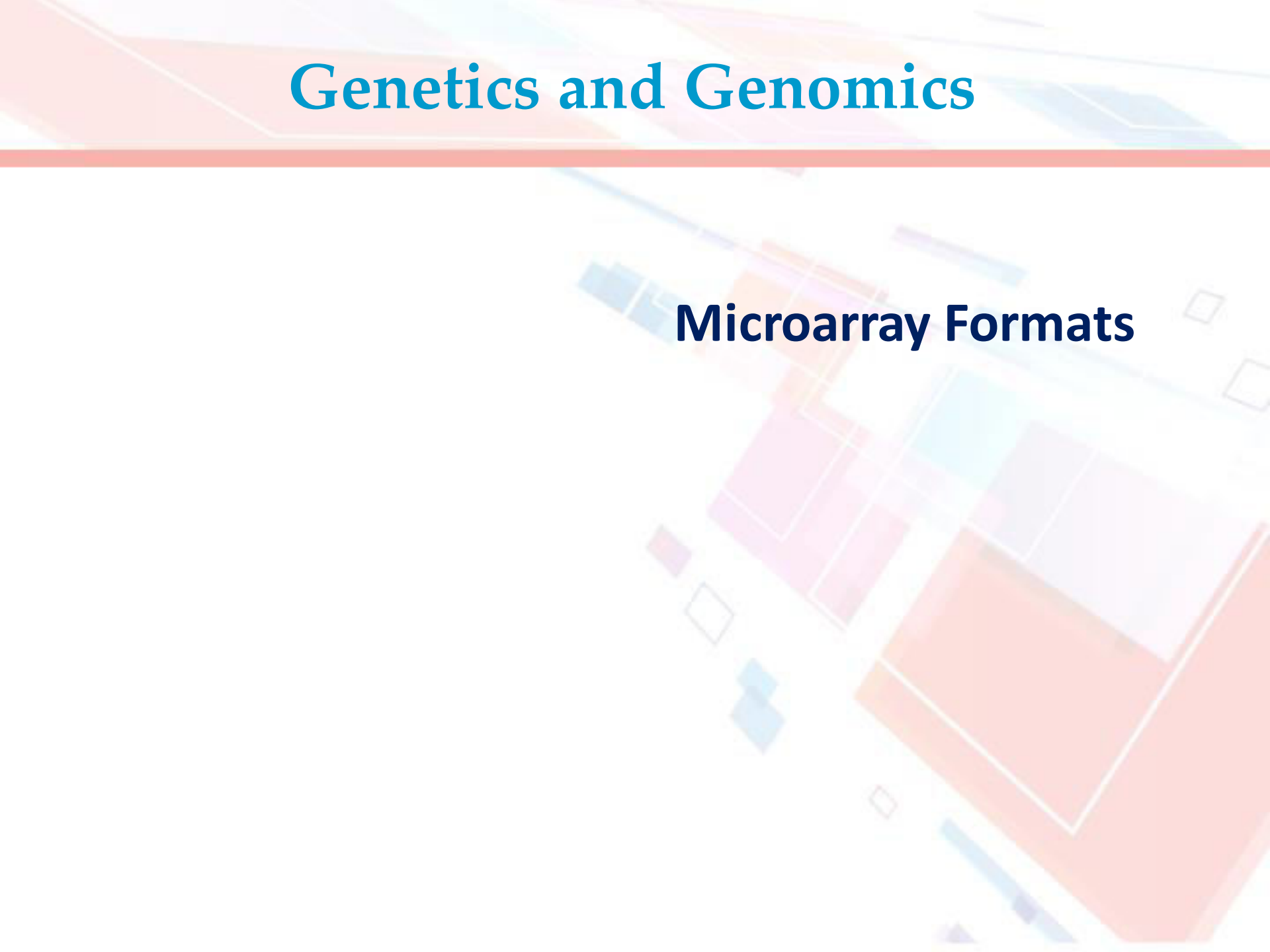
- SNP array:
Identifying single nucleotide polymorphism among alleles within or between populations

Types of Microarrays

Methylation Microarrays

- Methylation arrays determine methylated DNA
- Determining which regions of DNA are methylated ultimately to study epigenetics

Genetics and Genomics



Microarray Formats

Microarrays Formats

Main Formats

- Cartridge-based
Spotted
Electronic
- Spotted Glass Slide
- Tissue Section Slide

Microarrays Formats

Cartridge-Based Chips

- High density arrays of DNA oligos within a plastic housing
- One sample = One chip (Affymetrix, Agilent, Applied Biosystems etc)

Microarrays Formats

Cartridge-Based Chips

- Generally used with expression and DNA arrays

Microarrays Formats

Spotted Glass Arrays

- Uses cDNA, Oligonucleotide, protein, antibody
- Robotically spotted cDNAs or Oligonucleotides
- Printed on Nylon, Plastic, or Glass microscope slide

Microarrays Formats

Spotted cDNA and Oligo Glass Arrays

- Involves two dyes on the same slide

Red dye-Cy5

Green dye-Cy3

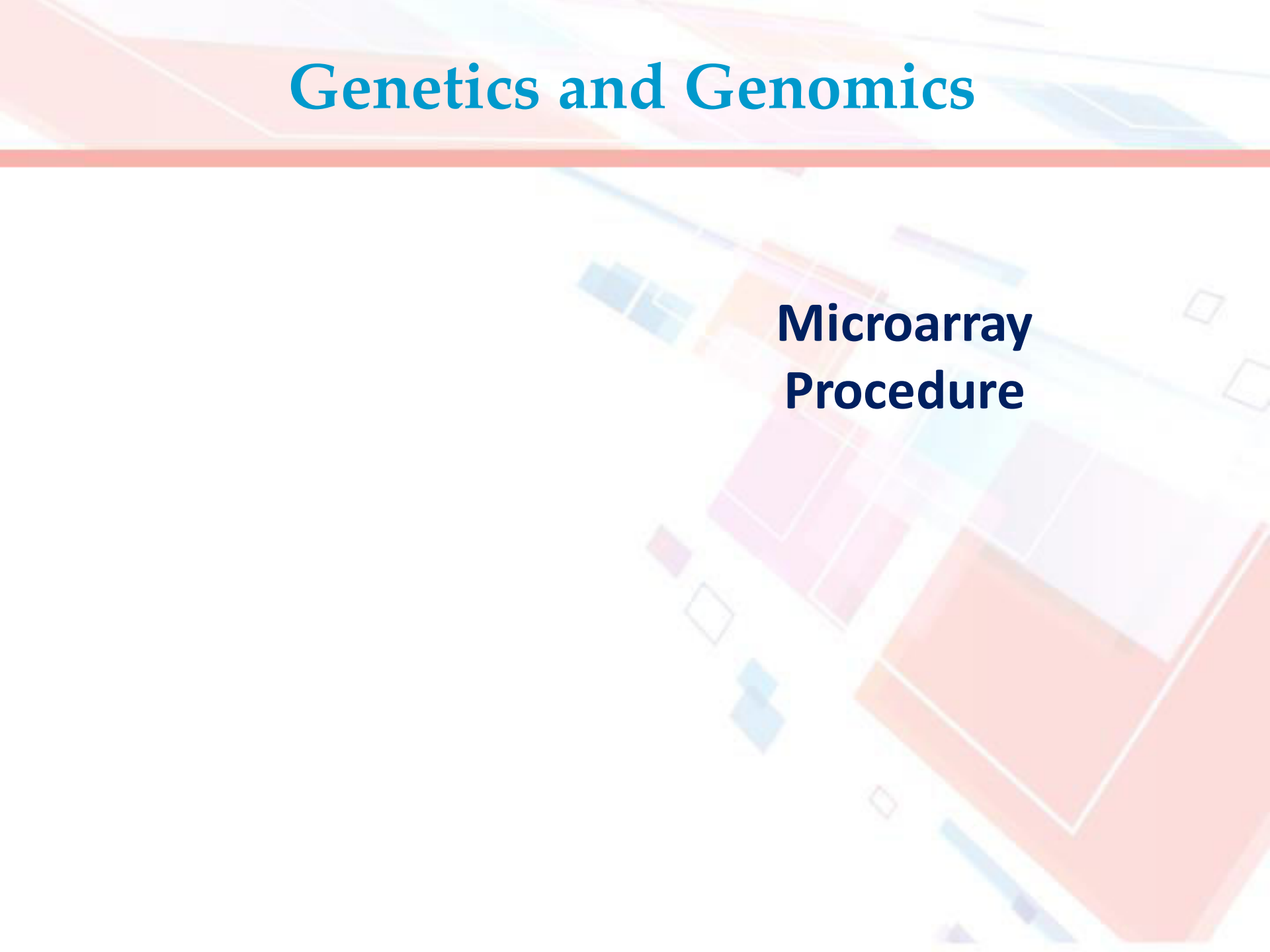
- Control and experimental cDNA

Microarrays Formats

Main Formats

- Cartridge-based
Spotted
Electronic
- Spotted Glass Slide
- Tissue Section Slide

Genetics and Genomics



Microarray Procedure

Microarray - Procedure

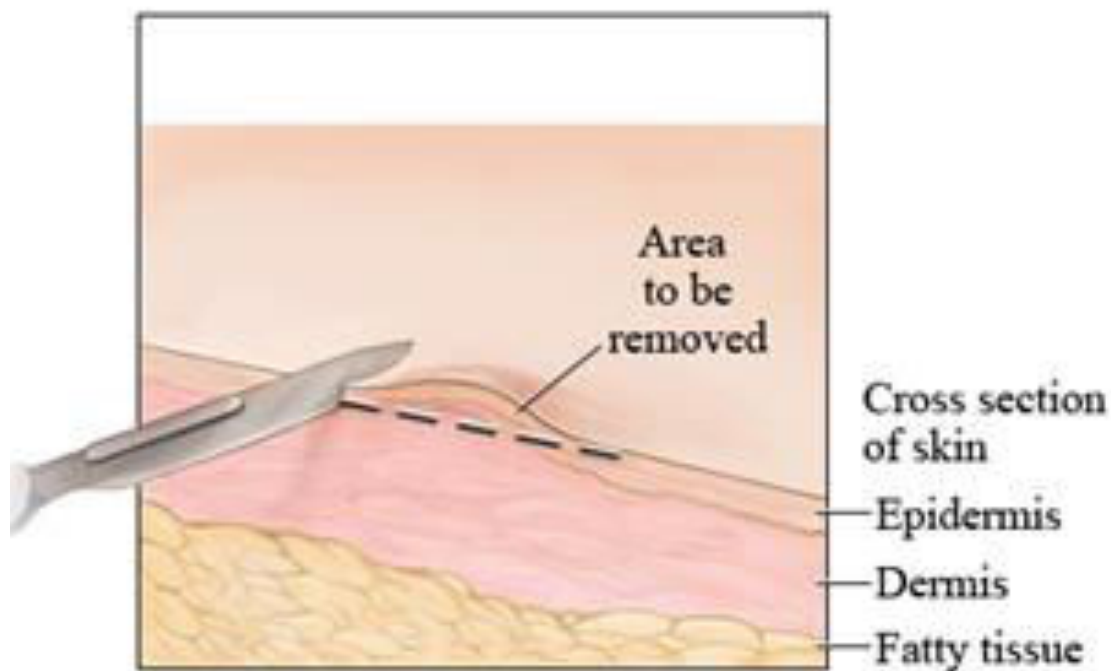
Microarray Procedure

- Collect Samples
- Isolate mRNA
- Create Labelled DNA
- Hybridization
- Microarray Scanner
- Analyze Data

Microarray - Procedure

Microarray Procedure

- Samples: This can be from a variety of organisms. Two samples – cancerous human skin tissue & healthy human skin tissue



Microarray - Procedure

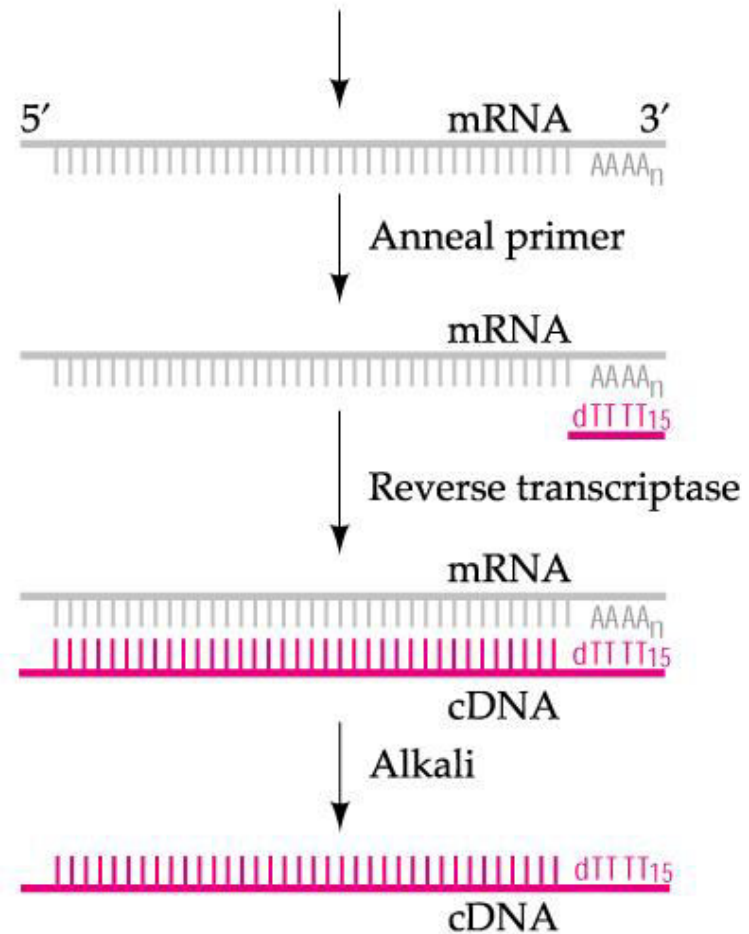
Microarray Procedure

- Extract the RNA from the samples.
- After isolating the RNA, isolate the mRNA from the rRNA and tRNA.
- mRNA has a poly-A tail, use a column containing beads with poly-T tails to bind the mRNA
- Rinse with buffer to release the mRNA from the beads. The buffer disrupts the pH, disrupting the hybrid bonds.

Microarray - Procedure

Microarray Procedure - Create Labeled cDNA

Add a labelling mix to the RNA. The labelling mix contains poly-T (oligo dT) primers, reverse transcriptase (to make cDNA), and fluorescently dyed nucleotides. Add cyanine 3 (fluoresces green) to the healthy cells and cyanine 5 (fluoresces red) to the cancerous cells.



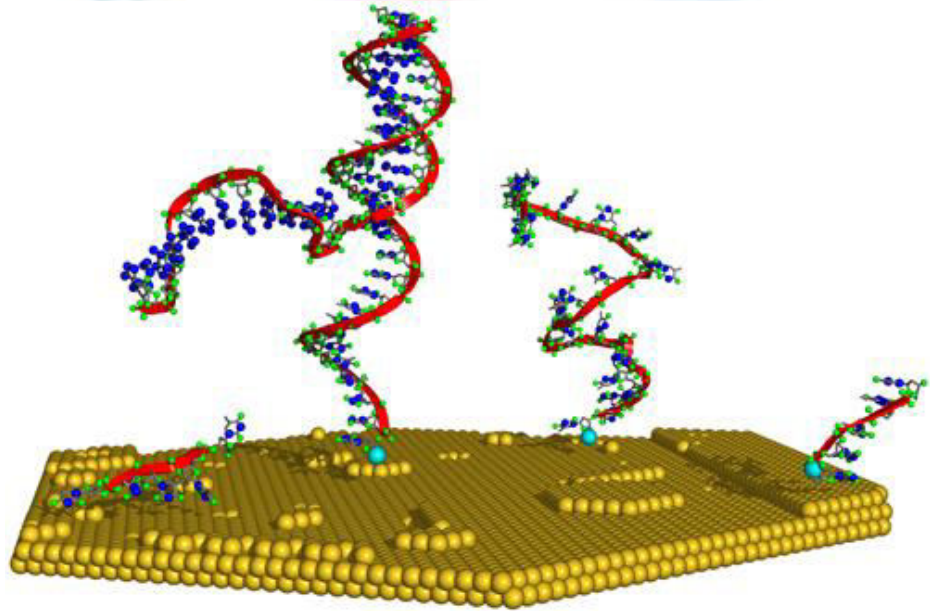
Microarray - Procedure

Microarray Procedure - Hybridization

Apply the cDNA to a microarray plate.

When comparing two samples, apply both samples to the same plate.

The ssDNA will bind to the cDNA already present on the plate.



Microarray - Procedure

Microarray Procedure – Lasers

- Laser scans array and produces images
- One laser for each color, e.g. one for green, one for red

Microarray - Procedure

Microarray Procedure – Microarray Scanner

The scanner has a laser, a computer, and a camera.

The laser causes the hybrid bonds to fluoresce.

The camera records the images produced when the laser scans the plate.

The computer allows to immediately view the results and it also stores the data.

Microarray - Procedure

Microarray Procedure – Microarray Scanner

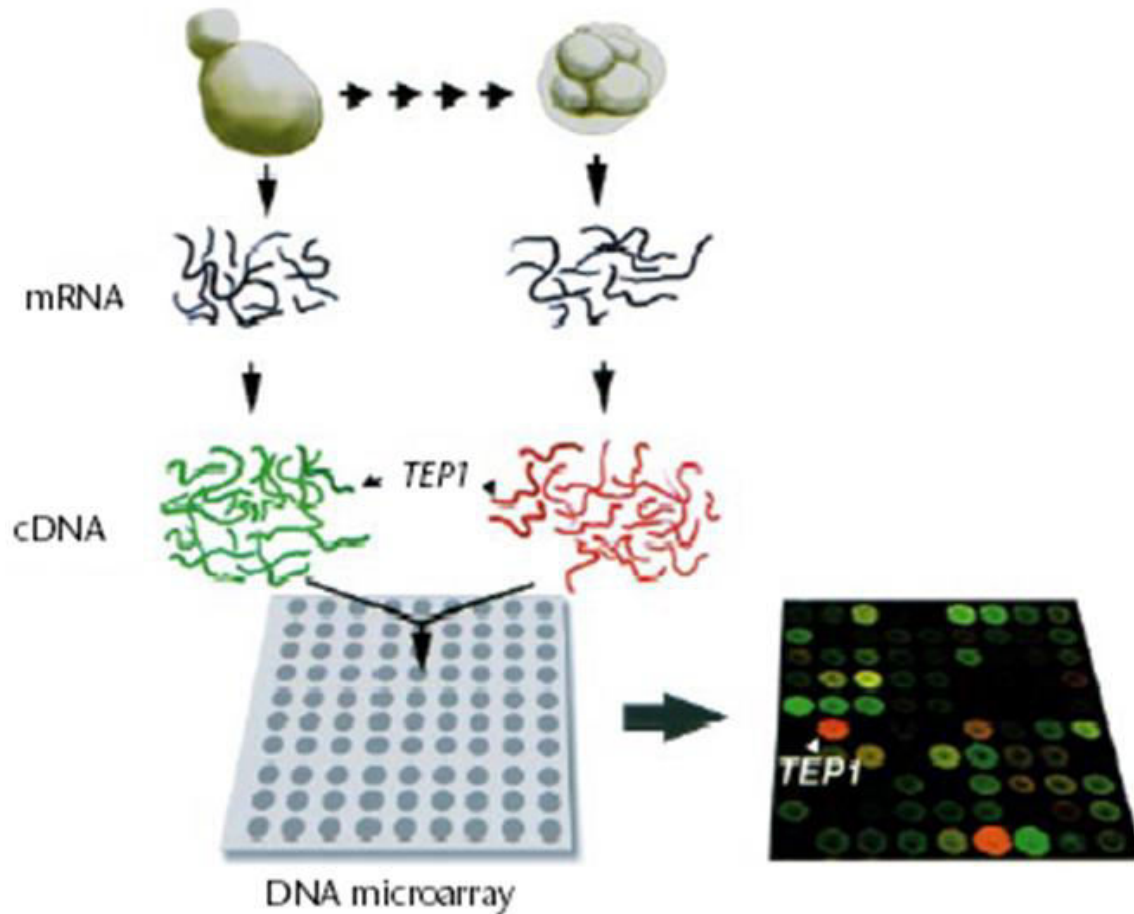
GREEN – the healthy sample hybridized more than the diseased sample.

RED – the diseased/cancerous sample hybridized more than the nondiseased sample.



Microarray - Procedure

Microarray Procedure – Microarray Scanner



Microarray - Procedure

Microarray Procedure – Microarray Scanner

YELLOW - both samples hybridized equally to the target DNA.

BLACK - areas where neither sample hybridized to the target DNA.



Microarray - Procedure

Microarray Procedure – Microarray Scanner

By comparing the differences in gene expression between the two samples, we can understand more about the genomics of a disease.

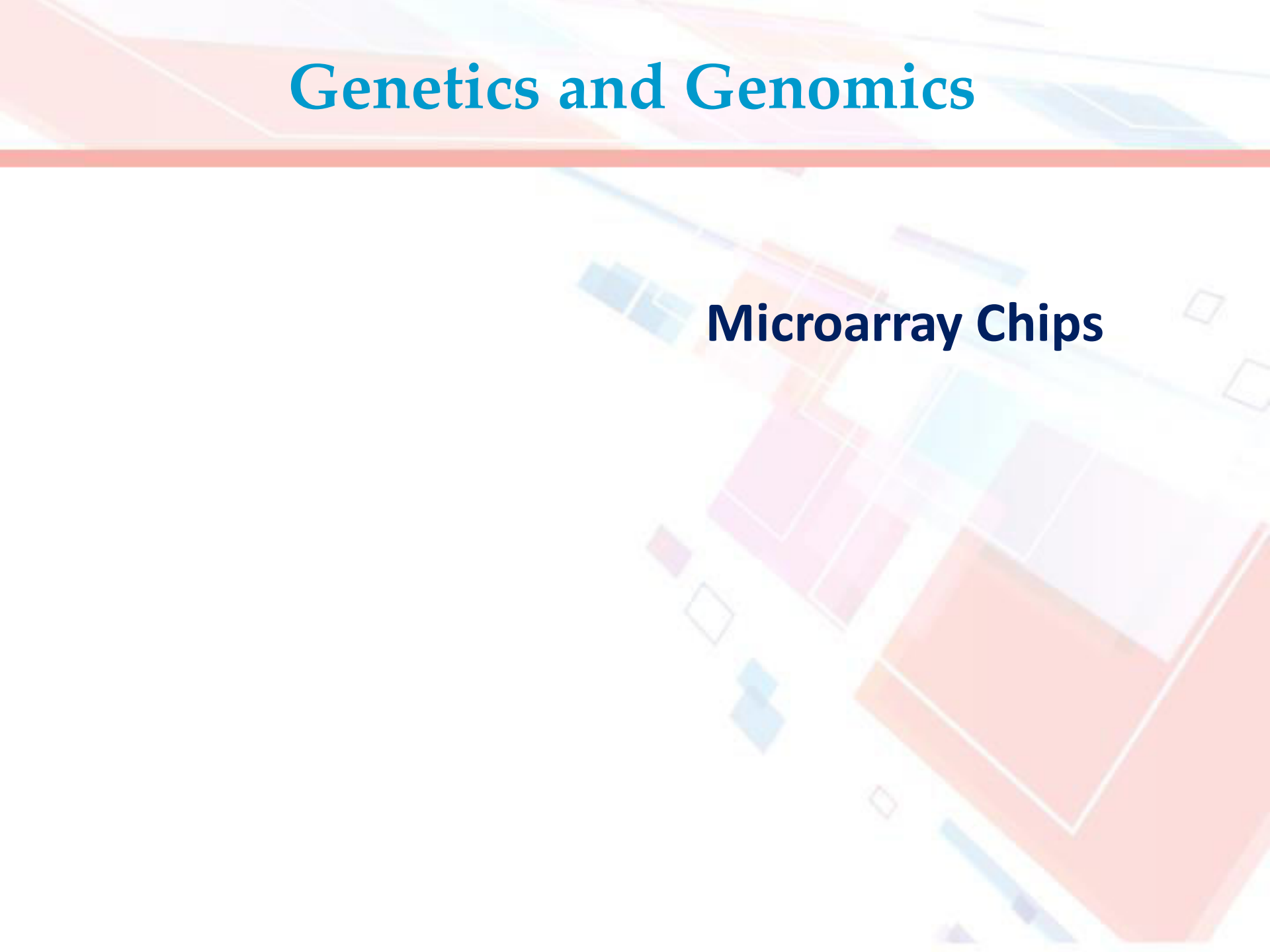


Microarray - Procedure

Microarray Procedure

- Collect Samples
- Isolate mRNA
- Create Labelled DNA
- Hybridization
- Microarray Scanner
- Analyze Data

Genetics and Genomics



Microarray Chips

Microarray Chips

Microarray Chips

- There are two types of DNA Chips/Microarrays:
- cDNA based microarray
- Oligonucleotide based microarray

Microarray Chips

cDNA Based Microarray Chips

- Chips are prepared by using cDNA- cDNA chips or cDNA microarray.
- The cDNAs are amplified. Then these are immobilized on a solid support made up of nylon filter or glass slide

Microarray Chips

cDNA Based Microarray Chips

- The probe DNA are loaded by capillary action.
- Small volume of this DNA is spotted on solid surface
- DNA is delivered mechanically or in a robotic manner.

Microarray Chips

cDNA Based Microarray Chips

- When one DNA spotting is done, the pin is washed and loaded with fresh DNA to start the second cycle

Microarray Chips

DNA Based Microarray Chips

- Almost similar type procedure
- Gene expression profiling
- Discovery of drugs
- Diagnostics and genetic engineering
- Alternative splicing detection

Microarray Chips

DNA Based Microarray Chips

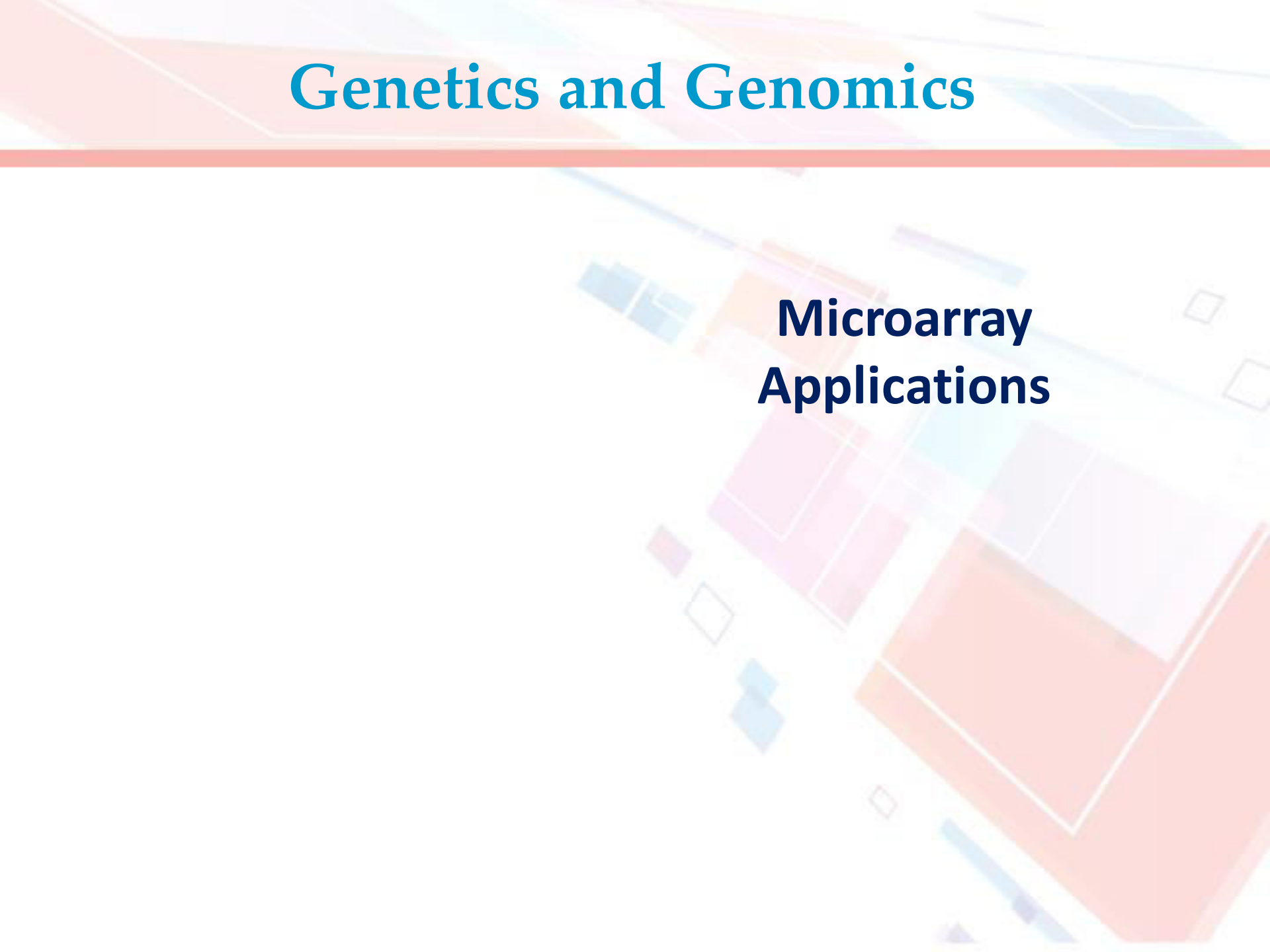
- Functional genomics
- DNA sequencing
- Toxicological research
(Toxicogenomics)

Microarray Chips

Microarray Chips

- There are 2 types of DNA Chips/Microarrays:
- cDNA based microarray
- Oligonucleotide based microarray

Genetics and Genomics



Microarray Applications

Microarray Applications

Microarray - Advantages

- Provides data for thousands of genes
- One experiment instead of many
- Fast and easy to obtain results
- Closer to discovering cures for diseases and cancer

Microarray Applications

Microarray Advantages

- Different parts of DNA can be used to study gene expression

Microarray Applications

Disadvantages

- The biggest disadvantage of DNA chips is that they are expensive to create.
- The production of too many results at a time requires long time for analysis, which is quite complex in nature.

Microarray Applications

Disadvantages

- The DNA chips do not have very long shelf life, which proves to be another major disadvantage of the technology.

Microarray Applications

Microarray - Limitations

- Cross-hybridization of sequences with high identity
- Chip to chip variation
- The real limitation is Bioinformatics
- Expensive – repeat experiments

Microarray Applications

Gene Discovery and Disease Diagnostics

- Gene discovery and disease diagnosis
- Classify the types of cancer on the basis of the patterns of gene activity in the tumor cells

Microarray Applications

Pharmacogenomics

- Study of correlations between therapeutic responses to drugs and the genetic profiles of the patients

Microarray Applications

Toxicogenomics

- Microarray technology allows us to research the impact of toxins on cells.
- Some toxins can change the genetic profiles of cells.

Microarray Applications

Biological Applications

- Many biological discovery
- New and better molecular diagnostics
- New molecular targets for therapy
- Finding and refining biological pathways

Genetics and Genomics

Types of RNA and RNomics

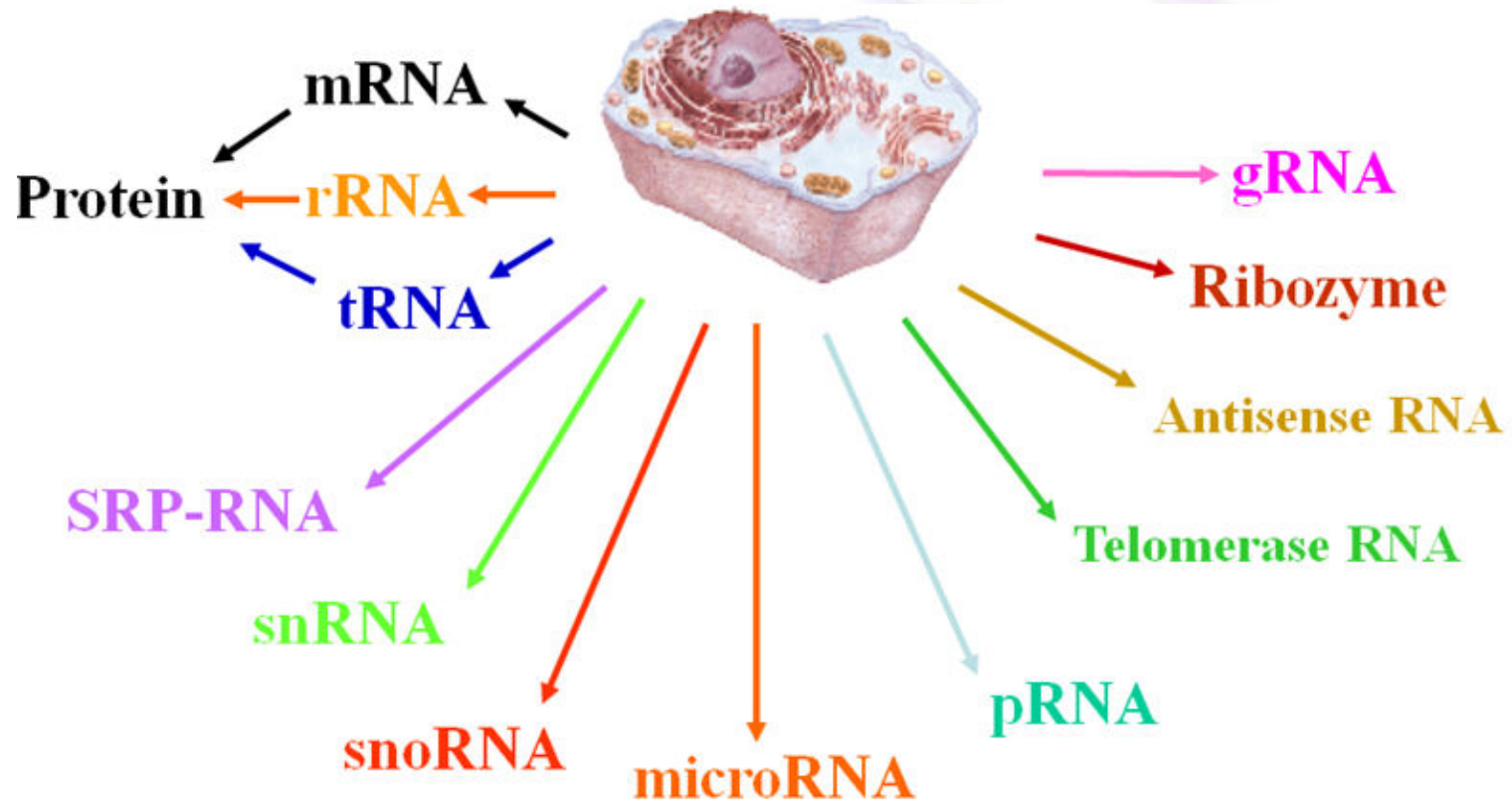
Types of RNA and RNomics

Types of RNA

- Wide range of RNAs
- Some of them are coding while others are noncoding

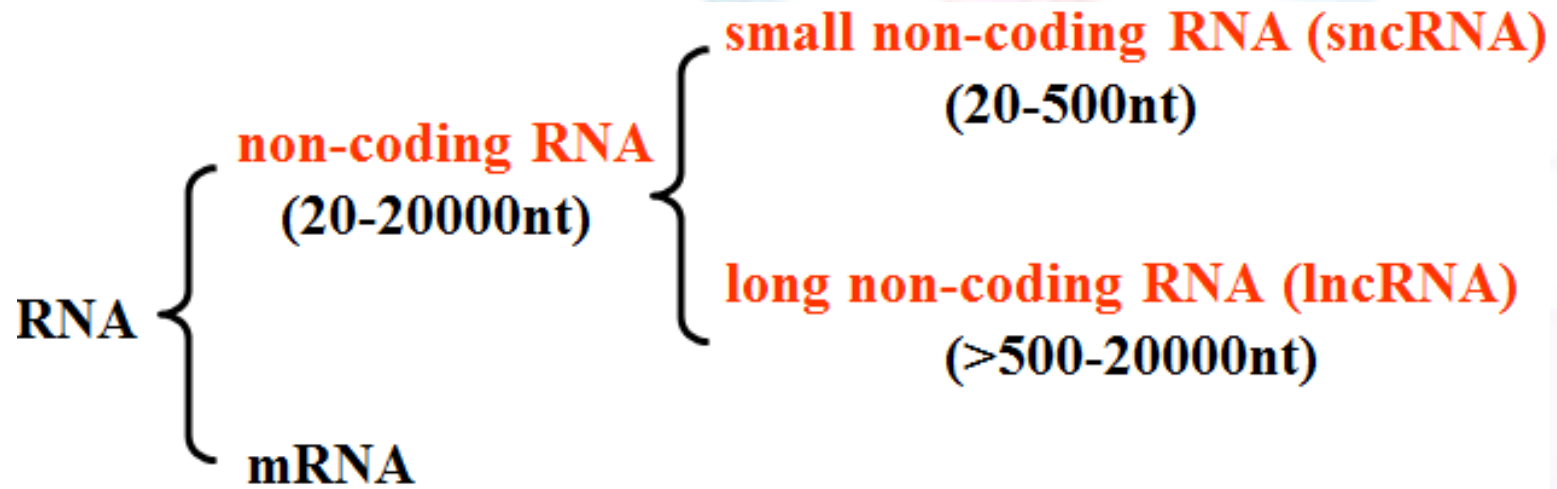
Types of RNA and RNomics

Types of RNA



Types of RNA and RNomics

Types of RNA - Classification



Types of RNA and RNomics

Types of RNA – Housekeeping RNAs

Category	Name
Housekeeping RNAs	Transfer RNAs
	Ribosomal RNAs
	Small nucleolar RNAs
	Small nuclear RNAs

Types of RNA and RNomics

Types of RNA – Small Non-coding RNAs

Category	Name
Small ncRNAs (200 bp or less in size)	MicroRNAs
	Tiny transcription initiation RNAs
	Repeat associated small interfering RNAs
	Promoter-associated short RNAs
	Termini-associated short RNAs
	Antisense termini associated short RNAs
	Transcription start site antisense RNAs
	Retrotransposon-derived RNAs
	3'UTR-derived RNAs
	Splice-site RNAs

Types of RNA and RNomics

Types of RNA – Long Non-coding RNAs

Category	Name
Long ncRNA (over 200 bp in size)	Long or large intergenic ncRNAs
	Transcribed ultraconserved regions
	Pseudogenes
	Enhancer RNAs
	Repeat-associated ncRNAs
	Long intronic ncRNAs
	Antisense RNAs
	Promoter-associated long RNAs
	Long stress-induced non-coding transcripts

Types of RNA and RNomics

RNomics is the study

- Identification
- Expression
- Biogenesis
- Structure
- Regulation of expression
- Targets
- Biological functions of RNAs

Types of RNA and RNomics

Computational RNomics

- Searching conserved intronic sequences by comparative analysis of introns
- Searching conserved intergenic sequences

Types of RNA and RNomics

Experimental RNomics

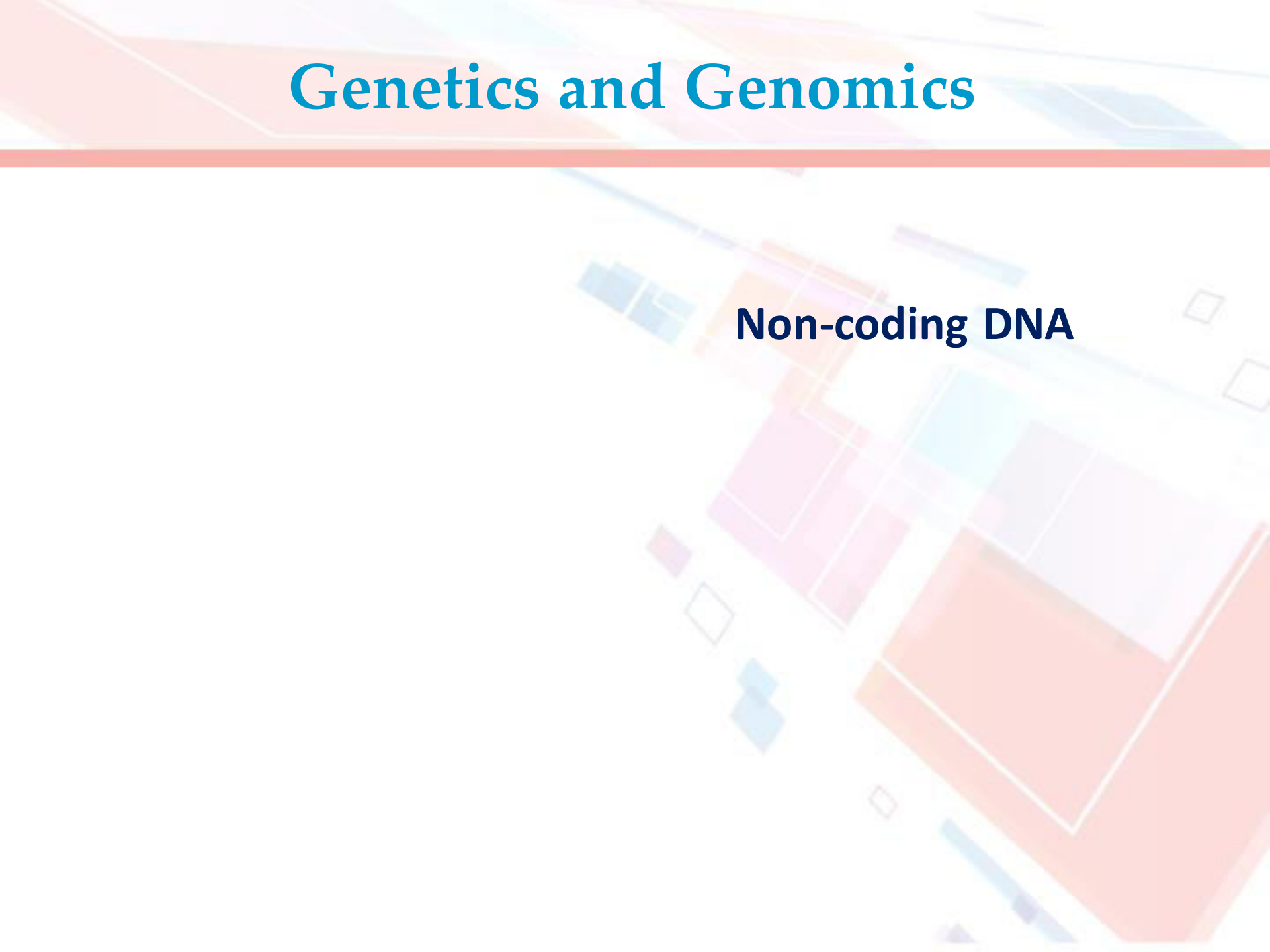
- PAGE separation of non-coding RNAs and sequencing
- Non-coding RNA enriched cDNA libraries and sequencing

Types of RNA and RNomics

Types of RNA

- Wide range of RNAs
- Some of them are coding RNA while others are noncoding.

Genetics and Genomics



Non-coding DNA

Non-coding DNA

Types of Non-coding DNA

- Non- coding functional RNA
- Cis- and Trans-regulatory elements
- Introns
- Pseudogenes

Non-coding DNA

Types of Non-coding DNA

- Repeat sequences, transposons and viral elements
- Telomeres

Non-coding DNA

Non-coding Functional RNA

- The RNA molecules which are not translated into proteins.
- Examples: Ribosomal RNA, transfer RNA & micro RNA

Non-coding DNA

Cis- and Trans-Regulatory Elements

- Those sequences that control the transcription of a nearby gene.
- Located within 5' or 3' untranslated regions or within introns.

Non-coding DNA

Introns

- They are non-coding sections of a gene.
- Transcribed in the precursor mRNA sequence but is ultimately removed by RNA splicing.
- Most of the introns appear to be mobile genetic elements.

Non-coding DNA

Pseudogenes

- They are related to known genes, that have lost their protein-coding ability or are otherwise no longer expressed in the cell.
- Arise from retrotransposition or genomic duplication of functional genes.

Non-coding DNA

Repeat Sequences, Transposons & Viral Elements

- Transposons and Retrotransposons are mobile genetic element
- Retrotransposons: LINEs, SINEs

Non-coding DNA

Telomeres

- Telomeres are regions of repetitive DNA
- Located at the end of a chromosome
- They provide protection from chromosomal deterioration during DNA replication

Non-coding DNA

Functions

- Essential for chromosome structure
- Genome Protection
- Enhancers, silencers, promoters, insulator
- Genetic switches
- Regulation of gene expression

Genetics and Genomics

Non-coding RNA

Non-coding RNA

Non-coding RNA

- Non-coding RNA is a RNA molecule that functions without being translated into a protein
- Large number of genes of noncoding RNA
- Noncoding RNA - different functions

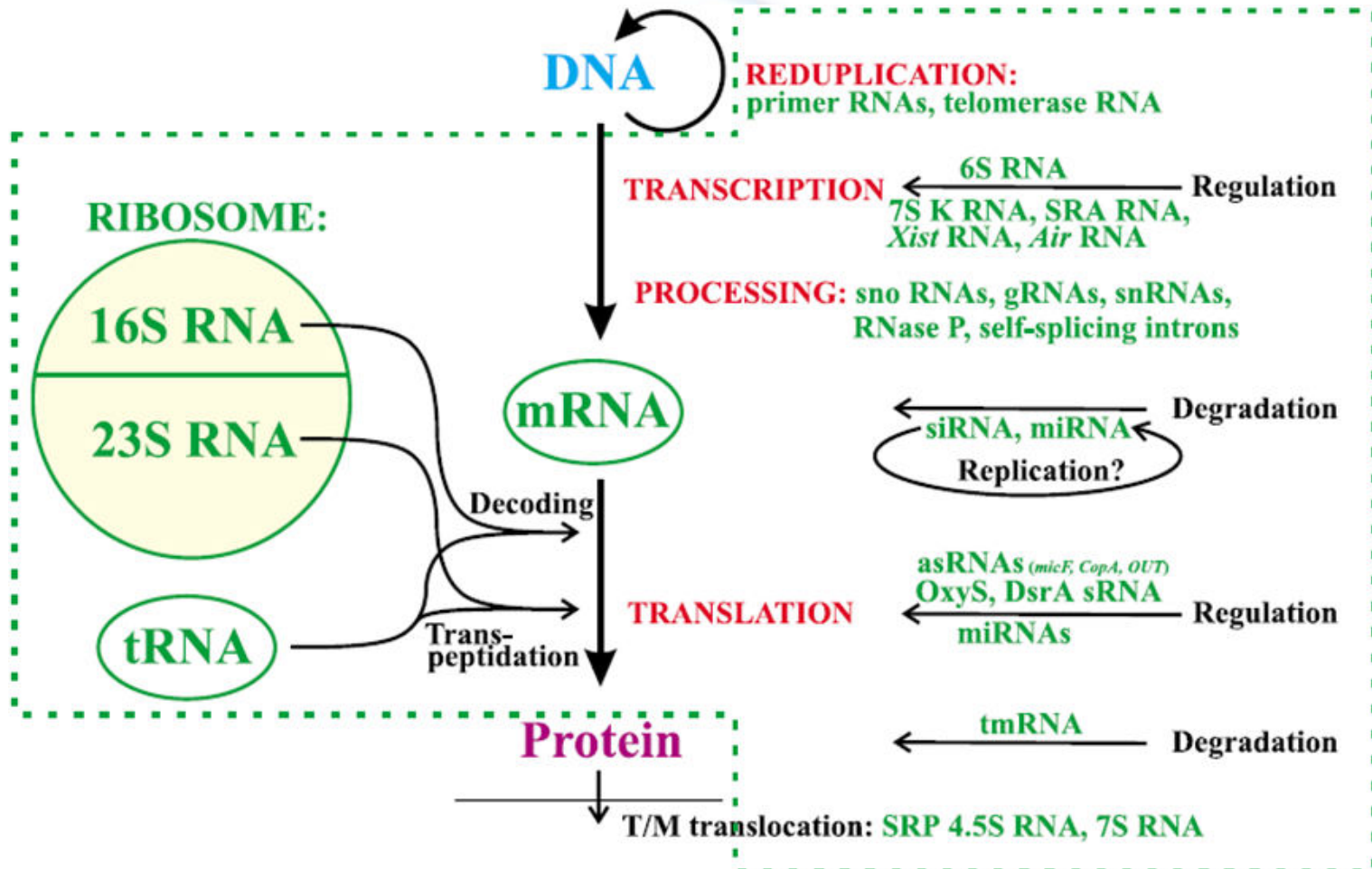
Non-coding RNA

Types of Non-coding RNA

- Ribosomal RNA
- Transfer RNA
- Small nuclear RNA
- Small nucleolar RNA
- Short interfering
- Micro RNA
- Long non-coding RNA

Non-coding RNA

Functional Diversity of Non-coding RNA



Non-coding RNA

Non-coding RNAs in Human Genome

tRNA	600	SRP RNA	1
18S rRNA	200	RNase P RNA	1
5.8S rRNA	200	Telomerase RNA	1
28S rRNA	200	RNase MRP	1
5S rRNA	200	Y RNA	5
snoRNA	300	Vault	4
miRNA	250	7SK RNA	1
U1	40	Xist	1
U2	30	H19	1
U4	30	BIC	1
U5	30		
U6	20		
U4atac	5	Antisense RNAs	1000s?
U6atac	5	Cis reg regions	100s?
U11	5	Others	?
U12	5		

Non-coding RNA

Non-coding RNA

- Non-coding RNA (ncRNA) is a RNA molecule that functions without being translated into a protein

Genetics and Genomics

MicroRNA (miRNA)

MicroRNA (miRNA)

MicroRNA (miRNA)

- Small non-coding double stranded RNAs
- Approximately 19-22 nucleotides long
- Repress activity of complementary mRNAs

MicroRNA (miRNA)

MicroRNA (miRNA)

- Regulate 30% of mammalian genes
- Described in invertebrates and vertebrates: worms, fungi, plants, and mammals
- Many are conserved between vertebrates and invertebrates

MicroRNA (miRNA)

MicroRNA (miRNA)

- miRNA originates from ssRNA that forms a hairpin secondary structure
- miRNA regulates post-transcriptional gene expression and is often not 100% complementary to the target

MicroRNA (miRNA)

MicroRNA (miRNA)

- Originate from capped & polyadenylated full length precursors (pri-miRNA)
- Hairpin precursor ~70 nt (pre-miRNA)
- Mature miRNA ~22 nt (miRNA)

MicroRNA (miRNA)

MicroRNA (miRNA)

- Drosha and Pasha are part of the “Microprocessor” protein complex (~600 - 650kDa)
- Drosha and Dicer are RNase III enzymes

MicroRNA (miRNA)

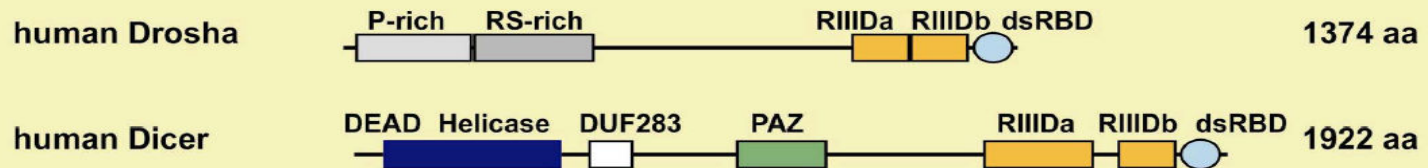
MicroRNA (miRNA)

- Pasha is a dsRNA binding protein
- Exportin 5 is a member of the karyopherin nucleocytoplasmic transport factors
- Argonautes are RNase H enzymes

MicroRNA (miRNA)

MicroRNA (miRNA)

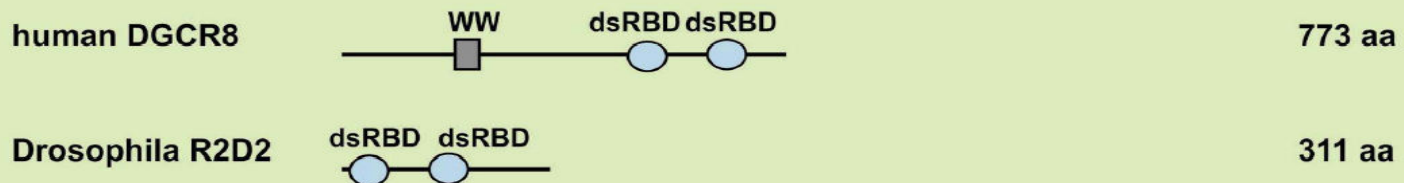
A. RNase III type proteins



B. Argonaute proteins



C. dsRNA-binding proteins



D. DEAD-box helicases



MicroRNA (miRNA)

MicroRNA (miRNA)

- ~1900 discovered in Homo sapiens
- Mouse and human have highly conserved miRNA genes

MicroRNA (miRNA)

MicroRNA (miRNA)

Conclusion

- Small non-coding double stranded RNAs
- Approximately 19-22 nucleotides long
- 30% of mammalian mRNAs are regulated by miRNAs

Genetics and Genomics

Biogenesis of MicroRNA (miRNA)

Biogenesis of MicroRNA (miRNA)

MicroRNA (miRNA)

- Small non-coding double stranded RNAs
- Approximately 19-22 nucleotides long
- miRNA originates with ssRNA that forms a hairpin secondary structure

Biogenesis of MicroRNA (miRNA)

MicroRNA (miRNA)

- Originate from capped & polyadenylated full length precursors (pri-miRNA)

Biogenesis of MicroRNA (miRNA)

Biogenesis of miRNA

- Primary-miRNA is transcribed in the nucleus, and is usually several kilobases long; having 5' cap and a poly-A tail.
- Cleaved in the nucleus by Drosha enzyme to 70nt hairpin transcript (pre-miRNA).
- Transported to the cytoplasm by Exportin 5 through nuclear pores.
- Cleaved by Dicer enzyme (RNase III enzyme) into 19-22nt ds-transcripts.

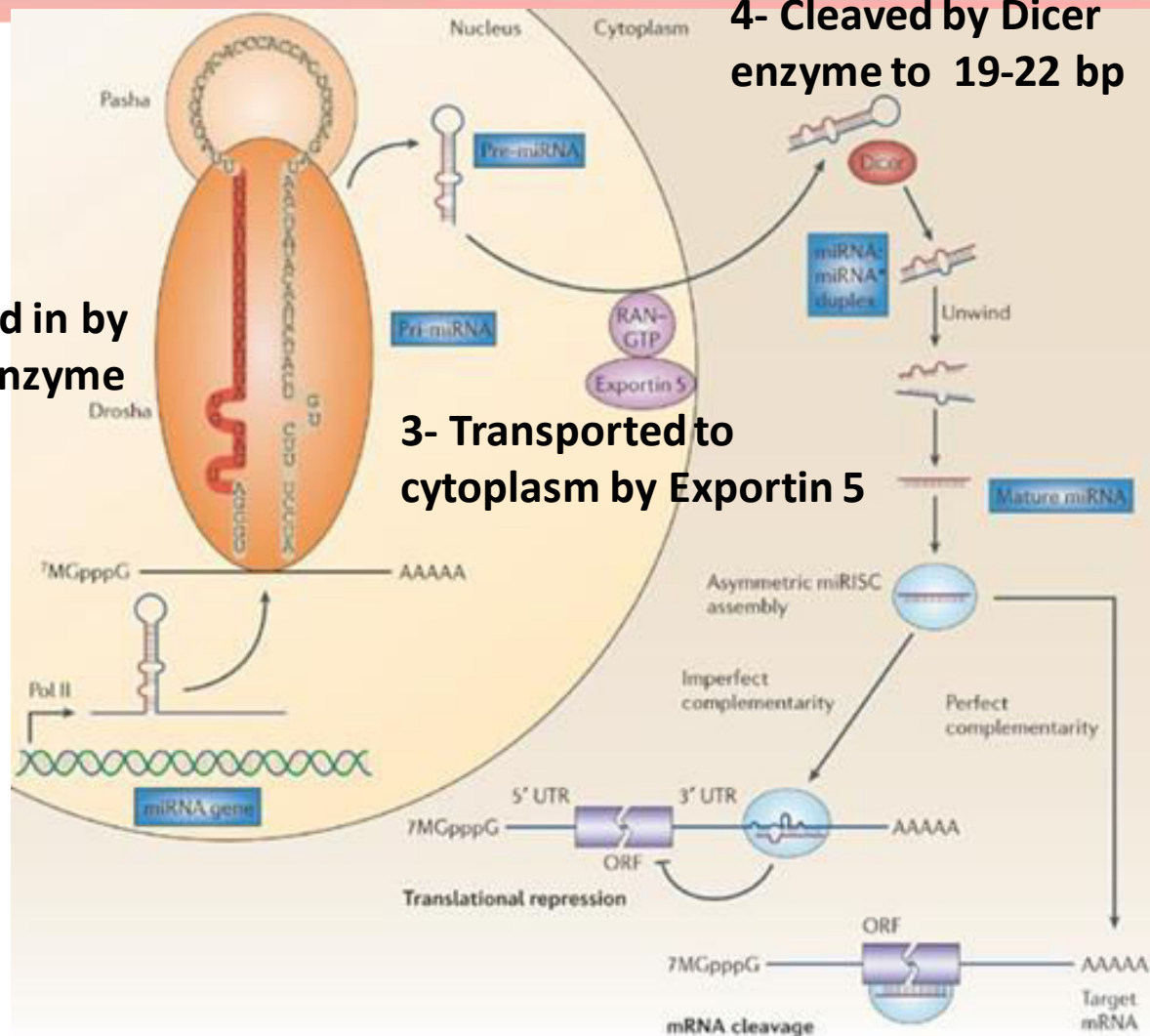
Biogenesis of MicroRNA (miRNA)

1- Transcribed in nucleus

2- Cleaved in by Drosha enzyme

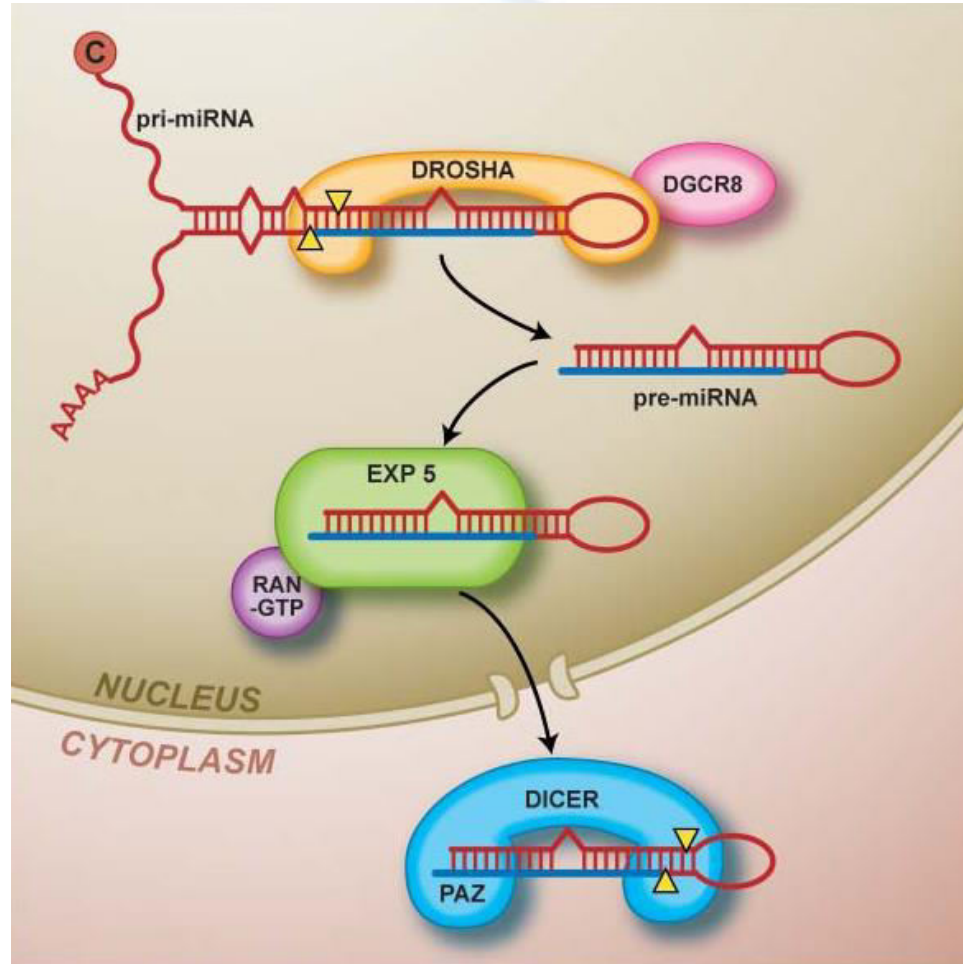
3- Transported to cytoplasm by Exportin 5

4- Cleaved by Dicer enzyme to 19-22 bp



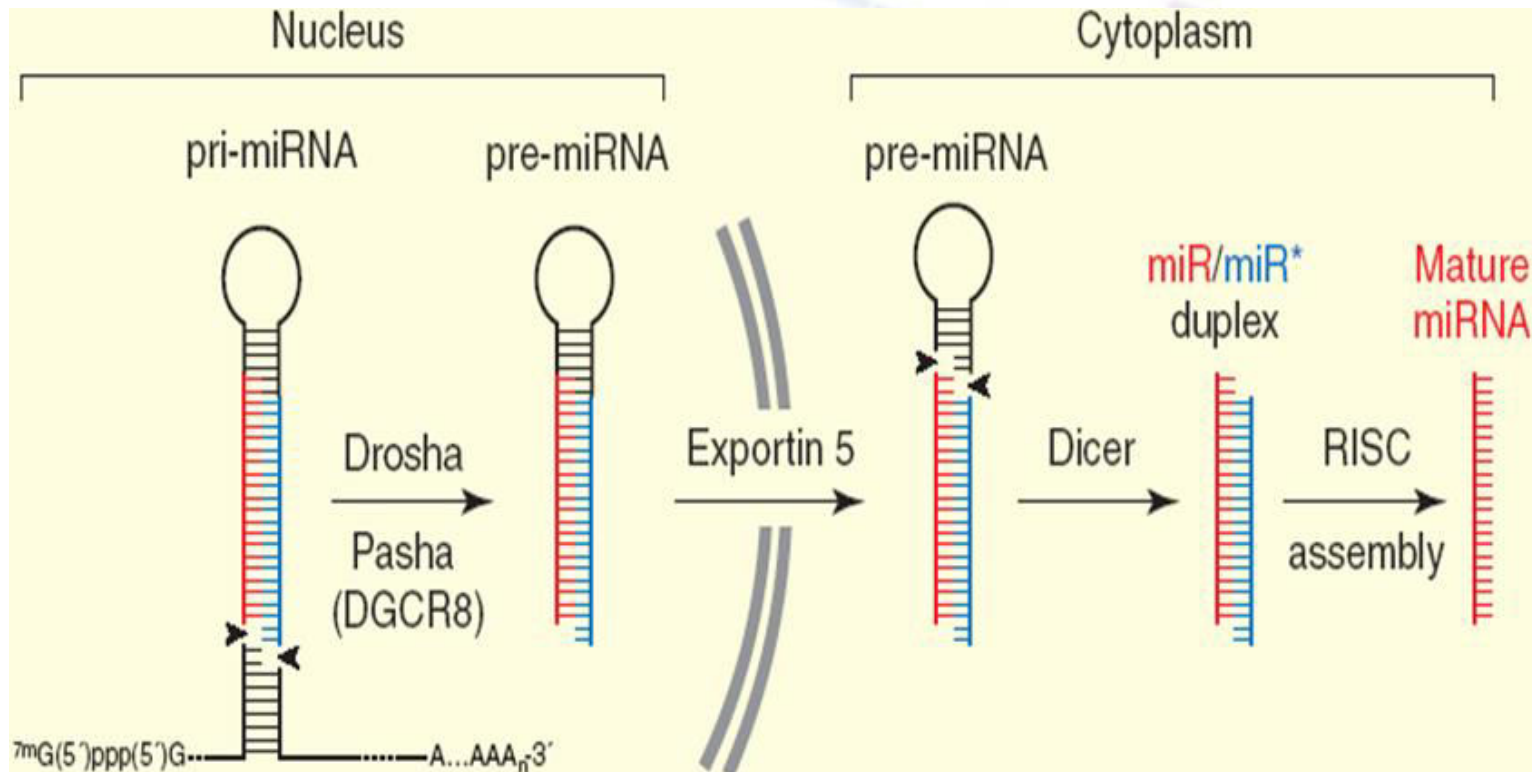
Biogenesis of MicroRNA (miRNA)

Illustration of miRNA Processing



Biogenesis of MicroRNA (miRNA)

Illustration of miRNA Processing - Another View



Biogenesis of MicroRNA (miRNA)

MicroRNA (miRNA)

- Small non-coding double stranded RNAs
- Approximately 19-22 nucleotides long

Genetics and Genomics

Functions of miRNA

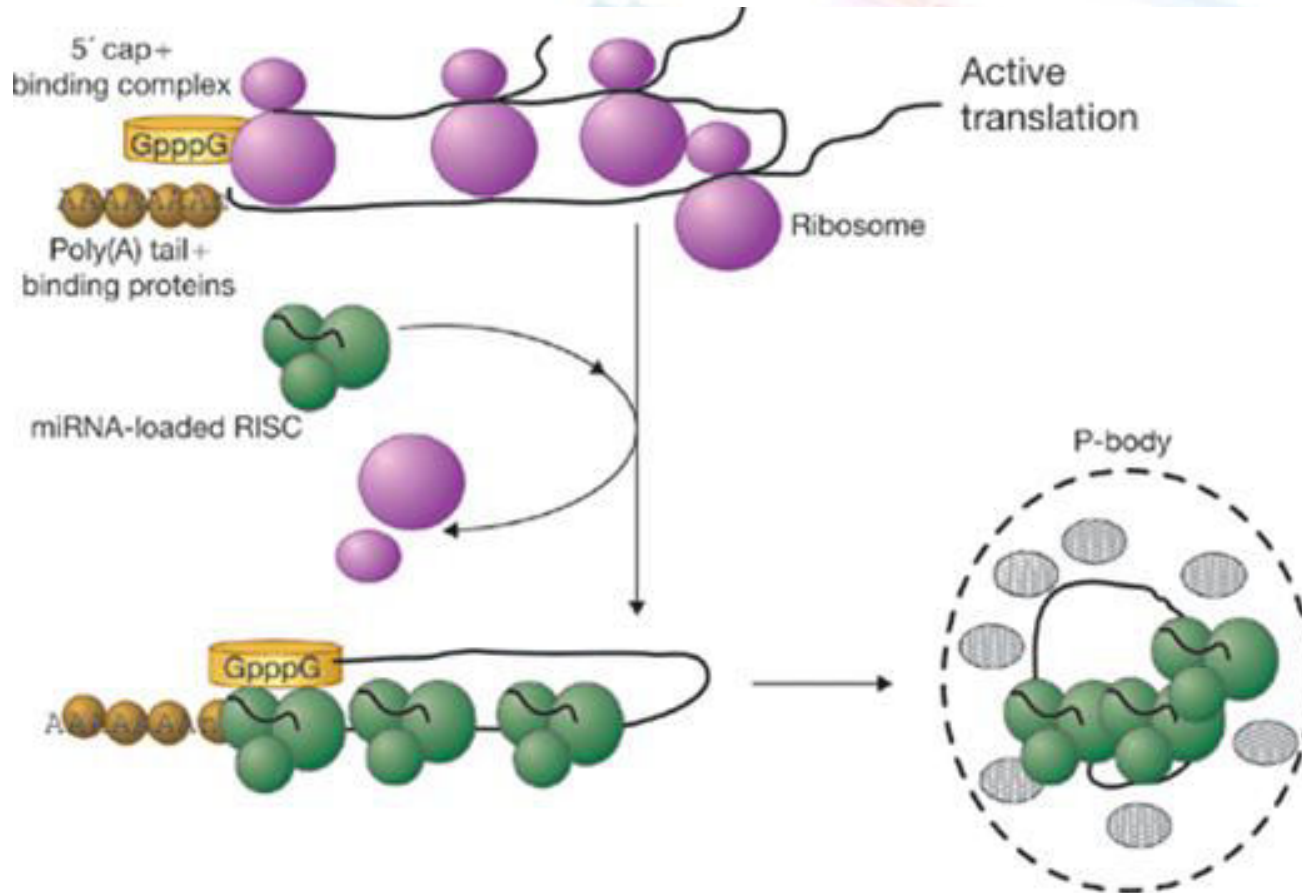
Functions of miRNA

Functions of miRNA

- Involved in the post-transcriptional regulation of gene expression
- Important in development
- Metabolic regulation (miR-375 & insulin secretion)

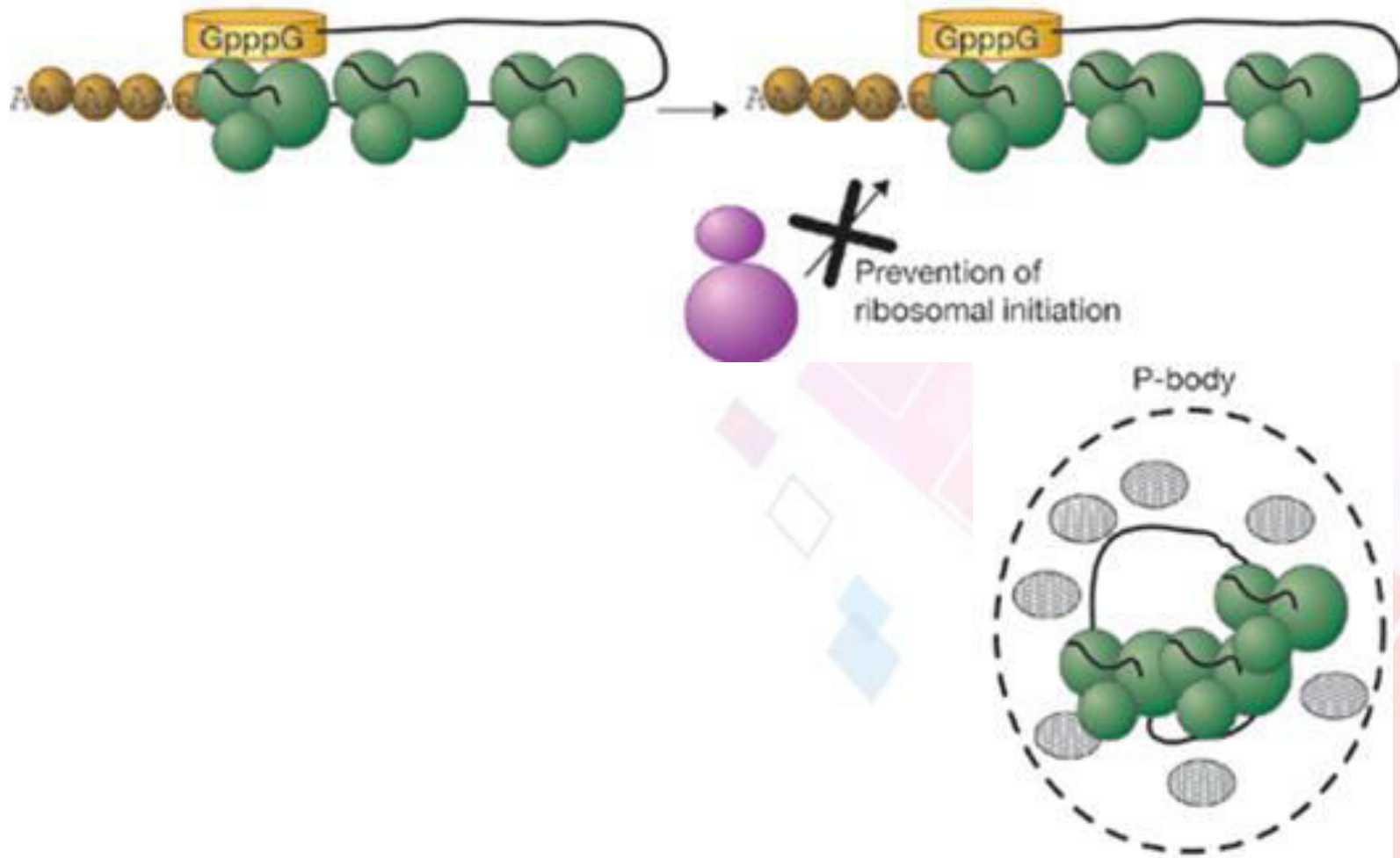
Functions of miRNA

Functions of miRNA: Processing bodies are sites of storage and/or degradation of mRNA



Functions of miRNA

Functions of miRNA



Functions of miRNA

Functions of miRNA

miRNA	Target	Function
miR-15/miR-16	Bcl2	Apoptosis
miR-1	GJA1/KCNJ2	Cardiac Arrhythmia
miR-146	IRAK1/TRAF6	Bacterial Infectious Response; TLR-NF κ B
miR-520h	ABCG2	Stem Cell Differentiation
miR-106a	Rb1	Cancer Pathogenesis
miR-let7	Multiple	Cell Cycle Regulation
miR-155	-	Adaptive Immunity
miR-223	-	Granulocyte Regulation
miR-208	-	Stress Response (Heart)

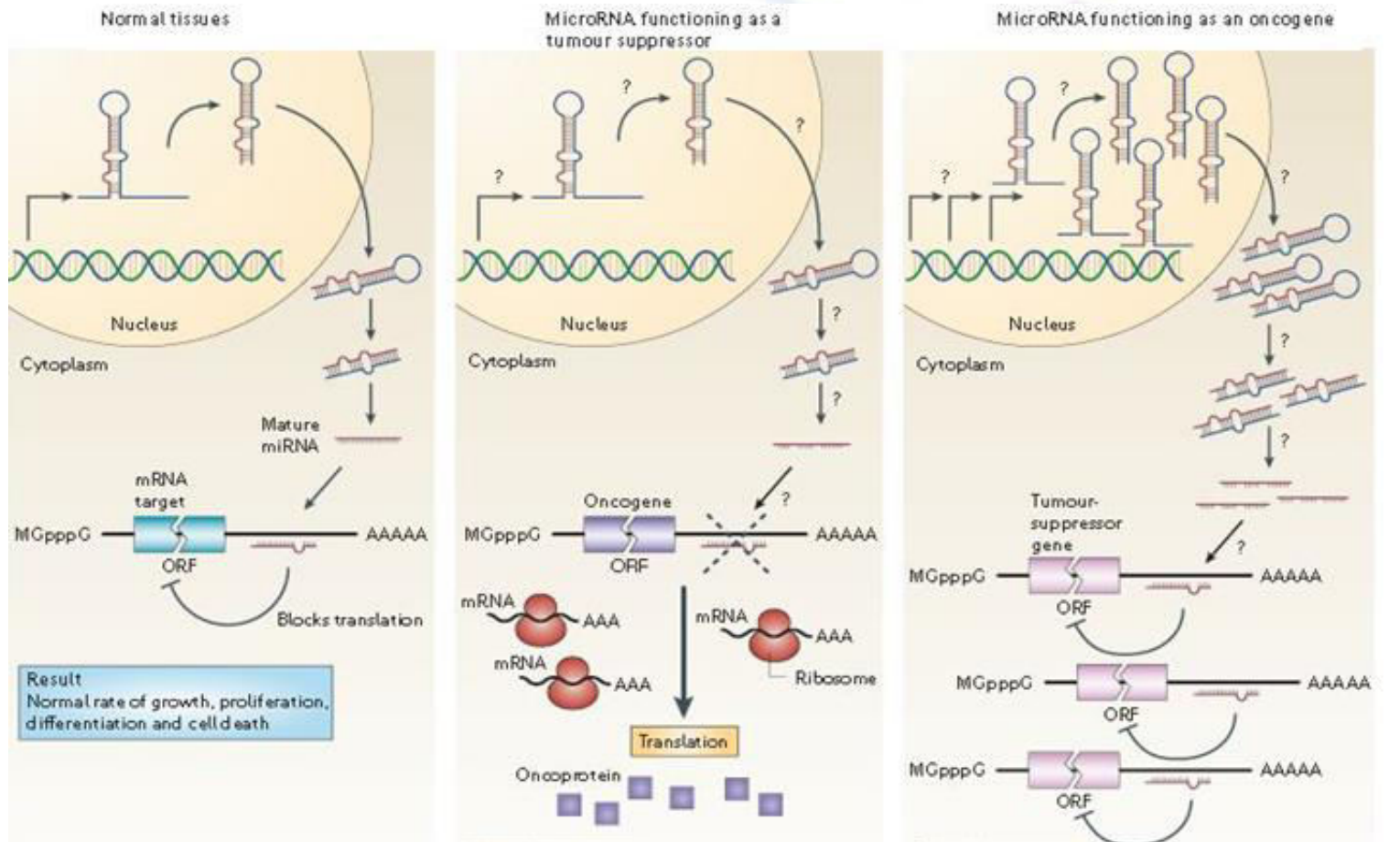
Functions of miRNA

Tumor Suppressors and Oncogenes

- miRNAs can function as tumour suppressors and oncogenes.
- Gene therapies that use miRNAs might be an effective approach to blocking tumour progression.
- miRNAs such as let-7, which has been shown to negatively regulate the Ras oncogenes, and miR-15 and miR-16, which negatively regulate BCL2, are promising candidates for cancer treatment.

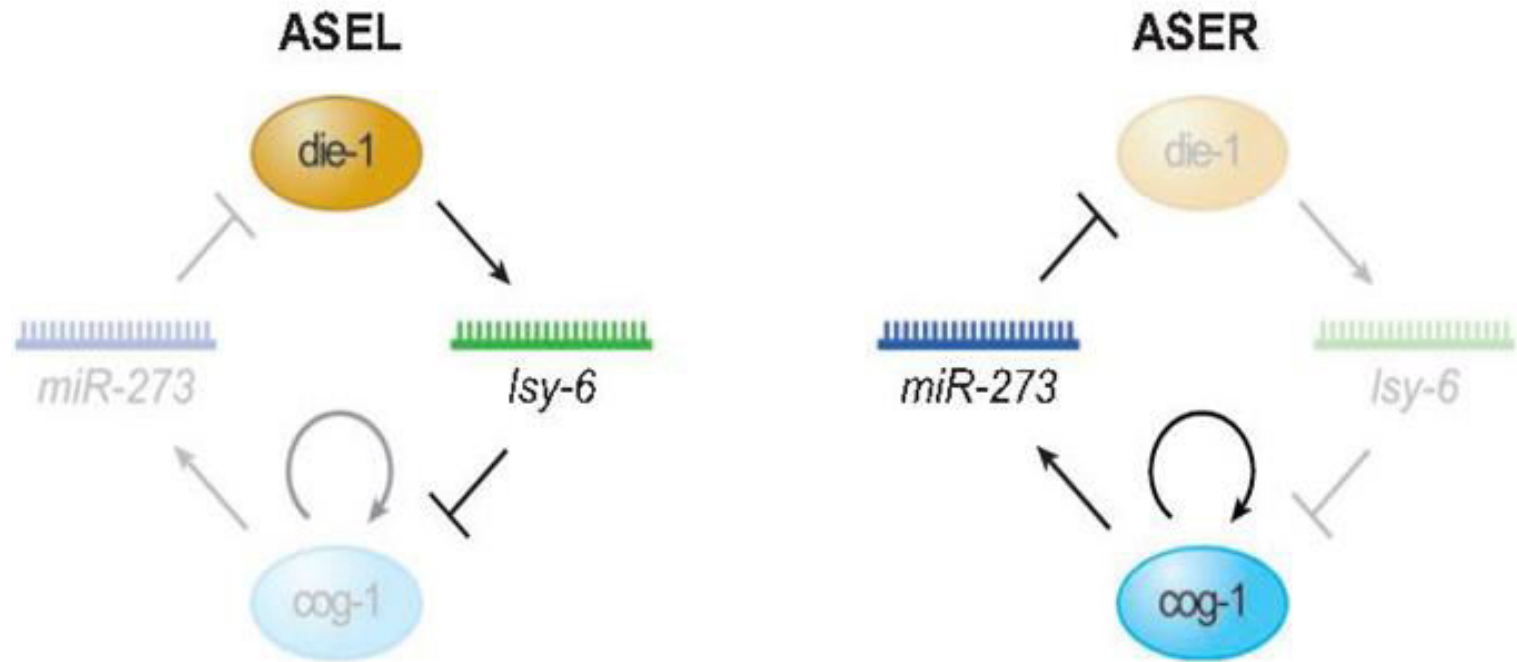
Functions of miRNA

Tumor Suppressors and Oncogenes



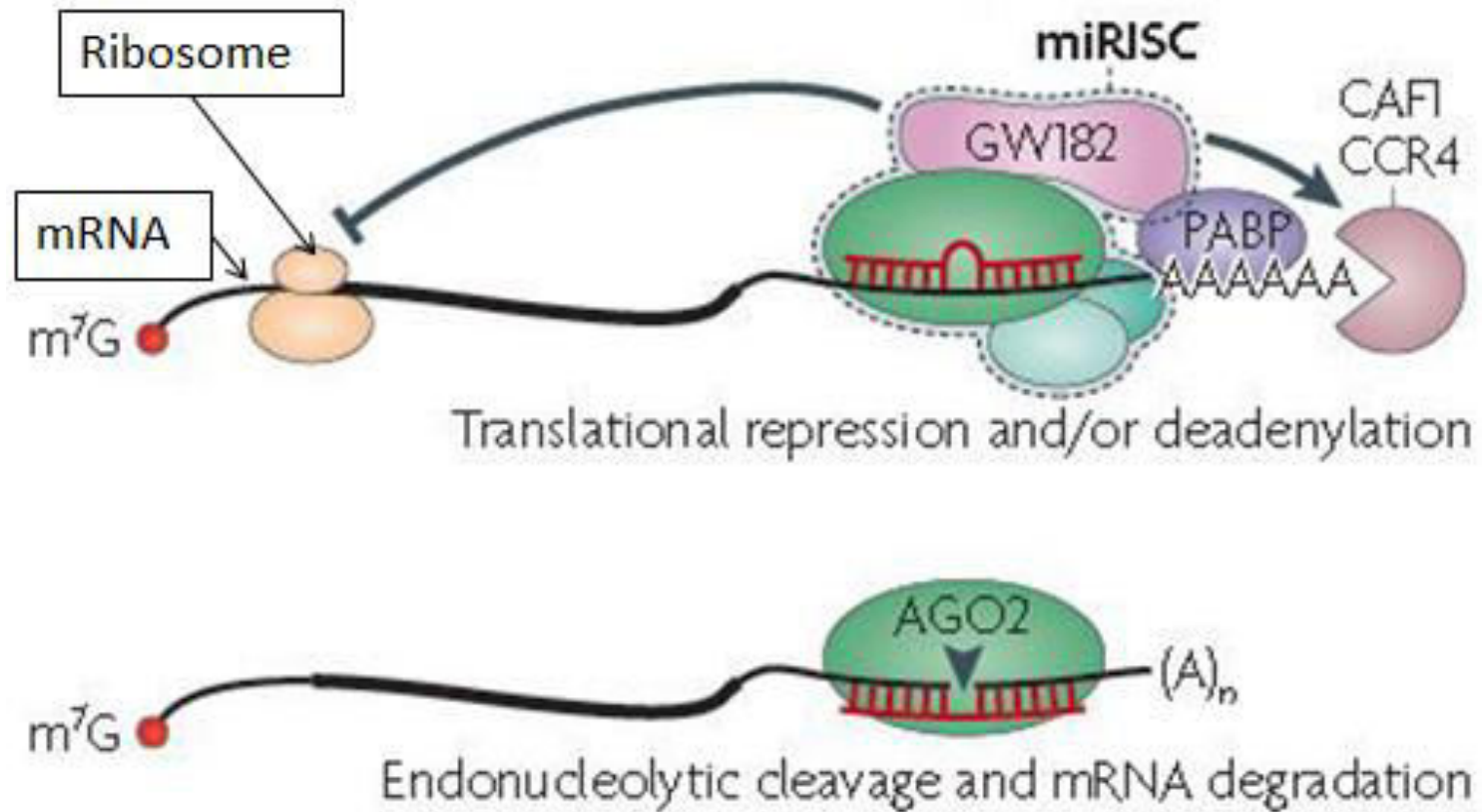
Functions of miRNA

Function as Developmental Switches



Functions of miRNA

Functions of miRNA: Gene Silencing in Plants and Animals



Functions of miRNA

Functions of miRNA

- Involved in the post-transcriptional/translational regulation of gene expression
- Metabolic regulation

Genetics and Genomics

miRNA Mode of Action

miRNA - Mode of Action

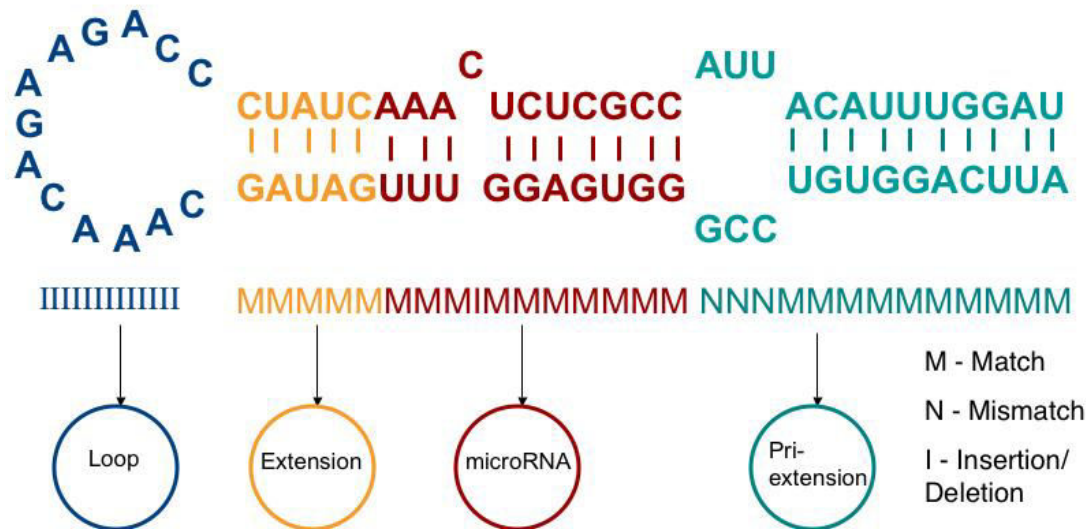
Mode of Action of miRNA

- Three different mode of action of miRNA
- mRNA degradation
- Transcriptional regulation
- Translational regulation

miRNA - Mode of Action

Structure of miRNA at Nucleotide level

a.

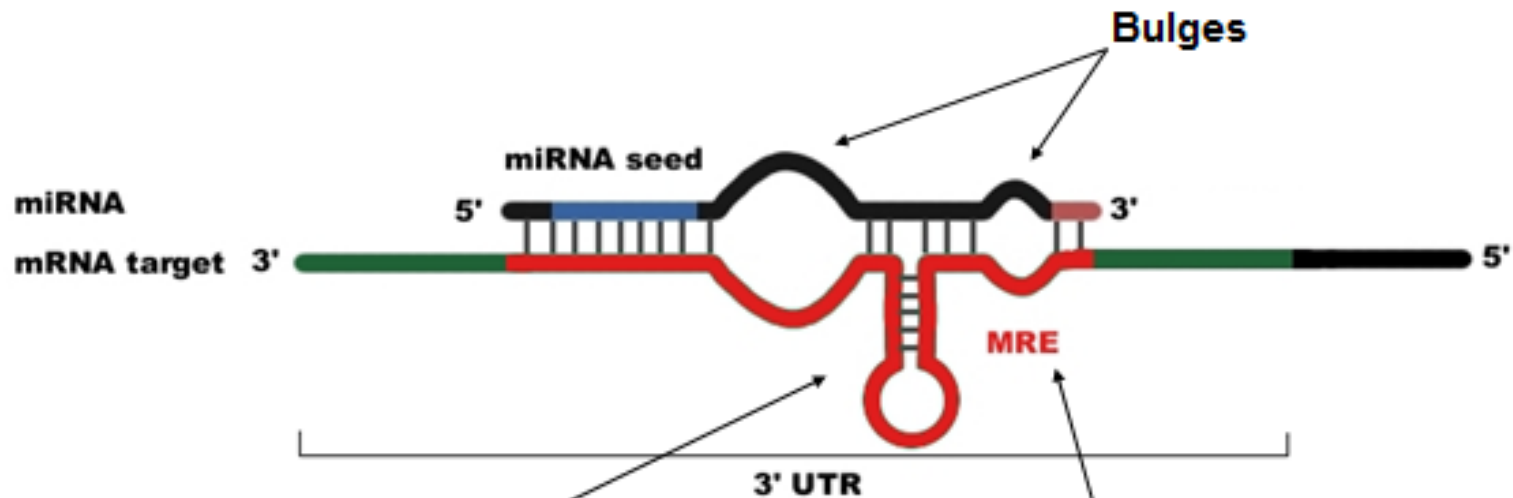


b.

CCAGAAGACAAACCUAUCAAACUCUCGCCAUUACAUUUGGAU
-----GAUAGUUU-GGAGUGGGCCUGUGGACUUA
LLLLLLLLLLLLLLLLEEEEERRRRRRRRRRRRPPPPPPPPPPPPPP

miRNA - Mode of Action

miRNA Binding to Target (mRNA)



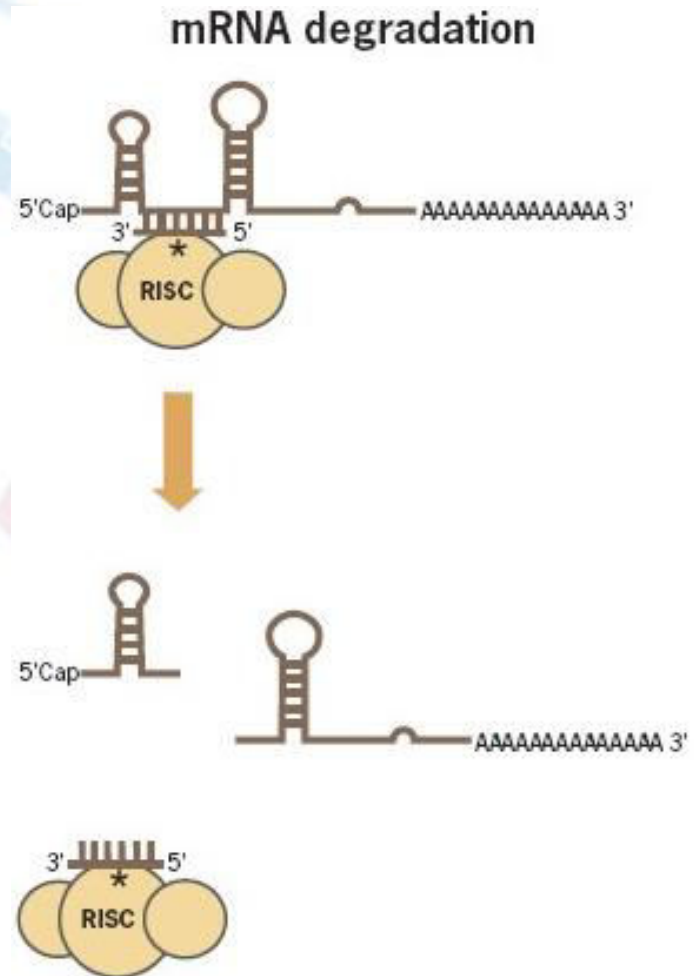
Hairpin is more stable than a simple bulge

The MRE is known as the “miRNA recognition element.” This is simply the sequence in the target that an miRNA binds to

miRNA - Mode of Action

Mode of Action of miRNA - mRNA Degradation

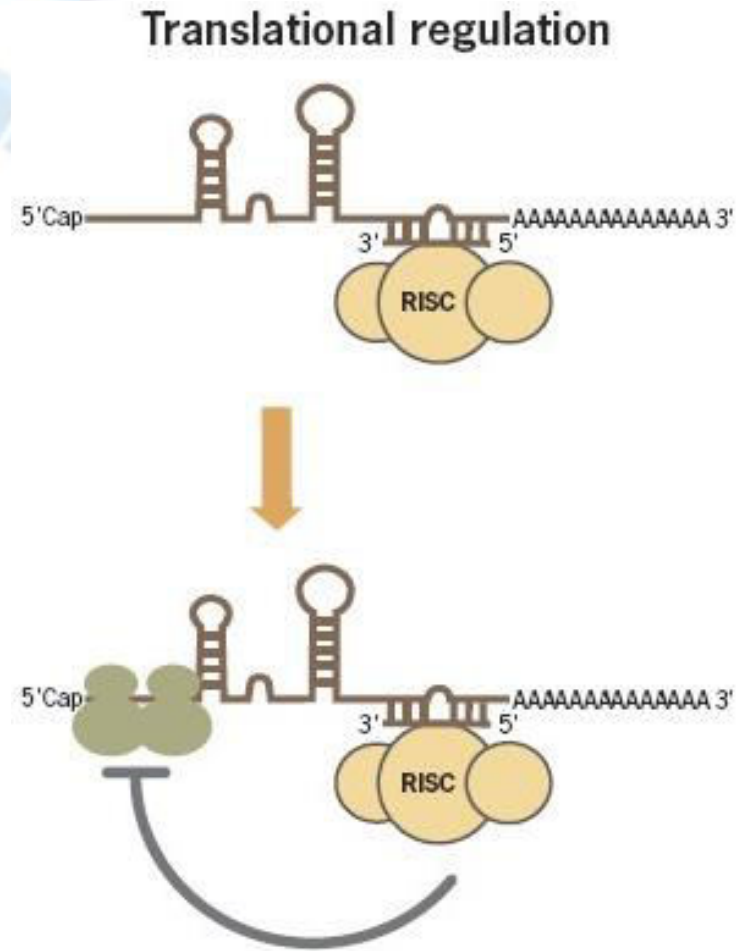
- Common in Plants



miRNA - Mode of Action

Mode of Action - Translational Regulation

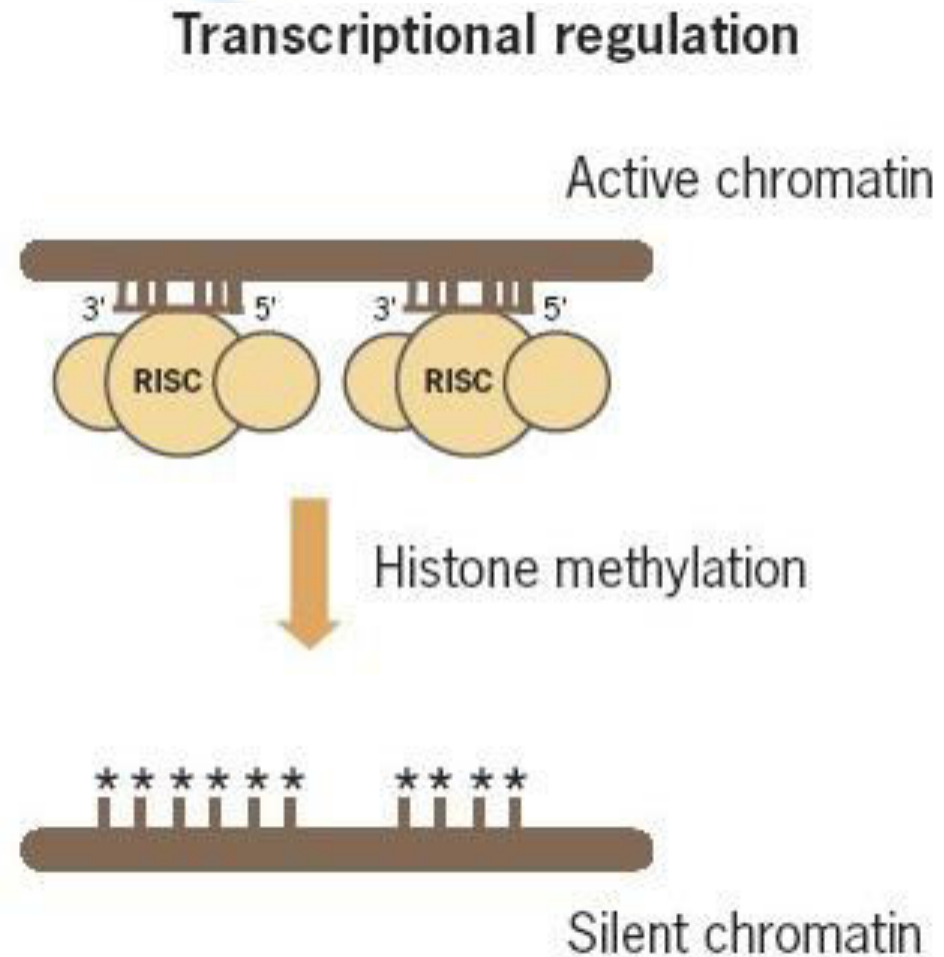
- Common in Animals



miRNA - Mode of Action

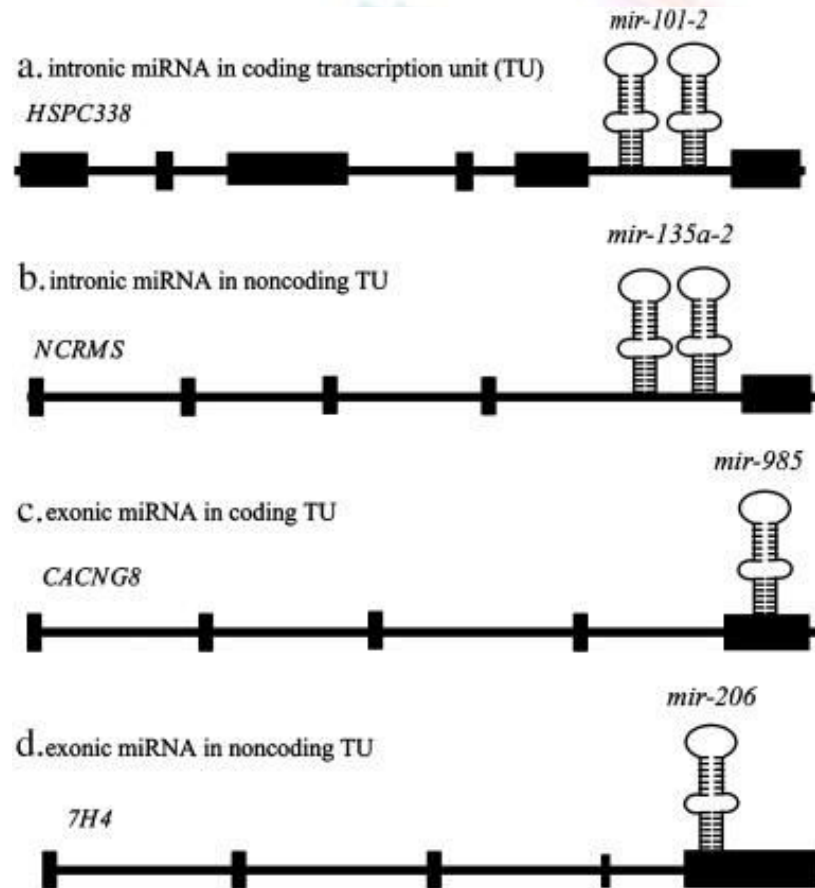
Mode of Action – Transcriptional Regulation

- Common in Yeast and Plants
- Sometimes in Animals



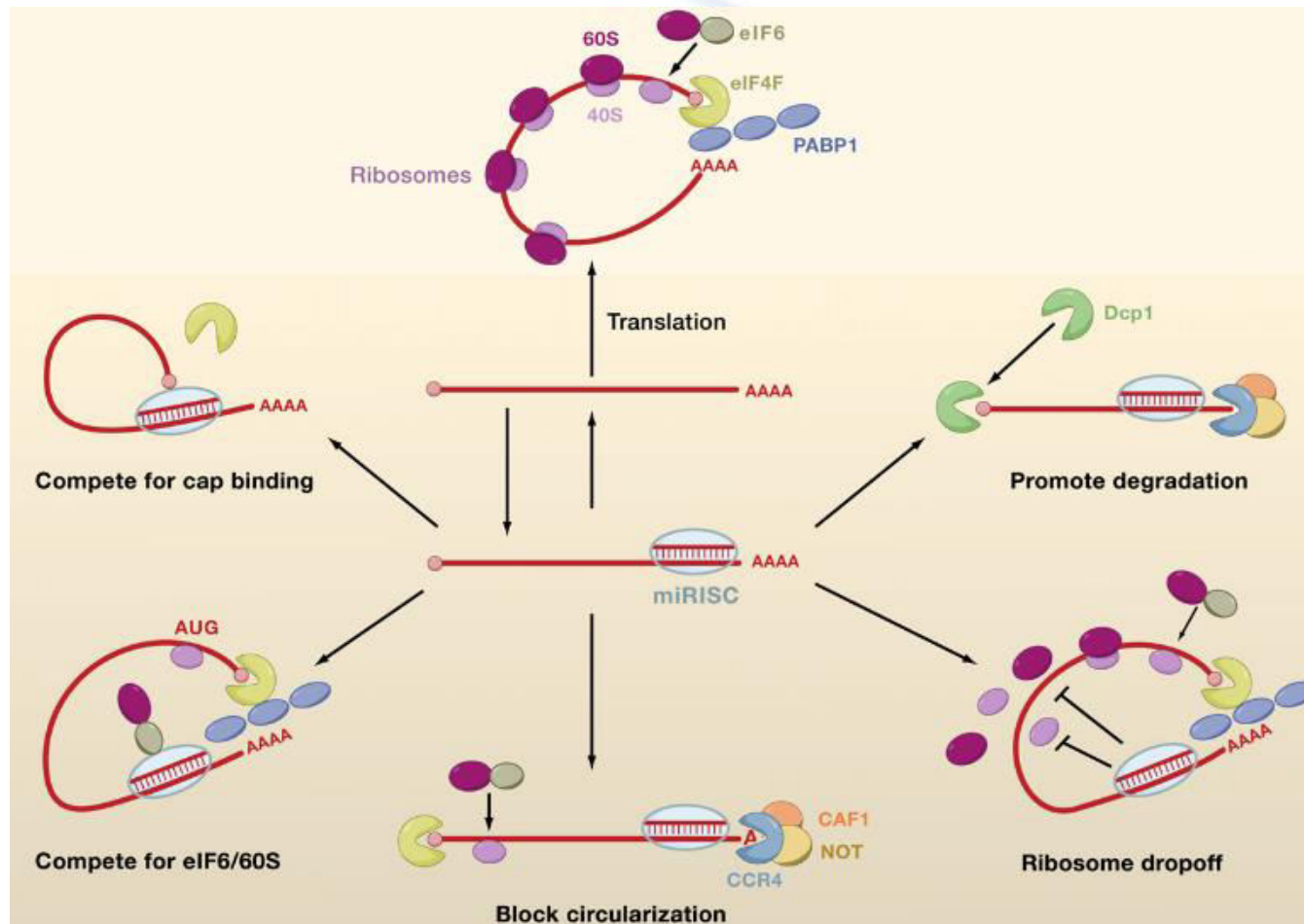
miRNA - Mode of Action

Schematic Illustration of the Genomic Organization and Structure of miRNA Genes



miRNA - Mode of Action

Mechanisms of miRISC-Mediated Repression



miRNA - Mode of Action

Mode of action of miRNA

- Three different mode of action of miRNA
- mRNA degradation
- Transcriptional regulation
- Translational regulation

Genetics and Genomics

Non-coding RNA and Silencing of X Chromosome

Non-coding RNA- X Chromosome Silencing

Silencing of one X chromosome

- X chromosome silencing is mediated by Xist – a 16,000 nt long ncRNA
- Xist ncRNA recruited complex has one entry site in X chromosome, corresponding to Xist gene itself

Non-coding RNA- X Chromosome Silencing

Silencing of one X chromosome

- Xist appears to recruit a specific histone isoform which maintains the chromosome in inactive state

Non-coding RNA- X Chromosome Silencing

Silencing of one X chromosome

- Additionally, Xist containing complexes recruit histone deacetylases and methylases

Non-coding RNA- X Chromosome Silencing

Silencing of one X chromosome

- Xist activity is regulated by another 40,000 nt long ncRNA – Tsix, which contains anti-sense sequence of Xist and therefore is able to regulate Xist activity by base-pairing to it

Non-coding RNA- X Chromosome Silencing

shRNA and Genetic Imprinting

- Activity of small heterochromatic RNAs (shRNAs) appear to be essential for establishing and maintaining the imprinted status of genes

Genetics and Genomics

RNA Induced Silencing Complex

RNA Induced Silencing Complex

RNA Induced Silencing Complex

- RISC is a large (~500-kDa) RNA multi-protein complex, which triggers mRNA degradation in response to siRNA

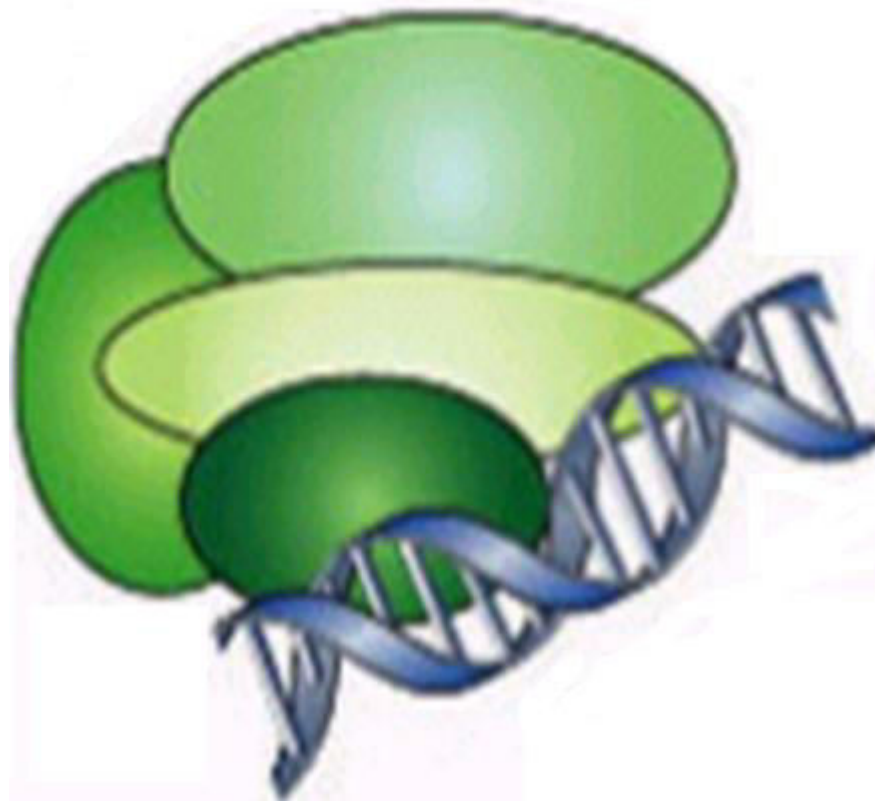
RNA Induced Silencing Complex

RISC

- Unwinding of double-stranded siRNA by ATP independent helicase.
- Active components of RISC are endonucleases called argonaute proteins which cleave the target mRNA strand.

RNA Induced Silencing Complex

RNA Induced Silencing Complex - Structure



RNA Induced Silencing Complex

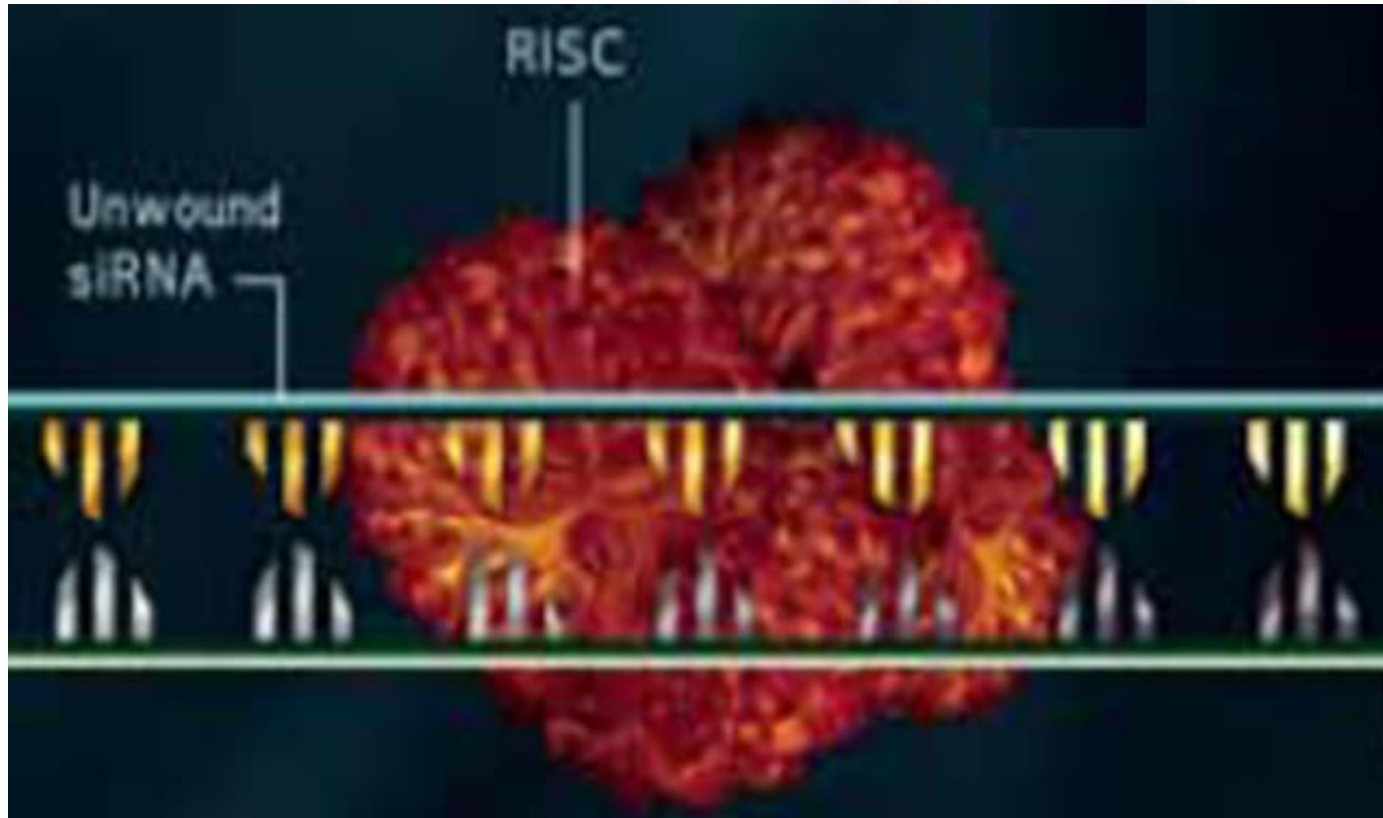
RNA Induced Silencing Complex

RISC is a large (~500-kDa) RNA-multiprotein complex, which triggers mRNA degradation

Some components have been defined by genetics, but function is unknown, e.g. -
unwinding of double-stranded siRNA (Helicase)
ribonuclease component cleaves mRNA (Nuclease).

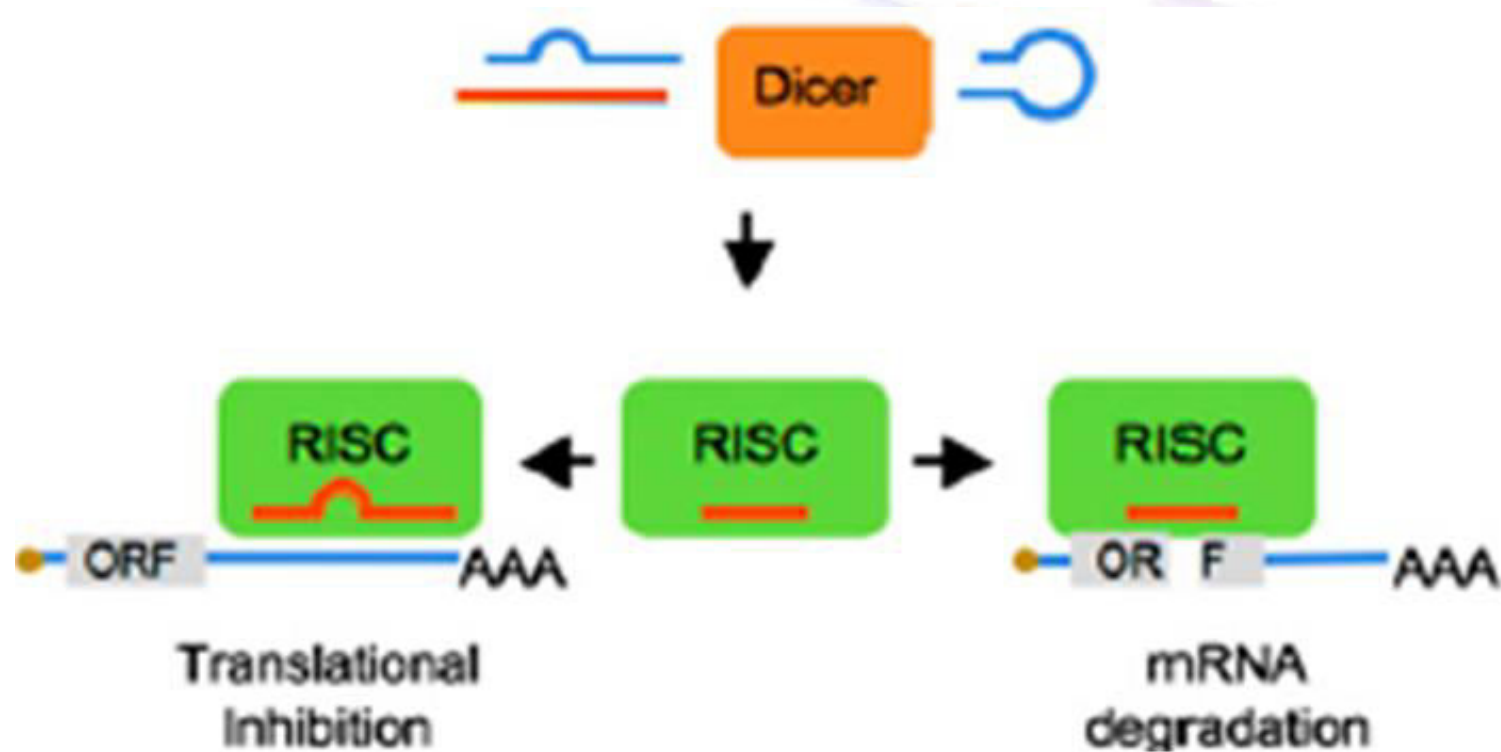
RNA Induced Silencing Complex

RNA Induced Silencing Complex



RNA Induced Silencing Complex

RNA Induced Silencing Complex



RNA Induced Silencing Complex

RNA Induced Silencing Complex

- RISC bound to partially complementary mRNA induces translational repression: results in destabilization of target mRNA
- The more RISC bound; greater the inhibitory effect
- RISC bound to perfectly complementary mRNA (to miRNA) are cleaved and degraded by RISC in a process similar to RNAi

RNA Induced Silencing Complex

RNA Induced Silencing Complex

- RISC is a large (~500-kDa) RNA-multiprotein complex, which triggers mRNA degradation in response to siRNA

Genetics and Genomics

Small Interfering RNA (siRNA)

Small Interfering RNA (siRNA)

siRNA

- Small interfering RNA (siRNA), sometimes known as short interfering RNA, are a class 20-25 nucleotide-long RNA molecules that interfere with the expression of genes.

Small Interfering RNA (siRNA)

siRNA

- They are naturally produced as part of the RNA interference (RNAi) pathway by the enzyme Dicer.
- Exogenously introduced by investigators to bring about knockdown of a particular gene.

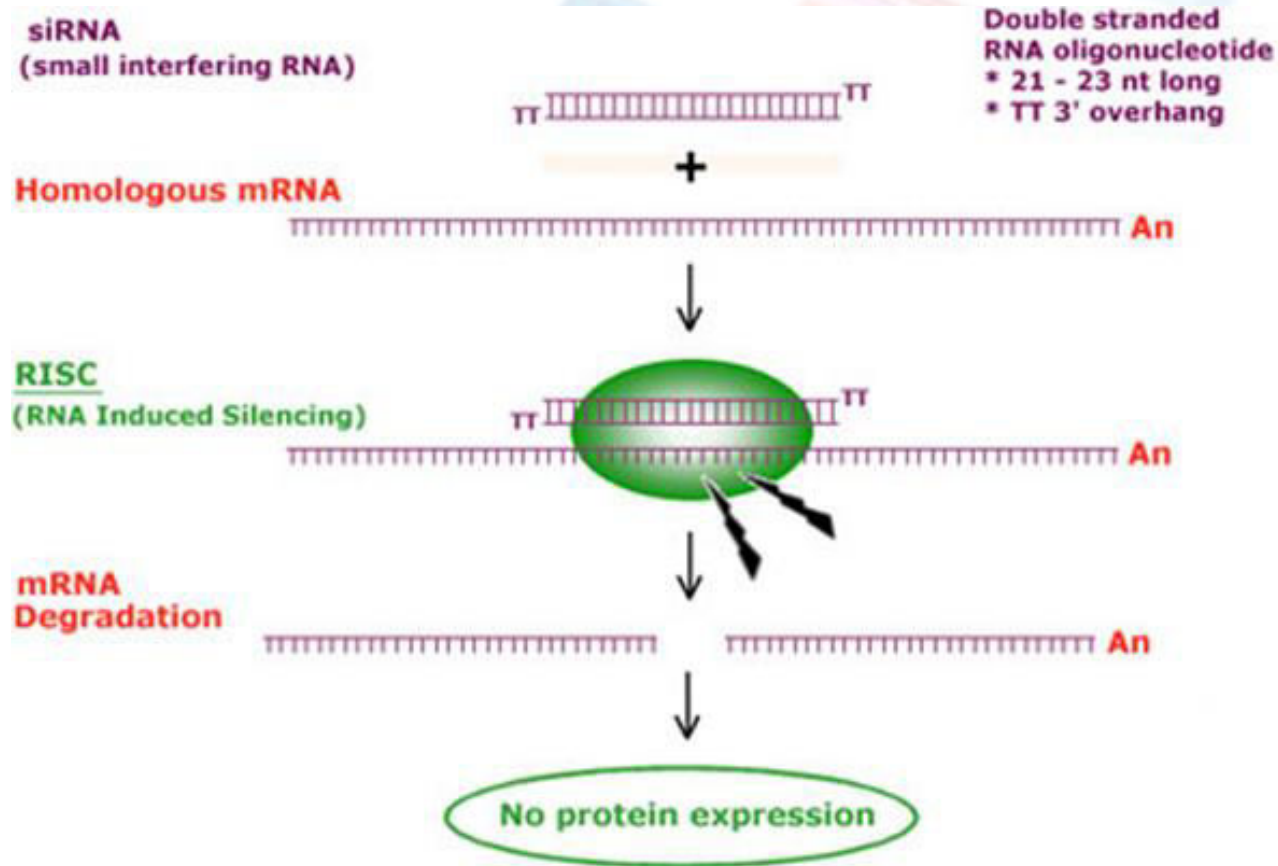
Small Interfering RNA (siRNA)

siRNA

- siRNA's have a well defined structure.
- A short (usually 21-nt) double-strand of RNA (dsRNA) with 2-nt overhangs on either end, including a 5' phosphate group and a 3' hydroxy (-OH) group.

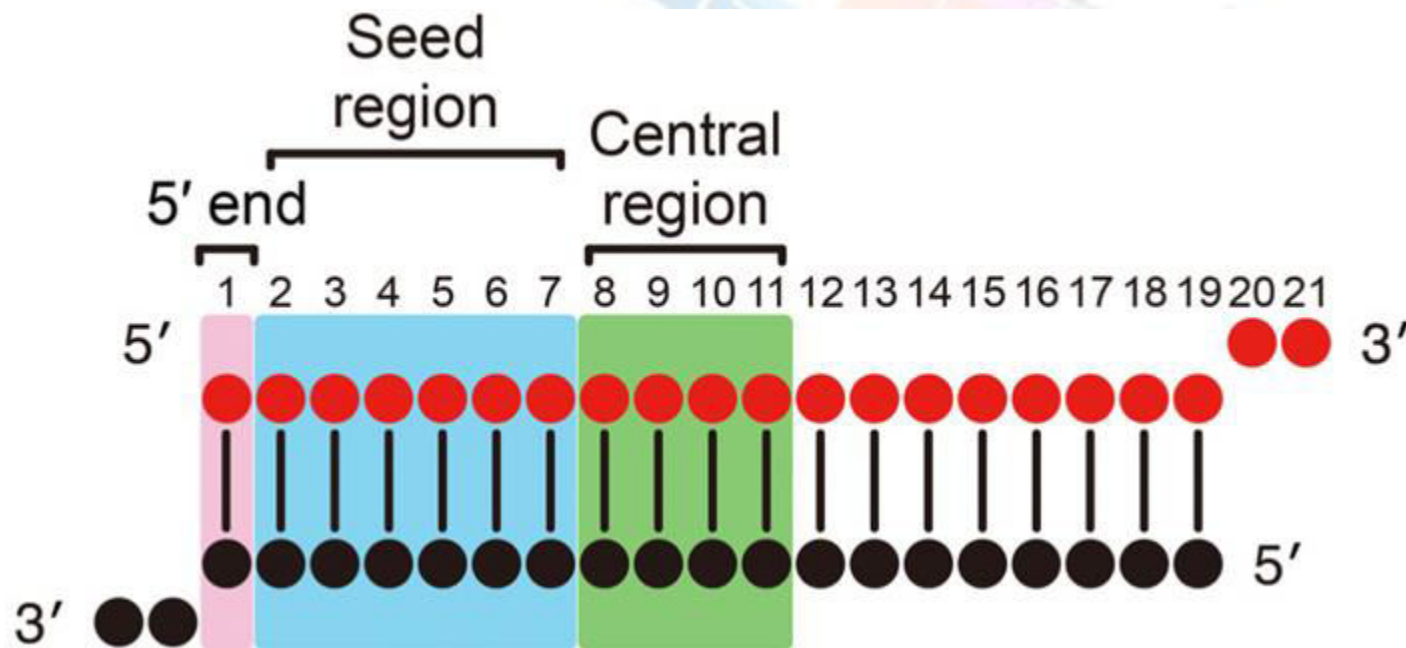
Small Interfering RNA (siRNA)

siRNA: short (usually 21-nt) double-strand of RNA (dsRNA) with 2-nt overhangs on either end.



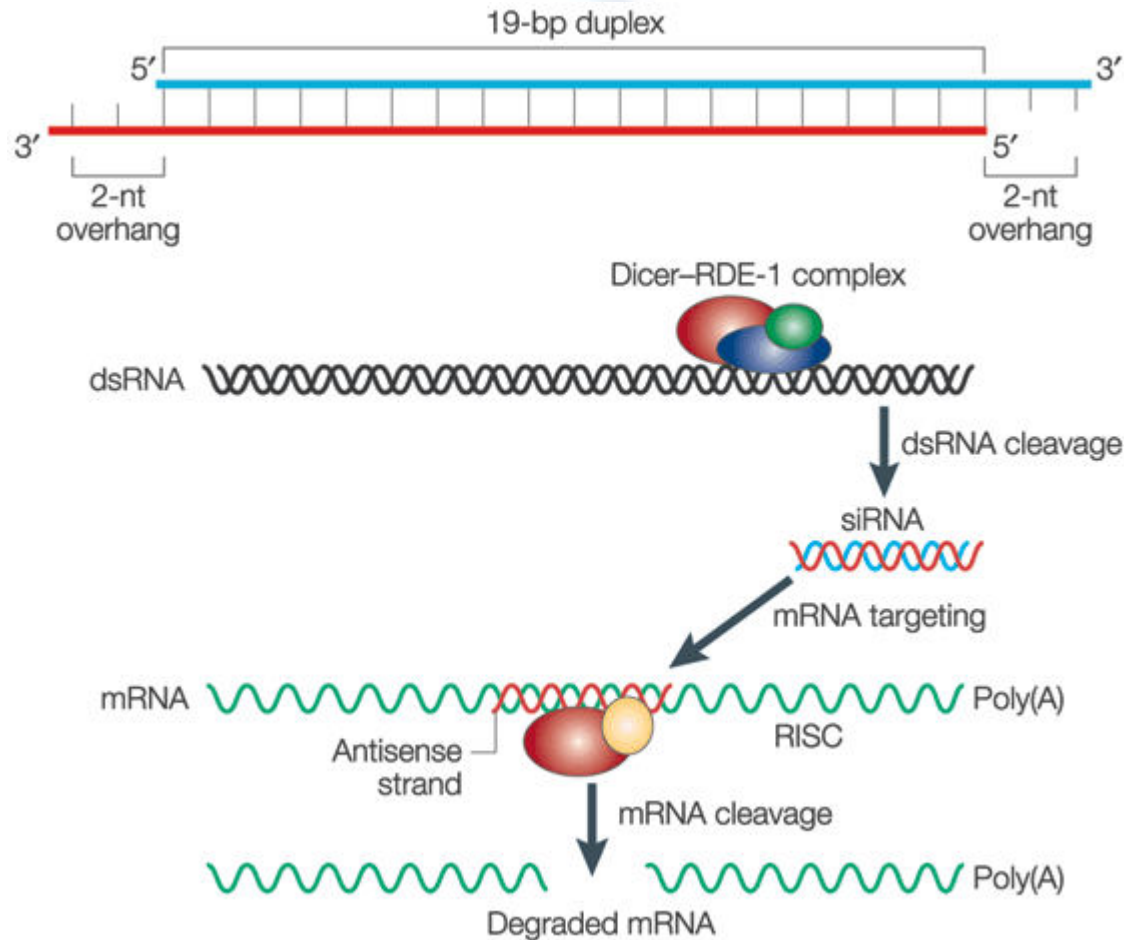
Small Interfering RNA (siRNA)

siRNA: Short double-strand of RNA (dsRNA) with 2-nt overhangs on either end.



Small Interfering RNA (siRNA)

siRNA:



Small Interfering RNA (siRNA)

siRNA

- Small interfering RNAs that have an integral role in the phenomenon of RNA interference (RNAi), a form of post-transcriptional gene silencing

Small Interfering RNA (siRNA)

siRNA

- A single base pair difference between the siRNA template and the target mRNA is enough to block the process.

Small Interfering RNA (siRNA)

Small interfering RNA

- Several different methods of expression
- Several different methods of delivery

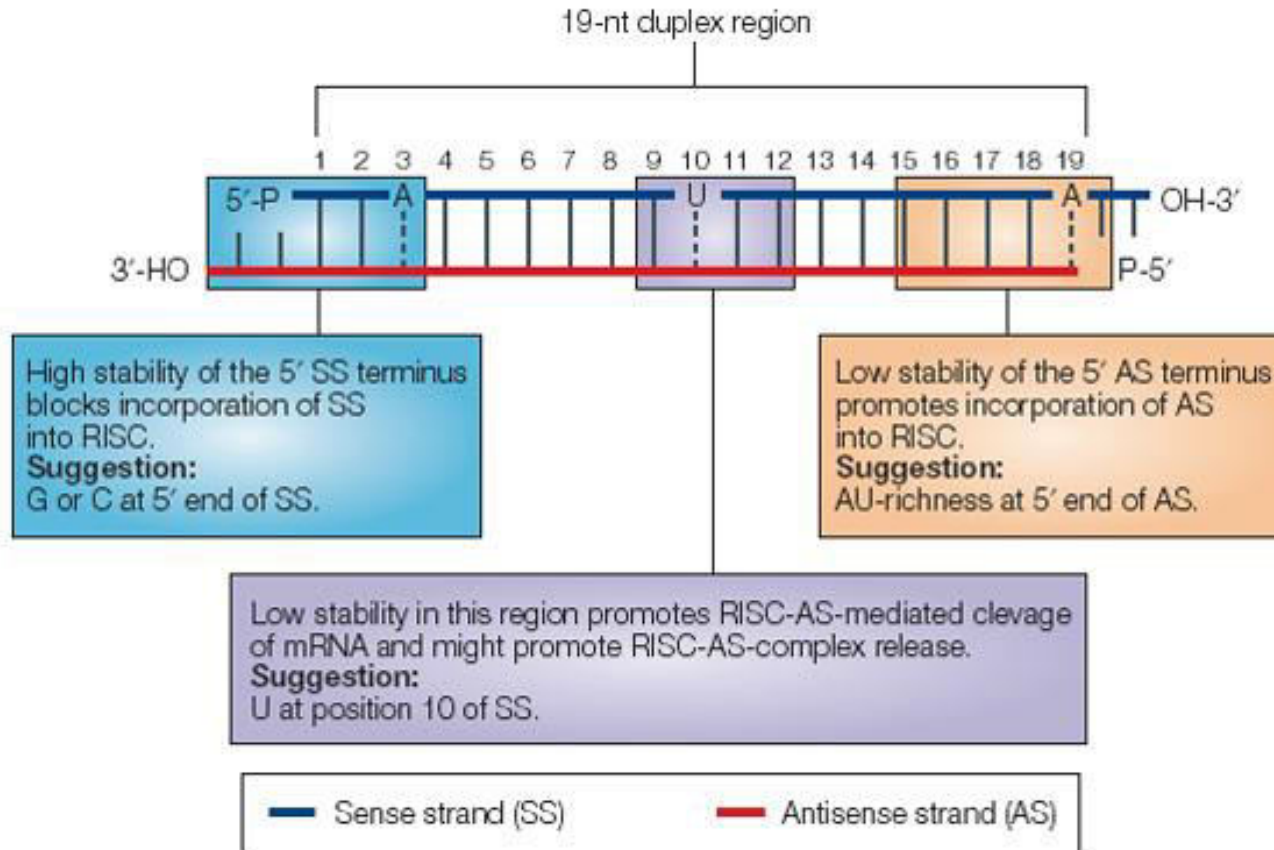
Small Interfering RNA (siRNA)

Small interfering RNA

- Dicer (type III RNase III) cleaves long dsRNA into siRNA
21-25nt dsRNA from exogenous sources

Small Interfering RNA (siRNA)

Small interfering RNA: siRNA design



Small Interfering RNA (siRNA)

siRNA

- Small interfering RNA (siRNA), sometimes known as short interfering RNA, are a class 20-25 nucleotide-long RNA molecules that interfere with the expression of genes.

Midterm Syllabus Finished

Effort By

Amaan Khan